

AK High-Dimensional Projection Structural Theory

Version 20.0.0: Finite Proof Compression, Higher Obstruction Bridges, and Problem Demonstration

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1 Chapter 1: Scope, Claim Discipline, and the v20 Finite-Certificate Contract

Scope Box (v20 conservative extension; finite proof compression, higher obstruction bridges, and problem demonstration).

All statements in this chapter, and all subsequent statements labeled as *Core*, are made under the following v20.0 guard-rails.

- **Release scope.** AK-HPDST v20.0 is the *Finite Proof Compression, Higher Obstruction Bridges, and Problem Demonstration Release*. Its central objective is to certify the absence of retained persistence or defect bars by finite, typed, reconstructible evidence, and then to transport such certified vanishing to higher derived obstruction groups under explicit eligibility and realization hypotheses.
- **Conservative-extension rule.** The repaired v19.0.0 Core is inherited as an immutable lower baseline. In particular, v20.0 does not weaken, replace, or silently reinterpret hard threshold deletion, two-sided threshold-gap stability, window-first evaluation, comparison modes, representative discipline, actual-morphism Type IV semantics, non-scalar obligations, or Claim/manifest/artifact governance. New v20 theorems are admissible only as conservative extensions whose complete hypotheses are displayed and whose stronger readings are explicitly excluded.
- **Core baseline and downstream discipline.** Chapters 1–8 and the Foundation Appendices determine the v20.0 Core baseline. Technical Extension, Problem Interface, Search/Platform, Claim Register, Micro-Theorem, Toy Bridge, and Companion materials may register, elaborate, instantiate, transport, implement, or test this baseline. They do not silently enlarge it.
- **Constructible one-parameter persistence over a field.** We work in

$$\text{Pers}_k^{\text{cons}} \subset [(\mathbb{R}, \leq), \text{Vect}_k]$$

over a field k . Every detector theorem is interpreted on a declared constructible one-parameter profile, a finite barcode profile, or a locally finite profile restricted to a declared audit range. Multiparameter, wild, uncontrolled infinite-rank, or nonconstructible data require a separate Interface theorem.

- **Standard barcode metric convention.** Barcode comparisons in the Core use the standard bottleneck distance d_B , unless another metric is explicitly declared outside the Core. Infinite bars have threshold

length $+\infty$. In finite-cost matchings, infinite bars match only infinite bars, with cost

$$\text{cost}([a, \infty), [a', \infty)) = |a - a'|.$$

A finite bar does not match an infinite bar at finite cost, and an infinite bar does not match the diagonal.

- **Hard finite-threshold deletion.** For $\tau > 0$, the Core uses

$$T_\tau^{\text{bar}},$$

which deletes finite bars of length at most τ , retains finite bars of length greater than τ , and retains all infinite bars. It is endpoint-independent, idempotent, infinite-bar preserving, and threshold-monotone. These algebraic properties do not imply global non-expansiveness, exactness, a Serre localization, or a global functor on all morphisms.

- **Threshold-gap-safe replacement for global stability.** If

$$d_B(B, C) \leq \varepsilon$$

and the finite bars of both B and C avoid

$$[\tau - \varepsilon, \tau + \varepsilon],$$

then

$$d_B(T_\tau^{\text{bar}} B, T_\tau^{\text{bar}} C) \leq \varepsilon.$$

The hypothesis is two-sided, and the half-width ε is sharp for this two-sided certificate. Every metric detector verdict that crosses hard deletion must consume this repaired v19 transfer chain or a separately proved stronger theorem.

- **Windowed evaluation first.** Windows are right-open intervals

$$W = [u, u'), \quad u' \in \mathbb{R} \cup \{\infty\},$$

and

$$\text{Eval}_{i, W, \tau}(F) := T_\tau^{\text{bar}}(W_W \mathbf{P}_i(F)).$$

No unconditional commutation

$$W_W T_\tau^{\text{bar}} \cong T_\tau^{\text{bar}} W_W$$

belongs to the Core. On a bounded window, an unwindowed infinite bar may become a finite clipped bar and must be classified by its window-relative length.

- **Finite observation is not finite proof.** A finite list of point values, ranks, sampled maps, numerical evaluations, or critical parameters is not a finite proof certificate merely because it is finite. A finite detector certificate requires, as applicable, a declared profile stage, endpoint convention, soundness theorem, completeness theorem, birth/germ exhaustiveness or stencil coverage, family membership, excess lower-bound source, metric transport, and immutable artifact identity. Missing required evidence yields undefined, not zero.

- **Detector stage discipline.** Every detector record declares exactly one source stage among

windowed_prethreshold, repaired_postthreshold,
kernel_defect, cokernel_defect.

The standard finite stencil detector reads a windowed pre-threshold profile and certifies the zero/nonzero predicate of its repaired post-threshold profile. A detector on a defect profile has a different source object and does not inherit topological-profile conclusions without an explicit theorem.

- **Soundness and completeness separation.** Detector nonzero soundness and detector zero completeness are distinct claims. A nonzero sampled image may witness a retained bar under the relevant typing conditions. A zero sampled detector proves repaired-profile zero only after a complete coverage or characterization theorem is discharged. No zero-reflection conclusion is inferred from soundness alone.
- **Endpoint and cofinal discipline.** Birth-point detectors use a fixed endpoint convention. Decorated intervals require the canonical germ formulation or an equivalent proved endpoint-aware theorem. On a bounded window, a bar reaching the right endpoint is window-cofinal but finite for threshold-length purposes. It is included in every completeness and excess calculation unless the governing theorem explicitly places it outside scope.
- **Finite proof compression.** Certificate size and minimal stencil size are defined relative to a declared family or information class, not retrospectively from a single already-known zero object. A family-relative MinCert statement records the family, window, threshold, endpoint convention, critical-data policy, and the source of every uniform excess or coverage bound. Optimization scores and compression ratios remain advisory or Technical Extension material unless separately proved as Core theorems.
- **Right-unbounded no-go boundary.** No uniform complete finite detector exists on a right-unbounded window with unbounded possible birth positions and no additional stabilization, finite-birth, global-bound, or finite-to-infinite reduction theorem. A finite detector on such a scope must display the additional structure that defeats the no-go theorem.
- **Repaired evaluation, exact comparison, and metric certification.** A repaired evaluation is definitionally meaningful once the displayed barcode expression exists. A threshold-gap certificate is mandatory whenever an inference uses a metric perturbation, overlap transport, Restart transport, representative comparison by a metric bound, tower comparison by a metric bound, or detector transport through hard deletion. An exact equality or profile isomorphism is not automatically a positive-radius robustness certificate.
- **Metric zero-verdict separation.** Pointwise detector rank is not asserted to be bottleneck-stable. A metric transfer of exact vanishing requires: a certified pre-threshold discrepancy, the two-sided threshold-gap theorem, and a post-threshold separation theorem. In the standard hard-threshold regime, the latter uses $2\varepsilon < \tau$. Detector completeness must hold on every profile whose zero predicate is inferred.
- **Typed scalarization.** Metric ledger values live in a declared commutative quantale

$$(\mathbb{V}, \oplus, 0_{\mathbb{V}}, \preceq),$$

and comparison with real-valued clearance passes through

$$\text{val}_B : V \longrightarrow [0, +\infty].$$

The shorthand

$$\Sigma\delta < \text{gap}_\tau$$

means

$$\text{val}_B(\mathbf{d}_{\text{met}}) < \text{gap}_\tau(\mathfrak{B}),$$

together with evidence that the scalarized value upper-bounds the actual bottleneck discrepancy consumed by the proof.

- **Threshold-clearance discipline.** For a barcode B ,

$$\text{clear}_\tau(B) := \inf_{I \in B_{\text{fin}}} |\ell(I) - \tau|,$$

with $\text{clear}_\tau(B) = +\infty$ if B has no finite bars. For a finite comparison family,

$$\text{gap}_\tau(B_1, \dots, B_m) := \min_j \text{clear}_\tau(B_j).$$

Thus $\text{gap}_\tau = \text{clear}_\tau$ on one profile; no factor $1/2$ is present. Clearance is measured after windowing and before deletion.

- **Non-scalar obligations.** Representative existence, actual comparison morphisms, realization, eligibility, edge naturality, detector applicability, endpoint convention, birth/germ completeness, stencil coverage, family membership, excess-bound provenance, terminal tameness, threshold reconciliation, higher Čech coherence, Restart transition certification, finite-to-infinite reduction, claim status, manifest consistency, and artifact identity are not numerical ledger components and cannot be repaired by numerical slack.
- **Representative discipline.** A filtered representative $C_{\tau, W}F$ is not a global collapse functor by default. It represents repaired profiles only in declared monitored degrees and at a declared equivalence strength. Persistence-profile equivalence does not automatically imply chain isomorphism, chain-homotopy equivalence, derived equivalence, detector preservation, or realization equivalence.
- **Higher obstruction bridge policy.** For an invoked degree n , a higher Ext discharge requires an n -eligible realization, a declared edge extractor, and an artifact-backed identification of H^{-n} with that extractor. The default Core direction is

$$\text{Eval}_{n, W, \tau}(F) = 0 \implies \text{Ext}^n(A_n(F), k) = 0.$$

A reverse implication, and hence a vanishing equivalence, is available only after a separately proved natural zero-reflection or faithfulness theorem for the declared extractor and profile class.

- **No synthetic promotion.** Shifting a detector value into a derived object may produce a valid algebraic identity in a synthetic or toy model. It does not by itself prove a natural PH_n -to- Ext^n Bridge, a realization theorem, or representative-independent categorical meaning. Such constructions are labeled synthetic, toy, or specification-level unless an independent natural realization theorem is supplied.

- **Type IV discipline and transient-defect separation.** Type IV diagnostics require an actual comparison artifact

$$\phi_{i,W,\tau} : \text{ColimProfile}_{i,W,\tau}(F_\bullet) \longrightarrow \text{ApexProfile}_{i,W,\tau}(F_\infty).$$

The terminal-generic diagnostics ($\mu_{\text{Collapse}}, \nu_{\text{Collapse}}$) are distinct from finite interior or transient defect detection. Zero terminal diagnostics do not imply full isomorphism. A v20 transient-defect verdict requires the full kernel/cokernel profiles and its own complete detector certificate.

- **Problem-Bridge and Return-Theorem discipline.** A Core certificate is not an external-domain theorem. Every problem demonstration separates: external-to-AK realization, AK-internal finite certification, any finite-to-infinite reduction, higher-obstruction transport, and the AK-to-external Return or soundness theorem. The external conclusion is available only when every invoked arrow is proved at the required strength.
- **Audit-first doctrine.** Search scores, scalar potentials, AI confidence values, rankings, advisory indicators, candidate stencils, and proposed realization routes may guide proof search. They do not change a Core verdict, establish a theorem, or repair a missing artifact.
- **Manifest discipline.** A certified proof object carries immutable scope, metric and threshold policies, evaluation order, detector records, completeness sources, threshold-gap records, representative and eligibility records, higher-edge records, ledgers and scalarizations, non-scalar obligations, Type IV and transient-defect scope, bridge and finite-to-infinite records when invoked, typed gate verdicts, claim references, repair ancestry, and artifact identities. Manifest completeness is necessary for certification but is not sufficient for mathematical truth.
- **Claim-register, micro-theorem, and toy-bridge boundary.** Appendix CR-v20 governs claim use, Appendix CM-v20 records reusable Core micro-theorems, and Appendix TB-v20 records deliberately restricted toy bridge theorems and counterexamples. Registration is not proof, and none of these appendices silently converts a detector, toy realization, or Bridge Program into an external-domain theorem.

Accordingly, this chapter fixes the admissible language for v20.0 and the boundaries among inherited metric repair, finite detector certification, higher obstruction discharge, proof compression, Type IV defect analysis, problem interfaces, external Return Theorems, Search artifacts, specifications, and explicit non-claims.

1.1 Claim taxonomy and document discipline

Definition 1.1 (Claim Taxonomy). Every mathematically meaningful statement is assigned a document role. The roles are:

- (i) **Core definition.** Typed language internal to the Core; not by itself an existence, uniqueness, exactness, or external-domain theorem.
- (ii) **Core theorem.** A proved statement inside the standing v20 Core regime, valid only under its complete hypotheses.
- (iii) **Core lemma.** A proved supporting statement used as an internal step.
- (iv) **Core micro-theorem.** A small reusable Core theorem recorded in Appendix CM-v20 at its exact strength.

- (v) **Metric-repair theorem.** A theorem replacing, constraining, or rejecting an earlier metric statement.
- (vi) **Operational policy.** A binding protocol for admissible certification; not an external mathematical theorem.
- (vii) **Interface definition.** A definition preparing external data for Core readability; it does not prove realization faithfulness or external transport.
- (viii) **Interface theorem.** A proved conditional theorem connecting declared external data and a Core-visible statement, or a Core certificate and an external conclusion.
- (ix) **Bridge candidate.** A precisely typed but not fully proved cross-layer implication.
- (x) **Bridge program.** A structured proof program for promoting a Bridge candidate.
- (xi) **Bridge theorem.** A proved cross-layer theorem with declared source and target semantics, realization, artifacts, defects, hypotheses, and failure modes.
- (xii) **Toy bridge theorem.** A theorem proved only over a deliberately small toy source category and recorded in Appendix TB-v20.
- (xiii) **Search artifact.** A proposal, map, ranking, heuristic, or AI-assisted output; never by itself proof.
- (xiv) **Companion template.** An operational or illustrative template; not a theorem or certificate.
- (xv) **Spec.** A speculative, schematic, conjectural, programmatic, or design-level statement.
- (xvi) **Successor-release candidate.** Material intentionally excluded from v20.0 theorem claims and reserved for a later release unless separately promoted.
- (xvii) **Non-claim.** A statement explicitly excluded from the asserted content.
- (xviii) **Deprecated or rejected claim.** A historical or proposed claim that is no longer admissible because it is false, unsupported, overstrong, ill-typed, or incompatible with v20 guard-rails.

Definition 1.2 (Three-axis claim typing). Every registered statement is typed along three independent axes:

- (i) its document role from Definition 1.1;
- (ii) its Claim Register evidence status, chosen from

proved, conditional, operational, spec,
deprecated, rejected, unknown;

- (iii) its invocation role, chosen from

theorem_evidence, operational_rule,
specification_target, provenance_only.

The admissible status–role combinations are fixed before verdict computation:

- (a) theorem evidence requires status proved, or status conditional with every registered hypothesis discharged;

- (b) an operational rule uses an operational item only as a typed procedure;
- (c) a specification target uses a spec item only as a schema, target, or acceptance criterion;
- (d) provenance-only use contributes no passing theorem clause.

The three axes are not interchangeable. A document role does not prove a statement, an evidence status does not change its mathematical type, and an invocation role cannot promote a weaker item to theorem evidence.

Remark 1.3 (Interpretive rule). The taxonomy separates mathematical, operational, interface, bridge, Search, and evidentiary authority. Whenever a statement’s role, status, scope, or hypotheses are ambiguous, the weaker safe interpretation prevails.

Remark 1.4 (Claim-register sovereignty). A claim may be used only at the strength jointly supported by its underlying proof or artifact and its Appendix CR-v20 entry. If the source and the register disagree, the weaker classification governs until the inconsistency is repaired.

Remark 1.5 (Registration is not proof). A statement does not become proved merely because it receives a ClaimID. A proved entry must point to an exact proof location; a conditional entry must record all hypotheses; an operational entry must not be read as an external theorem.

Remark 1.6 (Relation to Appendix CR-v20). Appendix CR-v20 records Claim IDs, exact statements, document roles, evidence statuses, source labels, hypotheses, permitted uses, prohibited stronger readings, proof locations, artifact references, and repair ancestry. It governs use but does not create proof by registration.

Remark 1.7 (Relation to Appendix CM-v20). Appendix CM-v20 records reusable Core micro-theorems after metric repair. They are valid only under their registered hypotheses and do not imply external-domain conclusions by themselves.

Remark 1.8 (Relation to Appendix TB). Appendix TB-v20 records theorem-level toy bridge models over deliberately small source categories. A toy bridge does not replace a problem-specific Bridge Theorem.

Remark 1.9 (Relation to successor-release candidates). Successor-release candidate material may organize future extensions, but it may not discharge a v20.0 theorem obligation unless separately promoted, reproved under the v20 guard-rails, registered, and artifact-bound.

1.2 v20 finite-certificate and higher-obstruction release contract

Definition 1.10 (Conservative v20 extension). The v20.0 Core is a conservative extension of the repaired v19.0.0 Core when all of the following hold:

- (i) every v19 theorem retained by v20 is used with the same statement, hypotheses, metric convention, evaluation order, and prohibited stronger readings;
- (ii) each new detector, compression, higher-edge, defect, reduction, or Return theorem has an explicit source and target type;
- (iii) no v20 theorem retroactively changes a v19 verdict or artifact;
- (iv) no v20 theorem converts a previously non-claim, specification, candidate, toy theorem, or operational rule into theorem evidence without a new proof and registration;

(v) every new dependency is recorded in the claim graph and immutable manifest.

Definition 1.11 (Four-pillar v20 contract). The v20 mathematical and interface program has four distinct pillars:

- (A) **Finite zero-reflecting detector calculus:** finite evidence for the exact zero/nonzero predicate of a repaired persistence or defect profile;
- (B) **Higher obstruction edge calculus:** conditional degree- n transport from repaired persistence vanishing to derived Ext^n -vanishing;
- (C) **Finite proof compression:** family-relative certificate size, completeness bounds, lower bounds, sharpness, and no-go results;
- (D) **Problem-Bridge demonstration:** a problem-specific chain containing realization, Core certification, any finite-to-infinite reduction, and a Return Theorem.

The four pillars are logically distinct and non-compensating. No implication between them is asserted without a separate theorem, and progress in one pillar does not silently discharge another.

Definition 1.12 (Finite observation and finite proof certificate). A *finite observation record* is any finite collection of sampled values, ranks, maps, endpoints, critical values, numerical approximations, or computed artifacts.

A *finite proof certificate* is a finite observation record together with all theorem, typing, completeness, provenance, and artifact data required to derive the declared mathematical predicate.

For a detector-based zero verdict, the required data include, as applicable:

- (i) the exact source profile and detector profile stage;
- (ii) monitored degree, window, threshold, and endpoint convention;
- (iii) a soundness theorem;
- (iv) a completeness, characterization, or coverage theorem;
- (v) a complete birth/germ set, coverage witness, or family-relative mesh certificate;
- (vi) the source and proof of any uniform threshold-excess lower bound;
- (vii) exact or metric comparison records used in transport;
- (viii) artifact identity, theorem identity, and claim-use preflight.

If a required item is absent, the detector-based zero conclusion is undefined.

Definition 1.13 (Detector profile stage). Every detector record declares exactly one source stage:

windowed_prethreshold,	repaired_postthreshold,
kernel_defect,	cokernel_defect.

The stage determines the detector's source object and permitted conclusion. A theorem proved for one stage is not transferred to another stage without an explicit stage-transport theorem.

Definition 1.14 (Finite detector certificate record). A finite detector certificate record contains at least:

- (i) a detector identifier and theorem identifier;
- (ii) the source profile identifier and stage;
- (iii) (n, W, τ) , endpoint convention, and window-cofinal policy;
- (iv) detector mode, chosen from a registered set such as

$$\text{birth, germ, certified_stencil, family_mesh;}$$
- (v) the finite sample or critical-value set;
- (vi) soundness evidence;
- (vii) completeness evidence and its non-circular source;
- (viii) exact or metric transport evidence when the source profile is not the profile whose predicate is consumed;
- (ix) certificate size and, when invoked, the family-relative MinCert record;
- (x) immutable source, computation, proof, and replay artifacts.

The record proves only the predicate stated by its governing theorem.

Definition 1.15 (Two-axis detector status). Detector certification and detector mathematical value are recorded on separate axes.

The certificate-status axis is:

$$\text{detector_certified, detector_failed, detector_pending.}$$

It maps to the canonical atomic status calculus by

$$\begin{aligned} \text{detector_certified} &\mapsto \text{pass,} \\ \text{detector_failed} &\mapsto \text{reject,} \\ \text{detector_pending} &\mapsto \text{undefined.} \end{aligned}$$

where `detector_failed` means that a well-typed required detector certificate condition is evidenced to fail.

The value axis is:

$$\text{zero, nonzero, undefined.}$$

A value is mathematically consumable only when the certificate status is `detector_certified`. A certified nonzero value rejects a required zero clause; a certified zero value discharges that zero clause only at the strength of the governing completeness theorem. An uncertified zero-looking output remains undefined. When detector semantics are outside the claim scope, the canonical atomic status is `not_invoked`; no specialized certificate or value token is imputed.

Policy 1 (Detector admission and non-compensation). *A detector may discharge a Core zero clause only when:*

- (i) *the detector is applicable to the exact declared profile stage;*
- (ii) *the certificate status is `detector_certified`;*
- (iii) *its zero-reflection theorem covers the declared object or family;*

- (iv) every endpoint, birth/germ, coverage, mesh, cofinal, and family-membership obligation passes;
- (v) every exact or metric transport obligation passes;
- (vi) all referenced artifacts are immutable and reconstructible.

No favorable numerical reserve, detector score, compression ratio, or independent passing clause compensates for a failed or undefined detector obligation.

Definition 1.16 (Metric-safe detector transport chain). A metric-safe transfer of a detector-based zero predicate from a windowed pre-threshold profile B to a compared profile C has the typed chain

$$\begin{aligned}
d_B(B, C) \leq \varepsilon &+ \text{two-sided threshold-gap certificate} \\
&\Downarrow \\
d_B(T_\tau^{\text{bar}} B, T_\tau^{\text{bar}} C) \leq \varepsilon &+ 2\varepsilon < \tau \\
&\Downarrow \\
T_\tau^{\text{bar}} B = 0 &\iff T_\tau^{\text{bar}} C = 0.
\end{aligned}$$

If the zero statement on either side is established by a detector, the detector must be complete on that side. This definition records the required theorem chain; Chapter 2 and the Foundation metric appendix supply its mathematical components.

Definition 1.17 (Higher obstruction contract). Fix an invoked degree $n \geq 1$. A standard n -eligible higher obstruction record consists of:

- (i) an admissible representative $C_{\tau, W}F$;
- (ii) a declared realized object
$$A_n(F) := R_n(C_{\tau, W}F) \in D^b(k\text{-mod});$$
- (iii) the declared standard amplitude regime, normally $A_n(F) \in D^{[-n, 0]}(k\text{-mod})$, or a separately proved admissible replacement;
- (iv) a fixed edge extractor E_n on the declared repaired-profile class;
- (v) an artifact-backed edge isomorphism

$$H^{-n}(A_n(F)) \xrightarrow{\sim} E_n(\text{Eval}_{n, W, \tau}(F));$$

- (vi) zero compatibility

$$P = 0 \implies E_n(P) = 0.$$

Under these data, Chapter 4 may prove the one-way discharge

$$\text{Eval}_{n, W, \tau}(F) = 0 \implies \text{Ext}^n(A_n(F), k) = 0.$$

Definition 1.18 (Natural zero-reflection upgrade). A higher obstruction record admits a zero-reflection upgrade on a declared profile class $\mathcal{P}_n$ only when:

- (i) E_n is fixed independently of the desired verdict;
- (ii) E_n is natural on the declared morphism class;

(iii) the edge-realization isomorphism is natural on the declared scope;

(iv) a theorem proves

$$E_n(P) = 0 \implies P = 0 \quad (P \in \mathcal{P}_n);$$

(v) representative-change and realization-change compatibility are separately discharged.

Only under these hypotheses may the vanishing equivalence

$$\text{Eval}_{n,W,\tau}(F) = 0 \iff \text{Ext}^n(A_n(F), k) = 0$$

be asserted on the declared scope. No object-level isomorphism $\text{PH}_n \cong \text{Ext}^n$ follows from this vanishing equivalence.

Remark 1.19 (Synthetic detector duality). For a vector space V , the shifted object $V[n]$ satisfies a tautological derived duality with Ext^n . Applying this construction to a detector output can be mathematically valid, but it is synthetic unless a separately defined natural realization identifies that shifted object with an independently meaningful $A_n(F)$. Synthetic duality may be used as a toy theorem, calibration, or lower-level algebraic component; it is not promoted to the principal higher Bridge.

Definition 1.20 (Family-relative proof compression). Let

$$\Theta = (\mathcal{F}, W, \tau, \text{endpt}, \text{critical}, \text{coverage})$$

be a declared information class. Define

$$\text{MinCert}(\Theta) := \inf \{|T| \mid T \text{ is a complete finite detector for every } B \in \mathcal{F}\},$$

with $\text{MinCert}(\Theta) = +\infty$ when no complete finite detector exists.

A family-relative proof-compression theorem must distinguish:

- (i) an upper bound produced by births, germs, mesh, or external structure;
- (ii) a lower bound or sharpness counterexample;
- (iii) endpoint and boundary conventions;
- (iv) the source of every family-wide excess or separation bound;
- (v) the unbounded-window no-go regime.

A single already-classified zero barcode does not define a nontrivial family-relative minimal certificate merely because the empty stencil suffices retrospectively.

Definition 1.21 (Problem-Bridge proof chain). A problem-specific v20 proof chain is a composable sequence of separately typed statements:

$$\begin{array}{c} \text{external source statement} \\ \Downarrow \text{realization/soundness-in} \\ \text{AK-readable filtered or persistence data} \\ \Downarrow \text{finite detector and Core certification} \\ \text{AK-side vanishing or defect conclusion} \\ \Downarrow \text{higher obstruction and/or finite-to-infinite reduction} \\ \text{AK-side theorem at the required infinite or derived scope} \\ \Downarrow \text{Return/soundness-out theorem} \\ \text{external mathematical conclusion.} \end{array}$$

Each arrow has its own role, hypotheses, artifacts, permitted use, and failure route.

Policy 2 (Return-Theorem and finite-to-infinite discipline). *No external-domain conclusion is inferred from a detector or Core pass alone. When the external target is infinite, asymptotic, analytic, tower-level, or global, the proof record must include:*

- (i) *a proved external-to-AK realization theorem;*
- (ii) *a certified finite detector or a separately proved non-finite Core route;*
- (iii) *a finite-to-infinite reduction theorem whenever finite data are used to establish an infinite statement;*
- (iv) *every invoked higher-edge and Type IV theorem;*
- (v) *a proved AK-to-external Return or soundness theorem.*

Known-theorem recovery is valid only at the external strength actually returned by the final theorem.

Remark 1.22 (Chapter allocation). Chapter 2 retains the repaired metric calculus and supplies detector transport dependencies. Chapter 3 supplies finite detector characterization, completeness, sharpness, family-relative minimal certificates, and no-go results. Chapter 4 supplies the higher Ext-edge and its zero-reflection boundary. Chapter 5 separates terminal-generic and transient defects. Chapter 6 supplies detector-aware record-level gluing. Chapter 7 supplies canonical detector and bridge audit semantics. Chapter 8 supplies realization, finite-to-infinite, and Return-Theorem interfaces.

Policy 3 (v20 release acceptance contract). *The designation AK-HPDST v20.0 complete release is reserved for a document set in which the following obligations have been discharged and registered at their exact strengths.*

Mandatory Core mathematics:

- (i) *birth-detector characterization;*
- (ii) *germ or equivalent decorated-endpoint characterization;*
- (iii) *certified-stencil and family-mesh completeness;*
- (iv) *strict-completeness bounds, boundary analysis, sharpness, and explicit miss counterexamples;*
- (v) *right-unbounded finite-detector no-go theorem;*
- (vi) *metric zero-verdict composition through the inherited threshold-gap calculus;*
- (vii) *family-relative MinCert definition and nontrivial bounds;*
- (viii) *higher Extⁿ-edge and one-way discharge;*
- (ix) *at least one non-synthetic natural zero-reflection theorem on a declared nontrivial profile class if a higher vanishing equivalence is advertised as a release result;*
- (x) *transient kernel and cokernel defect detection with terminal/interior separation.*

Mandatory verification and governance:

- (i) *CM registration at exact theorem strength;*
- (ii) *TB examples and counterexamples covering endpoint, cofinal, mesh, metric, higher-edge, and transient-defect cases;*

- (iii) *manifest, non-scalar obligation, failure-route, and artifact closure;*
- (iv) *no unresolved contradiction between Main, Foundation, CR, CM, TB, and machine-readable records.*

Mandatory external demonstration:

- (i) *at least one Known-Theorem Recovery in the Navier–Stokes or Iwasawa interface at the exact recovered strength;*
- (ii) *at least one proved natural realization theorem or one proved finite-to-infinite reduction theorem;*
- (iii) *a Return-Theorem analysis showing precisely which external conclusion does and does not follow.*

Proof-assistant formalization, computational certificate reduction, and comparison with full-barcode verification are strongly recommended but do not replace any mandatory item.

1.3 Inherited metric repair and the v20 Unified Collapse Contract

Theorem 1.23 (Threshold-gap-safe stability under the standard bottleneck metric). *Let B and C be finite barcodes, possibly including infinite bars, under the standard bottleneck metric and the inherited v19 infinite-bar convention. Let $\tau > 0$, $\varepsilon \geq 0$, and assume*

$$d_B(B, C) \leq \varepsilon.$$

Assume that no finite bar of either barcode has length in

$$[\tau - \varepsilon, \tau + \varepsilon].$$

Then

$$d_B(T_\tau^{\text{bar}} B, T_\tau^{\text{bar}} C) \leq \varepsilon.$$

This inherited v19 metric-repair theorem remains the mandatory transport component for every v20 metric inference through hard deletion.

Remark 1.24 (Proof location for the metric-repair theorem). The proof is supplied in Chapter 2, Theorem 2.34, and independently in Appendix A, Theorem 1.33 (the metric-repair theorem). Chapter 1 fixes the exact statement later chapters may consume.

Definition 1.25 (v20 Unified Collapse Contract: inherited and extended mathematical data). A v20 UCC mathematical contract contains the inherited repaired metric clauses and the following conservative extensions.

- (i) **Hard deletion.**

$$T_\tau^{\text{bar}}(B) = \{I \in B \mid \ell(I) > \tau \text{ or } \ell(I) = +\infty\}.$$

- (ii) **Barcode algebra.** The operation is endpoint-independent and idempotent, and

$$T_\sigma^{\text{bar}} T_\tau^{\text{bar}}(B) = T_{\max\{\sigma, \tau\}}^{\text{bar}}(B).$$

- (iii) **Infinite-bar invariance.** If $I = [a, \infty)$, then

$$T_\tau^{\text{bar}}(I) = I.$$

(iv) **Rejected global non-expansiveness.** No inequality

$$d_B(T_\tau^{\text{bar}} B, T_\tau^{\text{bar}} C) \leq d_B(B, C)$$

is asserted for arbitrary barcodes.

(v) **Windowed repaired evaluation.**

$$\text{Eval}_{i,W,\tau}(F) = T_\tau^{\text{bar}}(W_W \mathbf{P}_i(F)).$$

(vi) **Metric-use discipline.** Threshold-gap evidence is mandatory whenever a metric discrepancy is transported through hard deletion. Exact comparison uses a separately proved equality or isomorphism and is not automatically a positive-radius robustness statement.

(vii) **No unconditional crop-collapse commutation.** Such commutation may be used only under the exact criterion of Appendix A or a separately declared exact-admissible theorem.

(viii) **Representative compatibility.** If $C_{\tau,W}F$ is used, then in every monitored degree

$$\mathbf{P}_i(C_{\tau,W}F) \cong \text{Eval}_{i,W,\tau}(F),$$

at the explicitly declared equivalence strength.

(ix) **Non-scalar obligations.** Representative existence, actual morphisms, eligibility, terminal tameness, higher coherence, claim status, and artifact identity remain independent proof obligations.

(x) **Exact-admissible strengthening.** Strict functoriality, exactness, or canonical tower comparison may be used only under a separately proved and registered strengthening.

(xi) **Finite-detector strengthening.** A finite detector may discharge a repaired-profile zero clause only through Definition 1.14, Policy 1, and the applicable Chapter 3 characterization or completeness theorem.

(xii) **Higher-obstruction strengthening.** An invoked degree- n categorical clause uses Definition 1.17. The one-way discharge is the default strength. A vanishing equivalence requires Definition 1.18.

(xiii) **External-theorem separation.** A Core pass has no external-domain force without the complete Problem-Bridge chain of Definition 1.21 and Policy 2.

Policy 4 (v20 Unified Collapse Contract: audit policy). *A certified v20 run declares at least:*

(i) *the barcode metric and infinite-bar convention;*

(ii) *the windows, thresholds, monitored degrees, and deletion convention;*

(iii) *the collapse policy*

barcode_level, tau_split, exact_admissible;

(iv) *the evaluation order;*

(v) *exactly one comparison mode for every consumed comparison:*

exact, metric_gap_certified, not_compared;

- (vi) in mode *exact*, the actual equality or persistence-profile isomorphism, its source and target, scope, equivalence strength, and immutable proof artifact;
- (vii) in mode *metric_gap_certified*, every required crop, window-edge, discrepancy, scalarization, and two-sided threshold-gap record;
- (viii) mode *not_compared* only when the governing claim consumes no comparison inference; it cannot discharge an invoked overlap, transport, coherence, representative-comparison, or tower-comparison obligation;
- (ix) representative and eligibility policies when invoked;
- (x) the non-scalar obligation registry;
- (xi) for each metric-mode comparison, the metric quantale, bottleneck scalarization, finite protected profile family $\mathfrak{B}$, actual discrepancy upper-bound evidence, and strict safety inequality

$$\text{val}_{\mathfrak{B}}(\mathfrak{d}_{\text{met}}) < \text{gap}_{\tau}(\mathfrak{B});$$

- (xii) the Type IV scope and, when invoked, the comparison artifact, terminal-tameness evidence, fixed $(W, \tau, I_{\text{mon}})$, and defect record;
- (xiii) Claim, CM, and TB references together with evidence status and invocation role;
- (xiv) repair ancestry and immutable artifact references;
- (xv) for every invoked detector, the detector ID, profile stage, mode, endpoint convention, sample set, completeness source, two-axis status, certificate size, and artifact identity;
- (xvi) for every invoked higher Ext clause, n -eligibility, the realized object, amplitude policy, edge extractor, edge-realization artifact, naturality scope, and zero-reflection status;
- (xvii) for every invoked transient-defect clause, the full kernel/cokernel profile artifacts and the complete defect-detector record;
- (xviii) for every external conclusion, realization, finite-to-infinite, higher-obstruction, and Return-Theorem records with their exact external source and target statements.

No metric ledger, positive threshold reserve, or gap certificate is fabricated for a purely exact comparison. Conversely, an exact comparison does not by itself certify positive-radius robustness. Non-scalar obligations cannot be discharged by numerical slack.

Proof and status. Theorem 1.23 is the inherited metric theorem in the opening contract block; no detector rank-stability theorem is asserted. Definition 1.25 records mathematical data and admissibility clauses. Policy 4 records operational audit discipline.

1.4 UCC records, threshold-gap records, detectors, and repaired evaluation

Definition 1.26 (v20 UCC-Contract). A v20 UCC-Contract consists of:

- (1) a metric-ledger quantale and, when required, a bottleneck scalarization;
- (2) a barcode metric and infinite-bar convention;

- (3) right-open windows and separate partition or overlap data;
- (4) a threshold policy or threshold sweep;
- (5) the hard deletion and evaluation rules;
- (6) a comparison-mode policy;
- (7) threshold-gap, crop, and window-edge policies;
- (8) representative and eligibility policies when invoked;
- (9) metric and non-scalar obligation policies;
- (10) gate and tower-comparison policies;
- (11) Claim, CM, TB, repair, and artifact-identity policies;
- (12) detector applicability, completeness, endpoint, coverage, and two-axis status policies;
- (13) higher-degree eligibility, edge, naturality, and zero-reflection policies;
- (14) finite-to-infinite reduction and Return-Theorem policies when an external conclusion is invoked.

Definition 1.27 (Right-open windows, partition covers, and overlap covers). A window is a right-open interval

$$W = [u, u'), \quad u' \in \mathbb{R} \cup \{\infty\}.$$

A right-open partition cover is pairwise disjoint and ordered when sequential continuation is invoked. An overlap-admitting cover records every nonempty finite intersection used by the certificate logic and carries the required nerve, simplex, and face-compatibility data. The two cover structures are not silently identified.

Definition 1.28 (Window restriction). Define

$$W_W : \text{Pers}_k^{\text{cons}} \longrightarrow \text{Pers}_k^{\text{cons}}$$

by restriction to W , with extension by zero outside W when a globally indexed object is required. At barcode level, interval summands are clipped to W and empty intersections are discarded. If $u' < \infty$, the infinite bar $[a, \infty)$ becomes $[\max\{a, u\}, u')$ whenever the intersection is nonempty.

Definition 1.29 (Windowed pre-threshold profile). For $F \in \text{FiltCh}(()k)$, define

$$B_{i,W}(F) := W_W \mathbf{P}_i(F).$$

All threshold clearance used for certification is measured on this profile before hard deletion.

Definition 1.30 (Windowed repaired profile). Define

$$\mathbf{P}_{i,W,\tau}(F) := \text{Eval}_{i,W,\tau}(F) = T_\tau^{\text{bar}}(B_{i,W}(F)).$$

This is a repaired evaluation record. It is metric-gap-certified only after the relevant two-sided record is supplied.

Definition 1.31 (Canonical comparison mode). Every comparison consumed by a v20 claim records exactly one mode:

`exact,      metric_gap_certified,      not_compared.`

The meanings are:

- (i) `exact`: an actual equality or persistence-profile isomorphism between the declared source and target is proved and artifact-backed;
- (ii) **Metric-gap-certified mode**. A real pre-threshold bottleneck upper bound and a two-sided threshold-gap certificate are supplied at the same protected budget;
- (iii) `not_compared`: the governing claim consumes no comparison conclusion between the declared objects.

An exact comparison establishes only the stated exact relation for the stated profiles; it is not automatically a positive-radius robustness certificate. The mode `not_compared` cannot discharge a comparison obligation that the claim actually invokes.

Definition 1.32 (Exact comparison record). An exact comparison record contains:

- (i) immutable identifiers for the source and target profiles;
- (ii) the monitored degree, window, threshold, crop/restriction convention, and declared scope;
- (iii) an actual equality or persistence-profile isomorphism at the declared equivalence strength;
- (iv) the proof object and immutable artifact identifiers needed to reconstruct the exact comparison;
- (v) status `pass`, `reject`, or `undefined` under the Chapter 7 atomic status calculus.

No scalarized metric budget or positive threshold reserve is inferred merely from this record.

Definition 1.33 (Threshold clearance). For a barcode B ,

$$\text{clear}_\tau(B) := \inf_{I \in B_{\text{fin}}} |\ell(I) - \tau|.$$

If $B_{\text{fin}} = \emptyset$, set $\text{clear}_\tau(B) = +\infty$. If a finite bar has length exactly τ , then $\text{clear}_\tau(B) = 0$.

Definition 1.34 (Threshold safety margin). For a finite family $B_1, \dots, B_m$, define

$$\text{gap}_\tau(B_1, \dots, B_m) := \min_j \text{clear}_\tau(B_j).$$

For one barcode, $\text{gap}_\tau(B) = \text{clear}_\tau(B)$. The notation is a ledger-facing alias and introduces no second invariant.

Definition 1.35 (Window-edge clearance and crop budget). A window-edge clearance record, denoted $\text{clear}_{\partial W}(B)$, measures the separation of relevant endpoints from active window boundaries. A crop budget $\delta_{\text{crop}}(W)$ bounds discrepancy introduced by restriction, clipping, or moving boundaries before hard deletion. A source-level perturbation bound cannot be used as a windowed pre-threshold bound without a theorem or artifact controlling this crop step.

Definition 1.36 (Metric threshold-gap record). A threshold-gap record is used only in metric-gap-certified mode. Its canonical mode label is

`metric_gap_certified.`

The record contains at least:

- (i) immutable identifiers for the compared pre-threshold profiles;
- (ii) the two profiles, window, threshold, and the fixed mode

metric_gap_certified;

- (iii) a declared ε and evidence for $d_B(B, C) \leq \varepsilon$;
- (iv) crop and window-edge data when required;
- (v) $\text{clear}_\tau(B)$, $\text{clear}_\tau(C)$, and $\text{gap}_\tau(B, C)$;
- (vi) confirmation that neither finite-bar family meets $[\tau - \varepsilon, \tau + \varepsilon]$;
- (vii) the scalarization and its discrepancy upper-bound evidence when the budget is ledger-derived;
- (viii) reconstructible artifact references;
- (ix) one specialized status:

gap_certified, gap_failed, gap_pending.

A one-sided record is not a v20 threshold-gap certificate; v20 inherits the same two-sided requirement. This record is not required for a purely exact comparison.

Remark 1.37 (Specialized status mapping). The specialized gap statuses map to the canonical atomic statuses by

gap_certified $\mapsto$ pass,
gap_failed $\mapsto$ reject,
gap_pending $\mapsto$ undefined.

They do not create an independent audit-status algebra.

Remark 1.38 (Exact comparison versus robustness certification). An exact equality or proved profile isomorphism may establish equality of specific repaired outputs without a positive metric budget. It does not prove robustness under positive perturbation. A profile with a finite bar of length exactly τ may be exactly compared with itself, but it is not threshold-gap certified at τ .

Remark 1.39 (Bottleneck and interleaving metrics in the constructible one-parameter regime). In the constructible one-parameter persistence regime over a field, the isometry theorem identifies

$$d_{\text{int}}(P, Q) = d_B(\mathcal{B}(P), \mathcal{B}(Q)).$$

The identification is read under the inherited v19 infinite-bar convention. Interleaving-distance records may be used only when their barcode profiles are precisely those entering the threshold-gap record.

Definition 1.40 (Typed pipeline ledger and scalarized budget). For one declared metric-gap-certified comparison, let $\delta_j \in \mathbb{V}$ be its metric contributions. Define

$$\mathfrak{d}_{\text{met}} := \bigoplus_j \delta_j.$$

A certified scalarization satisfies

$$\text{val}_B(0_V) = 0, \quad \text{val}_B(q \oplus q') \leq \text{val}_B(q) + \text{val}_B(q').$$

Set

$$\varepsilon_B(\mathfrak{d}_{\text{met}}) := \text{val}_B(\mathfrak{d}_{\text{met}}).$$

The record must also establish

$$d_B(B, C) \leq \varepsilon_B(\mathfrak{d}_{\text{met}})$$

for each comparison to which the budget is applied. Non-scalar obligations do not belong to $\mathfrak{d}_{\text{met}}$.

Remark 1.41 (Sharpness of the ε -band). The sharp closed forbidden band for the two-sided theorem is

$$[\tau - \varepsilon, \tau + \varepsilon].$$

A half-width 2ε policy may be used only as an explicitly non-sharp conservative policy and may not be described as the sharp Core constant.

Remark 1.42 (Core sovereignty over Search). The Core accepts or rejects through declared mathematical conditions, typed obligations, diagnostics, budget checks, and artifact-supported records. No Search-side quantity upgrades a reject or undefined result.

1.5 Overlap Gate and B-Gate⁺

Definition 1.43 (Overlap Gate). Let

$$W_{\alpha\beta} := W_\alpha \cap W_\beta.$$

For every monitored degree, both local objects are evaluated directly on $W_{\alpha\beta}$ at one declared common threshold, with an explicit threshold-reconciliation record when their local thresholds differ. Exactly one comparison mode is then recorded.

An invoked pairwise Overlap Gate passes only under one of the following two branches.

- (i) **Exact branch.** Mode `exact` is declared and an exact overlap record supplies the directly evaluated common-window profiles and an actual equality or persistence-profile isomorphism

$$\text{Eval}_{i, W_{\alpha\beta}, \tau}(F_\alpha) \xrightarrow{\sim} \text{Eval}_{i, W_{\alpha\beta}, \tau}(F_\beta),$$

together with immutable source, target, proof, crop/restriction, and threshold-reconciliation artifacts.

- (ii) **Metric branch.** Mode `metric_gap_certified` is declared; the directly evaluated pre-threshold profiles

$$B_{i, W_{\alpha\beta}}^\alpha := W_{W_{\alpha\beta}} \mathbf{P}_i(F_\alpha), \quad B_{i, W_{\alpha\beta}}^\beta := W_{W_{\alpha\beta}} \mathbf{P}_i(F_\beta)$$

satisfy an artifact-supported estimate

$$d_B(B_{i, W_{\alpha\beta}}^\alpha, B_{i, W_{\alpha\beta}}^\beta) \leq \varepsilon_{\alpha\beta, i};$$

a two-sided metric threshold-gap record protects both profiles; crop, window-edge, comparison, and threshold-reconciliation contributions are declared; and

$$\varepsilon_{\alpha\beta, i} < \text{gap}_\tau(B_{i, W_{\alpha\beta}}^\alpha, B_{i, W_{\alpha\beta}}^\beta).$$

Theorem 1.23 then yields

$$d_{\text{int}}(\text{Eval}_{i, W_{\alpha\beta}, \tau}(F_\alpha), \text{Eval}_{i, W_{\alpha\beta}, \tau}(F_\beta)) \leq \varepsilon_{\alpha\beta, i}$$

in the declared constructible one-parameter regime.

Mode `not_compared` records that no overlap inference is consumed; it does not constitute a passing Overlap Gate and cannot discharge pairwise or higher Čech coherence. In either passing branch, every invoked representative, realization, and Type IV clause has compatible provenance; Type IV is either explicitly `not_invoked` by scope or is `certified_zero`; and every required artifact is present. If a detector-based zero statement is consumed on the overlap, the detector is applied to the directly evaluated common-window source profile and has a complete overlap-scope certificate. A zero detector on either unreduced local window does not substitute for this common-window record.

A passing pairwise gate does not by itself produce higher-simplex coherence, a global certificate atlas, a global filtered object, a finite-to-infinite reduction, or an external theorem.

Remark 1.44 (Exact and metric layers of the Overlap Gate). The metric branch has two barcode levels: it consumes a discrepancy bound on the directly windowed pre-threshold profiles and transports that bound to the repaired outputs. A post-threshold distance estimate alone does not certify pre-threshold classification stability.

The exact branch instead proves the declared equality or profile isomorphism directly. It may bypass metric transport for that exact claim, but it supplies no positive-radius threshold robustness unless a separate metric certificate is also proved.

Definition 1.45 (B-Gate⁺ (v20 finite-certificate repaired gate)). Fix a finite monitored degree set $I_{\text{mon}} \subset \mathbb{Z}_{\geq 0}$ containing degree 1 whenever the baseline degree-one gate is invoked. B-Gate⁺ passes on $(F; W, \tau, I_{\text{mon}})$ if every required clause passes and no required clause is undefined.

- (1) **Topological clauses.** For every invoked $n \in I_{\text{mon}}$,

$$\text{Eval}_{n,W,\tau}(F) = 0.$$

The zero statement is supported either by the full repaired-profile artifact or by a finite detector satisfying Definitions 1.14 and 1.15 together with Policy 1. A sampled zero without certified completeness is undefined. If an admissible compatible representative is supplied, the degree-one statement may also be recorded as

$$\text{PH}_1(C_{\tau,W}F) = 0.$$

- (2) **Comparison-mode clause.** Every consumed comparison declares exactly one mode from Definition 1.31. In mode `exact`, a valid exact comparison record is present. In mode `metric_gap_certified`, a valid two-sided metric threshold-gap record is present. Mode `not_compared` is admissible only when the governing scope consumes no comparison inference and cannot discharge an invoked comparison obligation.
- (3) **Detector clause.** Every detector used by the gate is applicable to the exact source stage and has certificate status `detector_certified`. Its value status is zero for a required zero clause. Birth/germ exhaustiveness, stencil coverage, family membership, excess-bound provenance, endpoint convention, cofinal policy, and artifact identity pass independently. A detector not used by the gate is explicitly `not_invoked` by scope.
- (4) **Categorical clauses.** For every invoked degree n , an admissible realization-compatible representative exists, the corresponding n -eligible regime holds, and

$$\text{Ext}^n(A_n(F), k) = 0.$$

The clause may be discharged by the Chapter 4 one-way theorem after exact repaired-profile vanishing has been established. A reverse inference requires Definition 1.18. Missing eligibility, edge realization, or naturality required by the consumed claim is undefined, not pass.

- (5) **Type IV and transient-defect clause.** Type IV is explicitly `not_invoked` or is `certified_zero` with a valid artifact and

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$$

at one fixed $(W, \tau, I_{\text{mon}})$. If the governing claim additionally requires absence of transient or interior defects, the full kernel and cokernel profiles are supplied and the applicable complete defect detector is certified zero. Terminal-generic zero alone does not discharge that stronger clause.

- (6) **Metric-budget and zero-transport clause.** For every comparison whose mode is `metric_gap_certified`, the protected family $\mathfrak{B}$ is finite, the scalarized value is proved to upper-bound the actual discrepancy, and

$$\text{val}_{\mathfrak{B}}(\mathfrak{d}_{\text{met}}) < \text{gap}_{\tau}(\mathfrak{B}).$$

If exact zero is transported across a positive bottleneck radius, the post-threshold separation condition, normally $2\varepsilon < \tau$, is separately discharged. No numerical budget or positive reserve is fabricated for a purely exact comparison.

- (7) **Non-scalar obligations.** Every required representative, morphism, eligibility, edge, naturality, detector applicability, endpoint, coverage, family, terminal-tameness, finite-to-infinite, coherence, claim-status, invocation-role, and artifact-identity obligation passes.
- (8) **Manifest and claim-use clause.** The record is complete and consistent under Chapter 7; every consumed claim has an admissible evidence-status–invocation-role combination; and no rejected, deprecated, unknown, unproved, operational, synthetic, toy, or spec item is promoted to theorem evidence.
- (9) **External conclusion clause.** This clause is outside the internal Core gate unless an external conclusion is actually invoked. In that case, the complete realization, finite-to-infinite, and Return chain of Policy 2 passes. An internal B-Gate⁺ pass alone never supplies this clause.

Remark 1.46 (Typed conjunction and no cancellation). B-Gate⁺ is a typed conjunction. A passing clause cannot cancel a reject or required undefined clause, and numerical reserve cannot repair a missing non-scalar artifact.

Remark 1.47 (Interpretation of B-Gate⁺). B-Gate⁺ is a Core acceptance predicate for a declared object, window, threshold, monitored scope, and proof-object record. It is not a Search score, probabilistic judgment, external-domain oracle, or transport theorem.

Remark 1.48 (No representative, no higher Ext-clause). A persistence zero clause may be evaluated from the full repaired profile or a certified complete detector. A degree- n Ext clause may not be invoked unless a realization-compatible admissible representative, realized object, amplitude policy, edge extractor, edge-realization artifact, and every applicable eligibility condition are supplied.

1.6 Failure landscape and the Type IV canon

Definition 1.49 (Failure landscape). The five failure types are preserved, with v20 detector and higher-degree specialization:

- **Type I: certified topological or persistence obstruction.** For at least one invoked monitored degree n ,

$$\text{Eval}_{n,W,\tau}(F) \neq 0.$$

A certified complete detector with value nonzero supplies a Type I witness for the corresponding required zero clause. An incomplete or inapplicable detector does not supply Type I; it leaves the clause undefined.

- **Type II: certified categorical obstruction in an eligible regime.** For at least one invoked n -eligible realized object,

$$\text{Ext}^n(A_n(F), k) \neq 0.$$

Missing eligibility, edge realization, or required naturality makes the clause undefined rather than categorically obstructed.

- **Type III: detector, operational, metric, comparison, or functorial failure.** This includes a well-typed failed detector applicability or completeness condition, failed birth/germ exhaustiveness, failed stencil coverage, invalid excess-bound source, failed exact or metric comparison, failed threshold-gap certification, invalid scalarization, insufficient discrepancy evidence, failed crop control, failed post-threshold zero-separation, missing representative or comparison morphism data, misuse of `not_compared`, failed threshold reconciliation, failed Restart transition control, or failed finite-to-infinite reduction. Missing evidence is normally undefined; an evidenced false required condition is reject/Type III.
- **Type IV: tower-level terminal-generic obstruction.** A valid comparison artifact exists and

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) \neq (0, 0).$$

A transient interior defect detected with zero terminal diagnostics is recorded separately and does not become terminal-generic Type IV by renaming. Missing artifact data are undefined, not obstructed.

- **Type V: claim-status, manifest, provenance, or promotion failure.** This includes use of a rejected claim, promotion of an operational, synthetic, toy, Search, or spec item to theorem evidence, concealment of missing detector or Bridge evidence, identifying undefined with zero or pass, fabricating `not_invoked` or `not_compared`, treating a finite numerical observation as an infinite theorem, artifact-identity or hash mismatch, incompatible versions, or inconsistent repair ancestry. A Type III undefined clause may coexist with Type V when its absence is misreported or governance data are inconsistent.

Definition 1.50 (Type IV scope status at the Chapter 1 contract level). Every run records exactly one Type IV status:

`not_invoked, undefined, certified_zero, obstructed.`

They mean, respectively: tower semantics are outside scope; tower semantics are in scope but required data are missing or ill-typed; a valid diagnostic exists and is zero; or a valid diagnostic exists and is nonzero.

Definition 1.51 (Tower comparison artifact). A tower comparison artifact is actual declared morphism data

$$\phi_{i,W,\tau} : \text{ColimProfile}_{i,W,\tau}(F_\bullet) \longrightarrow \text{ApexProfile}_{i,W,\tau}(F_\infty).$$

The source, target, morphism, kernel and cokernel profiles, and terminal-tameness evidence must be typed before Type IV diagnostics are computed. A bottleneck matching or expected limit correspondence is not automatically such a morphism.

Definition 1.52 (Terminal-generic Type IV diagnostics at contract level). When the artifact is defined and terminal tame, set

$$\begin{aligned} \mu_{i,W,\tau} &:= \text{tgdim}_W(\ker \phi_{i,W,\tau}), \\ \nu_{i,W,\tau} &:= \text{tgdim}_W(\text{coker } \phi_{i,W,\tau}). \end{aligned}$$

Fix a declared window–threshold pair (W, τ) and a declared finite set of monitored degrees I_{mon} . Define

$$\mu_{\text{Collapse}} := \sum_{i \in I_{\text{mon}}} \mu_{i,W,\tau},$$

$$\nu_{\text{Collapse}} := \sum_{i \in I_{\text{mon}}} \nu_{i,W,\tau}.$$

An aggregate over infinitely many monitored degrees, or across multiple window–threshold pairs, requires a separately declared aggregation and summability theorem. Chapter 5 supplies the precise terminal-generic definitions.

Remark 1.53 (Type IV canonical convention). The equality

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$$

means that no monitored terminal-generic kernel or cokernel defect is present. It does not imply full isomorphism, absence of finite interior defects, or object-level equivalence. The legacy Type IV symbols $u_{i,W,\tau}$ and u_{Collapse} are deprecated; $\nu_{i,W,\tau}$ and ν_{Collapse} are canonical. The legacy symbols create no normative alias in v20.0.

Remark 1.54 (Normative ν -migration rule). In theorem statements, hypotheses, gate clauses, diagnostic records, and manifest semantics, the cokernel diagnostic is written with ν . The Latin symbol u may occur only in explicit historical deprecation or migration prose and carries no v20.0 diagnostic meaning.

Definition 1.55 (Transient-defect detector contract). Assume a valid tower comparison artifact $\phi_{i,W,\tau}$ and full constructible defect profiles

$$K_{i,W,\tau} := \ker \phi_{i,W,\tau}, \quad Q_{i,W,\tau} := \text{coker } \phi_{i,W,\tau}$$

are supplied. A transient-defect detector is a finite detector applied at source stage `kernel_defect` or `cokernel_defect` to certify whether

$$T_{\tau_{\text{def}}}^{\text{bar}}(K_{i,W,\tau}) \quad \text{or} \quad T_{\tau_{\text{def}}}^{\text{bar}}(Q_{i,W,\tau})$$

is zero, for a separately declared defect threshold $\tau_{\text{def}} > 0$.

The terminal-generic pair (μ, ν) and the transient-defect verdict answer different questions:

- (i) (μ, ν) detects right-cofinal terminal defects;
- (ii) the transient detector detects retained finite noncofinal defects above τ_{def} ;
- (iii) zero results in both systems still imply full profile isomorphism only when every remaining shorter defect is excluded by the governing threshold or a separate no-residual-defect theorem.

Remark 1.56 (Terminal zero and transient zero). The record distinguishes at least the following cases:

- (i) if $(\mu, \nu) \neq (0, 0)$, a terminal or cofinal defect is present, independently of the transient-detector value;
- (ii) if $(\mu, \nu) = (0, 0)$ and the certified transient detector is `nonzero`, no monitored terminal defect is present but a retained interior defect is present;
- (iii) if $(\mu, \nu) = (0, 0)$ and the certified transient detector is `zero`, no monitored terminal defect and no interior defect longer than τ_{def} is present.

The final case is not a full-isomorphism theorem unless the shorter-defect regime is separately excluded.

Definition 1.57 (Monitored Windowed Collapse Zone). Fix a declared finite degree set I_{mon} . Define

$$\mathfrak{C}_{I_{\text{mon}}, W, \tau} := \{F \mid \text{Eval}_{n, W, \tau}(F) = 0 \text{ for every invoked } n \in I_{\text{mon}}, \\ \text{Ext}^n(A_n(F), k) = 0 \text{ for every invoked degree-}n \text{ Ext clause}\}.$$

The degree-one notation $\mathfrak{C}_{W, \tau}$ abbreviates $\mathfrak{C}_{\{1\}, W, \tau}$. Every Ext condition is meaningful only in its eligible realization regime. No converse from Ext-vanishing to repaired-profile vanishing is asserted without a registered natural zero-reflection theorem.

Remark 1.58 (Collapse Zone is not a complete gate). Membership in $\mathfrak{C}_{I_{\text{mon}}, W, \tau}$ does not by itself imply detector completeness, threshold-gap certification, metric-ledger pass, Type IV or transient-defect cleanliness, manifest completeness, B-Gate⁺ pass, finite-to-infinite validity, or an external-domain conclusion.

1.7 Minimal Manifest Axiom and proof-object discipline

Axiom 1.1 (Minimal Manifest Axiom). A proof-object candidate submitted for Core certification contains at least:

- (i) immutable RunID, theory version, schema version, Claim Register version, claim scope, repair ancestry, and artifact hashes;
- (ii) source and realized objects, windows, monitored degrees, finite I_{mon} , and overlap structure;
- (iii) metric, infinite-bar, endpoint, threshold, deletion, collapse, and evaluation policies;
- (iv) exactly one comparison mode for every consumed comparison;
- (v) in mode `exact`, the actual equality or profile-isomorphism proof, equivalence strength, and crop/restriction artifacts;
- (vi) in mode `metric_gap_certified`, every required threshold-gap, crop, window-edge, quantale, scalarization, actual discrepancy upper-bound, protected-family, strict-reserve, and post-threshold zero-separation record;
- (vii) for mode `not_compared`, the explicit scope exclusion proving that no comparison conclusion is consumed;
- (viii) for every invoked detector: detector ID, source profile ID, profile stage, detector mode, (n, W, τ) , endpoint convention, sample or critical set, cofinal policy, soundness theorem, completeness source, family-membership evidence, excess-bound provenance when used, metric mode, certificate size, two-axis detector status, and immutable artifacts;
- (ix) for every family-relative compression claim: information class Θ , admissible detector class, upper and lower bounds, boundary convention, no-go applicability, and MinCert evidence;
- (x) representative existence, compatibility, equivalence strength, actual morphism data, detector-preservation data when invoked, and realization compatibility;
- (xi) for every invoked degree- n Ext clause: the realized object $A_n(F)$, amplitude policy, edge extractor, edge-realization isomorphism, zero compatibility, eligibility data, and the status of any natural zero-reflection upgrade;

- (xii) *a non-scalar obligation registry with invocation state, dependency set, atomic status, evidence, and failure route;*
- (xiii) *Type IV scope and, when invoked, comparison, source semantics, provenance authority, terminal-tameness, fixed $(W, \tau, I_{\text{mon}})$, degreewise and total μ, ν diagnostics, and interior-defect status;*
- (xiv) *when transient defects are invoked: the full kernel/cokernel profiles, defect threshold, complete detector records, and the exact relationship to terminal-generic diagnostics;*
- (xv) *when an infinite, global, asymptotic, or external conclusion is invoked: realization, finite-to-infinite reduction, higher-obstruction, Return-Theorem, source-domain, target-domain, and external claim records;*
- (xvi) *atomic clause statuses*

pass, reject, undefined, not_invoked;

- (xvii) *manifest state*

complete, incomplete, inconsistent;

- (xviii) *an aggregate typed-conjunction verdict and complete failure-routing record;*

- (xix) *every Claim, CM, TB, synthetic, toy, candidate-only, and external-Bridge reference with its document role, evidence status, invocation role, exact source location, permitted strength, and prohibited stronger reading;*
- (xx) *a complete finite or countable clause registry, every justified scope exclusion, and immutable transitive artifact references sufficient for independent reconstruction and replay.*

Remark 1.59 (Why the manifest axiom belongs to the Core). Concrete storage may live in a Companion or implementation layer, but the requirement that certification be supported by typed, reconstructible, artifact-backed records belongs to the Core audit discipline.

Remark 1.60 (Canonical v20 manifest integration). Axiom 1.1 gives the foundational minimum. Chapter 3 extends repaired-evaluation and representative records. Chapter 7 supplies the canonical manifest, immutable identity, typed dependency closure, verdict reconstruction, replayability, and repair-as-new-record semantics.

Remark 1.61 (Appendix F and Appendix G boundary). Appendix F elaborates obligation types, invocation semantics, atomic status algebra, dependency closure, non-compensation, failure routing, and certificate readiness. Appendix G elaborates the manifest field schema, immutable identifiers, reference slots, artifact links, and serialization-neutral proof-object structure. Neither appendix establishes a mathematical claim merely by recording it.

Remark 1.62 (Claim, micro-theorem, and repair traceability). Every claim, micro-theorem, toy bridge, and repaired rule must be identified by its exact registered item and strength. This prevents silent reuse of rejected claims and silent promotion of toy or Search material.

Definition 1.63 (Manifest-complete proof-object candidate). A manifest-complete proof-object candidate satisfies the required field, typing, identity, and consistency conditions of Axiom 1.1 and Chapter 7. Manifest completeness means that the required record is present and internally consistent; it does not establish the truth of the recorded mathematical claims.

Definition 1.64 (Certified proof object). A certified proof object is a manifest-complete candidate such that every required mathematical claim has valid proof support, every computational claim has auditable artifact support, every required non-scalar obligation passes, every required gate clause passes, no required clause is reject or undefined, every registered claim is used within its permitted strength, and the aggregate verdict is pass.

Remark 1.65 (Manifest completeness is not proof). A complete manifest may faithfully record a failed proof, an obstructed tower, or a rejected claim. Completeness supports auditability; it does not replace mathematical validity.

1.8 Forward contract for countable certificate atlases

Policy 5 (Countable atlas forward contract). *Chapter 1 does not assert that countability, definability, finite Čech depth, and ordinary summability alone produce a global certificate. A countable local-to-global claim is admissible only through the typed record-level theorem of Chapter 6, in particular Theorem 6.65, or through a separately proved stronger theorem.*

A countable atlas record declares at least:

- (i) *an audit-locally finite right-open overlap cover;*
- (ii) *a finite Čech-depth bound;*
- (iii) *complete local records;*
- (iv) *when local detector certificates are used, each local record has a certified profile stage, endpoint convention, completeness theorem, sample set, and coverage artifact;*
- (v) *overlap-compatible birth/germ conventions and direct common-window evaluation of every detector source consumed on overlaps;*
- (vi) *every nonempty nerve simplex required by the declared cover and claim;*
- (vii) *face compatibility on every consumed restriction;*
- (viii) *a common-threshold or threshold-reconciliation policy;*
- (ix) *a local-to-global detector-coverage theorem proving that every globally retained bar is detected by at least one certified local or overlap stencil; local zero verdicts alone do not imply global zero;*
- (x) *every consumed face or simplex comparison uses one of*

`exact,      metric_gap_certified.`

The mode `not_compared` cannot discharge coherence;

- (xi) *exact proof artifacts for exact-mode simplices;*
- (xii) *scalarization, actual discrepancy upper-bound evidence, and positive sharp threshold reserve for metric-mode simplices only;*
- (xiii) *summable scalarized exposure over exactly the metric-mode simplices when a countable exposure policy is invoked;*

- (xiv) *correctly typed Type IV scope;*
- (xv) *all required non-scalar obligations;*
- (xvi) *a complete countable clause registry and audit-readable artifact closure.*

The conclusion is a Čech-coherent certificate atlas at record level. When detector certificates are invoked, the conclusion includes only the global zero predicate expressly proved by the local-to-global coverage theorem. No global filtered object, object-level descent, finite-to-infinite theorem, or external theorem follows without an additional theorem.

Remark 1.66 (Why the former countable-cover theorem is not retained). The earlier statement that a definable countable cover alone implies bounded global overlap error omitted typed scalarization, audit-local finiteness, higher-simplex coherence, face compatibility, Type IV scope discipline, and non-scalar obligations. The inherited repair delegates the valid mathematical theorem to Chapter 6.

1.9 Explicit non-claims of Chapter 1

The following exclusions are part of the meaning of the v20 Core.

- No unconditional global equivalence

$$\text{PH}_n \iff \text{Ext}^n$$

is claimed for arbitrary n , objects, realizations, coefficients, or profile classes.

- No reverse implication from Ext^n -vanishing to repaired-profile vanishing is available without a separately proved natural zero-reflection or faithfulness theorem.
- No vanishing equivalence is read as an object-level isomorphism $\text{PH}_n \cong \text{Ext}^n$.
- No detector obtained merely by shifting its output into a derived degree is promoted from synthetic or toy duality to a natural higher-obstruction Bridge.
- No adaptive choice of edge extractor, realization, detector, window, threshold, or sample set is made after inspecting the desired verdict unless the governing theorem explicitly permits and audits that adaptivity.
- No finite observation is a finite proof certificate without the applicable soundness, completeness, endpoint, coverage, family, metric, and artifact obligations.
- No detector zero verdict is asserted from missing or incomplete birth, germ, coverage, mesh, family-membership, excess-bound, endpoint, or profile-stage data.
- No sound detector is assumed complete merely because every observed sample is zero.
- No complete detector for one profile stage is used on another stage without a stage-transport theorem.
- No birth-point detector is endpoint-decoration invariant by default. Decorated endpoints require the germ theorem or another proved endpoint-aware formulation.
- No bounded-window cofinal bar is ignored merely because it arose by clipping an unbounded bar.
- No family-relative excess lower bound is certified by fully reading the same target barcode when the theorem claims non-circular proof compression, unless that full read is explicitly included in the certificate cost.

- No single-instance retrospective empty stencil is promoted to a meaningful family-relative MinCert theorem.
- No uniform complete finite detector exists on a right-unbounded scope with unbounded possible births and no additional finite-birth, stabilization, global-bound, cofinal-reduction, or finite-to-infinite theorem.
- No detector rank, pointwise image dimension, sample count, or certificate size is globally bottleneck-stable by default.
- No metric detector verdict follows from a bottleneck bound alone. The inherited two-sided threshold-gap transport, post-threshold zero-separation, detector completeness, and all crop/window-edge data remain required.
- No use of $2\varepsilon < \tau$ replaces the two-sided threshold-gap hypothesis; the two conditions solve different transport obligations.
- No positive-threshold Serre subcategory, exact reflector, abelian localization, or universal quotient is claimed in general.
- No global non-expansiveness of hard deletion under the standard bottleneck metric is claimed.
- No one-sided threshold-gap record is a v20 certificate.
- No finite bar of length exactly τ is gap-safe.
- No exact comparison is automatically a positive robustness certificate.
- No metric ledger, positive gap, reserve, detector score, or compression score is fabricated merely to encode a purely exact comparison.
- No record whose comparison mode is `not_compared` discharges an invoked overlap, transport, coherence, representative comparison, detector transport, or tower-comparison obligation.
- No exact overlap certificate is automatically a positive-radius overlap-robustness certificate.
- No unconditional crop–collapse commutation is claimed.
- No crop–collapse criterion may ignore an infinite bar that becomes finite after bounded-window clipping.
- No global strict filtered lift C_τ is asserted by default.
- No identity $C_{\tau,W}F = (C_\tau F)|_W$ is asserted without a proved compatibility theorem.
- No persistence-profile equivalence automatically preserves detector sample maps, birth/germ sets, realization, or higher-edge data.
- No bottleneck matching is automatically a persistence morphism.
- No tower comparison morphism is automatic outside a proved exact-admissible regime.
- No missing Type IV artifact is interpreted as zero.
- No no-tower claim is represented by a fabricated

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0).$$

- No zero terminal-generic diagnostic implies absence of finite interior or transient defects.
- No zero terminal-generic diagnostic together with a thresholded transient zero verdict implies full isomorphism unless the shorter residual-defect regime is separately excluded.
- No aggregate Type IV or transient diagnostic is defined over an undeclared or non-summable monitored scope.
- No legacy Type IV u -notation has normative force outside explicit historical deprecation or migration prose.
- No unscalarized quantale element is compared directly with real-valued clearance.
- No scalarized value is used without evidence that it upper-bounds the actual discrepancy consumed by the proof.
- No numerical budget repairs a missing non-scalar obligation.
- No local detector atlas proves global detector completeness without a local-to-global coverage theorem.
- No certificate-record gluing implies object-level descent.
- No finite detector certificate proves an infinite tower, continuum, asymptotic, global PDE, or p -adic analytic statement without a finite-to-infinite reduction theorem.
- No AK-side vanishing or B-Gate⁺ pass is an arithmetic, PDE, geometric, categorical, physical, or cryptographic theorem without a separately proved Return or soundness theorem.
- No external-to-AK realization theorem is inferred from similarity, analogy, dimensional matching, shared notation, or numerical correlation.
- No Return Theorem is inferred from the realization theorem; the two directions are logically distinct.
- No Known-Theorem Recovery result proves more than the exact external theorem and hypotheses returned by its final Bridge.
- No Search, AI, advisory, spectral, ranking, or candidate-stencil artifact is a Core proof object or gate verdict.
- No scalar potential or agent consensus replaces the typed verifier.
- No manifest completeness is identified with mathematical truth.
- No Claim Register entry is proof merely because it is registered.
- No Claim Register entry may promote a statement beyond its underlying proof.
- No operational rule, specification target, synthetic construction, toy theorem, or companion template is used as theorem evidence merely because it is registered or procedurally satisfied.
- No Core micro-theorem is an external-domain theorem.
- No toy bridge is an arithmetic, PDE, geometric, categorical, or cryptographic theorem.
- No Bridge candidate or Bridge Program is a Bridge theorem.

- No successor-release candidate is a v20.0 theorem claim.
- No reference to Appendix CR-v20, Appendix CM-v20, Appendix TB-v20, Appendix F, Appendix G, a Problem Interface appendix, or a Companion enlarges the Core by citation alone.
- No concrete YAML, JSON, database, or proof-store serialization format is fixed by Chapter 1.

1.10 Terminology and notation

Fix a base field k . We write Vect_k for finite-dimensional vector spaces and $\text{FiltCh}(()k)$ for finite-type filtered chain complexes. The degree-wise persistence functor is $\mathbf{P}_i$, window restriction is W_W , the windowed pre-threshold profile is

$$B_{i,W}(F) := W_W \mathbf{P}_i(F),$$

and repaired evaluation is

$$\text{Eval}_{i,W,\tau}(F) = T_\tau^{\text{bar}}(B_{i,W}(F)).$$

For a finite stencil $T \subseteq W$, the canonical finite-map detector namespace is

$$D_{T,\tau}^{\text{stc}}(B) := \bigoplus_{t \in T} \text{im}(B(t) \longrightarrow B(t + \tau)),$$

when every displayed map is defined. Chapter 3 fixes the exact birth, germ, stencil, mesh, and boundary conventions.

The detector profile-stage vocabulary is

windowed_prethreshold, repaired_postthreshold,
kernel_defect, cokernel_defect.

The detector certificate-status vocabulary is

detector_certified, detector_failed,
detector_pending.

and the detector value vocabulary is

zero, nonzero, undefined.

The two axes are not identified.

For a family information class Θ , $\text{MinCert}(\Theta)$ denotes the minimum complete finite certificate size, or $+\infty$ when no complete finite detector exists. For a windowed barcode B , $\text{exc}_{\tau,W}(B)$ denotes the window-relative threshold excess in the exact form fixed by Chapter 3.

The standard bottleneck distance is d_B , and the interleaving distance is d_{int} . Threshold clearance and safety margin are

$$\begin{aligned} \text{clear}_\tau(B) &= \inf_{I \in B_{\text{fin}}} |\ell(I) - \tau|, \\ \text{gap}_\tau(B_1, \dots, B_m) &= \min_j \text{clear}_\tau(B_j). \end{aligned}$$

Metric ledger values are compared with clearance through val_B . The canonical comparison modes are

exact, metric_gap_certified, not_compared.

For an invoked degree n , $A_n(F)$ denotes the declared realized derived object and E_n the declared higher edge extractor. Neither notation supplies eligibility, naturality, or zero reflection by itself.

The Type IV scope vocabulary is

`not_invoked, undefined, certified_zero, obstructed.`

The local diagnostics are

$$\mu_{i,W,\tau} = \text{tgdim}_W(\ker \phi_{i,W,\tau}), \quad \nu_{i,W,\tau} = \text{tgdim}_W(\text{coker } \phi_{i,W,\tau}).$$

For a fixed declared (W, τ) and finite monitored degree set I_{mon} , the aggregate pair is

$$\left(\sum_{i \in I_{\text{mon}}} \mu_{i,W,\tau}, \sum_{i \in I_{\text{mon}}} \nu_{i,W,\tau} \right) = (\mu_{\text{Collapse}}, \nu_{\text{Collapse}}).$$

The legacy Type IV symbols $u_{i,W,\tau}$ and u_{Collapse} are deprecated; $\nu_{i,W,\tau}$ and ν_{Collapse} are canonical.

For traceability, ClaimID, CMID, TBID, and RunID denote Claim, Core Micro-Theorem, Toy Bridge, and immutable run identifiers.

1.11 Provenance and status labels

Constructible interval decomposition, the standard bottleneck/interleaving isometry, ordinary barcode stability, and the field-coefficient derived identification used by the higher Ext-edge belong to standard mathematical background at their exact hypotheses.

Hard threshold deletion, repaired window-first evaluation, threshold-clearance ledgering, finite detector admission, family-relative proof compression, representative discipline, higher-edge eligibility, terminal/transient defect separation, Problem-Bridge decomposition, and typed audit architecture belong to the present AK-HPDST framework.

The v20 discipline retains the v19 metric repairs and additionally excludes overstrong language involving detector completeness from sampling alone, synthetic Ext promotion, finite evidence as an infinite theorem, automatic zero reflection, automatic finite-to-infinite passage, and external conclusions without Return Theorems.

Status labels for this chapter. *Inherited Core mathematical theorem / metric-repair theorem:* Theorem 1.23.

Core definitions and mathematical contracts: Definitions 1.1, 1.2, 1.10, 1.11, 1.12, 1.13, 1.14, 1.15, 1.16, 1.17, 1.18, 1.20, 1.21, 1.31, 1.32, 1.25, 1.26, 1.27, 1.28, 1.29, 1.30, 1.33, 1.34, 1.35, 1.36, 1.40, 1.43, 1.45, 1.49, 1.50, 1.51, 1.52, 1.55, 1.57, 1.63, 1.64.

Operational policies / audit axiom: Policy 1, Policy 2, Policy 3, Policy 4, Axiom 1.1, Policy 5.

Core doctrine and interpretive controls: Remark 1.3, Remark 1.4, Remark 1.5, Remark 1.6, Remark 1.7, Remark 1.8, Remark 1.9, Remark 1.19, Remark 1.22, Remark 1.24, Remark 1.37, Remark 1.38, Remark 1.39, Remark 1.41, Remark 1.42, Remark 1.44, Remark 1.46, Remark 1.47, Remark 1.48, Remark 1.53, Remark 1.54, Remark 1.56, Remark 1.58, Remark 1.59, Remark 1.60, Remark 1.61, Remark 1.62, Remark 1.65, Remark 1.66.

Explicit non-claims: Section 1.9.

Status labels for this chapter. *Inherited Core mathematical theorem / metric-repair theorem:* Theorem 1.23.

Core definitions and mathematical contracts: Definitions 1.1, 1.2, 1.10, 1.11, 1.12, 1.13, 1.14, 1.15, 1.16, 1.17, 1.18, 1.20, 1.21, 1.31, 1.32, 1.25, 1.26, 1.27, 1.28, 1.29, 1.30, 1.33, 1.34, 1.35, 1.36, 1.40, 1.43, 1.45, 1.49, 1.50, 1.51, 1.52, 1.55, 1.57, 1.63, 1.64.

Operational policies / audit axiom: Policy 1, Policy 2, Policy 3, Policy 4, Axiom 1.1, Policy 5.

Core doctrine and interpretive controls: Remark 1.3, Remark 1.4, Remark 1.5, Remark 1.6, Remark 1.7, Remark 1.8, Remark 1.9, Remark 1.19, Remark 1.22, Remark 1.24, Remark 1.37, Remark 1.38, Remark 1.39, Remark 1.41, Remark 1.42, Remark 1.44, Remark 1.46, Remark 1.47, Remark 1.48, Remark 1.53, Remark 1.54, Remark 1.56, Remark 1.58, Remark 1.59, Remark 1.60, Remark 1.61, Remark 1.62, Remark 1.65, Remark 1.66.

Explicit non-claims: Section 1.9.

2 Chapter 2: Barcode Metrics, Threshold Deletion, and Gap-Safe Stability

2.1 Scope of the inherited metric repair and the v20 zero-transport extension

This chapter supplies the persistence-theoretic metric foundation of the v20 Core. It preserves the repaired v19 threshold-deletion calculus at exactly its proved strength and adds only conservative consequences needed by the v20 finite-certificate architecture.

The inherited rule is:

For a positive threshold $\tau > 0$, hard barcode threshold deletion T_τ^{bar} deletes finite bars of length at most τ . It is endpoint-independent, idempotent, and threshold-monotone, but it is not globally non-expansive under the standard bottleneck metric. Metric transport through hard deletion requires the sharp two-sided threshold-gap theorem.

The new v20 rule is:

A bottleneck bound between repaired profiles does not preserve arbitrary pointwise detector values. It preserves the exact zero/nonzero predicate only inside the τ -reduced image and only under the sharp zero-separation condition $2\varepsilon \leq \tau$. The operational v20 contract uses the strict reserve $2\varepsilon < \tau$.

Accordingly, Chapter 2 supplies the mathematical components of the metric-safe detector chain declared in Chapter 1:

$$d_B(B, C) \leq \varepsilon + \text{two-sided threshold-gap certification} \implies d_B(T_\tau^{\text{bar}} B, T_\tau^{\text{bar}} C) \leq \varepsilon,$$

followed by

$$2\varepsilon \leq \tau \implies (T_\tau^{\text{bar}} B = 0 \iff T_\tau^{\text{bar}} C = 0).$$

Chapter 3 supplies detector characterization and completeness; this chapter supplies only the metric transport of the certified zero predicate.

Remark 2.1 (Conservative extension relative to v19). The v19 repair rejected global non-expansiveness and replaced it by the sharp two-sided threshold-gap-safe theorem. v20 retains that theorem, its hypotheses, the standard infinite-bar convention, window-first evaluation, and every prohibited stronger reading. The repaired-zero separation results below are consequences on the image of hard deletion; they do not turn hard deletion into a globally non-expansive, exact, or functorial operation.

Remark 2.2 (Division of responsibility among Chapters 1–3). Chapter 1 fixes detector admission, two-axis detector status, and the typed metric-safe detector chain. Chapter 2 proves the metric and zero-separation steps. Chapter 3 proves detector soundness, completeness, characterization, coverage, mesh, sharpness, and proof-compression results. No Chapter 2 metric statement supplies detector completeness.

Remark 2.3 (What this chapter does not do). This chapter does not prove a detector characterization theorem, a higher Ext^n -edge theorem, an arithmetic theorem, a PDE theorem, a geometric theorem, a cryptographic theorem, or an external Return Theorem. It also does not assert stability of pointwise stencil ranks, birth sets, germ data, or certificate sizes under bottleneck perturbation.

2.2 Constructible one-parameter persistence

Fix a field k . Let

$$\text{Pers}_k := [(\mathbb{R}, \leq), \text{Vect}_k]$$

be the category of one-parameter persistence modules over k , indexed by the ordered real line.

Definition 2.4 (Constructible persistence module). A persistence module $M \in \text{Pers}_k$ is called *constructible* if:

- (i) $M(t)$ is finite-dimensional over k for every $t \in \mathbb{R}$;
- (ii) on every bounded interval $J \subset \mathbb{R}$, there is a finite critical set outside which all structure maps are isomorphisms;
- (iii) there is no accumulation of critical parameters inside a bounded interval.

The full subcategory of constructible persistence modules is denoted by

$$\text{Pers}_k^{\text{cons}} \subset \text{Pers}_k.$$

Remark 2.5 (Working regime). All Core statements in this chapter are made inside the pointwise finite-dimensional constructible one-parameter regime over k , or inside finite barcode profiles obtained from declared windowed Core evaluations. Wild, nonconstructible, infinite-rank, or uncontrolled multi-parameter persistence data are outside the scope of this chapter unless a separate interface reduces them to declared constructible one-parameter profiles.

Definition 2.6 (Finite-type filtered chain complex). Let $\text{Ch}^b(k)$ be the category of bounded chain complexes of finite-dimensional k -vector spaces. A *finite-type filtered chain complex* is a pair

$$F = (C_\bullet, \{F^t C_\bullet\}_{t \in \mathbb{R}})$$

such that:

- (i) $F^t C_\bullet \subseteq F^{t'} C_\bullet$ whenever $t \leq t'$;
- (ii) the filtration is exhaustive and left-bounded;
- (iii) for each homological degree i , the persistence module

$$\mathbf{P}_i(F) : \mathbb{R} \rightarrow \text{Vect}_k, \quad t \mapsto H_i(F^t C_\bullet)$$

belongs to $\text{Pers}_k^{\text{cons}}$.

The category of such objects, with filtration-preserving chain maps, is denoted by

$$\text{FiltCh}(\cdot)k).$$

Remark 2.7 (Persistence layer first). The Core reads a filtered object F through monitored persistence profiles

$$\mathbf{P}_i(F).$$

Any filtered-level or derived-level use requires an additional representative policy. The barcode-level collapse operation by itself does not supply a global strict filtered lift.

2.3 Barcode normal forms and bar lengths

Definition 2.8 (Interval module). For an interval $I \subseteq \mathbb{R}$, the interval module k_I is the persistence module defined by

$$k_I(t) = \begin{cases} k, & t \in I, \\ 0, & t \notin I, \end{cases}$$

with structure maps equal to the identity whenever both source and target lie in I , and zero otherwise.

Definition 2.9 (Bar and barcode). An interval summand k_I is called a *bar*. A barcode of a constructible persistence module M is an interval decomposition

$$M \cong \bigoplus_{\lambda \in \Lambda} k_{I_\lambda},$$

with locally finite barcode support on bounded intervals. The corresponding multiset of bars is denoted by

$$\mathcal{B}(M) = \{I_\lambda\}_{\lambda \in \Lambda}.$$

Theorem 2.10 (Constructible interval decomposition). *Every object $M \in \text{Pers}_k^{\text{cons}}$ admits a locally finite interval decomposition*

$$M \cong \bigoplus_{\lambda \in \Lambda} k_{I_\lambda}.$$

The barcode is unique up to permutation of interval summands.

Proof sketch. This is the standard interval-decomposition theorem for pointwise finite-dimensional constructible one-parameter persistence modules over a field. On each bounded interval, constructibility gives a finite representation of a linearly ordered finite poset, hence a decomposition into interval representations. Local finiteness assembles the global barcode, and Krull–Schmidt uniqueness gives uniqueness up to permutation. $\square$

Definition 2.11 (Length of a bar). Let $I \subset \mathbb{R}$ be a finite bar with endpoint values $a < b$. Its length is

$$\ell(I) := b - a.$$

If $I = [a, \infty)$, or differs from this only by endpoint bookkeeping at a , it is called an infinite bar and is assigned length

$$\ell(I) := +\infty$$

for thresholding purposes.

Remark 2.12 (Endpoint independence). The length $\ell(I)$ depends only on endpoint values, not on whether finite endpoints are open, closed, or half-open. All threshold rules in this chapter are therefore independent of harmless endpoint bookkeeping conventions.

2.4 Standard bottleneck metric and infinite-bar convention

Definition 2.13 (Standard bottleneck distance with infinite bars). Let B and C be finite barcode profiles, or locally finite constructible barcode profiles restricted to a declared bounded audit range. The bottleneck distance $d_B(B, C)$ is computed by matchings between bars and, for finite bars, the diagonal.

The convention for infinite bars is fixed as follows.

- (i) A finite bar may be matched to the diagonal with cost $\ell(I)/2$.
- (ii) An infinite bar is never matched to the diagonal at finite cost.
- (iii) A finite bar and an infinite bar are never matched at finite cost.
- (iv) Two infinite bars $[a, \infty)$ and $[a', \infty)$ may be matched with cost

$$\text{cost}([a, \infty), [a', \infty)) := |a - a'|.$$

Endpoint bookkeeping at finite endpoints is ignored according to the endpoint-independence convention.

Lemma 2.14 (Infinite-bar threshold invariance). *For every finite threshold $\tau > 0$, infinite bars are invariant under hard finite-threshold deletion:*

$$T_\tau^{\text{bar}}([a, \infty)) = [a, \infty).$$

Moreover, infinite bars are never contained in a finite threshold band

$$[\tau - \varepsilon, \tau + \varepsilon].$$

Proof. An infinite bar has length $+\infty$. Since T_τ^{bar} deletes only finite bars of length at most τ , an infinite bar is retained. The finite interval $[\tau - \varepsilon, \tau + \varepsilon]$ contains only finite real values, so $+\infty$ does not lie in it. $\square$

Remark 2.15 (Bottleneck–interleaving isometry). For q -tame or constructible one-parameter persistence modules over k , the usual isometry theorem identifies the interleaving distance and the bottleneck distance on barcodes:

$$d_{\text{int}}(M, N) = d_B(\mathcal{B}(M), \mathcal{B}(N)).$$

When infinite bars are present, this statement is read under the infinite-bar convention of Definition 2.13; namely infinite bars match infinite bars by birth-time displacement $|a - a'|$, and finite-to-infinite or infinite-to-diagonal matchings have infinite cost. Under this convention, a d_B -formulated stability statement may be transferred to a d_{int} -formulated comparison only when the persistence objects and barcode profiles being compared are exactly those appearing in the certified record.

2.5 Positive-threshold short bars are not Serre in general

The following observation remains one of the categorical guard-rails of the repaired Core.

Proposition 2.16 (Positive-threshold short-bar objects are not Serre in general). *Let $\tau > 0$. The class of constructible persistence modules whose barcode summands all have finite length at most τ is not extension-closed in general. Consequently, it is not a Serre subcategory of $\text{Pers}_k^{\text{cons}}$ in general.*

Proof. It suffices to give a counterexample. Take $\tau = 1$ and consider the short exact sequence of interval modules

$$0 \longrightarrow k_{[1,2)} \longrightarrow k_{[0,2)} \longrightarrow k_{[0,1)} \longrightarrow 0.$$

The subobject $k_{[1,2)}$ has length 1, and the quotient $k_{[0,1)}$ also has length 1. However, the middle object $k_{[0,2)}$ has length 2. Thus an extension of two short-bar objects can contain a longer bar. Therefore the class of bars of length at most τ is not extension-closed in general. $\square$

Corollary 2.17 (No global exact reflector claim). *For $\tau > 0$, hard threshold deletion is not asserted in the general Core to arise from an exact Serre localization, an abelian exact reflector, or a global exact endofunctor on all of $\text{Pers}_k^{\text{cons}}$.*

Remark 2.18 (Why this guard-rail matters). If positive-threshold deletion were treated as an exact Serre reflector without additional hypotheses, then exactness, functoriality, tower comparison maps, and filtered lifts would inherit unsupported categorical strength. The repaired formulation keeps the barcode-level operation while requiring stronger claims to be explicitly justified by τ -split or exact-admissible hypotheses.

2.6 Hard barcode threshold deletion

Definition 2.19 (Hard barcode threshold deletion). Let B be a barcode profile. For $\tau > 0$, define hard barcode threshold deletion by

$$T_\tau^{\text{bar}}(B) := \{ I \in B \mid \ell(I) > \tau \text{ or } \ell(I) = +\infty \}.$$

Equivalently, T_τ^{bar} deletes precisely the finite bars of length at most τ . Thus:

$$\begin{aligned} \ell(I) < \tau &\implies I \text{ is deleted,} \\ \ell(I) = \tau &\implies I \text{ is deleted,} \\ \ell(I) > \tau &\implies I \text{ is retained,} \\ \ell(I) = +\infty &\implies I \text{ is retained.} \end{aligned}$$

For a constructible persistence module M , the notation $T_\tau^{\text{bar}}(M)$ means threshold deletion applied to $\mathcal{B}(M)$, read at the barcode-profile level.

Remark 2.20 (Status of T_τ^{bar}). In the general Core regime, T_τ^{bar} is an operation on barcode normal forms and hence on isomorphism classes of constructible persistence modules. It is not asserted, without additional hypotheses, to be a strict exact endofunctor on all morphisms of $\text{Pers}_k^{\text{cons}}$.

Proposition 2.21 (Basic properties of hard threshold deletion). *For every $\tau > 0$, the operation T_τ^{bar} satisfies:*

(i) *endpoint independence;*

(ii) *idempotence:*

$$T_\tau^{\text{bar}} T_\tau^{\text{bar}}(B) = T_\tau^{\text{bar}}(B);$$

(iii) *infinite-bar preservation;*

(iv) *threshold monotonicity: if $0 < \tau \leq \sigma$, then every bar retained by $T_\sigma^{\text{bar}}(B)$ is retained by $T_\tau^{\text{bar}}(B)$.*

Proof. All assertions follow directly from the definition. Deleting all finite bars of length at most τ twice has the same effect as deleting them once. Infinite bars have length $+\infty$, hence are never deleted by a finite threshold. Endpoint choices do not affect length. Finally, increasing the threshold can only delete additional finite bars. $\square$

Proposition 2.22 (Threshold composition). *For $\tau, \sigma > 0$,*

$$T_\sigma^{\text{bar}} T_\tau^{\text{bar}}(B) = T_{\max\{\tau, \sigma\}}^{\text{bar}}(B)$$

at the barcode-profile level.

Proof. Applying T_τ^{bar} deletes bars of finite length at most τ . Applying T_σ^{bar} afterwards deletes bars of finite length at most σ among those remaining. The total effect is deletion of all finite bars of length at most $\max\{\tau, \sigma\}$. $\square$

2.7 Counterexample to global non-expansiveness

The next example is the metric repair point of v19.

Example 2.23 (Failure of global non-expansiveness). Fix $\tau > 0$, and let $0 < \eta < \tau/4$. Consider the one-bar barcodes

$$B = \{[0, \tau + \eta)\}, \quad C = \{[0, \tau - \eta)\}.$$

Before threshold deletion, matching the two bars to each other has cost 2η . Matching both bars to the diagonal has cost

$$\max \left\{ \frac{\tau + \eta}{2}, \frac{\tau - \eta}{2} \right\} = \frac{\tau + \eta}{2} > 2\eta,$$

where the strict inequality follows from $\eta < \tau/4$. Therefore

$$d_B(B, C) = 2\eta.$$

After hard threshold deletion,

$$T_\tau^{\text{bar}}(B) = \{[0, \tau + \eta)\}, \quad T_\tau^{\text{bar}}(C) = \emptyset.$$

Therefore the remaining bar of $T_\tau^{\text{bar}}(B)$ must be matched to the diagonal, giving

$$d_B(T_\tau^{\text{bar}}(B), T_\tau^{\text{bar}}(C)) = \frac{\tau + \eta}{2}.$$

Moreover,

$$\frac{\tau + \eta}{2} > 2\eta = d_B(B, C),$$

again because $\eta < \tau/4$. Consequently,

$$d_B(T_\tau^{\text{bar}}(B), T_\tau^{\text{bar}}(C)) \leq d_B(B, C)$$

does not hold in general.

Remark 2.24 (Meaning of the counterexample). The failure occurs exactly at the threshold boundary. A small endpoint perturbation can move a finite bar from length $> \tau$ to length $\leq \tau$, thereby changing whether it is retained or deleted. Thus the correct stability statement must exclude bars near the threshold.

Corollary 2.25 (Deprecated v18 metric claim). *The statement that hard positive-threshold deletion T_τ^{bar} is globally non-expansive under the standard bottleneck metric is rejected in v19. Any occurrence of that claim in earlier formulations is deprecated and may not be used in v20 proof construction.*

2.8 Threshold clearance and gap

Definition 2.26 (Threshold clearance). Let B be a barcode profile. Define the finite-bar threshold clearance by

$$\text{clear}_\tau(B) := \inf_{I \in B_{\text{fin}}} |\ell(I) - \tau|,$$

where B_{fin} denotes the finite bars of B . If B has no finite bars, set

$$\text{clear}_\tau(B) := +\infty.$$

If B contains a finite bar of length exactly τ , then

$$\text{clear}_\tau(B) = 0.$$

Definition 2.27 (Threshold gap as ledger alias). For a barcode profile B , define the ledger safety margin

$$\text{gap}_\tau(B) := \text{clear}_\tau(B).$$

For a finite family of barcode profiles $B_1, \dots, B_m$, define

$$\text{gap}_\tau(B_1, \dots, B_m) := \min_{1 \leq j \leq m} \text{clear}_\tau(B_j).$$

Thus gap_τ is not a second invariant distinct from threshold clearance; it is the ledger-facing alias for the minimum finite-bar clearance at the deletion threshold.

Definition 2.28 (Bottleneck scalarization and typed metric budget). Let $(V, \oplus, 0_V, \preceq)$ be the declared ledger quantale. A *bottleneck scalarization* is a monotone, subadditive map

$$\text{val}_B : V \longrightarrow [0, +\infty], \quad \text{val}_B(0_V) = 0, \quad \text{val}_B(q \oplus q') \leq \text{val}_B(q) + \text{val}_B(q').$$

For a metric ledger element $\mathfrak{d}_{\text{met}} \in V$, define the real-valued bottleneck budget

$$\varepsilon_B(\mathfrak{d}_{\text{met}}) := \text{val}_B(\mathfrak{d}_{\text{met}}).$$

Whenever the established shorthand

$$\Sigma\delta < \text{gap}_\tau$$

is used in this chapter, it means the typed real inequality

$$\varepsilon_B(\mathfrak{d}_{\text{met}}) < \text{gap}_\tau$$

for the declared protected comparison family, together with theorem-level, verified-computation, or artifact evidence that the scalarized value upper-bounds the actual bottleneck discrepancy consumed by the proof. Only metric defects belong to $\mathfrak{d}_{\text{met}}$. Representative existence, comparison morphisms, eligibility, terminal tameness, claim status, and manifest completeness are non-scalar proof obligations and cannot be repaired by numerical slack.

Remark 2.29 (Sharp threshold-safety margin versus scalarization tightness). Under an ε -bottleneck matching, the length of a finite bar may change by at most 2ε . For a *two-sided* threshold-gap certificate, however, the sharp forbidden band is

$$[\tau - \varepsilon, \tau + \varepsilon],$$

not $[\tau - 2\varepsilon, \tau + 2\varepsilon]$. Consequently, for a scalarized budget

$$\varepsilon = \varepsilon_B(\mathfrak{d}_{\text{met}}),$$

the condition

$$\varepsilon < \text{gap}_\tau(B) = \text{clear}_\tau(B)$$

means that all finite bars of the relevant pre-threshold profile avoid

$$[\tau - \varepsilon, \tau + \varepsilon].$$

The former factor $1/2$ is not part of the v20.0 two-sided Core certificate; v20 inherits this correction from v19.0.0. The word *sharp* refers to the optimal half-width of the threshold band for a given true bottleneck bound; it does not assert that a chosen scalarization val_B is numerically tight.

Definition 2.30 (Windowed pre-threshold profile). For $F \in \text{FiltCh}(\cdot)k$, monitored degree i , and declared window W , define the *windowed pre-threshold profile*

$$B_{i,W}(F) := W_W \mathbf{P}_i(F).$$

This is the barcode profile on which threshold clearance and threshold gap are measured. The repaired evaluation is then

$$\text{Eval}_{i,W,\tau}(F) = T_\tau^{\text{bar}}(B_{i,W}(F)).$$

Remark 2.31 (Clearance is measured before deletion, after windowing). The quantity clear_τ is never measured on the already deleted profile $\text{Eval}_{i,W,\tau}(F)$. It is measured on the windowed pre-threshold profile

$$B_{i,W}(F) = W_W \mathbf{P}_i(F).$$

This order matters: the threshold gap certifies that hard deletion will not change classification under the declared perturbation budget. Once a bar has already been deleted, its distance from the threshold can no longer be audited.

Definition 2.32 (Threshold-gap comparison record). A threshold-gap comparison record for two windowed pre-threshold profiles

$$B := B_{i,W}(F), \quad C := B_{i,W}(G)$$

at one declared monitored degree i , common window W , and threshold τ contains:

- (i) immutable identifiers for B and C ;
- (ii) the two pre-threshold profiles, the monitored degree, the common window, and the threshold;
- (iii) the finite-bar length data or certified summaries for both profiles;
- (iv) $\text{clear}_\tau(B)$, $\text{clear}_\tau(C)$, and

$$\text{gap}_\tau(B, C) = \min\{\text{clear}_\tau(B), \text{clear}_\tau(C)\};$$

- (v) the metric ledger element $\mathfrak{d}_{\text{met}}$, the declared scalarization val_B , and

$$\varepsilon_B := \text{val}_B(\mathfrak{d}_{\text{met}});$$

- (vi) theorem-level, verified-computation, or artifact evidence that

$$d_B(B, C) \leq \varepsilon_B;$$

- (vii) confirmation that neither finite-bar family meets the closed band

$$[\tau - \varepsilon_B, \tau + \varepsilon_B],$$

equivalently the strict two-sided condition

$$\varepsilon_B < \text{gap}_\tau(B, C);$$

- (viii) every crop, window-edge, endpoint, or threshold-reconciliation component used to obtain the windowed comparison bound;
- (ix) reconstructible artifact references;

(x) one specialized status

`gap_certified, gap_failed, gap_pending,`

mapping respectively to pass, reject, and undefined.

The record is two-sided by definition. Here `gap_failed` means that a well-typed required gap condition is proved or artifact-evidenced to fail. Missing, incomplete, stale, or pending evidence is recorded as `gap_pending`, never as `gap_failed`. A single-profile clearance record may be retained as a readiness record, but it is not a metric comparison certificate and does not establish transport through hard deletion.

Remark 2.33 (Length exactly τ is not gap-safe). A finite bar of length exactly τ gives $\text{clear}_\tau = 0$. Such a profile may still be evaluated by T_τ^{bar} , and the bar is deleted because deletion uses the rule $\ell(I) \leq \tau$. However, the profile is not positively gap-ready at threshold τ . It cannot support positive-radius metric transport through hard deletion at that threshold. A separately proved exact identity may still be recorded in exact comparison mode, but it is not a robustness certificate.

2.9 Threshold-gap-safe stability

Theorem 2.34 (Metric-repair theorem: sharp threshold-gap-safe stability). *Let B and C be barcodes with finitely many bars, possibly including infinite bars, interpreted under the bottleneck convention of Definition 2.13. Let $\tau > 0$ and $\varepsilon \geq 0$. Assume*

$$d_B(B, C) \leq \varepsilon.$$

Assume also that no finite bar of either B or C has length in the closed threshold band

$$[\tau - \varepsilon, \tau + \varepsilon].$$

Then

$$d_B(T_\tau^{\text{bar}}(B), T_\tau^{\text{bar}}(C)) \leq \varepsilon.$$

Proof. First consider the degenerate case $\varepsilon = 0$. Then $d_B(B, C) = 0$, so B and C are identical as barcode profiles under the stated barcode convention. The closed threshold band is the singleton $\{\tau\}$, and the hypothesis says that no finite bar has length exactly τ . Applying the same deterministic deletion rule to identical barcodes gives

$$T_\tau^{\text{bar}}(B) = T_\tau^{\text{bar}}(C),$$

hence the conclusion follows.

Now assume $\varepsilon > 0$. Choose a bottleneck matching between B and C of cost at most ε . Such a matching exists for the finite barcode profiles under consideration. We construct a matching between the retained barcodes after applying T_τ^{bar} by modifying the chosen matching only by deleting matched pairs that are removed on both sides.

Let $I \in B$ be a bar retained by T_τ^{bar} . If I is infinite, then by Definition 2.13 it cannot be matched at finite cost to the diagonal or to a finite bar. Hence it is matched to an infinite bar of C , which is also retained.

Suppose next that I is finite. Since I is retained, $\ell(I) > \tau$. Because no finite bar of B has length in $[\tau - \varepsilon, \tau + \varepsilon]$, this implies

$$\ell(I) > \tau + \varepsilon.$$

There are two cases.

If I is matched to the diagonal with cost at most ε , then the same diagonal match may be kept after threshold deletion. It contributes cost at most ε to the retained-barcode matching. This case is possible when

$\ell(I)/2 \leq \varepsilon$, and it is the reason the proof must not assume that every retained finite bar has a finite matched partner.

If I is matched to a finite bar $J \in C$, then the endpoint displacement bound gives

$$|\ell(I) - \ell(J)| \leq 2\varepsilon.$$

If J were deleted, then $\ell(J) \leq \tau$. Since C also avoids $[\tau - \varepsilon, \tau + \varepsilon]$, this would force

$$\ell(J) < \tau - \varepsilon.$$

Together with $\ell(I) > \tau + \varepsilon$, this gives

$$\ell(I) - \ell(J) > 2\varepsilon,$$

contradicting $|\ell(I) - \ell(J)| \leq 2\varepsilon$. Therefore J is retained. Finite-to-infinite matching is impossible at finite cost by Definition 2.13.

The same argument, with B and C interchanged, shows that every retained finite bar of C that is not kept on the diagonal is matched to a retained finite bar of B , while every retained infinite bar is matched to a retained infinite bar. Hence the chosen matching restricts, with the permitted retained-to-diagonal matches, to a bottleneck matching between $T_\tau^{\text{bar}}(B)$ and $T_\tau^{\text{bar}}(C)$ of cost at most ε . Thus

$$d_B(T_\tau^{\text{bar}}(B), T_\tau^{\text{bar}}(C)) \leq \varepsilon.$$

□

Example 2.35 (Sharpness of the ε -band). The constant ε in the two-sided closed threshold band is sharp. Let $0 < c < \varepsilon$, choose $0 < \eta < \varepsilon - c$, and take τ large enough that the diagonal costs below exceed ε . Set

$$L_+ := \tau + c + \eta, \quad L_- := \tau - c - \eta,$$

and consider the centered finite bars

$$I = \left[-\frac{L_+}{2}, \frac{L_+}{2}\right), \quad J = \left[-\frac{L_-}{2}, \frac{L_-}{2}\right).$$

Then both lengths avoid the smaller closed band $[\tau - c, \tau + c]$, while

$$d_B(\{I\}, \{J\}) = c + \eta < \varepsilon.$$

However I is retained and J is deleted by T_τ^{bar} . Hence

$$T_\tau^{\text{bar}}(\{I\}) = \{I\}, \quad T_\tau^{\text{bar}}(\{J\}) = \emptyset.$$

For τ sufficiently large, $L_+/2 > \varepsilon$, so

$$d_B(T_\tau^{\text{bar}}(\{I\}), T_\tau^{\text{bar}}(\{J\})) = \frac{L_+}{2} > \varepsilon.$$

Thus no two-sided closed band of half-width $c < \varepsilon$ suffices in general. The sharp two-sided half-width is exactly ε .

Remark 2.36 (Two-sided certificate). The threshold-gap certificate is two-sided. It is not enough for B alone to avoid the threshold band. If C contains a finite bar in $[\tau - \varepsilon, \tau + \varepsilon]$, then a matched pair may cross the deletion threshold and change retained/deleted status. For one-sided certificates, a 2ε -type margin is a natural conservative constant; it is not the sharp constant for the two-sided v19 Core theorem.

Corollary 2.37 (Scalarized-ledger-safe threshold classification). *Let B and C be finite barcode profiles, possibly with infinite bars. Let $\mathfrak{d}_{\text{met}} \in \mathcal{V}$ be a declared metric ledger and set*

$$\varepsilon_B := \text{val}_B(\mathfrak{d}_{\text{met}}).$$

Suppose

$$d_B(B, C) \leq \varepsilon_B$$

and

$$\varepsilon_B < \text{gap}_\tau(B, C).$$

Then

$$d_B(T_\tau^{\text{bar}}(B), T_\tau^{\text{bar}}(C)) \leq \varepsilon_B.$$

Proof. The strict inequality

$$\varepsilon_B < \text{gap}_\tau(B, C) = \min\{\text{clear}_\tau(B), \text{clear}_\tau(C)\}$$

implies that no finite bar of either barcode has length in

$$[\tau - \varepsilon_B, \tau + \varepsilon_B].$$

Apply Theorem 2.34 with $\varepsilon = \varepsilon_B$. □

2.10 Sharp zero separation inside the repaired image

Definition 2.38 (τ -reduced barcode profile). A barcode profile P is τ -reduced if

$$T_\tau^{\text{bar}}(P) = P.$$

Equivalently, every finite bar of P has length strictly greater than τ , while every infinite bar is retained. Every repaired profile $T_\tau^{\text{bar}}(B)$ is τ -reduced.

Definition 2.39 (Zero radius of a finite barcode profile). For a finite barcode profile P , define

$$\text{zrad}(P) := d_B(P, 0),$$

where 0 denotes the empty barcode. Under the standard convention,

$$\text{zrad}(P) = \begin{cases} 0, & P = 0, \\ \frac{1}{2} \max_{I \in P} \ell(I), & P \neq 0 \text{ and every bar is finite,} \\ +\infty, & P \text{ contains an infinite bar.} \end{cases}$$

Lemma 2.40 (Distance-to-zero formula). *Definition 2.39 gives the exact bottleneck distance from P to the zero profile.*

Proof. When P has only finite bars, every bar must be matched to the diagonal, and the matching cost is the maximum of the diagonal costs $\ell(I)/2$. If an infinite bar is present, it cannot be matched to the diagonal at finite cost. The empty profile has distance zero from itself. □

Theorem 2.41 (Sharp zero separation for τ -reduced profiles). *Let P and Q be finite τ -reduced barcode profiles, possibly including infinite bars. Let $\varepsilon \geq 0$, and suppose*

$$d_B(P, Q) \leq \varepsilon \quad \text{and} \quad 2\varepsilon \leq \tau.$$

Then

$$P = 0 \iff Q = 0.$$

The constant is sharp: if $2\varepsilon > \tau$, the conclusion can fail.

Proof. Assume $P = 0$. If Q contained an infinite bar, then $d_B(P, Q) = +\infty$, contrary to the hypothesis. If $Q \neq 0$ and all its bars were finite, τ -reducedness would give

$$\ell(I) > \tau \geq 2\varepsilon$$

for every bar $I \in Q$. Lemma 2.40 would then imply

$$d_B(0, Q) = \frac{1}{2} \max_{I \in Q} \ell(I) > \varepsilon,$$

again a contradiction. Hence $Q = 0$. The converse is symmetric.

For sharpness, suppose $2\varepsilon > \tau$. Choose L with $\tau < L \leq 2\varepsilon$, let $P = 0$, and let Q consist of one centered finite bar of length L . Then Q is τ -reduced, $d_B(P, Q) = L/2 \leq \varepsilon$, but $P = 0$ and $Q \neq 0$. $\square$

Remark 2.42 (Strict operational reserve). The theorem is valid at the closed boundary $2\varepsilon = \tau$, because repaired finite bars have length *strictly* greater than τ . The Chapter 1 audit contract uses $2\varepsilon < \tau$ to retain a positive reserve and to avoid equality-sensitive scalarization or rounding policies. The strict policy is conservative, not mathematically sharp.

Proposition 2.43 (Exact zero transport). *If $P \cong Q$ as persistence profiles, then*

$$P = 0 \iff Q = 0.$$

No metric ledger or threshold-gap certificate is required for this exact claim. The actual isomorphism and its immutable proof artifact are required.

Proof. The zero object is invariant under isomorphism. $\square$

Theorem 2.44 (Metric-safe repaired-zero transport). *Let B and C be finite barcode profiles, possibly including infinite bars. Let $\tau > 0$, $\varepsilon \geq 0$, and assume*

$$d_B(B, C) \leq \varepsilon.$$

Assume that no finite bar of either profile has length in

$$[\tau - \varepsilon, \tau + \varepsilon],$$

and assume

$$2\varepsilon \leq \tau.$$

Then

$$T_\tau^{\text{bar}}(B) = 0 \iff T_\tau^{\text{bar}}(C) = 0.$$

Proof. Theorem 2.34 gives

$$d_B(T_\tau^{\text{bar}}(B), T_\tau^{\text{bar}}(C)) \leq \varepsilon.$$

Both repaired profiles are τ -reduced by Definition 2.38. Apply Theorem 2.41. □

Corollary 2.45 (Scalarized metric-safe repaired-zero transport). *Let $\mathfrak{d}_{\text{met}} \in \mathbb{V}$, set*

$$\varepsilon_B := \text{val}_B(\mathfrak{d}_{\text{met}}),$$

and suppose the record proves

$$d_B(B, C) \leq \varepsilon_B, \quad \varepsilon_B < \text{gap}_\tau(B, C), \quad 2\varepsilon_B < \tau.$$

Then

$$T_\tau^{\text{bar}}(B) = 0 \iff T_\tau^{\text{bar}}(C) = 0.$$

This is the strict operational form consumed by the Chapter 1 metric-safe detector chain.

Definition 2.46 (Post-threshold zero-separation record). A post-threshold zero-separation record contains:

- (i) immutable identifiers for the repaired profiles $P = T_\tau^{\text{bar}}(B)$ and $Q = T_\tau^{\text{bar}}(C)$;
- (ii) the common threshold τ , the proved bound $d_B(P, Q) \leq \varepsilon$, and its derivation;
- (iii) evidence that P and Q are τ -reduced;
- (iv) the zero-separation inequality $2\varepsilon \leq \tau$, or the stricter operational inequality $2\varepsilon < \tau$;
- (v) the theorem reference Theorem 2.41;
- (vi) exact source, scalarization, computation, proof, and artifact identities.

Missing evidence yields `undefined`. An evidenced false required inequality yields `reject`. No new independent status algebra is created.

2.11 Detector-zero transport and the rank-instability boundary

Definition 2.47 (Metric-safe detector-zero transport record). A metric-safe detector-zero transport record from B to C contains:

- (i) the complete Chapter 3 detector certificate establishing $T_\tau^{\text{bar}}(B) = 0$, including its exact profile stage;
- (ii) the two-sided threshold-gap comparison record for B and C ;
- (iii) the repaired-profile distance bound produced by Theorem 2.34;
- (iv) the post-threshold zero-separation record of Definition 2.46;
- (v) the conclusion $T_\tau^{\text{bar}}(C) = 0$;
- (vi) immutable detector, comparison, theorem, and replay artifacts.

The record transports only the exact zero predicate. It does not transport the detector sample values, ranks, birth set, germ set, sample count, or certificate size.

Corollary 2.48 (Detector-certified metric zero transport). *Assume a complete detector certificate proves*

$$T_{\tau}^{\text{bar}}(B) = 0.$$

If the hypotheses of Corollary 2.45 hold for (B, C) , then

$$T_{\tau}^{\text{bar}}(C) = 0.$$

The converse holds after exchanging B and C . Detector completeness is required only for the profile whose zero statement is obtained from detector data; the metric theorem transports the already certified zero predicate.

Proof. The detector certificate supplies the source zero statement. Apply Corollary 2.45. $\square$

Definition 2.49 (Pointwise span-rank observation). For a persistence module M , a point t , and a span $s > 0$, define the pointwise span-rank observation

$$r_{t,s}(M) := \dim_k \text{im}(M(t) \longrightarrow M(t+s)),$$

when the displayed map is defined. This is a finite observation, not by itself a complete detector certificate.

Example 2.50 (Arbitrarily small perturbation can change a pointwise detector rank). Fix $s > 0$ and $L > 0$. For any $0 < \delta < \min\{L, (s+L)/2\}$, let

$$B_{\delta} = \{[0, s+L]\}, \quad C_{\delta} = \{[\delta, s+L+\delta]\}.$$

Matching the two bars gives

$$d_B(B_{\delta}, C_{\delta}) = \delta.$$

Both bars have the same length $s+L$, so they have the same clearance from any threshold depending only on that length. Nevertheless, at $t = 0$,

$$r_{0,s}(B_{\delta}) = 1, \quad r_{0,s}(C_{\delta}) = 0.$$

Thus pointwise span-rank observations may change under arbitrarily small bottleneck perturbations, even without any bar crossing a length threshold.

Proposition 2.51 (No bottleneck stability theorem for raw pointwise detector ranks). *There is no general implication of the form*

$$d_B(B, C) \leq \varepsilon \implies r_{t,s}(B) = r_{t,s}(C)$$

for a fixed sample point t and span s , even for one-bar finite barcodes and arbitrarily small positive ε .

Proof. Apply Example 2.50 with $0 < \delta \leq \varepsilon$. $\square$

Remark 2.52 (What metric transport preserves). The v20 metric chain preserves the repaired zero/nonzero predicate under the sharp separation hypotheses. It does not preserve the raw output of a fixed-point detector. Birth, germ, coverage, mesh, endpoint, and stage conditions remain Chapter 3 non-scalar obligations and cannot be replaced by metric slack.

2.12 Window restriction and window-edge budget

Definition 2.53 (Right-open window). A domain window is a right-open interval

$$W = [u, u'),$$

where $u \in \mathbb{R}$, $u' \in \mathbb{R} \cup \{+\infty\}$, and $u < u'$. If $u' < +\infty$, W is called a bounded audit window. If $u' = +\infty$, W is called a terminal audit window.

Definition 2.54 (Window restriction). For $W = [u, u')$, define the window restriction

$$W_W(M)$$

by restricting M to W , and, when a globally indexed persistence module is required, extending by zero outside the declared window.

At the barcode level, W_W clips each bar I to

$$I \cap W,$$

discarding empty intersections.

Remark 2.55 (Infinite bars under window restriction). The infinite-bar convention of Definition 2.13 applies before windowing and on terminal windows $W = [u, \infty)$. On a bounded audit window $W = [u, u')$ with $u' < \infty$, an infinite bar $[a, \infty)$ is clipped to

$$[\max\{a, u\}, u')$$

whenever the intersection is nonempty. Thus it becomes a finite bar inside the bounded window and is then thresholded according to its clipped finite length. On a terminal audit window $W = [u, \infty)$, it remains infinite and is never deleted by T_τ^{bar} .

Proposition 2.56 (No unconditional crop–collapse commutation). *In general,*

$$W_W T_\tau^{\text{bar}}(B) \neq T_\tau^{\text{bar}} W_W(B).$$

Proof. Let $B = \{[0, 10)\}$, let $W = [0, 1)$, and take $\tau = 2$. Since the global bar has length $10 > \tau$, it survives T_τ^{bar} . Therefore

$$W_W T_\tau^{\text{bar}}(B) = \{[0, 1)\}.$$

On the other hand,

$$W_W(B) = \{[0, 1)\},$$

whose length is $1 \leq \tau$. Thus

$$T_\tau^{\text{bar}} W_W(B) = \emptyset.$$

Hence the two operations do not commute in general. $\square$

Definition 2.57 (Window-edge clearance). A window-edge clearance record for a source-level perturbation problem records a declared margin between the bar endpoints or critical parameters relevant to $\mathbf{P}_i(F)$ and the window boundary points u, u' . Its purpose is to certify that passage from source-level perturbation to windowed pre-threshold profile comparison does not silently gain or lose bars at the crop boundary.

Definition 2.58 (Crop budget). A crop budget is a declared ledger component

$$\delta_{\text{crop}}(i, W)$$

bounding the discrepancy introduced when moving from a source-level comparison to a comparison of windowed pre-threshold profiles:

$$B_{i,W}(F) = W_W \mathbf{P}_i(F).$$

Whenever a proof uses a source-level perturbation statement to infer

$$d_B(B_{i,W}(F), B_{i,W}(G)) \leq \varepsilon,$$

the contribution of window restriction, boundary clipping, endpoint uncertainty, and crop instability must either be proved zero by a window-edge clearance record or included in the ledger as $\delta_{\text{crop}}(i, W)$.

Remark 2.59 (Gap-safe stability consumes windowed profile distance). Theorem 2.34 does not assert that a raw perturbation of filtered objects automatically gives a bottleneck bound after window restriction. It consumes the hypothesis

$$d_B(B, C) \leq \varepsilon$$

for the barcode profiles to which threshold deletion is applied. In AK applications, those profiles are the windowed pre-threshold profiles

$$B_{i,W}(F) = W_W \mathbf{P}_i(F).$$

Therefore window-edge stability belongs to the ledger and is separate from threshold-gap stability.

Example 2.60 (Infinite-bar obstruction to the former finite-only condition). Let

$$B = \{[0, +\infty)\}, \quad W = [0, 1), \quad \tau = 2.$$

The infinite bar is retained by global threshold deletion, so

$$W_W T_\tau^{\text{bar}}(B) = \{[0, 1)\}.$$

After windowing first, however, the bar has finite clipped length $1 \leq \tau$, and hence

$$T_\tau^{\text{bar}} W_W(B) = \emptyset.$$

Thus any commutation criterion that checks only globally retained finite bars is insufficient: globally retained infinite bars must also be tested after clipping to a bounded window.

Definition 2.61 (τ -window-adapted condition). A pair (B, W) is called τ -window-adapted if every bar $I \in B$ retained by T_τ^{bar} , including every infinite bar, satisfies one of the following:

- (i) $I \cap W = \emptyset$;
- (ii) the clipped bar $I \cap W$ is retained by threshold deletion on the window, equivalently

$$\ell(I \cap W) > \tau \quad \text{or} \quad \ell(I \cap W) = +\infty.$$

For a bounded window, clause (ii) requires every nonempty crop of a globally retained infinite bar to have finite clipped length strictly greater than τ . For a terminal window, a nonempty crop of an infinite bar remains infinite and automatically satisfies clause (ii).

Proposition 2.62 (Exact crop–collapse commutation criterion). *For a barcode profile B , a right-open window W , and $\tau > 0$, the following are equivalent:*

(i) (B, W) is τ -window-adapted;

(ii)

$$\mathbb{W}_W T_\tau^{\text{bar}}(B) = T_\tau^{\text{bar}} \mathbb{W}_W(B)$$

as barcode multisets, including multiplicities.

Proof. Assume first that (B, W) is τ -window-adapted. If a finite bar I is deleted globally, then $\ell(I) \leq \tau$; clipping cannot increase length, so every nonempty crop $I \cap W$ is also deleted after windowing. If a bar I is retained globally, the adaptedness condition says that it is either disjoint from W or its nonempty crop is retained after windowing. Thus every bar contributes with the same multiplicity in both orders.

Conversely, suppose the two barcode multisets are equal and let I be retained globally with $I \cap W \neq \emptyset$. If $I \cap W$ were a finite bar of length at most τ , then the occurrence of $I \cap W$ contributed by I would appear in $\mathbb{W}_W T_\tau^{\text{bar}}(B)$ but would be deleted from $T_\tau^{\text{bar}} \mathbb{W}_W(B)$. The multiplicities would differ, contradicting equality. Hence the crop is retained, and (B, W) is τ -window-adapted. $\square$

Remark 2.63 (Default policy and audit use). The default Core policy is not to assume τ -window-adaptedness. Gate-relevant quantities are evaluated by the fixed order

$$\mathbf{P}_i(F) \mapsto \mathbb{W}_W \mathbf{P}_i(F) \mapsto T_\tau^{\text{bar}}(\mathbb{W}_W \mathbf{P}_i(F)).$$

Conditional commutation may be used only when Definition 2.61 is verified for all globally retained bars, including infinite bars, and the verification is recorded in the manifest.

2.13 Core windowed evaluation

Definition 2.64 (Windowed collapsed persistence evaluation). Let $F \in \text{FiltCh}(\cdot)k$, let i be a monitored degree, let W be a declared right-open window, and let $\tau > 0$. The Core windowed collapsed evaluation is

$$\text{Eval}_{i,W,\tau}(F) := T_\tau^{\text{bar}}(\mathbb{W}_W \mathbf{P}_i(F)) = T_\tau^{\text{bar}}(B_{i,W}(F)).$$

Remark 2.65 (Fixed order). The expression

$$\text{Eval}_{i,W,\tau}(F) = T_\tau^{\text{bar}}(\mathbb{W}_W \mathbf{P}_i(F))$$

means the following ordered pipeline:

$$F \mapsto \mathbf{P}_i(F) \mapsto B_{i,W}(F) = \mathbb{W}_W \mathbf{P}_i(F) \mapsto T_\tau^{\text{bar}}(B_{i,W}(F)).$$

The Core first restricts to the declared window and only then applies hard threshold deletion.

Definition 2.66 (Repaired evaluation record). A repaired evaluation record for $(F; i, W, \tau)$ contains:

- (i) immutable source-object and profile identifiers;
- (ii) the monitored degree i , declared window W , and threshold τ ;
- (iii) the windowed pre-threshold profile $B_{i,W}(F)$;
- (iv) the repaired profile

$$\text{Eval}_{i,W,\tau}(F) = T_\tau^{\text{bar}}(B_{i,W}(F));$$

(v) the comparison mode, chosen from

`exact, metric_gap_certified, not_compared;`

(vi) reconstructible artifact references.

Definition 2.67 (Threshold-gap-certified repaired evaluation comparison). A metric comparison between two repaired evaluation records is *threshold-gap-certified* only when the compared windowed pre-threshold profiles are exactly the profiles appearing in a valid two-sided record of Definition 2.32. The record must establish both

$$d_B(B, C) \leq \varepsilon_B$$

and

$$\varepsilon_B < \text{gap}_\tau(B, C).$$

The associated comparison mode is `metric_gap_certified`. A separately proved equality or profile isomorphism is recorded instead with comparison mode `exact`; it does not require a fabricated positive-radius gap certificate and does not by itself assert robustness under perturbation.

Remark 2.68 (Evaluation is definable without certification). The profile $\text{Eval}_{i, W, \tau}(F)$ is definitionally available once the windowed barcode profile is given. Whenever a metric discrepancy is transported through hard deletion—including metric overlap, Restart, representative, or tower comparisons—the stronger two-sided threshold-gap comparison record is required. A separately proved exact equality or profile isomorphism may instead be used in `exact` mode; such an exact record is not automatically a positive-radius robustness certificate.

2.14 Overlap Gate: post-threshold comparison and pre-threshold gap

The Overlap Gate has two different metric layers. This distinction is essential in v19.

Definition 2.69 (Overlap pre-threshold profiles). For two local inputs F_α, F_β and an overlap window

$$W_{\alpha\beta} := W_\alpha \cap W_\beta,$$

define the overlap pre-threshold profiles by

$$B_{i, W_{\alpha\beta}}^\alpha := W_{W_{\alpha\beta}} \mathbf{P}_i(F_\alpha), \quad B_{i, W_{\alpha\beta}}^\beta := W_{W_{\alpha\beta}} \mathbf{P}_i(F_\beta).$$

The overlap repaired profiles are

$$\text{Eval}_{i, W_{\alpha\beta}, \tau}(F_\alpha) = T_\tau^{\text{bar}}(B_{i, W_{\alpha\beta}}^\alpha), \quad \text{Eval}_{i, W_{\alpha\beta}, \tau}(F_\beta) = T_\tau^{\text{bar}}(B_{i, W_{\alpha\beta}}^\beta).$$

Remark 2.70 (Two-level structure of the Overlap Gate). The metric comparison clause of the Overlap Gate compares the *post-threshold* repaired profiles:

$$d_{\text{int}}(\text{Eval}_{i, W_{\alpha\beta}, \tau}(F_\alpha), \text{Eval}_{i, W_{\alpha\beta}, \tau}(F_\beta)).$$

By contrast, the threshold-gap clause is computed on the *pre-threshold* profiles:

$$B_{i, W_{\alpha\beta}}^\alpha, \quad B_{i, W_{\alpha\beta}}^\beta.$$

Thus the gate deliberately uses two barcode levels:

gap is measured before deletion, agreement is checked after deletion.

Proposition 2.71 (Overlap transfer under a scalarized threshold-gap certificate). *Let $\mathfrak{d}_{\alpha\beta,i}$ be the declared overlap metric ledger and set*

$$\varepsilon_{\alpha\beta,i} := \text{val}_B(\mathfrak{d}_{\alpha\beta,i}).$$

Suppose

$$d_B(B_{i,W_{\alpha\beta}}^\alpha, B_{i,W_{\alpha\beta}}^\beta) \leq \varepsilon_{\alpha\beta,i}.$$

If the two-sided threshold-gap record satisfies

$$\varepsilon_{\alpha\beta,i} < \text{gap}_\tau(B_{i,W_{\alpha\beta}}^\alpha, B_{i,W_{\alpha\beta}}^\beta),$$

then

$$d_B(\text{Eval}_{i,W_{\alpha\beta},\tau}(F_\alpha), \text{Eval}_{i,W_{\alpha\beta},\tau}(F_\beta)) \leq \varepsilon_{\alpha\beta,i}.$$

Equivalently, by the bottleneck–interleaving isometry in the declared constructible regime, the same bound may be read using d_{int} .

Proof. This is Corollary 2.37 applied to the two overlap pre-threshold profiles and the overlap metric ledger $\mathfrak{d}_{\alpha\beta,i}$. The final sentence follows from Remark 2.15. $\square$

Corollary 2.72 (Overlap repaired-zero equivalence). *Under the hypotheses of Proposition 2.71, assume in addition*

$$2\varepsilon_{\alpha\beta,i} \leq \tau.$$

Then

$$\text{Eval}_{i,W_{\alpha\beta},\tau}(F_\alpha) = 0 \iff \text{Eval}_{i,W_{\alpha\beta},\tau}(F_\beta) = 0.$$

For audit consumption, the strict reserve $2\varepsilon_{\alpha\beta,i} < \tau$ may be required.

Proof. Proposition 2.71 supplies the repaired-profile bottleneck bound. Both repaired profiles are τ -reduced. Apply Theorem 2.41. $\square$

2.15 Exact-admissible regimes

The general Core does not treat T_τ^{bar} as a global exact functor. Nevertheless, stronger exact calculus may be available under separately declared hypotheses.

Definition 2.73 (τ -split object). An object $M \in \text{Pers}_k^{\text{cons}}$ is called τ -split if it is equipped with a declared decomposition

$$M \cong M_{\leq \tau} \oplus M_{> \tau}$$

such that:

- (i) every finite bar of $M_{\leq \tau}$ has length at most τ ;
- (ii) every finite bar of $M_{> \tau}$ has length greater than τ ;
- (iii) all infinite bars, if any, lie in $M_{> \tau}$;
- (iv) the decomposition is part of the declared data, not merely inferred informally.

Definition 2.74 (τ -split morphism). A morphism

$$f : M \rightarrow N$$

between τ -split objects is called τ -split if it respects the declared decompositions:

$$f(M_{\leq \tau}) \subseteq N_{\leq \tau}, \quad f(M_{> \tau}) \subseteq N_{> \tau}.$$

Definition 2.75 (τ -split regime). A τ -split regime is a declared working subcategory or workflow domain in which all objects and all morphisms under consideration are τ -split in the above sense.

Proposition 2.76 (Functorial deletion in a τ -split regime). *Inside a declared τ -split regime, the assignment*

$$M \mapsto M_{> \tau}$$

is functorial and agrees with $T_{\tau}^{\text{bar}}(M)$ on barcode profiles.

Proof. Since every morphism respects the declared short/long decomposition, it restricts to a morphism

$$M_{> \tau} \rightarrow N_{> \tau}.$$

Therefore $M \mapsto M_{> \tau}$ is functorial in the τ -split regime. By construction, $M_{> \tau}$ consists exactly of the bars retained by T_{τ}^{bar} . $\square$

Definition 2.77 (Exact-admissible collapse regime). An exact-admissible collapse regime is a declared τ -split regime in which the retained-long-bar assignment

$$M \mapsto M_{> \tau}$$

is additionally verified to preserve the exact sequences used in the relevant Core argument.

Remark 2.78 (Exactness is conditional). Exactness of positive-threshold collapse is not a global Core theorem. It is a conditional property available only in an exact-admissible regime. Whenever such a regime is used, the proof object must declare:

- (i) the τ -split decomposition policy;
- (ii) the morphism compatibility policy;
- (iii) the exact sequences being tested;
- (iv) the artifact or proof establishing exact preservation.

2.16 Filtered representatives of windowed collapse

Because T_{τ}^{bar} is treated in the general Core as a barcode-level operation, filtered-level collapse must be phrased in terms of representatives.

Definition 2.79 (Windowed collapsed persistence profile). For $F \in \text{FiltCh}(()k)$, monitored degree i , window W , and threshold τ , define the monitored windowed collapsed profile by

$$\mathcal{P}_{i,W,\tau}(F) := \text{Eval}_{i,W,\tau}(F) = T_{\tau}^{\text{bar}}(B_{i,W}(F)).$$

Definition 2.80 (Admissible filtered representative). Let $F \in \text{FiltCh}(\langle \rangle k)$, let W be a declared window, and let $\tau > 0$. A filtered object

$$C_{\tau, W}F$$

is called an admissible filtered representative of the windowed collapse of F if, for every monitored degree i ,

$$\mathbf{P}_i(C_{\tau, W}F) \cong \mathcal{P}_{i, W, \tau}(F)$$

at the persistence barcode level, up to the declared representative equivalence policy.

Remark 2.81 (No global strict lift by default). The general Core does not assert a strict global endofunctor

$$C_\tau : \text{FiltCh}(\langle \rangle k) \rightarrow \text{FiltCh}(\langle \rangle k)$$

realizing positive-threshold collapse in every degree and on every morphism. The default object is the windowed representative $C_{\tau, W}F$, whose existence and properties must be declared in the proof object or implementation record.

Definition 2.82 (Representative equivalence). Two admissible representatives $C_{\tau, W}F$ and $C'_{\tau, W}F$ are considered equivalent for Core gate purposes if their monitored persistence profiles agree:

$$\mathbf{P}_i(C_{\tau, W}F) \cong \mathbf{P}_i(C'_{\tau, W}F)$$

for all monitored degrees i , under the declared barcode equivalence policy.

Remark 2.83 (Filtered quasi-isomorphism). When filtered quasi-isomorphism data are available, representative equivalence may be strengthened to filtered quasi-isomorphism. However, filtered quasi-isomorphism is not required merely to define the windowed collapsed barcode profile. It becomes necessary only when the argument uses filtered-level or derived-level constructions.

2.17 Consequences for $B\text{-Gate}^+$

The repaired Chapter 2 affects how the Core gate is read.

Definition 2.84 (Windowed B -side input). The B -side input for a monitored degree i , window W , and threshold τ is the collapsed profile

$$\text{Eval}_{i, W, \tau}(F) = T_\tau^{\text{bar}}(B_{i, W}(F)),$$

or an admissible filtered representative $C_{\tau, W}F$ realizing this profile.

Definition 2.85 (Windowed topological clause). The topological clause of $B\text{-Gate}^+$ on $(F; W, \tau)$ is read as

$$\text{Eval}_{1, W, \tau}(F) = 0$$

at the monitored persistence-profile level. Equivalently, if an admissible representative $C_{\tau, W}F$ is supplied and realizes the declared profile, the clause may be read as

$$\text{PH}_1(C_{\tau, W}F) = 0.$$

Definition 2.86 (Windowed categorical clause). The categorical clause of $B\text{-Gate}^+$ may be invoked only when an admissible filtered representative $C_{\tau, W}F$ is supplied and the eligible realization condition holds:

$$R(C_{\tau, W}F) \in D^{[-1, 0]}(k\text{-mod}).$$

Only then is the Ext-clause

$$\text{Ext}^1(R(C_{\tau, W}F), k) = 0$$

well-typed for the Core bridge.

Definition 2.87 (Comparison-typed B -Gate⁺ input). A B -Gate⁺ input records:

- (i) the post-threshold repaired profile $\text{Eval}_{1,W,\tau}(F)$;
- (ii) the pre-threshold profile $B_{1,W}(F)$;
- (iii) the comparison mode for every comparison involving that profile;
- (iv) for `metric_gap_certified` mode, the two compared pre-threshold profiles, their two-sided threshold-gap comparison record, and every ledger component needed to justify the actual bottleneck bound, including any crop budget;
- (v) for `exact` mode, the equality or profile-isomorphism proof and its artifact reference.

A single-profile clearance value is readiness information only and is not substituted for a two-sided metric comparison record.

Remark 2.88 (PH before Ext). The barcode-level profile $\text{Eval}_{1,W,\tau}(F)$ is the primary persistence datum. The Ext-clause is not a replacement for it. Ext is read only through an eligible representative and the one-way bridge established later.

2.18 Tower comparison and typed Type IV input

Since T_τ^{bar} is not a global strict exact functor in the general Core, tower comparison maps must be declared rather than assumed. The full terminal-generic diagnostic is defined in Chapter 5; the present section fixes only the well-typed input contract.

Definition 2.89 (Windowed collapsed tower objects). Let

$$F_0 \rightarrow F_1 \rightarrow F_2 \rightarrow \dots$$

be a directed tower in the declared filtered or persistence regime. For a monitored degree i , window W , and threshold τ , write

$$M_{n;i,W,\tau} := \text{Eval}_{i,W,\tau}(F_n).$$

The sequence of objects $M_{n;i,W,\tau}$ is not by itself a directed system: actual profile morphisms and their composition laws must be declared, unless supplied by a τ -split or exact-admissible regime.

Definition 2.90 (Tower comparison artifact). A tower comparison artifact for (i, W, τ) is declared data specifying:

- (i) a repaired tower-profile system with actual persistence-profile morphisms;
- (ii) a source profile $\text{ColimProfile}_{i,W,\tau}(F_\bullet)$ with declared provenance;
- (iii) an apex profile $\text{ApexProfile}_{i,W,\tau}(F_\infty)$;
- (iv) an actual comparison morphism

$$\phi_{i,W,\tau} : \text{ColimProfile}_{i,W,\tau}(F_\bullet) \longrightarrow \text{ApexProfile}_{i,W,\tau}(F_\infty);$$

- (v) the typing, kernel/cokernel, terminal-tameness, and artifact data required by Chapter 5.

A sequence of barcode multisets or bottleneck matchings without actual profile morphisms is not a tower comparison artifact.

Remark 2.91 (No automatic tower comparison). The comparison morphism $\phi_{i,W,\tau}$ is not produced automatically by hard threshold deletion, by a barcode matching, or by the existence of a source tower and an apex object. It must be explicitly constructed and typed, or supplied canonically by a declared exact-admissible regime.

Definition 2.92 (Typed Type IV scope status). Every Chapter 2 run records exactly one Type IV scope status:

`not_invoked,      undefined,      certified_zero,      obstructed.`

They mean respectively:

- (i) no tower, apex, limit, or colimit comparison belongs to the claim;
- (ii) tower semantics are invoked, but the artifact or a required terminal-generic input is missing or ill-typed;
- (iii) a valid artifact exists and the Chapter 5 terminal-generic kernel/cokernel diagnostic is zero;
- (iv) a valid artifact exists and that diagnostic is nonzero.

A missing artifact is `undefined`, never a numerical zero and never `not_invoked` unless the scope policy explicitly excludes tower semantics.

Definition 2.93 (Collapse diagnostics notation contract). For a valid tower comparison artifact whose kernel and cokernel defect profiles are terminal-tame on W , Chapter 5 defines

$$\mu_{i,W,\tau} := \text{tgdim}_W(\ker \phi_{i,W,\tau}), \quad \nu_{i,W,\tau} := \text{tgdim}_W(\text{coker } \phi_{i,W,\tau}).$$

On a right-unbounded window, tgdim_W is the eventual generic dimension and counts defect bars cofinal to $+\infty$. On a bounded window, it is the left limit of the dimension at the right endpoint and counts defect bars cofinal to that endpoint; it must not be replaced by the dimension after zero extension to $+\infty$. For a fixed declared pair (W, τ) and a declared finite monitored degree set I_{mon} , define

$$\mu_{\text{Collapse}} := \sum_{i \in I_{\text{mon}}} \mu_{i,W,\tau}, \quad \nu_{\text{Collapse}} := \sum_{i \in I_{\text{mon}}} \nu_{i,W,\tau}.$$

An infinite monitored scope requires a separately declared aggregation or summability theorem. The legacy symbols $u_{i,W,\tau}$ and u_{Collapse} are deprecated; $\nu_{i,W,\tau}$ and ν_{Collapse} remain the sole canonical cokernel-diagnostic notation in v20.0.

Remark 2.94 (Type IV remains Core, but zero is typed). Type IV remains part of the Core gate discipline, but its comparison map is not silently inherited from threshold deletion. The shorthand

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$$

may be used only under status `certified_zero`. Under `obstructed`, a valid diagnostic is nonzero. Under `undefined` or `not_invoked`, no numerical pair is assigned. In particular, missing or excluded tower data are never represented by a fabricated zero pair. The zero pair means absence of monitored terminal-generic kernel and cokernel defect relative to the declared artifact; it does not imply that the comparison morphism is an isomorphism on finite interior bars.

Remark 2.95 (Object-distance control does not automatically control defect profiles). A bottleneck bound between the source and target objects of two comparison morphisms does not, by itself, produce bottleneck bounds between their kernels or cokernels. Metric transport of a detector on `kernel_defect` or `cokernel_defect` therefore requires actual comparison morphisms between the defect profiles, or a separately proved kernel/cokernel stability theorem with the required commutative data. Object proximity cannot manufacture Type IV or transient-defect evidence.

2.19 Manifest consequences

A certified proof object using Chapter 2 must record the following data.

Definition 2.96 (Collapse policy manifest entry). A collapse policy manifest entry contains:

- (i) the threshold τ , monitored degrees i , and window W ;
- (ii) the window type: bounded audit window or terminal audit window;
- (iii) the evaluation order

$$\mathbf{P}_i \rightarrow W_W \mathbf{P}_i \rightarrow T_\tau^{\text{bar}}(W_W \mathbf{P}_i);$$
- (iv) the collapse status

$$\text{barcode_level}, \quad \text{tau_split}, \quad \text{exact_admissible};$$
- (v) the repaired evaluation record and comparison mode for every comparison;
- (vi) for each metric discrepancy transported through hard deletion, the two-sided threshold-gap comparison record; for each exact comparison, the exact equality or profile-isomorphism proof;
- (vii) for every metric transport of a repaired zero predicate, the post-threshold zero-separation record, including the bound $2\varepsilon \leq \tau$ or the stricter operational reserve;
- (viii) for every detector-derived zero statement, the complete Chapter 3 detector certificate and, when transported, the metric-safe detector-zero transport record;
- (ix) the metric ledger $\mathfrak{d}_{\text{met}}$, bottleneck scalarization val_B , and scalarized budget ε_B whenever metric mode is invoked;
- (x) window-edge clearance or crop-budget data when source-level perturbations are transported through window restriction;
- (xi) any τ -window-adapted commutation certificate, including explicit verification of globally retained infinite bars on bounded windows;
- (xii) whether an admissible filtered representative $C_{\tau, W}F$ is supplied;
- (xiii) the typed Type IV status and, if invoked, the tower comparison artifact, terminal-generic inputs, monitored degree set, and canonical μ/ν diagnostics;
- (xiv) the repair record documenting the replacement of global non-expansiveness by threshold-gap-safe stability, the repair of the former finite-only crop condition, and the migration from legacy u to canonical ν notation.

Definition 2.97 (Metric and crop repair record). A Chapter 2 repair record contains:

- (i) the deprecated claim being avoided: global bottleneck non-expansiveness of T_τ^{bar} ;
- (ii) the replacement theorem: Theorem 2.34;
- (iii) the counterexample reference: Example 2.23;
- (iv) the relevant pre-threshold profiles and two-sided threshold-gap certificate;

(v) the metric ledger, scalarization, and typed inequality

$$\text{val}_B(\mathfrak{d}_{\text{met}}) < \text{gap}_\tau;$$

(vi) any crop budget required to obtain the windowed pre-threshold profile comparison;

(vii) if crop–collapse commutation is used, the exact criterion of Proposition 2.62;

(viii) explicit confirmation that globally retained infinite bars were included in that criterion;

(ix) the Type IV scope status and artifact reference when tower semantics are invoked;

(x) every post-threshold zero-separation record consumed by the proof;

(xi) explicit confirmation that no raw pointwise detector rank or certificate-size stability was inferred from bottleneck proximity;

(xii) confirmation that $\nu_{i,W,\tau}$ and ν_{Collapse} are used canonically and that legacy u -notation is confined to deprecation metadata.

Remark 2.98 (Audit consequence). A proof object that simply states “apply T_τ ” is incomplete under v20. It must state whether T_τ means barcode-level hard threshold deletion, a τ -split functorial model, or an exact-admissible collapse regime. If a metric discrepancy is transported through hard deletion, it must cite a two-sided scalarized threshold-gap comparison record rather than global non-expansiveness. If the transported conclusion is a repaired zero predicate, it must additionally cite a post-threshold zero-separation record; threshold-gap stability alone bounds distance but does not identify zero and nonzero profiles. A separately proved exact comparison is recorded in exact mode with its proof and is not misrepresented as a positive-radius gap certificate. If crop–collapse commutation is used, it must verify all globally retained bars, including infinite bars clipped by bounded windows. If Type IV is invoked, it must provide an actual comparison morphism and the typed terminal-generic diagnostic input of Chapter 5.

2.20 Operating summary

For each monitored object F , degree i , window W , and threshold τ , the repaired Chapter 2 pipeline remains

$$F \mapsto \mathbf{P}_i(F) \mapsto B_{i,W}(F) = W_W \mathbf{P}_i(F) \mapsto T_\tau^{\text{bar}}(B_{i,W}(F)) = \text{Eval}_{i,W,\tau}(F).$$

Threshold clearance is measured on $B_{i,W}(F)$, not on the repaired profile. The metric-safe repaired-zero chain is

$$\begin{aligned} \mathfrak{d}_{\text{met}} &\xrightarrow{\text{val}_B} \varepsilon_B \geq d_B(B, C), & \varepsilon_B < \text{gap}_\tau(B, C) \\ &\Downarrow \\ &d_B(T_\tau^{\text{bar}} B, T_\tau^{\text{bar}} C) \leq \varepsilon_B \\ &\Downarrow \quad 2\varepsilon_B \leq \tau \\ &T_\tau^{\text{bar}} B = 0 \iff T_\tau^{\text{bar}} C = 0. \end{aligned}$$

The mathematically sharp zero-separation boundary is $2\varepsilon_B \leq \tau$; the v20 operational contract uses the strict reserve $2\varepsilon_B < \tau$.

A complete detector may establish the source zero predicate. Chapter 2 then transports that predicate through the displayed metric chain. It does not transport raw detector ranks, sample values, birth/germ sets, coverage, certificate size, or completeness. Those remain Chapter 3 obligations.

The default evaluation order is window first and threshold second. Crop–collapse commutation is available only under Proposition 2.62, including the bounded-window treatment of globally infinite bars. Source-level perturbations require crop and window-edge control before they become bounds on the actual pre-threshold profiles.

Filtered or derived arguments require an admissible representative $C_{\tau, W}F$. Tower and transient-defect arguments require actual comparison morphisms. Metric proximity of tower objects does not automatically control kernel or cokernel defect profiles. Exactness or strict functoriality requires a declared τ -split or exact-admissible regime.

2.21 Explicit non-claims

The following are not asserted in this chapter.

- (i) No global non-expansiveness of hard positive-threshold deletion under the standard bottleneck metric is asserted.
- (ii) No claim is made that positive-threshold short-bar objects form a Serre subcategory in general.
- (iii) No claim is made that T_{τ}^{bar} is a global exact endofunctor on all of $\text{Pers}_k^{\text{cons}}$.
- (iv) No claim is made that T_{τ}^{bar} is a global abelian reflector or Serre localization.
- (v) No unconditional commutation
$$W_W T_{\tau}^{\text{bar}} = T_{\tau}^{\text{bar}} W_W$$
is asserted.
- (vi) No finite-only window-adapted condition is sufficient: globally retained infinite bars must also be checked after clipping.
- (vii) No raw source-level perturbation implies a windowed pre-threshold profile bound without a crop-stability argument or crop budget.
- (viii) No unscalarized quantale element is compared directly with the real-valued clearance $\text{gap}_{\tau} = \text{clear}_{\tau}$.
- (ix) No numerical budget repairs a missing representative, comparison morphism, eligibility certificate, terminal-tameness certificate, claim status, or manifest record.
- (x) No threshold-gap certificate is one-sided.
- (xi) No single-profile clearance or gap-readiness record is treated as a two-sided metric comparison certificate.
- (xii) No finite bar of length exactly τ is treated as gap-safe at threshold τ .
- (xiii) No 2ε -band is claimed to be sharp for the two-sided theorem, and no factor convention $\text{gap}_{\tau} = \frac{1}{2} \text{clear}_{\tau}$ is used.
- (xiv) No global strict filtered lift $C_{\tau} : \text{FiltCh}(()k) \rightarrow \text{FiltCh}(()k)$ is asserted by default.
- (xv) No tower comparison map after threshold deletion is automatic unless an actual artifact or exact-admissible regime is declared.
- (xvi) No barcode matching by itself defines the kernel or cokernel of a tower comparison.

- (xvii) No undefined Type IV diagnostic is interpreted as zero, and no no-tower claim fabricates a numerical zero instead of recording `not_invoked`.
- (xviii) No equality $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$ implies full comparison-map isomorphism without a no-interior-defect theorem.
- (xix) No legacy symbol $u_{i,W,\tau}$ or u_{Collapse} has normative diagnostic meaning in v20.0.
- (xx) No bounded-window cofinal bar is called a global infinite bar.
- (xxi) No Ext-clause is well-typed without an admissible filtered representative and an eligible realization condition.
- (xxii) No Search artifact, advisory invariant, spectral indicator, or AI-generated ranking replaces the windowed collapsed persistence evaluation or any typed proof obligation.
- (xxiii) No repaired-zero equivalence is inferred from threshold-gap stability alone; the zero-separation inequality is an additional theorem hypothesis.
- (xxiv) No metric zero transport is asserted when $2\varepsilon > \tau$; counterexamples exist even inside the τ -reduced image.
- (xxv) No raw pointwise detector rank, birth set, germ set, sample count, coverage witness, or certificate size is bottleneck-stable by default.
- (xxvi) No detector soundness statement is promoted to completeness by metric proximity or by repeated zero observations.
- (xxvii) No metric-safe detector transport changes the detector profile stage; a stage-transport theorem is required.
- (xxviii) No object-level bottleneck bound automatically gives a bound on kernel or cokernel defect profiles.
- (xxix) No finite detector certificate supplied by Chapter 3 is proved in this chapter merely because its zero predicate can be transported metrically.

2.22 Provenance and status

The constructible persistence setting, interval decomposition, barcode uniqueness, bar-length convention, and bottleneck/interleaving isometry are standard persistence-theoretic facts in the one-parameter constructible regime over a field, interpreted here under Definition 2.13.

The inherited v19 Core content of this chapter is the metric and crop repair: hard positive-threshold deletion is not globally non-expansive under the standard bottleneck metric; its admissible stability use is governed by sharp two-sided threshold-gap-safe stability; and crop-collapse commutation is governed by the exact all-retained-bars criterion of Proposition 2.62, including infinite bars clipped by bounded windows. The v20 conservative extension adds the sharp zero-separation theorem inside the τ -reduced image, the resulting metric-safe repaired-zero transport theorem, and the explicit boundary showing that pointwise detector ranks are not bottleneck-stable. Ledger use remains typed through a bottleneck scalarization and a comparison-specific discrepancy proof. Type IV input is forwarded to Chapter 5 using canonical μ/ν notation and actual comparison artifacts.

Status labels for this chapter. *Standard persistence theorem:* Theorem 2.10.

Metric-repair and repaired-zero transport theorems: Theorem 2.34, Corollary 2.37, Theorem 2.41, Theorem 2.44, Corollary 2.45, Corollary 2.48, Corollary 2.72.

Core correction results: Proposition 2.16, Corollary 2.17, Example 2.23, Corollary 2.25, Proposition 2.56, Example 2.60, Proposition 2.62, Example 2.50, Proposition 2.51.

Core definitions: Definitions 2.4, 2.6, 2.9, 2.13, 2.19, 2.26, 2.27, 2.28, 2.30, 2.61, 2.64, 2.80, 2.90, 2.92, 2.93, 2.38, 2.39, 2.49.

Operational policy and records: Definitions 2.32, 2.57, 2.58, 2.66, 2.67, 2.46, 2.47, 2.96, 2.97; Remarks 2.65, 2.63, 2.91, 2.94.

Exact-admissible optional regime: Definitions 2.73, 2.74, 2.75, 2.77; Proposition 2.76.

Explicit exclusions: Subsection 2.21.

3 Chapter 3: Windowed Evaluation, Finite Detector Calculus, and Representative Discipline

Chapter 3 contract. This chapter is the finite-certificate layer between the inherited metric repair of Chapter 2 and Appendix A and the eligible realization and higher Ext-edge of Chapter 4. It preserves the complete v19 repaired evaluation, comparison, crop, and representative discipline and adds the v20 finite zero-reflecting detector calculus.

The standard detector reads the declared windowed pre-threshold profile

$$B_{n,W}(F) = W_W \mathbf{P}_n(F)$$

through finitely many structure maps

$$B(t) \longrightarrow B(t + \tau).$$

A finite observation becomes a finite proof certificate only after its profile stage, endpoint convention, soundness, completeness or coverage theorem, family membership, excess-bound source, and immutable artifacts are certified. Missing completeness evidence yields an undefined zero verdict, never a fabricated vanishing theorem.

The chapter proves exact birth and right-germ characterizations, externally certified stencil completeness, family-mesh completeness with augmented boundary gaps, its strict sharpness boundary, family-relative minimal certificate bounds, and a right-unbounded-window no-go theorem. It also records the exact composition with the metric-safe repaired-zero transport of Chapter 2. It does not claim pointwise detector-rank stability under bottleneck perturbation.

The chapter does *not* prove a higher categorical bridge

$$\mathbf{PH}_n \Longrightarrow \mathbf{Ext}^n.$$

That bridge belongs to Chapter 4. Nor does this chapter identify a finite detector with a natural Ext-edge. It supplies only the repaired profile and finite-certificate predicates that a separately eligible natural edge may consume.

All numerical metric budgets are typed through a declared bottleneck scalarization

$$\text{val}_B : V \longrightarrow [0, +\infty].$$

Every metric transport through hard deletion uses either the two-sided threshold-gap and repaired-zero chain of Chapter 2 or a separately proved exact comparison. Coverage, endpoint, birth/germ exhaustiveness, family-wide excess, representative existence, realization, morphism, claim-status, manifest, and artifact obligations are non-scalar.

The Chapter 3 record extension is subordinate to the canonical audit manifest of Chapter 7. The specialized detector statuses refine, but do not replace, the Chapter 7 atomic status calculus.

3.1 Scope and standing discipline

All statements in this chapter are made in the v20 conservative-extension regime fixed by Chapters 1–2 and Appendix A. The repaired v19 metric and record contracts are inherited without weakening.

In particular:

- (i) persistence data live in constructible one-parameter persistence over a field k ;
- (ii) barcode-level threshold deletion T_τ^{bar} deletes precisely finite bars of length $\leq \tau$ and preserves infinite bars before bounded-window clipping;
- (iii) global non-expansiveness of T_τ^{bar} under the standard bottleneck metric is not asserted;
- (iv) the replacement stability principle is sharp two-sided threshold-gap-safe stability with closed band

$$[\tau - \varepsilon, \tau + \varepsilon];$$
- (v) the ledger margin is

$$\text{gap}_\tau = \text{clear}_\tau,$$
 with gap_τ retained as the ledger-facing alias of clear_τ ;
- (vi) right-open windows $W = [u, u')$ are part of the certified data;
- (vii) no unconditional crop–collapse commutation is used;
- (viii) a filtered object $C_{\tau, W}F$ is an admissible representative of a repaired windowed barcode profile, not a global collapse functor;
- (ix) the metric ledger lives in a declared commutative quantale

$$(\mathbb{V}, \oplus, 0_{\mathbb{V}}, \preceq),$$

and is compared with real-valued clearance only after application of a declared monotone subadditive bottleneck scalarization $\text{val}_{\mathbb{B}}$;

- (x) no numerical budget repairs a missing representative, realization, comparison morphism, eligibility certificate, claim-status entry, manifest field, or artifact identity;
- (xi) if crop–collapse commutation is invoked, the Chapter 2 / Appendix A condition is checked on every globally retained bar, including infinite bars clipped by bounded windows;
- (xii) each comparison is explicitly typed as

exact, metric_gap_certified, not_compared;

- (xiii) every detector declares exactly one profile stage and may be consumed only at the strength proved for that stage;
- (xiv) exact point detectors use a declared left-closed right-open barcode convention; decorated endpoints use the right-germ theorem or another separately proved decoration-safe theorem;
- (xv) detector soundness and detector completeness are separate mathematical obligations, while Chapter 1 governs their joint admission as a Core zero certificate;
- (xvi) family-mesh completeness uses an augmented mesh containing both boundary gaps of the effective detector domain;
- (xvii) a family-wide threshold-excess lower bound must come from an independently declared model, separation theorem, filtration rule, or external artifact rather than retrospective full reading of the target barcode when proof compression is claimed.

Remark 3.1 (Why Chapter 3 moved before the Ext-edge). Earlier drafts placed eligibility and the Ext-edge in Chapter 3. The v19 metric repair made the record discipline logically prior. The v20 extension adds a second logically prior layer: one must know whether a finite detector is sound, complete, stage-correct, endpoint-correct, and artifact-backed before its vanishing predicate is passed to a higher edge. For this reason all eligible realization and Ext-edge statements remain deferred to Chapter 4.

3.2 Pre-threshold windowed profiles

Definition 3.2 (Pre-threshold windowed profile). Let $F \in \text{FiltCh}(\cdot)k$, let i be a monitored degree, and let

$$W = [u, u')$$

be a declared right-open window. The *pre-threshold windowed profile* is the barcode profile

$$B_{i,W}(F) := W_W \mathbf{P}_i(F).$$

It is the object on which threshold clearance is measured and to which any metric discrepancy consumed by hard deletion must apply.

Definition 3.3 (Repaired windowed evaluation). For a threshold $\tau > 0$, the repaired windowed evaluation is

$$\text{Eval}_{i,W,\tau}(F) := T_\tau^{\text{bar}}(B_{i,W}(F)) = T_\tau^{\text{bar}}(W_W \mathbf{P}_i(F)).$$

Remark 3.4 (The two profiles are different). The pre-threshold profile $B_{i,W}(F)$ and the repaired evaluation $\text{Eval}_{i,W,\tau}(F)$ are not interchangeable. The former is used for threshold clearance and metric-gap records. The latter is used for post-collapse gate clauses and repaired-profile comparison. Thus a valid record keeps both:

$$\text{Pre}_{i,W}(F) := B_{i,W}(F), \quad \text{Post}_{i,W,\tau}(F) := \text{Eval}_{i,W,\tau}(F).$$

Remark 3.5 (Fixed order). The Core order is

$$F \mapsto \mathbf{P}_i(F) \mapsto W_W \mathbf{P}_i(F) \mapsto T_\tau^{\text{bar}}(W_W \mathbf{P}_i(F)).$$

No identity

$$W_W T_\tau^{\text{bar}} \cong T_\tau^{\text{bar}} W_W$$

is used without a separately declared window-adapted or exact-admissible hypothesis. The window-adapted condition is the necessary-and-sufficient condition of Chapter 2 and Appendix A: every globally retained bar, including an infinite bar, must be disjoint from the window or remain retained after nonempty clipping. A finite-only check is insufficient.

3.3 Threshold clearance and ledger gap

Definition 3.6 (Threshold clearance on a windowed profile). Let B be a barcode profile with finitely many bars in the monitored record, possibly including infinite bars. For $\tau > 0$, define

$$\text{clear}_\tau(B) := \inf_{I \in B_{\text{fin}}} |\ell(I) - \tau|,$$

where B_{fin} denotes the finite bars of B . If $B_{\text{fin}} = \emptyset$, set

$$\text{clear}_\tau(B) := +\infty.$$

For an AK evaluation record, write

$$\text{clear}_\tau(F; i, W) := \text{clear}_\tau(B_{i,W}(F)).$$

Definition 3.7 (Ledger gap). The ledger gap is the clearance itself:

$$\text{gap}_\tau(B) := \text{clear}_\tau(B).$$

For a finite family of monitored pre-threshold profiles $B_1, \dots, B_m$, define

$$\text{gap}_\tau(B_1, \dots, B_m) := \min_{1 \leq j \leq m} \text{clear}_\tau(B_j).$$

Thus gap_τ is not a half-clearance in v19.0.0; it is a ledger-facing alias for the actual threshold clearance.

Remark 3.8 (Why $\text{gap}_\tau = \text{clear}_\tau$). The sharp two-sided threshold-gap-safe theorem uses the closed band

$$[\tau - \varepsilon, \tau + \varepsilon].$$

Let $\mathfrak{d}_{\text{met}} \in \mathbb{V}$ be a gate-scoped metric ledger and let

$$\varepsilon_B(\mathfrak{d}_{\text{met}}) := \text{val}_B(\mathfrak{d}_{\text{met}}).$$

The strict readiness condition

$$\varepsilon_B(\mathfrak{d}_{\text{met}}) < \text{gap}_\tau(B) = \text{clear}_\tau(B)$$

ensures that no finite bar of B lies in

$$[\tau - \varepsilon_B(\mathfrak{d}_{\text{met}}), \tau + \varepsilon_B(\mathfrak{d}_{\text{met}})].$$

For metric transport, the same budget must also upper-bound the bottleneck discrepancy between two declared profiles, and the condition must hold on both. The shorthand $\Sigma\delta < \text{gap}_\tau$ is permitted only when the run manifest identifies $\Sigma\delta$ with this scalarized real budget and records the discrepancy theorem or artifact.

Remark 3.9 (Bars of length exactly τ). If B contains a finite bar of length exactly τ , then

$$\text{clear}_\tau(B) = \text{gap}_\tau(B) = 0.$$

Such a profile is not positively gap-ready. It may still be evaluated by T_τ^{bar} , and the bar is deleted because the convention is $\ell(I) \leq \tau$. A separately proved exact identity may still be used in exact mode, but it is not a positive-radius robustness certificate.

3.4 Typed scalarized metric ledger

Definition 3.10 (Bottleneck scalarization and metric ledger). Let

$$(V, \oplus, 0_V, \preceq)$$

be the declared commutative ledger quantale used by the run. A *bottleneck scalarization* is a monotone subadditive map

$$\text{val}_B : V \longrightarrow [0, +\infty]$$

satisfying

$$\text{val}_B(0_V) = 0, \quad \text{val}_B(q \oplus q') \leq \text{val}_B(q) + \text{val}_B(q').$$

For a gate-scoped metric ledger $\mathfrak{d}_{\text{met}} \in V$, define its real bottleneck budget by

$$\varepsilon_B(\mathfrak{d}_{\text{met}}) := \text{val}_B(\mathfrak{d}_{\text{met}}).$$

A scalarized ledger may be consumed only when a theorem, verified computation, or reconstructible artifact proves the relevant pre-threshold profile bound

$$d_B(B, C) \leq \varepsilon_B(\mathfrak{d}_{\text{met}}).$$

Monotonicity is understood relative to $\preceq$ on V and the usual order on $[0, +\infty]$.

Definition 3.11 (Non-scalar record obligations). The following are non-scalar proof obligations whenever they are in scope: representative existence, profile-isomorphism data, realization existence and compatibility, actual comparison morphisms, terminal-tameness certificates, eligibility, claim-status entries, manifest completeness and consistency, and artifact provenance and identity. They are not elements of V and cannot be discharged by increasing $\varepsilon_B(\mathfrak{d}_{\text{met}})$.

Remark 3.12 (Sharp band versus scalarization tightness). The adjective *sharp* refers to the optimal half-width ε of the forbidden threshold band for a given true bottleneck bound. It does not assert that a chosen scalarization val_B is numerically tight. A conservative scalarization may enlarge the certified band, but it does not change the sharp constant in the metric theorem.

3.5 Repaired evaluation and comparison records

Definition 3.13 (Repaired evaluation record). A *repaired evaluation record* for $(F; i, W, \tau)$ contains at least:

- (i) an immutable identifier for the filtered object or persistence source F ;
- (ii) the monitored degree i ;
- (iii) the right-open window $W = [u, u')$, including whether $u' = +\infty$;
- (iv) the threshold $\tau > 0$;
- (v) the pre-threshold profile

$$B_{i,W}(F) = W_W \mathbf{P}_i(F);$$

- (vi) the repaired post-threshold profile

$$\text{Eval}_{i,W,\tau}(F) = T_\tau^{\text{bar}}(B_{i,W}(F));$$

- (vii) the finite-bar deletion convention $\ell(I) \leq \tau$;
- (viii) the infinite-bar and endpoint conventions used by the barcode metric;
- (ix) artifact references sufficient to reconstruct the data.

Remark 3.14 (Definition, not certification). A repaired evaluation record is only a typed record. It does not by itself certify stability, transport, overlap compatibility, representative compatibility, or B-Gate⁺ pass. Those uses require a two-sided metric comparison record, a separately proved exact comparison, and every applicable non-scalar obligation.

Definition 3.15 (Single-profile threshold-gap readiness record). A *single-profile threshold-gap readiness record* for $(F; i, W, \tau)$, relative to a real test radius $\varepsilon_B \geq 0$, contains:

- (i) the immutable identifier of $B_{i,W}(F)$;
- (ii) the clearance

$$\text{clear}_\tau(B_{i,W}(F));$$

- (iii) the equality

$$\text{gap}_\tau(B_{i,W}(F)) = \text{clear}_\tau(B_{i,W}(F));$$

- (iv) the strict check

$$\varepsilon_B < \text{gap}_\tau(B_{i,W}(F));$$

- (v) the resulting closed-band exclusion

$$\ell(I) \notin [\tau - \varepsilon_B, \tau + \varepsilon_B]$$

for every finite bar $I \in B_{i,W}(F)$;

- (vi) reconstructible evidence for the finite-bar summary and clearance.

For $\varepsilon_B = 0$, the strict check means $\text{clear}_\tau(B_{i,W}(F)) > 0$.

This record certifies only that one profile avoids the threshold band at a declared radius. It is *not* a metric comparison certificate, does not supply a second profile, and does not establish transport through hard deletion. The historical label `def:ch3-gap-certified-record` is retained for reference stability, but the normative role is readiness rather than comparison certification.

Definition 3.16 (Comparison record and two-sided metric certificate). Let F and G be two filtered objects or certified profile sources. A comparison record on (i, W, τ) declares exactly one mode:

$$\text{exact}, \quad \text{metric_gap_certified}, \quad \text{not_compared}.$$

In mode `exact`, the record contains a separately proved equality or profile isomorphism between the relevant pre-threshold or repaired profiles, its exact source and target, and reconstructible proof or artifact evidence. No positive-radius robustness follows merely from exactness.

In mode `metric_gap_certified`, the record is a two-sided threshold-gap comparison certificate and contains:

- (i) immutable identifiers for

$$B := B_{i,W}(F), \quad C := B_{i,W}(G);$$

(ii) one metric ledger $\mathfrak{d}_{\text{met}}$, a declared scalarization val_B , and

$$\varepsilon_B := \text{val}_B(\mathfrak{d}_{\text{met}});$$

(iii) theorem-level, verified-computation, or artifact evidence that

$$d_B(B, C) \leq \varepsilon_B;$$

(iv) $\text{clear}_\tau(B)$, $\text{clear}_\tau(C)$, and

$$\text{gap}_\tau(B, C) = \min\{\text{clear}_\tau(B), \text{clear}_\tau(C)\};$$

(v) the strict two-sided reserve

$$\varepsilon_B < \text{gap}_\tau(B, C),$$

equivalently confirmation that neither finite-bar family meets $[\tau - \varepsilon_B, \tau + \varepsilon_B]$;

(vi) every declared metric component used to establish the windowed comparison bound, including crop, endpoint, measurement, realization-metric, computation, and threshold-reconciliation terms;

(vii) reconstructible artifact references;

(viii) one specialized status

$$\text{gap_certified}, \quad \text{gap_failed}, \quad \text{gap_pending},$$

mapping respectively to pass, reject, and undefined.

Here `gap_failed` means that a well-typed required gap condition is proved or artifact-evidenced to fail. Missing, incomplete, stale, or pending evidence is recorded as `gap_pending`, never as `gap_failed`.

The metric certificate is two-sided by definition. Two single-profile readiness records without the evidenced discrepancy bound do not constitute a comparison certificate. In mode `not_compared`, no comparison inference is permitted.

Proposition 3.17 (Scalarized gap-certified stability transfer). *A comparison record on $(F, G; i, W, \tau)$ whose mode is `metric_gap_certified`, with*

$$\varepsilon_B = \text{val}_B(\mathfrak{d}_{\text{met}}),$$

implies

$$d_B(\text{Eval}_{i, W, \tau}(F), \text{Eval}_{i, W, \tau}(G)) \leq \varepsilon_B.$$

Proof. Apply the sharp threshold-gap-safe stability theorem of Chapter 2 and Appendix A to the pre-threshold profiles

$$B_{i, W}(F), \quad B_{i, W}(G).$$

The comparison record supplies the real bottleneck bound and the two-sided closed-band exclusion at that same budget. The theorem therefore yields the asserted post-threshold bound. $\square$

Remark 3.18 (Isometry convention). In the constructible one-parameter regime, d_B and d_{int} are identified by the barcode isometry theorem under the same infinite-bar cost convention fixed in Chapter 2 and Appendix A. Infinite bars are compared by birth-time displacement and are not finitely matched to finite bars or to the diagonal. A certificate may record the comparison using d_B or d_{int} only when the modules and barcodes are precisely those named in the comparison record and the metric convention is declared.

3.6 Crop budget and window-edge clearance

Definition 3.19 (Window-edge clearance record). A *window-edge clearance record* for a comparison of F and G on (i, W) records the stability of passage from the source-level data to

$$B_{i,W}(F) = W_W \mathbf{P}_i(F), \quad B_{i,W}(G) = W_W \mathbf{P}_i(G)$$

near the window endpoints. It contains:

- (i) the window endpoints u, u' and the bounded/terminal policy;
- (ii) immutable identifiers for the source and resulting windowed profiles;
- (iii) the relevant bar endpoints or critical parameters near u and u' , when required;
- (iv) a declared metric crop component

$$\delta_{\text{crop}}(i, W) \in \mathcal{V};$$

- (v) the theorem, verified computation, or artifact supporting that component and its use in the windowed bottleneck bound.

Remark 3.20 (Threshold gap does not replace crop control). Threshold clearance controls movement across the length threshold τ after windowing. It does not prove that a perturbation of F and G produces a controlled perturbation of their windowed profiles. The latter is a crop-stability question and must be supported separately through a crop component or a declared window-edge theorem. This is the Chapter 3 record-level expansion of the crop-budget discipline fixed in Chapter 2 and Appendix A; it is not an independent replacement definition.

Definition 3.21 (Profile-level metric-ledger decomposition). A profile comparison ledger may be decomposed in $\mathcal{V}$ as

$$\mathbf{d}_{\text{met}} := \delta_{\text{source,met}} \oplus \delta_{\text{crop}} \oplus \delta_{\text{measurement}} \oplus \delta_{\text{realization,met}} \oplus \delta_{\text{computation}} \oplus \delta_{\text{other,met}}.$$

Every displayed component is a declared *metric discrepancy* component; in particular, $\delta_{\text{realization,met}}$ measures only an evidenced metric distortion induced by a realization step. Realization existence, compatibility, and eligibility remain non-scalar obligations.

The real bottleneck budget is

$$\varepsilon_B := \text{val}_B(\mathbf{d}_{\text{met}}).$$

When all components are nonnegative real upper bounds and $\oplus$ is ordinary addition, this specializes to the usual sum. The threshold-gap comparison record consumes ε_B only after the run establishes

$$d_B(B_{i,W}(F), B_{i,W}(G)) \leq \varepsilon_B.$$

Undeclared or non-metric quantities do not belong to this decomposition.

Remark 3.22 (No shortcut from source perturbation). A source-level perturbation estimate is not automatically a threshold-gap comparison certificate. The certified chain is

$$\begin{aligned} \text{source comparison} &\implies (\mathbf{d}_{\text{met}}, \text{val}_B) \\ &\implies d_B(B_{i,W}(F), B_{i,W}(G)) \leq \varepsilon_B \\ &\implies d_B(\text{Eval}_{i,W,\tau}(F), \text{Eval}_{i,W,\tau}(G)) \leq \varepsilon_B, \end{aligned}$$

where $\varepsilon_B = \text{val}_B(\mathbf{d}_{\text{met}})$, and the last implication additionally consumes the two-sided gap reserve. Each arrow requires its own theorem or artifact.

3.7 Infinite bars and window policy

Definition 3.23 (Infinite-bar policy for windowed evaluation). Infinite bars are treated as follows.

- (i) Before windowing, an infinite bar has length $+\infty$ for thresholding.
- (ii) Under the standard infinite-bar bottleneck convention,
$$[a, \infty) \text{ matches } [a', \infty) \text{ with cost } |a - a'|.$$
- (iii) A finite bar and an infinite bar are not matched at finite bottleneck cost.
- (iv) If $W = [u, u']$ is bounded, window restriction may clip an infinite bar to a finite bar.
- (v) If $W = [u, \infty)$, an infinite bar intersecting W remains infinite after windowing.

Remark 3.24 (Where infinite bars affect the gate). Hard finite-threshold deletion never deletes an unwindowed infinite bar. However, bounded windowing may turn it into a finite clipped bar before thresholding. Therefore the infinite-bar convention must be read together with the declared window policy. The Core never lets an unwindowed infinite-bar rule override the actual windowed profile $B_{i,W}(F)$. If crop-collapse commutation is claimed, every globally retained infinite bar must be checked after clipping; on a bounded window it may become a finite bar of length at most τ , which destroys commutation.

3.8 Representative discipline

Definition 3.25 (Admissible filtered representative). Let $F \in \text{FiltCh}(\langle \rangle k)$, let W be a declared window, let $\tau > 0$, and let I_{mon} be the declared set of monitored degrees. A filtered object

$$C_{\tau,W}F \in \text{FiltCh}(\langle \rangle k)$$

is an *admissible filtered representative* of the repaired windowed profile family if, for every $i \in I_{\text{mon}}$, it is equipped with a declared profile isomorphism

$$\rho_i : \mathbf{P}_i(C_{\tau,W}F) \xrightarrow{\cong} \text{Eval}_{i,W,\tau}(F).$$

The isomorphisms, their source and target identities, and their equivalence strength are part of the representative data. Profile admissibility alone does not imply chain isomorphism, filtered chain-homotopy equivalence, filtered quasi-isomorphism, or derived equivalence.

Definition 3.26 (Representative record). A representative record for $C_{\tau,W}F$ contains:

- (i) the repaired evaluation records it realizes;
- (ii) an immutable identifier for $C_{\tau,W}F$;
- (iii) the monitored set I_{mon} ;
- (iv) the actual profile isomorphisms

$$\rho_i : \mathbf{P}_i(C_{\tau,W}F) \xrightarrow{\cong} \text{Eval}_{i,W,\tau}(F), \quad i \in I_{\text{mon}};$$

- (v) the declared equivalence strength;

- (vi) artifact references for the representative construction and the isomorphisms;
- (vii) whether filtered, derived, realization, or Ext-level use is requested;
- (viii) the representative-record status

`not_invoked,      undefined,      certified;`

- (ix) the claim-status, dependency, manifest, and artifact-identity entries governing the assertion.

A missing in-scope representative or profile-isomorphism artifact yields `undefined`; it is never repaired by metric slack.

Remark 3.27 (Representative is not a global collapse functor). The notation $C_{\tau, W}F$ does not assert a global functor

$$C_{\tau} : \text{FiltCh}(()k) \longrightarrow \text{FiltCh}(()k).$$

It also does not assert

$$C_{\tau, W}F = (C_{\tau}F)|_W.$$

It means only that a filtered representative has been supplied for the repaired windowed profile family in the declared monitored degrees.

Definition 3.28 (Realization-compatible representative). A representative $C_{\tau, W}F$ is *realization-compatible* for a declared later use if it is supplied with:

- (i) a typed realization route $\mathcal{R}(C_{\tau, W}F)$;
- (ii) source and target categories and functors;
- (iii) the comparison or compatibility morphisms required by that route;
- (iv) artifact evidence and the additional hypotheses required by the eligible realization regime of Chapter 4.

Realization compatibility is not automatic from persistence-profile admissibility.

Remark 3.29 (Eligibility deferred). This chapter does not define or certify the full Ext-eligible regime. It only records whether a representative is intended for such use and whether the required realization data are present. The amplitude, edge-realization, and Ext-edge conditions are stated in Chapter 4.

3.9 Representative invariance at the profile level

Proposition 3.30 (Profile-level representative invariance). *Let $C_{\tau, W}F$ and $C'_{\tau, W}F$ be admissible representatives of the same repaired windowed profile family, and let i be a common monitored degree. Then*

$$\mathbf{P}_i(C_{\tau, W}F) = 0 \iff \mathbf{P}_i(C'_{\tau, W}F) = 0.$$

Moreover, each condition is equivalent to

$$\text{Eval}_{i, W, \tau}(F) = 0.$$

Thus the monitored topological vanishing clause is independent of the choice of admissible representative at profile-isomorphism strength.

Proof. Both monitored persistence profiles are isomorphic to $\text{Eval}_{i,W,\tau}(F)$. The zero object is preserved and reflected by an isomorphism, so the three vanishing statements are equivalent. $\square$

Remark 3.31 (Derived invariance is stronger). Profile-level representative invariance does not imply chain-level, derived, or Ext-level representative invariance. If a certificate invokes such constructions, it must supply realization-compatible representative data and, when required, a filtered quasi-isomorphism, derived isomorphism, or explicit comparison theorem at the consumed strength.

3.10 Evaluation use in B-Gate⁺

For invoked higher degrees $n \in I_{\text{mon}}$, the corresponding topological and categorical clauses are governed by the Chapter 1 gate contract and Chapter 4. The definitions below preserve the baseline degree-one reading inherited from v19. The finite detector calculus developed later in this chapter is degree-generic.

Definition 3.32 (Topological clause input). The topological input for B-Gate⁺ on $(F; W, \tau)$ is

$$\text{Eval}_{1,W,\tau}(F) = T_{\tau}^{\text{bar}}(B_{1,W}(F)).$$

The topological clause passes when

$$\text{Eval}_{1,W,\tau}(F) = 0.$$

If an admissible representative $C_{\tau,W}F$ and its degree-one profile isomorphism are supplied, the same clause may be read as

$$\text{PH}_1(C_{\tau,W}F) = 0.$$

Definition 3.33 (Scalarized gap-ready topological input). The topological input is *gap-ready at radius $\varepsilon_B \geq 0$* if its pre-threshold profile satisfies

$$\varepsilon_B < \text{gap}_{\tau}(B_{1,W}(F)).$$

Equivalently, every finite bar of $B_{1,W}(F)$ avoids

$$[\tau - \varepsilon_B, \tau + \varepsilon_B].$$

This is a single-profile readiness property. A stable perturbation claim also requires a second profile and a comparison record in mode `metric_gap_certified`, including evidence that the same ε_B upper-bounds their pre-threshold bottleneck discrepancy. The historical label is retained for reference compatibility.

Remark 3.34 (Gate value versus gate safety). The equality

$$\text{Eval}_{1,W,\tau}(F) = 0$$

is a post-threshold value statement. Single-profile gap readiness is a pre-threshold band-separation statement. A robustness claim is stronger: it requires the full two-sided comparison record. None of these numerical or profile statements supplies missing representative, realization, claim-status, manifest, or artifact evidence.

Remark 3.35 (Ext-clause not discharged here). This chapter does not discharge the Ext¹-clause. It prepares the representative and evaluation records needed for the later one-way bridge. No implication

$$\text{Eval}_{1,W,\tau}(F) = 0 \implies \text{Ext}^1(\mathcal{R}(C_{\tau,W}F), k) = 0$$

is asserted in Chapter 3.

3.11 Overlap comparison discipline

Definition 3.36 (Overlap pre-threshold profile). Let W_α and W_β be overlapping windows and write $W_{\alpha\beta} := W_\alpha \cap W_\beta$. The pre-threshold overlap profiles are

$$B_{i,W_{\alpha\beta}}^\alpha := W_{W_{\alpha\beta}} \mathbf{P}_i(F_\alpha), \quad B_{i,W_{\alpha\beta}}^\beta := W_{W_{\alpha\beta}} \mathbf{P}_i(F_\beta).$$

They are evaluated directly on the common overlap window, not obtained by silently reusing already-deleted local outputs.

Definition 3.37 (Overlap repaired evaluation). At one declared common threshold τ , the repaired overlap evaluations are

$$\begin{aligned} \text{Eval}_{i,W_{\alpha\beta},\tau}(F_\alpha) &:= T_\tau^{\text{bar}}(B_{i,W_{\alpha\beta}}^\alpha), \\ \text{Eval}_{i,W_{\alpha\beta},\tau}(F_\beta) &:= T_\tau^{\text{bar}}(B_{i,W_{\alpha\beta}}^\beta). \end{aligned}$$

Remark 3.38 (Two-layer structure of the Overlap Gate). In metric-gap mode, the Overlap Gate records one scalarized budget

$$\varepsilon_{\alpha\beta,i} := \text{val}_B(\mathfrak{d}_{\alpha\beta,i}),$$

together with the pre-threshold statements

$$d_B(B_{i,W_{\alpha\beta}}^\alpha, B_{i,W_{\alpha\beta}}^\beta) \leq \varepsilon_{\alpha\beta,i},$$

$$\varepsilon_{\alpha\beta,i} < \text{gap}_\tau(B_{i,W_{\alpha\beta}}^\alpha, B_{i,W_{\alpha\beta}}^\beta).$$

The metric-repair theorem then yields the post-threshold bound

$$d_{\text{int}}(\text{Eval}_{i,W_{\alpha\beta},\tau}(F_\alpha), \text{Eval}_{i,W_{\alpha\beta},\tau}(F_\beta)) \leq \varepsilon_{\alpha\beta,i}$$

in the declared constructible isometry regime. Thus the gap and consumed discrepancy are pre-threshold, while the resulting agreement is post-threshold. In exact mode, a separately proved equality or isomorphism is recorded instead and no positive-radius robustness is inferred.

Higher overlap coherence, simplex records, typed Type IV status, and countable-atlas conditions belong to Chapter 6 and are not created by a pairwise comparison record.

Proposition 3.39 (Scalarized overlap threshold safety). Let $\mathfrak{d}_{\alpha\beta,i}$ be a declared overlap metric ledger and set

$$\varepsilon_{\alpha\beta,i} := \text{val}_B(\mathfrak{d}_{\alpha\beta,i}).$$

Suppose

$$d_B(B_{i,W_{\alpha\beta}}^\alpha, B_{i,W_{\alpha\beta}}^\beta) \leq \varepsilon_{\alpha\beta,i}$$

and

$$\varepsilon_{\alpha\beta,i} < \text{gap}_\tau(B_{i,W_{\alpha\beta}}^\alpha, B_{i,W_{\alpha\beta}}^\beta).$$

Then

$$d_B(\text{Eval}_{i,W_{\alpha\beta},\tau}(F_\alpha), \text{Eval}_{i,W_{\alpha\beta},\tau}(F_\beta)) \leq \varepsilon_{\alpha\beta,i}.$$

Proof. The two assumptions supply a comparison record in mode `metric_gap_certified`. Apply Proposition 3.17 on the common overlap window. $\square$

3.12 Detector profile stage and effective domain

Definition 3.40 (Detector profile stage). Every detector record declares exactly one source stage:

$$\begin{array}{ll} \text{windowed_prethreshold}, & \text{repaired_postthreshold}, \\ \text{kernel_defect}, & \text{cokernel_defect}. \end{array}$$

The standard finite detector of this chapter is proved for `windowed_prethreshold`. A theorem proved for one stage is not reused at another stage without an explicit stage-transport theorem. The kernel and cokernel stages are consumed only after Chapter 5 supplies an actual comparison morphism and well-typed defect profiles.

Definition 3.41 (Effective detector domain). Let $W = [a, b)$, where $b \in \mathbb{R} \cup \{+\infty\}$, and let $\tau > 0$. Define

$$W_\tau := \{t \in W \mid t + \tau \in W\}.$$

Thus

$$W_\tau = \begin{cases} [a, b - \tau), & b < +\infty \text{ and } b - a > \tau, \\ \emptyset, & b < +\infty \text{ and } b - a \leq \tau, \\ [a, +\infty), & b = +\infty. \end{cases}$$

Only sample points in W_τ may be used by the exact τ -stencil detector.

Remark 3.42 (Window-relative length). All bars in the detector calculus are bars of the already windowed profile. A bar reaching the right endpoint of a bounded window is therefore a finite window-cofinal bar. Its length is its clipped window-relative length, not $+\infty$. A bar remains genuinely infinite only on a right-unbounded window.

3.13 Exact finite stencil and active-start regions

Definition 3.43 (Exact finite stencil detector). Let B be a constructible persistence module on W , let $T \subseteq W_\tau$ be finite, and fix the left-closed right-open interval registration convention. Define

$$D_{T,\tau}^{\text{stc}}(B) := \bigoplus_{t \in T} \text{im}(B(t) \longrightarrow B(t + \tau)).$$

Its detector value is zero precisely when every displayed image space is zero. The direct sum is finite and is counted with vector-space multiplicity.

Definition 3.44 (Retained bars and active-start region). For a windowed barcode B , let

$$\text{Ret}_{\tau,W}(B) := \{I \in \mathcal{B}(B) \mid \ell_W(I) > \tau\},$$

with multiplicity, where a genuinely infinite bar has $\ell_W(I) = +\infty$. For an interval summand I , define its τ -active-start region by

$$A_\tau(I) := \{t \in W_\tau \mid t, t + \tau \in I\}.$$

If $I = [p, q)$ in the left-closed right-open convention, then

$$A_\tau(I) = \begin{cases} [p, q - \tau), & q < +\infty \text{ and } q - p > \tau, \\ [p, +\infty), & q = +\infty, \\ \emptyset, & q - p \leq \tau. \end{cases}$$

In particular, $A_\tau(I) \neq \emptyset$ exactly when I is retained by T_τ^{bar} .

Proposition 3.45 (Interval decomposition of a stencil image). *Under the hypotheses of Definition 3.43,*

$$\dim_k \operatorname{im}(B(t) \rightarrow B(t + \tau))$$

equals the number, counted with multiplicity, of interval summands I of B for which $t \in A_\tau(I)$. Consequently,

$$D_{T,\tau}^{\text{stc}}(B) \neq 0 \iff \exists I \in \operatorname{Ret}_{\tau,W}(B) \text{ with } T \cap A_\tau(I) \neq \emptyset.$$

Proof. Decompose B into interval modules. On an interval summand k_I , the structure map from t to $t + \tau$ is the identity exactly when both parameters lie in I , and is zero otherwise. Direct sums preserve images componentwise in the finite-dimensional constructible regime. Summing over $t \in T$ gives the equivalence. $\square$

Theorem 3.46 (Finite-stencil soundness). *For every finite $T \subseteq W_\tau$,*

$$D_{T,\tau}^{\text{stc}}(B) \neq 0 \implies T_\tau^{\text{bar}}(B) \neq 0.$$

Equivalently,

$$T_\tau^{\text{bar}}(B) = 0 \implies D_{T,\tau}^{\text{stc}}(B) = 0.$$

No coverage or mesh hypothesis is required for this direction.

Proof. A nonzero stencil contains a nonzero component at some $t \in T$. Proposition 3.45 then supplies an interval summand whose active-start region contains t . Such an interval has window-relative length $> \tau$, hence survives hard deletion. $\square$

Remark 3.47 (Soundness is not completeness). A zero stencil may miss a retained bar whose active-start region contains no sample. Thus the converse of Theorem 3.46 requires a birth, germ, coverage, or family-mesh completeness theorem. Operational admission of a Core zero verdict follows the Chapter 1 detector-certificate contract.

3.14 Birth detector characterization

Definition 3.48 (Exhaustive birth set). For a finite left-closed right-open windowed barcode

$$\mathcal{B}(B) = \{[p_\lambda, q_\lambda)\}_{\lambda \in \Lambda},$$

define the set of distinct birth values and its effective τ -stencil by

$$T_{\text{birth}}(B) := \{p_\lambda \mid \lambda \in \Lambda\}, \quad T_{\text{birth},\tau}(B) := T_{\text{birth}}(B) \cap W_\tau.$$

A birth-set certificate is *exhaustive* when it proves that every interval summand of the declared windowed profile has its left endpoint in the recorded finite set. Only the effective subset is sampled. Every retained interval has its birth in W_τ , whereas a short interval born too near the right boundary may be omitted because its τ -map is not defined inside the window. Multiplicities belong to the profile artifact even though repeated birth values are sampled only once.

Theorem 3.49 (Birth detector characterization). *Let B be a finite windowed barcode in the left-closed right-open convention, and let its exhaustive birth set be certified. Then*

$$D_{T_{\text{birth},\tau}(B),\tau}^{\text{stc}}(B) \neq 0 \iff T_\tau^{\text{bar}}(B) \neq 0.$$

Equivalently,

$$D_{T_{\text{birth},\tau}(B),\tau}^{\text{stc}}(B) = 0 \iff T_\tau^{\text{bar}}(B) = 0.$$

Proof. Soundness is Theorem 3.46. Conversely, let $I = [p, q)$ be a retained interval. Then $q - p > \tau$, so $p + \tau < q$. The interval summand contributes nontrivially to

$$\text{im}(B(p) \rightarrow B(p + \tau)),$$

and $p \in T_{\text{birth}, \tau}(B)$. Hence the birth stencil is nonzero. $\square$

Remark 3.50 (Endpoint dependence of the point-birth theorem). The proof evaluates $B(p)$, so it consumes the left-closed registration convention. It is not decoration-invariant. Decorated intervals are handled canonically by the right-germ theorem below.

3.15 Right-germ detector characterization

Definition 3.51 (Right germ and shifted germ map). For a constructible persistence module B and $p \in \mathbb{R}$, define the right germ

$$B(p^+) := \lim_{\epsilon \downarrow 0} B(p + \epsilon).$$

For $\tau > 0$, the structure maps induce the shifted germ map

$$\gamma_{p, \tau} : B(p^+) \longrightarrow B((p + \tau)^+)$$

from the compatible family

$$B(p + \epsilon) \longrightarrow B(p + \tau + \epsilon).$$

This definition consumes a declared constructible right-germ convention.

Definition 3.52 (Germ detector). Let $T_{\text{gbirth}}(B)$ be the finite set of geometric lower endpoint values of the decorated interval summands of the declared windowed barcode, and put

$$T_{\text{gbirth}, \tau}(B) := T_{\text{gbirth}}(B) \cap W_\tau.$$

Define

$$D_\tau^{\text{germ}}(B) := \bigoplus_{p \in T_{\text{gbirth}, \tau}(B)} \text{im}(\gamma_{p, \tau}).$$

An exhaustive geometric-birth certificate is part of the detector record. Every retained interval has its geometric birth in the effective subset.

Theorem 3.53 (Decoration-invariant germ characterization). *Let B be a finite decorated windowed barcode with an exhaustive geometric birth set. Then, independently of open or closed endpoint decoration,*

$$D_\tau^{\text{germ}}(B) \neq 0 \iff T_\tau^{\text{bar}}(B) \neq 0.$$

Equivalently,

$$\boxed{D_\tau^{\text{germ}}(B) = 0 \iff T_\tau^{\text{bar}}(B) = 0.}$$

Proof. Consider one decorated interval with geometric endpoints $p < q$. Its contribution to the shifted germ map is nonzero exactly when points strictly to the right of both p and $p + \tau$ remain inside the interval for sufficiently small positive displacement. This is equivalent to $p + \tau < q$, hence to $q - p > \tau$. When $q - p = \tau$, the target right germ is taken strictly to the right of q and is zero even if the endpoint q itself is closed. Summing over the exhaustive geometric birth set gives the equivalence. $\square$

3.16 Externally certified stencil completeness

Definition 3.54 (Stencil coverage certificate). A stencil coverage certificate for (B, W, τ, T) is an artifact-backed proof of

$$\forall I \in \text{Ret}_{\tau, W}(B), \quad T \cap A_{\tau}(I) \neq \emptyset.$$

The record declares the source of coverage evidence and includes bounded-window cofinal bars. Coverage is a non-scalar obligation.

Theorem 3.55 (Certified stencil completeness). *Let $T \subseteq W_{\tau}$ be finite and suppose a valid stencil coverage certificate is supplied. Then*

$$D_{T, \tau}^{\text{stc}}(B) = 0 \iff T_{\tau}^{\text{bar}}(B) = 0.$$

Proof. The implication from right to left is Theorem 3.46. If the repaired profile is nonzero, there is a retained interval I . Coverage supplies $t \in T \cap A_{\tau}(I)$, and Proposition 3.45 makes the stencil nonzero. Taking contrapositives gives the other implication. $\square$

Remark 3.56 (Logical validity versus proof compression). The theorem is mathematically valid for any genuine coverage proof. A claim of finite proof compression additionally records a non-circular source for coverage, such as a family generation rule, finite critical-data theorem, quantized endpoint lattice, separation theorem, combinatorial filtration constraint, or external-domain theorem. Coverage reconstructed only by first enumerating the complete target barcode may still certify the theorem, but it does not by itself establish a meaningful compression gain.

3.17 Family threshold excess and augmented mesh

Definition 3.57 (Window-relative threshold excess). For a windowed barcode B , define

$$\text{exc}_{\tau, W}(B) := \inf_{I \in \text{Ret}_{\tau, W}(B)} (\ell_W(I) - \tau),$$

with $\inf \emptyset := +\infty$. A genuinely infinite retained interval contributes $+\infty$; a bounded-window cofinal interval contributes its finite clipped excess.

For a nonempty barcode family $\mathcal{F}$, a *uniform excess lower bound certificate* is a declared number

$$\underline{e}_{\tau, W}(\mathcal{F}) > 0$$

with artifact-backed proof that

$$\underline{e}_{\tau, W}(\mathcal{F}) \leq \text{exc}_{\tau, W}(B) \quad \text{for every } B \in \mathcal{F}.$$

Remark 3.58 (Independent source of excess). When family-mesh completeness is advertised as proof compression, the lower bound must be obtained independently of full retrospective reading of the target barcode. Permitted sources include a model-class theorem, endpoint lattice, Lipschitz or separation theorem, filtration constraint, external domain theorem, or certified family generator. An absent or unverifiable source makes detector completeness undefined.

Definition 3.59 (Augmented mesh). Assume $W = [a, b]$ is bounded and $b - a > \tau$. Put

$$c := b - \tau, \quad W_{\tau} = [a, c), \quad L_{\text{eff}} := c - a = b - a - \tau.$$

For

$$T = \{t_1 < \dots < t_m\} \subseteq [a, c),$$

define the augmented mesh

$$h_{W_\tau}(T) := \max\{t_1 - a, t_2 - t_1, \dots, t_m - t_{m-1}, c - t_m\}.$$

For $T = \emptyset$, set

$$h_{W_\tau}(\emptyset) := L_{\text{eff}}.$$

The endpoints a, c are virtual boundary points for gap measurement; they are not automatically detector samples.

Lemma 3.60 (Augmented-mesh hitting lemma). *Let $J = [p, r) \subseteq [a, c)$. If*

$$r - p > h_{W_\tau}(T),$$

then

$$J \cap T \neq \emptyset.$$

Proof. If $J \cap T = \emptyset$, then J lies in one component between two successive augmented boundary or sample points. Its length is at most the corresponding augmented gap, hence at most $h_{W_\tau}(T)$, a contradiction. $\square$

Theorem 3.61 (Family-mesh completeness). *Let $W = [a, b)$ be bounded, let $\mathcal{F}$ be a nonempty family of left-closed right-open windowed barcodes, and suppose a valid uniform excess lower bound*

$$\underline{e}_{\tau, W}(\mathcal{F}) > 0$$

is supplied. If a finite stencil $T \subseteq W_\tau$ satisfies

$$h_{W_\tau}(T) < \underline{e}_{\tau, W}(\mathcal{F}),$$

then T is complete for every $B \in \mathcal{F}$:

$$\boxed{\forall B \in \mathcal{F}, \quad D_{T, \tau}^{\text{stc}}(B) = 0 \iff T_\tau^{\text{bar}}(B) = 0.}$$

If $b - a \leq \tau$, then $W_\tau = \emptyset$ and every windowed finite bar has length $\leq \tau$, so the zero conclusion is trivial.

Proof. Let $I = [p, q)$ be any retained interval of any $B \in \mathcal{F}$. Its active-start region is

$$A_\tau(I) = [p, q - \tau)$$

and has length

$$q - p - \tau \geq \underline{e}_{\tau, W}(\mathcal{F}) > h_{W_\tau}(T).$$

The hitting lemma gives $T \cap A_\tau(I) \neq \emptyset$. Thus T satisfies the coverage hypothesis of Theorem 3.55. $\square$

Theorem 3.62 (Strict mesh sharpness and equality boundary). *Fix a bounded effective domain $W_\tau = [a, c)$ and a finite stencil T .*

(i) If $h_{W_\tau}(T) > e > 0$, then there exists a one-bar barcode B with

$$\text{exc}_{\tau, W}(B) > e, \quad T_\tau^{\text{bar}}(B) \neq 0, \quad D_{T, \tau}^{\text{stc}}(B) = 0.$$

- (ii) The strict hypothesis in Theorem 3.61 cannot be replaced uniformly by $h_{W_\tau}(T) \leq e$. For example, if the left boundary gap satisfies $t_1 - a = e$, the retained interval

$$I = [a, a + \tau + e)$$

has active region $[a, t_1)$, which contains no detector sample.

- (iii) At equality $h_{W_\tau}(T) = e$, completeness is boundary- and gap-location-sensitive. No universal pass or fail conclusion is asserted without additional endpoint or placement data.

Proof. For (i), choose an augmented gap of length $h_{W_\tau}(T)$, choose e' with $e < e' < h_{W_\tau}(T)$, and place a half-open interval $[p, p + e')$ strictly inside that gap, with the left-boundary case allowed to start at a . Set

$$I = [p, p + \tau + e').$$

Then $I \subseteq W$, its excess is $e' > e$, and its active-start region is the chosen sample-free interval. This gives the claimed barcode. Item (ii) is the displayed boundary construction. Item (iii) follows because an internal equality gap has different left-endpoint behavior from the left-boundary equality gap under the fixed half-open convention. $\square$

3.18 Detector certificate and two-axis status

Definition 3.63 (Finite detector certificate record). A finite detector certificate record contains at least:

- (i) detector, theorem, source-profile, and immutable run identifiers;
- (ii) the profile stage and the exact source object;
- (iii) the monitored degree n , window W , threshold τ , and endpoint convention;
- (iv) detector mode

`birth, germ, certified_stencil, family_mesh;`
- (v) the finite sample, birth, or geometric-birth set;
- (vi) soundness evidence;
- (vii) completeness, characterization, or coverage evidence;
- (viii) family membership and the independently sourced excess lower bound when family mesh is used;
- (ix) exact or metric transport records when the consumed predicate belongs to a compared profile;
- (x) certificate size, any family-relative MinCert bound, and replay artifacts;
- (xi) specialized certificate and value statuses.

Definition 3.64 (Two-axis detector status). Detector certificate status and detector mathematical value are independent fields.

The certificate axis is

`detector_certified, detector_failed, detector_pending,`

mapping respectively to

pass, reject, undefined.

Here `detector_failed` means that a well-typed required detector condition is evidenced false. Missing, incomplete, stale, or pending evidence is `detector_pending`, not failure.

The value axis is

zero, nonzero, undefined.

Under the Chapter 1 admission policy, a zero value discharges a zero clause only with certified completeness. A certified nonzero value rejects a required zero clause. A raw zero-looking computation with missing completeness remains undefined. The mathematical soundness theorem remains available at its exact strength, but an operational gate consumes only the registered status combination permitted by Chapter 1 and Chapter 7.

Proposition 3.65 (Detector invariance under profile isomorphism). *Let*

$$\eta : B \xrightarrow{\sim} C$$

be an isomorphism of persistence modules on the declared window. Then for every finite stencil $T \subseteq W_\tau$,

$$D_{T,\tau}^{\text{stc}}(B) \cong D_{T,\tau}^{\text{stc}}(C).$$

The corresponding right-germ detectors are also isomorphic when the same geometric-birth convention is used. Hence detector zero/nonzero is invariant under an actual Chapter 3 profile isomorphism.

Proof. Naturality of η gives a commutative square for each structure map $B(t) \rightarrow B(t + \tau)$. Vertical isomorphisms induce an isomorphism of image spaces, and finite direct sums preserve it. Passing to right-germ colimits gives the germ statement. $\square$

3.19 Metric-safe detector verdict composition

Theorem 3.66 (Metric zero-verdict composition). *Let B and C be windowed pre-threshold profiles. Suppose:*

(i) valid complete detector certificates give

$$D_{T_B,\tau}^{\text{stc}}(B) = 0 \iff T_\tau^{\text{bar}}(B) = 0$$

and

$$D_{T_C,\tau}^{\text{stc}}(C) = 0 \iff T_\tau^{\text{bar}}(C) = 0;$$

(ii) $d_B(B, C) \leq \varepsilon$;

(iii) the Chapter 2 two-sided threshold-gap hypotheses hold at the same ε ;

(iv) $2\varepsilon \leq \tau$.

Then

$$D_{T_B,\tau}^{\text{stc}}(B) = 0 \iff D_{T_C,\tau}^{\text{stc}}(C) = 0.$$

The operational v20 certificate uses the strict reserve $2\varepsilon < \tau$.

Proof. Chapter 2 gives

$$T_\tau^{\text{bar}}(B) = 0 \iff T_\tau^{\text{bar}}(C) = 0$$

under the stated metric, gap, and zero-separation hypotheses. Compose this equivalence with the two detector zero-reflection certificates. $\square$

Remark 3.67 (Zero predicate, not pointwise-rank stability). The theorem does not compare detector ranks at common sample points. It composes complete zero-reflection with repaired-zero metric transport. Chapter 2 supplies explicit counterexamples to pointwise detector-rank stability under arbitrarily small bottleneck perturbations.

Definition 3.68 (Boundary-safe robust stencil). Let B and C be equipped with actual ε -interleaving morphisms, and define the boundary-safe sample domain

$$W_{\tau, \varepsilon}^{\text{rob}} := \{t \mid t - \varepsilon, t + \tau + \varepsilon \in W\}.$$

For finite $T \subseteq W_{\tau, \varepsilon}^{\text{rob}}$, define

$$D_{T, \tau, \varepsilon}^{\text{rob}}(B) := \bigoplus_{t \in T} \text{im}(B(t - \varepsilon) \longrightarrow B(t + \tau + \varepsilon)).$$

For $W = [a, b)$, the boundary-safe domain is $[a + \varepsilon, b - \tau - \varepsilon)$ when nonempty.

Proposition 3.69 (Robust-to-exact sandwich). *Let B, C carry specified ε -interleaving morphisms on the required boundary-safe domain. Then*

$$D_{T, \tau, \varepsilon}^{\text{rob}}(B) \neq 0 \implies D_{T, \tau}^{\text{stc}}(C) \neq 0,$$

and symmetrically with B and C interchanged.

Proof. For each $t \in T$, the structure map

$$B(t - \varepsilon) \longrightarrow B(t + \tau + \varepsilon)$$

factors, by the interleaving identities and naturality, as

$$B(t - \varepsilon) \longrightarrow C(t) \longrightarrow C(t + \tau) \longrightarrow B(t + \tau + \varepsilon).$$

If the composite is nonzero, the middle exact structure map of C is nonzero. Taking finite direct sums proves the implication. $\square$

Remark 3.70 (Supporting status of the sandwich). The sandwich is one-way and requires actual interleaving morphisms, not merely a scalar distance value. It is a supporting detector theorem, not the main zero-equivalence theorem and not a substitute for completeness.

3.20 Family-relative minimal certificate

Definition 3.71 (Detector information class). A detector information class is a tuple

$$\Theta = (\mathcal{F}, W, \tau, \text{stage}, \text{endpt}, \text{critical}, \text{coverage}),$$

where $\mathcal{F}$ is a nonempty declared barcode family and the remaining fields fix the detector stage, endpoint, critical-data, and coverage policies. A finite set T is *complete* for Θ when the governing detector theorem proves, for every $B \in \mathcal{F}$,

$$D_{T, \tau}^{\text{stc}}(B) = 0 \iff T_{\tau}^{\text{bar}}(B) = 0.$$

Definition 3.72 (Family-relative minimal certificate). Define

$$\text{MinCert}(\Theta) := \min\{|T| \mid T \text{ is a complete finite stencil for } \Theta\},$$

when such a finite stencil exists, and set

$$\text{MinCert}(\Theta) := +\infty$$

otherwise. A certificate cost scheme may refine $|T|$, but weighted proof size and compression ratios are Technical quantities unless separately promoted.

Proposition 3.73 (Birth-set upper bound). Suppose the union of exhaustive birth sets

$$T_{\text{birth}, \tau}(\mathcal{F}) := \bigcup_{B \in \mathcal{F}} T_{\text{birth}, \tau}(B)$$

is finite and certified. Then

$$\text{MinCert}(\Theta) \leq |T_{\text{birth}, \tau}(\mathcal{F})|.$$

For an object-relative certificate, the corresponding upper bound is $|T_{\text{birth}, \tau}(B)|$.

Proof. The union contains every birth set, so the birth characterization detects every retained bar of every family member. $\square$

Definition 3.74 (Minimal augmented-mesh stencil number). For effective length $L_{\text{eff}} \geq 0$ and target augmented mesh $\gamma > 0$, define

$$m_{\text{mesh}}(L_{\text{eff}}, \gamma) := \max \left\{ 0, \left\lceil \frac{L_{\text{eff}}}{\gamma} \right\rceil - 1 \right\}.$$

Theorem 3.75 (Exact stencil count for an augmented-mesh target). Let $W_\tau = [a, c)$ have length $L_{\text{eff}} = c - a$. Then:

(i) there exists a stencil $T \subseteq [a, c)$ with

$$|T| = m_{\text{mesh}}(L_{\text{eff}}, \gamma), \quad h_{W_\tau}(T) \leq \gamma;$$

(ii) every stencil satisfying $h_{W_\tau}(T) \leq \gamma$ has

$$|T| \geq m_{\text{mesh}}(L_{\text{eff}}, \gamma).$$

Thus the displayed number is exact for the augmented-mesh design problem.

Proof. Let

$$N := \lceil L_{\text{eff}}/\gamma \rceil.$$

If $N = 0$, the domain is empty. If $N \geq 1$, divide the interval into N equal gaps and place $N - 1$ internal sample points. Each gap has length $L_{\text{eff}}/N \leq \gamma$, proving existence. Conversely, m samples determine $m + 1$ augmented gaps whose lengths sum to L_{eff} . If each is at most γ , then $L_{\text{eff}} \leq (m + 1)\gamma$, which gives the lower bound after taking ceilings. $\square$

Corollary 3.76 (Family mesh upper bound for MinCert). Under the hypotheses of Theorem 3.61, choose

$$0 < \gamma < \underline{e}_{\tau, W}(\mathcal{F}).$$

Then

$$\text{MinCert}(\Theta) \leq m_{\text{mesh}}(L_{\text{eff}}, \gamma).$$

This is an upper bound for the family-relative certificate problem; it need not be the exact MinCert for a family with additional structure.

3.21 Right-unbounded-window no-go theorem

Theorem 3.77 (Uniform finite-detector no-go on an unbounded birth family). *Let*

$$W = [a, +\infty), \quad \tau > 0, \quad e > 0,$$

and consider the translation family

$$\mathcal{U}_{a,\tau,e} := \{B_p = \{[p, p + \tau + e)\} \mid p \geq a\}.$$

For every finite stencil $T \subseteq W_\tau$, there exists $B_p \in \mathcal{U}_{a,\tau,e}$ such that

$$D_{T,\tau}^{\text{stc}}(B_p) = 0, \quad T_\tau^{\text{bar}}(B_p) \neq 0.$$

Let $\Theta_{\mathcal{U}}$ denote the detector information class whose barcode family is $\mathcal{U}_{a,\tau,e}$ and whose remaining fields are the displayed stage, window, threshold, and endpoint policies. Consequently,

$$\text{MinCert}(\Theta_{\mathcal{U}}) = +\infty.$$

More generally, the same conclusion holds for any family that admits retained active-start regions arbitrarily far to the right of every finite sample set.

Proof. A finite T is bounded above. Choose $p > \max T$, with the empty-stencil case arbitrary. Then the retained bar has active-start region $[p, p + e)$, which is disjoint from T . The stencil is zero by Proposition 3.45, while the bar has length $\tau + e > \tau$ and survives deletion. $\square$

Remark 3.78 (Admissible escape conditions). The theorem does not exclude finite detectors on a right-unbounded window when additional structure is proved. Possible escape conditions include a finite known birth set, a global birth bound, eventual zero, eventual stabilization, a finite-to-infinite reduction theorem, or a separately certified countable cofinal detector. Finite observed stability alone does not establish any of these conditions.

3.22 Legacy whole-barcode E_1 detector

Definition 3.79 (Residual degree-one bar mass). Let $\mathcal{B}_{1,W,\tau}^{\text{res}}(F)$ denote the residual degree-one barcode of $\text{Eval}_{1,W,\tau}(F)$, counted with multiplicity. Define

$$E_1(F; W, \tau) := \sum_{I \in (\mathcal{B}_{1,W,\tau}^{\text{res}}(F))_{\text{fin}}} \ell(I).$$

If an infinite residual degree-one bar is present, set

$$E_1(F; W, \tau) := +\infty,$$

unless a separately declared truncated detector is being used. Empty or zero-length intervals are not barcode summands in the monitored constructible profile.

Proposition 3.80 (Sound E_1 -vanishing detector). *If*

$$E_1(F; W, \tau) = 0,$$

then

$$\text{Eval}_{1,W,\tau}(F) = 0.$$

Proof. By definition, the presence of an infinite residual bar would give $E_1 = +\infty$. Every nonempty finite interval summand has strictly positive length and contributes positively, with multiplicity, to the displayed sum. Therefore $E_1 = 0$ implies that no residual degree-one bar is present, so the repaired degree-one profile is zero. $\square$

Corollary 3.81 (Whole-barcode E_1 characterization). *For a finite residual barcode under the stated convention,*

$$E_1(F; W, \tau) = 0 \iff \text{Eval}_{1,W,\tau}(F) = 0.$$

Proof. The forward implication is Proposition 3.80. The reverse implication is immediate because the zero residual barcode has empty length sum. $\square$

Remark 3.82 (Why E_1 is not finite proof compression). The scalar E_1 reads the already reconstructed residual barcode. It is a whole-profile vanishing functional, not a finite-map stencil certificate. It does not replace the exact profile statement, is not asserted to be bottleneck-Lipschitz, is not a threshold-gap budget, and does not by itself establish finite proof compression. A truncated variant carries its own policy and cannot silently inherit the characterization above.

3.23 Manifest consequences

Definition 3.83 (Chapter 3 evaluation and representative manifest extension). The Chapter 3 manifest is an extension of the foundational Chapter 1 Minimal Manifest Axiom. A proof object invoking Chapter 3 data records:

- (i) repaired evaluation records and immutable profile identities;
- (ii) pre-threshold profiles $B_{i,W}(F)$ and repaired evaluations $\text{Eval}_{i,W,\tau}(F)$;
- (iii) clearance and gap values and any single-profile readiness records;
- (iv) the comparison mode for every comparison;
- (v) for metric-gap mode, every metric ledger, its aggregation law, scalarization, real budget, evidenced discrepancy bound, two-sided gap reserve, specialized status, and artifact reference;
- (vi) for exact mode, the exact equality or profile-isomorphism proof and its precise source and target;
- (vii) crop budgets or window-edge records when source-level comparisons are transported to windowed profiles;
- (viii) any claimed τ -window-adapted commutation certificate, including checks for globally retained infinite bars on bounded windows;
- (ix) infinite-bar and bounded/terminal-window policies;
- (x) representative records, actual profile isomorphisms, monitored degrees, equivalence strengths, and typed statuses whenever $C_{\tau,W}F$ is in scope;
- (xi) realization-compatibility data when filtered, derived, or Ext-level use is intended;
- (xii) all in-scope non-scalar obligations, dependency states, atomic statuses, and artifact identities;

- (xiii) detector identifiers, source profile stages, monitored degrees, windows, thresholds, endpoint conventions, and detector modes;
- (xiv) finite stencil, birth, or germ sets and their immutable computation artifacts;
- (xv) detector soundness and completeness theorem identities;
- (xvi) birth/germ exhaustiveness, stencil coverage, family membership, augmented-mesh, and cofinal-window evidence;
- (xvii) every uniform excess lower bound and its independently declared source;
- (xviii) detector certificate status and detector value status on separate axes;
- (xix) certificate size, family-relative MinCert bounds, and any unbounded-window no-go or escape-condition record;
- (xx) actual interleaving morphisms and boundary-safe domains when the robust-to-exact sandwich is invoked;
- (xxi) claim-status references, schema/version identifiers, repair ancestry, and artifact hashes sufficient for audit and replay.

Remark 3.84 (Canonical manifest hierarchy). Chapter 1 supplies the foundational manifest axiom; this chapter expands it for repaired evaluation, comparison mode, metric-ledger, crop, window, and representative records; Chapter 4 adds eligibility records. Chapter 7 integrates these extensions into the canonical v20 audit manifest with immutable run identity, completeness/consistency status, typed clause verdicts, replayability, and repair-as-new-record semantics. For final serialization and verdict reconstruction, the Chapter 7 contract is canonical.

Remark 3.85 (Record completeness and failure semantics). A record containing only $\text{Eval}_{i,W,\tau}(F)$ can state a post-threshold value, but cannot support stability, overlap transfer, representative compatibility, detector completeness, or Ext eligibility. Missing required Chapter 3 data yield an `undefined` audit clause, not a passing clause and not a mathematical counterexample. A detector condition or inequality known by valid evidence to be false yields `reject`; missing evidence yields `undefined`. An internally inconsistent record is routed to Type V rejection under Chapter 7.

3.24 Explicit non-claims of Chapter 3

The following are not asserted in this chapter.

- (i) No global non-expansiveness of T_τ^{bar} is asserted.
- (ii) No band of half-width 2ε is claimed to be sharp for the two-sided theorem; the sharp half-width is ε .
- (iii) No half-clearance convention
$$\text{gap}_\tau = \frac{1}{2} \text{clear}_\tau$$
is used.
- (iv) No single-profile readiness record is a metric comparison certificate.
- (v) No exact equality or profile isomorphism is automatically a positive-radius robustness certificate.
- (vi) No unconditional crop–collapse commutation is asserted.

- (vii) No unscaled quantale element is compared directly with the real-valued clearance $\text{gap}_\tau = \text{clear}_\tau$.
- (viii) No numerical budget repairs a missing representative, profile isomorphism, realization, comparison morphism, eligibility certificate, claim-status entry, manifest field, or artifact identity.
- (ix) No realization existence or compatibility obligation is hidden inside $\delta_{\text{realization, met}}$.
- (x) No finite-only window-adapted certificate is sufficient; globally retained infinite bars must also be checked after clipping.
- (xi) No pairwise overlap record supplies higher Čech coherence or countable-atlas gluing without Chapter 6 hypotheses.
- (xii) No Chapter 3 manifest fragment supersedes the canonical Chapter 7 audit manifest.
- (xiii) No source-level perturbation is automatically converted into a windowed-profile perturbation.
- (xiv) No global strict filtered collapse functor C_τ is asserted.
- (xv) No identity $C_{\tau, W}F = (C_\tau F)|_W$ is asserted.
- (xvi) Persistence-profile isomorphism does not automatically imply chain isomorphism, homotopy equivalence, filtered quasi-isomorphism, or derived equivalence.
- (xvii) No derived or Ext-level conclusion follows from an admissible representative without the eligible realization regime of Chapter 4.
- (xviii) No reverse implication $\text{Ext}^1 \Rightarrow \text{PH}_1$ is asserted.
- (xix) No undefined or not-invoked Type IV state is represented by a fabricated numerical pair:

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0).$$
- (xx) No problem-specific arithmetic, PDE, geometric, categorical, or cryptographic theorem is proved.
- (xxi) No Search artifact, advisory invariant, scalar potential, ranking, or AI confidence value affects a Core repaired-evaluation verdict.
- (xxii) No finite observation set is a finite proof certificate without a governing soundness and completeness theorem.
- (xxiii) No zero stencil implies repaired-profile zero without exhaustive birth/germ data, certified coverage, family mesh, or another proved zero-reflection theorem.
- (xxiv) No missing birth, germ, endpoint, coverage, family-membership, excess, profile-stage, or artifact evidence is interpreted as detector zero.
- (xxv) No point-birth detector is decoration-invariant; decorated endpoints require the right-germ theorem or another proved replacement.
- (xxvi) No theorem proved for `windowed_prethreshold` is silently transferred to repaired, kernel-defect, or cokernel-defect stages.
- (xxvii) No scalarized metric reserve proves stencil coverage or family completeness.

- (xxviii) No family excess bound obtained only by retrospectively reading the complete target barcode is advertised as independent proof compression.
- (xxix) No family-mesh completeness is asserted from an ordinary internal sample gap that omits boundary gaps.
- (xxx) No universal conclusion is asserted at the equality $h_{W_\tau}(T) = \underline{e}_{\tau, W}$ without boundary-location data.
- (xxxi) No $h > \underline{e}$ stencil is complete for the unrestricted one-bar family covered by the sharpness theorem.
- (xxxii) No object-relative empty stencil for an already known zero barcode is promoted to a nontrivial family-relative MinCert theorem.
- (xxxiii) No exact augmented-mesh stencil count is identified with the exact MinCert of an arbitrary structured family.
- (xxxiv) No uniformly complete finite stencil exists for the unbounded translation family of Theorem 3.77.
- (xxxv) No finite prefix or observed stabilization proves an unbounded-window escape condition or finite-to-infinite reduction theorem.
- (xxxvi) No bottleneck bound by itself provides actual interleaving morphisms for the robust-to-exact sandwich.
- (xxxvii) No metric zero-verdict theorem asserts equality of pointwise detector ranks or equality of sample maps.
- (xxxviii) No whole-barcode E_1 functional is represented as a finite-map proof-compression certificate.

3.25 Provenance and status

This chapter depends on:

- (i) Chapter 1 for v20 claim discipline, finite detector admission, family-relative proof compression, B-Gate⁺, ledger discipline, Type IV status semantics, and manifest discipline;
- (ii) Chapter 2 for repaired windowed evaluation, comparison modes, crop control, sharp threshold-gap-safe stability, repaired-zero separation, metric detector-zero transport, and pointwise-rank guard-rails;
- (iii) Appendix A for the detailed proof of the metric repair and the exact crop-collapse criterion;
- (iv) Appendix B for fuller representative discipline when filtered or derived constructions are used;
- (v) Chapter 4 for n -eligible realization, the higher one-way discharge, and any separately natural zero-reflecting Ext theorem;
- (vi) Chapter 6 for higher overlap coherence and countable-atlas gluing;
- (vii) Chapter 7 for the canonical audit manifest and typed verdict semantics.

Status labels for this chapter. *Inherited v19 Core definitions and propositions.* Definitions 3.2, 3.3, 3.6, 3.7, 3.10, 3.11, 3.13, 3.15, 3.16, 3.23, 3.25, 3.26, 3.32, and 3.33; Propositions 3.17, 3.30, and 3.39.

New v20 finite-detector definitions. Definitions 3.40, 3.41, 3.43, 3.44, 3.48, 3.51, 3.52, 3.54, 3.57, 3.59, 3.63, 3.64, 3.68, 3.71, 3.72, and 3.74.

New v20 detector theorems and supporting results. Proposition 3.45, Theorem 3.46, Theorem 3.49, Theorem 3.53, Theorem 3.55, Lemma 3.60, Theorem 3.61, Theorem 3.62, Proposition 3.65, Theorem 3.66, Proposition 3.69, Proposition 3.73, Theorem 3.75, Corollary 3.76, and Theorem 3.77.

Inherited and strengthened whole-barcode detector. Definition 3.79, Proposition 3.80, and Corollary 3.81.

Operational policies and manifest records. Definitions 3.19, 3.21, 3.28, 3.36, 3.37, and 3.83.

Explicit exclusions. Subsection 3.24.

4 Chapter 4: Higher Eligibility, Ext-Edges, and Zero-Reflection Boundaries

Chapter 4 contract (v20 conservative higher-edge extension). Chapter 3 fixes repaired profiles, finite detector certificates, comparison modes, and representative records. This chapter preserves the complete v19 degree-one bridge and extends its architecture to every invoked degree $n \geq 1$.

The unconditional algebraic input is the canonical field-coefficient identification

$$\mathrm{Ext}^n(A, k) := \mathrm{Hom}_{D^b(k\text{-mod})}(A, k[n]) \xrightarrow{\sim} \mathrm{Hom}_k(H^{-n}(A), k),$$

valid naturally for bounded derived objects over the field k . The standard amplitude requirement

$$A_n(F) \in D^{[-n, 0]}(k\text{-mod})$$

is retained as a conservative eligibility and typing policy; it is not misstated as a necessary hypothesis for the algebraic isomorphism itself.

The persistence-to-derived identification

$$H^{-n}(A_n(F)) \xrightarrow{\sim} E_n(\mathrm{Eval}_{n, w, \tau}(F))$$

is an explicit, artifact-backed edge-realization interface. It is not manufactured from a barcode, a detector value, or a desired gate verdict. Under zero compatibility this interface yields the one-way discharge

$$\mathrm{Eval}_{n, w, \tau}(F) = 0 \implies \mathrm{Ext}^n(A_n(F), k) = 0.$$

A reverse implication is available only on a declared profile class under a separately proved natural zero-reflection theorem. The resulting statement is an equivalence of *vanishing predicates*, not an object-level isomorphism $\mathrm{PH}_n \cong \mathrm{Ext}^n$.

A shifted detector object may satisfy a tautological Ext-duality, but such an encoding is classified as synthetic unless an independently meaningful, natural realization theorem identifies it with the declared $A_n(F)$. Missing eligibility, edge realization, detector completeness, naturality, or artifact evidence is undefined; it is never interpreted as categorical vanishing.

4.1 Scope and placement in the v20 Core

All statements in this chapter are made in the v20.0 conservative-extension regime fixed by Chapters 1–3 and Appendix A. The degree-one package is inherited unchanged from v19.0.0; the higher-degree package is new

v20 Core mathematics. Thus:

- (i) persistence data live in the constructible one-parameter barcode regime over a field k ;
- (ii) the repaired evaluation is the post-threshold profile

$$\text{Eval}_{i,W,\tau}(F) = T_{\tau}^{\text{bar}}(W_W \mathbf{P}_i(F));$$

- (iii) threshold clearance is measured on the pre-threshold profile

$$B_{i,W}(F) := W_W \mathbf{P}_i(F), \quad \text{gap}_{\tau} = \text{clear}_{\tau};$$

- (iv) every comparison declares exactly one Chapter 3 mode

$$\text{exact}, \quad \text{metric_gap_certified}, \quad \text{not_compared};$$

a metric comparison transported through hard deletion requires the typed condition

$$\text{val}_B(\mathfrak{d}_{\text{met}}) < \text{gap}_{\tau}(\mathfrak{B})$$

for the declared finite family $\mathfrak{B}$, whereas an exact comparison carries its own equality or profile-isomorphism proof and is not automatically a positive-radius robustness certificate;

- (v) an object $C_{\tau,W}F$ is an admissible filtered representative of the repaired profile, not a global strict collapse functor;
- (vi) a categorical Ext^n -clause may be invoked only after the corresponding n -eligible realization record has been declared complete and consistent; the v19 degree-one clause is the case $n = 1$;
- (vii) realization and edge-extraction policies are fixed at declared scope level and may not be selected after inspecting the desired gate verdict;
- (viii) audited B-Gate⁺ use retains the independent threshold-gap, metric-ledger, Type IV, manifest, and claim-status clauses of Chapters 1, 5, and 7.

Remark 4.1 (Relocation of the previous failure-taxonomy chapter). Earlier drafts placed failure types, tower comparison artifacts, and Type IV diagnostics in Chapter 4. After the v19 Chapter 3 repair, that material is logically downstream of the one-way bridge and should be read as the next failure-diagnostics chapter. The present Chapter 4 is therefore reserved for eligibility, the inherited Ext^1 -edge, and its controlled higher-degree extension.

Remark 4.2 (Metric repair is consumed, not reproved). This chapter does not reprove threshold-gap-safe stability. It assumes that the repaired evaluation records consumed here have already been formed according to Chapter 3, with any required threshold-gap certification supplied separately. The bridge below is a categorical discharge result, not a stability theorem for threshold deletion.

4.2 Repaired degree-one input

Definition 4.3 (Degree-one repaired profile). For $F \in \text{FiltCh}(()k)$, a right-open window $W = [u, u')$, and a threshold $\tau > 0$, the degree-one repaired profile consumed by this chapter is

$$\text{Post}_{1,W,\tau}(F) := \text{Eval}_{1,W,\tau}(F) = T_{\tau}^{\text{bar}}(W_W \mathbf{P}_1(F)).$$

The statement $\text{Eval}_{1,W,\tau}(F) = 0$ means that this post-threshold barcode profile is the zero persistence profile.

Definition 4.4 (Degree-one admissible representative). An admissible representative $C_{\tau,W}F$ is *degree-one compatible* if its Chapter 3 representative record contains an actual profile isomorphism

$$\rho_{1,F,W,\tau} : \mathbf{P}_1(C_{\tau,W}F) \xrightarrow{\sim} \text{Eval}_{1,W,\tau}(F).$$

The source, target, equivalence strength, and reconstructible artifact for $\rho_{1,F,W,\tau}$ are part of the record. A bare symbol $\cong$ without such evidence is not an eligibility certificate. When this holds, the notation

$$\text{PH}_1(C_{\tau,W}F) = 0$$

is used only as shorthand for the vanishing of that realized repaired profile.

Remark 4.5 (No global collapse interpretation). The notation $C_{\tau,W}F$ does not mean $(C_{\tau}F)|_W$, and no global exact functor C_{τ} is used. The only profile entering this chapter is the repaired windowed profile

$$T_{\tau}^{\text{bar}}(W_W \mathbf{P}_1(F)).$$

4.3 Eligible realization records

Definition 4.6 (Realization datum). A *realization datum* for a degree-one compatible representative $C_{\tau,W}F$ consists of:

- (i) a declared eligible domain

$$\mathcal{E} \subseteq \text{FiltCh}(()k)$$

containing $C_{\tau,W}F$;

- (ii) a genuine functor, fixed at the declared scope level,

$$\mathcal{R} : \mathcal{E} \longrightarrow D^b(k\text{-mod});$$

- (iii) the realized object

$$A_{F,W,\tau} := \mathcal{R}(C_{\tau,W}F);$$

- (iv) immutable policy, version, and artifact identifiers for $\mathcal{E}$ and $\mathcal{R}$.

The functor may be partial through the declared domain $\mathcal{E}$, but it is not an objectwise rule chosen after the desired Ext verdict is known.

Definition 4.7 (Residual edge extractor). A *residual degree-one edge extractor* is a scope-level, isomorphism-invariant vector-space-valued policy

$$\text{Edge}_1 : \text{Prof}_1^{\text{rep}} \longrightarrow \text{Vect}_k$$

on the declared groupoid or category of repaired degree-one profiles and admissible profile isomorphisms, satisfying

$$\text{Edge}_1(0) = 0.$$

If no profile morphism category has been declared, this is read as an assignment invariant under barcode-profile isomorphism. The extractor is fixed independently of the objectwise gate verdict. It need not be faithful or zero-reflecting unless a stronger bridge theorem explicitly proves that property.

Definition 4.8 (Edge-realization certificate). An *edge-realization certificate* is attached to the declared data

$$(F; W, \tau, C_{\tau, W}F, \mathcal{R}, \text{Edge}_1).$$

It is an actual declared isomorphism

$$\epsilon_{F, W, \tau} : H^{-1}(\mathcal{R}(C_{\tau, W}F)) \xrightarrow{\sim} \text{Edge}_1(\text{Eval}_{1, W, \tau}(F)).$$

Its source, target, dependence on the representative isomorphism $\rho_{1, F, W, \tau}$, and reconstructible proof artifact are recorded. The only objectwise implication consumed by the Core bridge is

$$\text{Eval}_{1, W, \tau}(F) = 0 \implies H^{-1}(\mathcal{R}(C_{\tau, W}F)) = 0.$$

An objectwise certificate does not by itself establish naturality in F . Whenever naturality or transport across morphisms is claimed, the certificates must form a compatible natural family on the declared eligible domain.

Definition 4.9 (Eligible realization record). An *eligible realization record* for $(F; W, \tau)$ consists of:

- (i) a repaired evaluation record for $\text{Eval}_{1, W, \tau}(F)$;
- (ii) a degree-one compatible admissible representative $C_{\tau, W}F$, its Chapter 3 representative record, and the actual isomorphism $\rho_{1, F, W, \tau}$;
- (iii) a realization datum $(\mathcal{E}, \mathcal{R})$, membership $C_{\tau, W}F \in \mathcal{E}$, and the realized object $A_{F, W, \tau}$;
- (iv) the amplitude certificate
$$A_{F, W, \tau} \in D^{[-1, 0]}(k\text{-mod});$$
- (v) the fixed residual-edge policy Edge_1 and its policy identifier;
- (vi) an edge-realization certificate $\epsilon_{F, W, \tau}$, with objectwise or natural-family scope explicitly declared;
- (vii) a field coefficient declaration for k ;
- (viii) a non-circularity entry confirming that $\mathcal{E}$, $\mathcal{R}$, and Edge_1 were fixed independently of the desired pass/reject verdict;
- (ix) representative-equivalence and realization-equivalence audit data;
- (x) immutable artifact references sufficient to reconstruct the representative, realization, amplitude, and edge certificates;
- (xi) the applicable claim-status and manifest entries under the canonical Chapter 7 audit contract.

A windowed object $(F; W, \tau)$ is called *eligible for the Ext-edge* only when the required record is complete and consistent. An invoked but incomplete, circular, or ill-typed record yields an `undefined` categorical clause, not a passing clause and not a mathematical counterexample.

Remark 4.10 (Eligibility is stronger than having a representative). An admissible representative alone is not enough to invoke Ext^1 . The Ext -edge also needs a scope-level realization functor, an amplitude certificate, a fixed residual-edge policy, and an actual edge-realization certificate. The canonical derived-category edge proved below does not construct that interface certificate. Consequently, the one-way bridge is a conditional Core/interface theorem, not an unconditional identification of persistent homology with derived Ext .

Remark 4.11 (Gap certification and non-scalar obligations are separate). The eligible realization record does not replace the Chapter 3 comparison record. For audited B-Gate⁺ use, every `metric_gap_certified` comparison still requires an evidenced pre-threshold discrepancy bound and the typed inequality

$$\text{val}_B(\mathfrak{d}_{\text{met}}) < \text{gap}_\tau(\mathfrak{B})$$

measured on the relevant two-sided pre-threshold profile family $\mathfrak{B}$. A comparison in `exact` mode instead carries its exact proof and is not automatically a robustness certificate.

Conversely, numerical clearance cannot manufacture an admissible representative, a realization functor, an amplitude certificate, an edge-realization certificate, or a complete manifest. Those are non-scalar proof obligations. The present chapter discharges only the categorical Ext^1 -edge once exact repaired topological vanishing and full eligibility have been established.

4.4 Degree- n repaired input and standard eligibility

Definition 4.12 (Degree- n repaired profile). Fix an integer $n \geq 1$. The degree- n repaired profile is

$$P_{n,W,\tau}(F) := \text{Post}_{n,W,\tau}(F) := \text{Eval}_{n,W,\tau}(F) = T_\tau^{\text{bar}}(W_W \mathbf{P}_n(F)).$$

It is the exact profile whose vanishing may be established directly, by a complete Chapter 3 detector, or by a valid metric zero-transport chain.

Definition 4.13 (Degree- n compatible representative). An admissible representative $C_{\tau,W}F$ is *degree- n compatible* if its Chapter 3 record contains an actual profile isomorphism

$$\rho_{n,F,W,\tau} : \mathbf{P}_n(C_{\tau,W}F) \xrightarrow{\sim} \text{Eval}_{n,W,\tau}(F).$$

The monitored degree, source and target identities, equivalence strength, and immutable proof artifact are part of the data. Compatibility in degree n does not imply compatibility in another degree.

Definition 4.14 (Degree-indexed realization datum). A degree-indexed realization datum consists of a declared domain $\mathcal{E}_n \subseteq \text{FiltCh}((k))$, a genuine scope-level functor

$$\mathcal{R}_n : \mathcal{E}_n \longrightarrow D^b(k\text{-mod}),$$

and the independently meaningful realized object

$$A_{n,F,W,\tau} := \mathcal{R}_n(C_{\tau,W}F).$$

The domain, functor, version, coefficient field, and artifact identities are fixed before evaluation of the desired zero or nonzero verdict. Different values of n may use the same realization only when that common scope and its compatibility are explicitly declared.

Definition 4.15 (Higher residual edge extractor). For a declared repaired-profile class $\mathcal{P}_n$, a degree- n residual edge extractor is a scope-level assignment

$$E_n : \mathcal{P}_n \longrightarrow \text{Vect}_k^{\text{fd}}$$

that is invariant under the declared profile isomorphisms and satisfies

$$E_n(0) = 0.$$

When morphism-level naturality is invoked, E_n is a functor on the specified morphism category. Objectwise invariance is not silently promoted to naturality, faithfulness, conservativity, or zero-reflection.

Definition 4.16 (Higher edge-realization certificate). A degree- n edge-realization certificate is an artifact-backed isomorphism

$$\epsilon_{n;F,W,\tau} : H^{-n}(A_{n;F,W,\tau}) \xrightarrow{\sim} E_n(P_{n,W,\tau}(F)).$$

The record declares whether this certificate is objectwise or belongs to a natural family. If naturality is claimed, for every allowed morphism the square formed by $H^{-n}(\mathcal{R}_n(-))$, $E_n(\text{Eval}_{n,W,\tau}(-))$, and the corresponding ϵ_n commutes.

Definition 4.17 (Standard n -eligible realization record). For $n \geq 1$, a standard n -eligible realization record for $(F; W, \tau)$ contains:

- (i) the exact degree- n repaired profile and its Chapter 3 record;
- (ii) a degree- n compatible representative and the actual $\rho_{n;F,W,\tau}$;
- (iii) a degree-indexed realization datum and membership $C_{\tau,W}F \in \mathcal{E}_n$;
- (iv) the conservative standard amplitude certificate

$$A_{n;F,W,\tau} \in D^{[-n,0]}(k\text{-mod}),$$

or a separately proved admissible replacement explicitly registered at its exact strength;

- (v) a fixed higher edge extractor E_n and its policy identifier;
- (vi) the higher edge-realization certificate $\epsilon_{n;F,W,\tau}$;
- (vii) zero compatibility $P = 0 \Rightarrow E_n(P) = 0$;
- (viii) the objectwise or natural-family scope of every item;
- (ix) non-circularity, representative-change, and realization-change records;
- (x) complete claim, artifact, and manifest references.

An invoked but incomplete, circular, stale, or ill-typed record yields `undefined`. It does not yield a false theorem, a passing clause, or a zero Ext group.

Remark 4.18 (Degree-one specialization). Definitions 4.6–4.9 are retained as the canonical v19 degree-one package. They agree with the preceding definitions at $n = 1$, with

$$\mathcal{R}_1 = \mathcal{R}, \quad E_1 = \text{Edge}_1, \quad A_{1;F,W,\tau} = A_{F,W,\tau}.$$

No change in the strength of the inherited degree-one theorem is intended.

Remark 4.19 (Amplitude is an eligibility policy). The amplitude interval $[-n, 0]$ makes the realization route auditable and isolates the intended negative edge. Over a field, however, the canonical algebraic identification of $\text{Ext}^n(A, k)$ with the dual of $H^{-n}(A)$ holds for every bounded A . The amplitude policy and the algebraic theorem must therefore not be conflated.

4.5 Field-coefficient higher Ext-edges

In this chapter, $k\text{-mod}$ denotes finite-dimensional k -vector spaces. The superscript in $D^{[-1,0]}$ refers to the standard cohomological t -structure.

Theorem 4.20 (Canonical higher Ext-edge over a field). *Let $n \geq 0$ and let $A \in D^b(k\text{-mod})$, where k is a field. There is a canonical isomorphism, natural contravariantly in A ,*

$$\text{Ext}^n(A, k) := \text{Hom}_{D^b(k\text{-mod})}(A, k[n]) \xrightarrow{\sim} \text{Hom}_k(H^{-n}(A), k).$$

Proof. The standard hyper-Ext spectral sequence gives

$$E_2^{p,q} = \text{Ext}_k^p(H^{-q}(A), k) \implies \text{Ext}^{p+q}(A, k).$$

Because $k\text{-mod}$ is semisimple, $\text{Ext}_k^p(V, k) = 0$ for every finite-dimensional V and every $p > 0$. Hence the spectral sequence is concentrated in the column $p = 0$, has no nonzero differential or extension ambiguity, and its total-degree n term is canonically

$$E_2^{0,n} = \text{Hom}_k(H^{-n}(A), k).$$

The spectral sequence and its edge identification are functorial in A , which gives the stated naturality. $\square$

Corollary 4.21 (Higher edge vanishing and reflection). *For $A \in D^b(k\text{-mod})$ and $n \geq 0$,*

$$\text{Ext}^n(A, k) = 0 \iff H^{-n}(A) = 0.$$

Proof. Theorem 4.20 identifies the Ext group with the linear dual of the finite-dimensional vector space $H^{-n}(A)$. A finite-dimensional vector space over a field has zero dual if and only if it is zero. $\square$

Lemma 4.22 (Formality over a field; noncanonical splitting). *Let $A \in D^b(k\text{-mod})$, where k is a field. Then there exists an isomorphism*

$$A \cong \bigoplus_{j \in \mathbb{Z}} H^j(A)[-j]$$

in $D^b(k\text{-mod})$. In general this direct-sum splitting depends on choices and is not used to prove the naturality of the Ext-edge identification below.

Proof. Every short exact sequence of k -vector spaces splits, so every bounded complex is quasi-isomorphic to its cohomology complex with zero differential. Choosing splittings gives the stated derived isomorphism. The choice-dependence is precisely why the natural edge theorem is proved instead from the canonical truncation triangle. $\square$

Theorem 4.23 (Canonical two-term Ext¹-edge identification). *Let $A \in D^{[-1,0]}(k\text{-mod})$. Then there is a canonical isomorphism, natural in A ,*

$$\text{Ext}^1(A, k) := \text{Hom}_{D^b(k\text{-mod})}(A, k[1]) \xrightarrow{\sim} \text{Hom}_k(H^{-1}(A), k).$$

Proof. The standard t -structure gives the canonical truncation triangle

$$H^{-1}(A)[1] \longrightarrow A \longrightarrow H^0(A) \xrightarrow{+1} H^{-1}(A)[2].$$

Apply the contravariant cohomological functor $\mathrm{Hom}_{D^b(k\text{-mod})}(-, k[1])$. The resulting long exact sequence contains the natural exact segment

$$\begin{aligned} \mathrm{Hom}_{D^b}(H^0(A), k[1]) &\longrightarrow \mathrm{Hom}_{D^b}(A, k[1]) \\ &\longrightarrow \mathrm{Hom}_{D^b}(H^{-1}(A)[1], k[1]) \longrightarrow \mathrm{Hom}_{D^b}(H^0(A), k[2]). \end{aligned}$$

Because $k\text{-mod}$ is semisimple,

$$\mathrm{Hom}_{D^b}(H^0(A), k[1]) = \mathrm{Ext}_k^1(H^0(A), k) = 0$$

and

$$\mathrm{Hom}_{D^b}(H^0(A), k[2]) = \mathrm{Ext}_k^2(H^0(A), k) = 0.$$

Hence the middle arrow is an isomorphism. Finally,

$$\mathrm{Hom}_{D^b}(H^{-1}(A)[1], k[1]) \cong \mathrm{Hom}_k(H^{-1}(A), k).$$

The truncation triangle and the long exact sequence are functorial in A , so the resulting isomorphism is natural. $\square$

Corollary 4.24 (Naturality square). *For every morphism $f : A \rightarrow A'$ in $D^{[-1,0]}(k\text{-mod})$, the square*

$$\begin{array}{ccc} \mathrm{Ext}^1(A', k) & \longrightarrow & \mathrm{Ext}^1(A, k) \\ \downarrow \sim & & \downarrow \sim \\ \mathrm{Hom}_k(H^{-1}(A'), k) & \longrightarrow & \mathrm{Hom}_k(H^{-1}(A), k) \end{array}$$

commutes, where the horizontal maps are induced contravariantly by f and by $H^{-1}(f)$, respectively.

Proof. This is the naturality assertion of Theorem 4.23 written as a commutative square. $\square$

Corollary 4.25 (Edge vanishing criterion). *If $A \in D^{[-1,0]}(k\text{-mod})$ and $H^{-1}(A) = 0$, then*

$$\mathrm{Ext}^1(A, k) = 0.$$

Proof. Apply Theorem 4.23. $\square$

Remark 4.26 (Why the field hypothesis matters). Over a general coefficient ring, the truncation argument yields an exact sequence whose outer terms include $\mathrm{Ext}^1(H^0(A), k)$ and $\mathrm{Ext}^2(H^0(A), k)$. These groups need not vanish, so the displayed natural isomorphism is not a theorem in that generality without additional projectivity, injectivity, or homological-dimension hypotheses.

Remark 4.27 (Splitting versus naturality). Lemma 4.22 explains the semisimple shape of the answer, but a chosen direct-sum decomposition of A need not be natural. Theorem 4.23 obtains the same group canonically from truncation and therefore supports representative and realization functoriality.

4.6 Higher one-way discharge and zero-reflection boundary

Theorem 4.28 (Eligible higher Ext-edge identification). *Let $n \geq 1$ and suppose $(F; W, \tau)$ has a complete standard n -eligible realization record. Then there is an artifact-backed isomorphism*

$$\mathrm{Ext}^n(A_{n;F,W,\tau}, k) \xrightarrow{\sim} \mathrm{Hom}_k(E_n(\mathrm{Eval}_{n,W,\tau}(F)), k).$$

It is natural on exactly the morphism scope on which both the realization family and the edge-realization certificates are natural.

Proof. Apply Theorem 4.20 to $A_{n;F,W,\tau}$, then compose with the dual of the declared isomorphism

$$H^{-n}(A_{n;F,W,\tau}) \xrightarrow{\sim} E_n(\text{Eval}_{n,W,\tau}(F)).$$

The claimed naturality is the composite of the naturality of the derived edge and the declared naturality of the edge-realization family. $\square$

Theorem 4.29 (Higher one-way repaired discharge). *Let $n \geq 1$ and let $(F; W, \tau)$ be standard n -eligible. Then*

$$\text{Eval}_{n,W,\tau}(F) = 0 \implies \text{Ext}^n(A_{n;F,W,\tau}, k) = 0.$$

Proof. Set $P = \text{Eval}_{n,W,\tau}(F)$. Zero compatibility gives $E_n(P) = E_n(0) = 0$. Theorem 4.28 then identifies the Ext group with $\text{Hom}_k(0, k) = 0$. $\square$

Definition 4.30 (Natural higher zero-reflection package). A standard n -eligible record has a *natural zero-reflection package* on a declared repaired-profile class $\mathcal{P}_n$ when:

- (i) E_n is fixed independently of the desired verdict;
- (ii) E_n is natural on the declared morphism class;
- (iii) the edge-realization certificates form a natural family;
- (iv) a proved theorem establishes

$$E_n(P) = 0 \implies P = 0 \quad (P \in \mathcal{P}_n);$$

- (v) representative-change and realization-change compatibility pass;
- (vi) every completeness, stage, endpoint, family-membership, and artifact obligation used by the zero-reflection theorem passes independently.

This definition instantiates the Chapter 1 natural zero-reflection upgrade.

Theorem 4.31 (Zero-reflecting higher vanishing equivalence). *Let $n \geq 1$. Suppose $(F; W, \tau)$ is standard n -eligible, its repaired profile $P = \text{Eval}_{n,W,\tau}(F)$ belongs to $\mathcal{P}_n$, and the record has a natural higher zero-reflection package on $\mathcal{P}_n$. Then*

$$\boxed{\text{Eval}_{n,W,\tau}(F) = 0 \iff \text{Ext}^n(A_{n;F,W,\tau}, k) = 0}.$$

This is an equivalence of vanishing predicates on the declared scope.

Proof. The forward implication is Theorem 4.29. Conversely, if the Ext group vanishes, Theorem 4.28 and finite-dimensional dual reflection imply $E_n(P) = 0$. The separately proved zero-reflection theorem then gives $P = 0$. $\square$

Proposition 4.32 (Functoriality of a fixed stencil detector). *Fix a window, a threshold τ , and a finite stencil T in the valid Chapter 3 detector domain. On any declared category of persistence modules and actual persistence morphisms on that domain, the assignment*

$$P \longmapsto D_{T,\tau}^{\text{stc}}(P) = \bigoplus_{t \in T} \text{im}(P(t) \rightarrow P(t + \tau))$$

defines a functor to $\text{Vect}_k^{\text{fd}}$.

Proof. For a persistence morphism $f : P \rightarrow Q$, naturality of the structure maps gives, for each $t \in T$, a commutative square

$$\begin{array}{ccc} P(t) & \longrightarrow & P(t + \tau) \\ \downarrow f_t & & \downarrow f_{t+\tau} \\ Q(t) & \longrightarrow & Q(t + \tau). \end{array}$$

Hence $f_{t+\tau}$ sends the image of the upper horizontal map into the image of the lower horizontal map and induces a linear map between those image spaces. Taking the finite direct sum over T gives $D_{T,\tau}^{\text{stc}}(f)$. Identity and composition are inherited componentwise from the persistence morphisms. $\square$

Corollary 4.33 (Complete fixed-stencil higher vanishing equivalence). *Let $n \geq 1$ and let $(F; W, \tau)$ be standard n -eligible on a declared profile class $\mathcal{P}_n$. Assume:*

(i) *the independently meaningful edge extractor is the fixed-stencil functor*

$$E_n(P) = D_{T,\tau}^{\text{stc}}(P);$$

(ii) *the higher edge-realization certificates form a natural family;*

(iii) *a Chapter 3 completeness theorem, with all stage, endpoint, coverage, family, and artifact obligations discharged, proves*

$$D_{T,\tau}^{\text{stc}}(P) = 0 \iff P = 0 \quad (P \in \mathcal{P}_n);$$

(iv) *representative-change and realization-change compatibility pass.*

Then

$$\boxed{D_{T,\tau}^{\text{stc}}(\text{Eval}_{n,W,\tau}(F)) = 0 \iff \text{Ext}^n(A_{n;F,W,\tau}, k) = 0.}$$

Proof. Proposition 4.32 supplies the required naturality of the fixed detector. The Chapter 3 completeness theorem supplies zero-reflection. The remaining hypotheses complete the natural zero-reflection package, so Theorem 4.31 applies. $\square$

Remark 4.34 (Detector completeness is not edge realization). A Chapter 3 completeness theorem proves that a detector reflects zero on a specified profile class. It does not construct the independently meaningful realized object $A_n(F)$, nor the isomorphism $H^{-n}(A_n(F)) \cong E_n(P)$. Conversely, an edge-realization certificate does not prove detector completeness. Both obligations are required for the zero-reflecting higher equivalence.

Remark 4.35 (No object-level PH_n -to- Ext^n isomorphism). Theorem 4.31 identifies only the truth values of two zero predicates. A persistence profile, its homology, an extracted finite-dimensional edge, and an Ext group live in different typed layers. No isomorphism $\text{PH}_n \cong \text{Ext}^n$ is asserted.

4.7 The one-way bridge $\text{PH}_1 \Rightarrow \text{Ext}^1$

Theorem 4.36 (Conditional one-way repaired bridge). *Let $(F; W, \tau)$ be eligible for the Ext -edge in the sense of Definition 4.9. If*

$$\text{Eval}_{1,W,\tau}(F) = 0,$$

then

$$\text{Ext}^1(\mathcal{R}(C_{\tau,W}F), k) = 0.$$

Equivalently, when the representative realizes the repaired degree-one profile,

$$\text{PH}_1(C_{\tau,W}F) = 0 \implies \text{Ext}^1(\mathcal{R}(C_{\tau,W}F), k) = 0.$$

Proof. Set

$$A_{F,W,\tau} := \mathcal{R}(C_{\tau,W}F).$$

By eligibility,

$$A_{F,W,\tau} \in D^{[-1,0]}(k\text{-mod}).$$

The edge-realization certificate gives

$$H^{-1}(A_{F,W,\tau}) \cong \text{Edge}_1(\text{Eval}_{1,W,\tau}(F)).$$

If $\text{Eval}_{1,W,\tau}(F) = 0$, then by zero-compatibility of the edge extractor,

$$\text{Edge}_1(\text{Eval}_{1,W,\tau}(F)) = \text{Edge}_1(0) = 0.$$

Hence

$$H^{-1}(A_{F,W,\tau}) = 0.$$

Corollary 4.25 now gives

$$\text{Ext}^1(A_{F,W,\tau}, k) = 0,$$

which is the asserted conclusion. $\square$

Remark 4.37 (Discharge theorem, not equivalence). Theorem 4.36 is a conditional discharge theorem for the categorical clause of B-Gate⁺. Its unconditional mathematical input is the canonical two-term Ext-edge; its interface hypothesis is the eligible realization record. It states that repaired topological vanishing is sufficient to discharge the eligible Ext¹-edge. It does not state that categorical vanishing reconstructs the repaired degree-one barcode profile, nor does it prove the edge-realization certificate from persistence data.

Remark 4.38 (No use of search, spectral, or CCC reverse principles). The bridge uses only the eligible realization record and the two-term field-coefficient Ext-edge. The realization and residual-edge policies must be fixed before the objectwise verdict is evaluated. The proof does not use CCC authorization, spectral-gap hypotheses, Bridge Programs, validity maps, advisory invariants, Search-layer outputs, or AI-generated confidence values.

4.8 Gate integration

Definition 4.39 (Higher Ext-edge discharge record). For an invoked degree $n \geq 1$, a higher Ext-edge discharge record contains:

- (i) the exact repaired-profile zero statement, including the detector or comparison theorem by which it was established;
- (ii) the complete standard n -eligible record;
- (iii) the theorem identifier for Theorem 4.29;
- (iv) the higher derived edge and edge-realization artifacts;
- (v) the categorical conclusion $\text{Ext}^n(A_{n;F,W,\tau}, k) = 0$;
- (vi) when a reverse inference is consumed, the complete natural zero-reflection package and the theorem identifier for Theorem 4.31;
- (vii) immutable Claim, detector, representative, realization, and artifact references sufficient for replay.

Metric transport, when used, is cited through the Chapter 2–3 repaired-zero chain; the higher Ext theorem itself does not create a metric certificate.

Definition 4.40 (Typed degree- n Ext-clause status). For every invoked $n \geq 1$, the categorical clause has exactly one atomic status:

`not_invoked, undefined, pass, reject.`

Here `undefined` means that an invoked typing, representative, realization, amplitude, edge, naturality, completeness, claim, or artifact obligation is missing or ill-typed. The status `reject` requires evidence that the well-typed Ext group is nonzero; missing evidence is never reported as rejection.

Proposition 4.41 (Higher B-Gate⁺ categorical discharge). *For every invoked degree $n \geq 1$, a complete standard n -eligible record and*

$$\text{Eval}_{n,W,\tau}(F) = 0$$

discharge the corresponding categorical clause:

$$\text{Ext}^n(A_{n;F,W,\tau}, k) = 0.$$

Proof. This is Theorem 4.29 read as an atomic gate clause. □

Definition 4.42 (Aggregate higher categorical gate status). For a finite monitored degree set I_{mon} , the higher categorical gate passes exactly when every invoked degree- n Ext clause passes and every excluded degree is justified as `not_invoked`. One rejected or required undefined degree prevents aggregate pass. No cancellation or numerical compensation is permitted.

Definition 4.43 (Ext-edge discharge record). An *Ext-edge discharge record* for $(F; W, \tau)$ contains:

- (i) the repaired evaluation artifact establishing the exact post-threshold statement $\text{Eval}_{1,W,\tau}(F) = 0$;
- (ii) the comparison mode and, only when that exact vanishing was transported from another object or run, the corresponding exact or two-sided metric comparison record;
- (iii) the complete eligible realization record of Definition 4.9;
- (iv) the theorem identifier for Theorem 4.36;
- (v) the canonical Ext-edge identification and its naturality data;
- (vi) the resulting categorical conclusion

$$\text{Ext}^1(\mathcal{R}(C_{\tau,W}F), k) = 0;$$

- (vii) immutable artifact and claim-register references sufficient for replay.

If repaired topological vanishing is asserted under perturbation, the record also cites the relevant two-sided threshold-gap certificate, evidenced pre-threshold discrepancy bound, and scalarized metric budget. Exact mode instead cites the exact equality or profile-isomorphism proof.

Definition 4.44 (Typed Ext-edge clause status). The categorical clause attached to $(F; W, \tau)$ has one of the following audit meanings:

- (i) `not_invoked`: the run manifest explicitly places the Ext-clause outside scope;

- (ii) **undefined**: the clause is invoked but a required representative, eligibility, amplitude, edge, artifact, claim-status, or manifest record is absent or ill-typed;
- (iii) **pass**: the clause is well-typed and $\text{Ext}^1(\mathcal{R}(C_{\tau,WF}), k) = 0$, whether by Theorem 4.36 or by a separately verified direct computation;
- (iv) **reject**: the clause is well-typed and the eligible Ext^1 -group is evidenced to be nonzero.

The statuses **undefined** and **not_invoked** are distinct; neither is a fabricated zero group.

Proposition 4.45 (B-Gate⁺ categorical discharge). *Let $(F; W, \tau)$ be eligible for the Ext-edge. If the repaired topological clause*

$$\text{Eval}_{1,W,\tau}(F) = 0$$

holds, then the categorical Ext¹-clause of B-Gate⁺ is satisfied:

$$\text{Ext}^1(\mathcal{R}(C_{\tau,WF}), k) = 0.$$

Proof. This is exactly Theorem 4.36 read as a gate-discharge statement. □

Remark 4.46 (What remains after categorical discharge). The Ext-edge discharge is only one atomic B-Gate⁺ clause. A full audited pass still requires, independently, the exact topological statement

$$\text{Eval}_{1,W,\tau}(F) = 0,$$

complete and consistent manifest/claim-status data, and every comparison obligation in its declared mode. In particular, a metric transport through hard deletion requires

$$\text{val}_B(\mathfrak{d}_{\text{met}}) < \text{gap}_{\tau}(\mathfrak{B})$$

together with an evidenced discrepancy bound, while exact mode requires its exact proof. The pass also requires a Type IV scope status equal to

$$\text{not_invoked} \quad \text{or} \quad \text{certified_zero}.$$

If tower semantics are invoked, the comparison morphism, terminal-tameness, and defect diagnostics must be present; **undefined** and **obstructed** fail the gate. No passing categorical clause cancels another rejected or required undefined clause.

4.9 Representative and realization invariance

Definition 4.47 (Degree- n realization-equivalent representatives). Two standard n -eligible representatives are degree- n realization-equivalent when a declared derived isomorphism

$$\alpha_n : A_{n;F,W,\tau} \xrightarrow{\sim} A'_{n;F,W,\tau}$$

is supplied and the two higher edge-realization certificates commute with $H^{-n}(\alpha_n)$ and the declared profile identification.

Proposition 4.48 (Higher representative invariance). *For degree- n realization-equivalent eligible representatives, the Ext groups are identified contravariantly along α_n , and both the one-way discharge and any declared natural zero-reflecting equivalence have the same truth value for the two representatives.*

Proof. Apply $\text{Hom}_{D^b}(-, k[n])$ to α_n . Naturality of Theorem 4.20 identifies this map with dualizing $H^{-n}(\alpha_n)$. Compatibility of the edge certificates and, when invoked, of the zero-reflection package transports all hypotheses and conclusions. $\square$

Definition 4.49 (Realization-equivalent representatives). Two eligible representatives $C_{\tau, W}F$ and $C'_{\tau, W}F$ are called *realization-equivalent* if their realized objects are isomorphic in the derived category:

$$\mathcal{R}(C_{\tau, W}F) \cong \mathcal{R}(C'_{\tau, W}F) \quad \text{in } D^b(k\text{-mod}),$$

and the edge-realization certificates are compatible with that isomorphism.

Proposition 4.50 (Representative invariance of Ext discharge). *Let the realization-equivalent eligible representatives be*

$$C_{\tau, W}F \quad \text{and} \quad C'_{\tau, W}F.$$

Then the declared realization isomorphism induces an identification of the categorical groups, and the truth value of

$$\text{Ext}^1(\mathcal{R}(C_{\tau, W}F), k) = 0$$

is independent of the representative choice.

Proof. The declared isomorphism

$$\alpha : \mathcal{R}(C_{\tau, W}F) \xrightarrow{\sim} \mathcal{R}(C'_{\tau, W}F)$$

induces an isomorphism under the contravariant functor $\text{Hom}_{D^b}(-, k[1])$. This identification is canonical relative to the recorded α ; no independence from an unrecorded choice of realization is asserted. By Corollary 4.24, it agrees with dualizing the induced map on H^{-1} . Compatibility of the edge-realization certificates therefore transports the discharge data, and vanishing is invariant. $\square$

Remark 4.51 (Profile invariance is weaker). Profile-level representative invariance from Chapter 3 does not by itself imply realization-level invariance. For Ext-use, the proof object must record realization-equivalence or must commit to a specific representative. This prevents the Ext^1 -verdict from depending on an untracked representative choice.

4.10 Interaction with the E_1 detector

Proposition 4.52 (Complete detector higher Ext discharge). *Let $n \geq 1$, suppose $(F; W, \tau)$ is standard n -eligible, and let a Chapter 3 detector certificate on the exact profile stage prove*

$$D(\text{Eval}_{n, W, \tau}(F)) = 0 \implies \text{Eval}_{n, W, \tau}(F) = 0.$$

If the detector value is certified zero, then

$$\text{Ext}^n(A_{n, F, W, \tau}, k) = 0.$$

If the detector is also the declared natural zero-reflecting edge E_n , then its zero value is equivalent to Ext vanishing under Theorem 4.31.

Proof. Detector completeness gives exact repaired-profile vanishing. Apply Theorem 4.29. The final assertion is the zero-reflecting theorem specialized to the declared detector edge. $\square$

Proposition 4.53 (Operational E_1 -detector discharge). *Suppose $(F; W, \tau)$ is eligible for the Ext-edge and the canonical, non-truncated operational detector of Chapter 3 satisfies*

$$E_1(F; W, \tau) = 0.$$

Then

$$\text{Ext}^1(\mathcal{R}(C_{\tau, W}F), k) = 0.$$

Proof. By the canonical detector rule of Chapter 3, $E_1 = 0$ already excludes both finite and infinite residual bars and therefore implies

$$\text{Eval}_{1, W, \tau}(F) = 0.$$

The conclusion follows from Theorem 4.36. $\square$

Remark 4.54 (No stability or budget role for E_1). The preceding proposition uses the canonical E_1 only as a vanishing detector. It does not make E_1 a bottleneck-stable quantity, a threshold-gap certificate, a metric-ledger component, or a replacement for

$$\text{val}_B(\mathfrak{d}_{\text{met}}) < \text{gap}_{\tau}(\mathfrak{B}).$$

A truncated detector has its own policy and does not inherit the proposition without a separate proof.

4.11 Non-converse discipline

Proposition 4.55 (Higher reverse inference requires zero-reflection). *Under standard n -eligibility alone,*

$$\text{Ext}^n(A_{n; F, W, \tau}, k) = 0$$

implies only

$$E_n(\text{Eval}_{n, W, \tau}(F)) = 0.$$

It implies repaired-profile vanishing only when a separate zero-reflection theorem for E_n is available on the declared profile class.

Proof. The first implication follows from Theorem 4.28 and finite-dimensional dual reflection. The passage from $E_n(P) = 0$ to $P = 0$ is exactly the additional zero-reflection hypothesis and is not contained in eligibility. $\square$

Definition 4.56 (Synthetic detector object). For a finite-dimensional detector output $D(P)$ and $n \geq 1$, define the shifted encoded object

$$A_{D, n}^{\text{det}}(P) := D(P)[n] \quad \text{in } D^b(k\text{-mod}).$$

This object is called *synthetic* unless it is independently identified, through a natural realization theorem, with a purpose-faithful external or Core realization.

Proposition 4.57 (Encoded detector duality). *For every finite-dimensional detector output $D(P)$,*

$$\text{Ext}^n(A_{D, n}^{\text{det}}(P), k) \cong \text{Hom}_k(D(P), k).$$

Consequently the encoded Ext group vanishes if and only if $D(P) = 0$.

Proof. The shifted object has $H^{-n}(D(P)[n]) = D(P)$ and no other cohomology. Apply Theorem 4.20 and finite-dimensional dual reflection. $\square$

Remark 4.58 (Synthetic duality non-promotion rule). Proposition 4.57 is a valid algebraic encoding theorem, but by itself it merely rewrites detector information in a derived category. It is classified as *Detector Duality* or *Encoded Ext Theorem*; it is not the principal higher Bridge, does not prove a realization theorem, and does not establish $\mathrm{PH}_n \cong \mathrm{Ext}^n$.

Remark 4.59 (General coefficient rings are outside the v20 Core). Over a general coefficient ring R , the spectral sequence

$$E_2^{p,q} = \mathrm{Ext}_R^p(H^{-q}(A), R) \implies \mathrm{Ext}_R^{p+q}(A, R)$$

need not collapse. Higher differentials, extension data, and nonzero outer terms may obstruct an edge isomorphism. Any ring-valued higher bridge belongs to a separately proved Technical Extension or successor release.

Proposition 4.60 (Failure of the reverse bridge for a natural nonfaithful edge). *Under Definitions 4.6–4.9, the implication*

$$\mathrm{Ext}^1(\mathcal{R}(C_{\tau,W}F), k) = 0 \implies \mathrm{Eval}_{1,W,\tau}(F) = 0$$

is false in general, even when the realization and edge extractor are fixed functorially at scope level.

Proof. Choose $0 < \tau < L$, choose $\delta > 0$, set

$$W = [0, L + \delta), \quad t_* = L + \frac{\delta}{2},$$

and let F be a finite filtered complex with a degree-one generator born at 0 and a degree-two generator born at L whose boundary kills the degree-one generator. Its degree-one persistence barcode on W is the single interval $[0, L)$. Since $L > \tau$, hard threshold deletion retains that interval, so

$$\mathrm{Eval}_{1,W,\tau}(F) \cong k_{[0,L)} \neq 0.$$

Take this filtered model as the degree-one compatible representative.

Fix, independently of the verdict, the point-evaluation edge policy and realization functor

$$\mathrm{Edge}_1(P) := P(t_*), \quad \mathcal{R}_{t_*}(C) := H_1(C^{t_*})[1].$$

Both are functorial on the evident declared domain, and $\mathcal{R}_{t_*}(C) \in D^{[-1,-1]}(k\text{-mod}) \subset D^{[-1,0]}(k\text{-mod})$. The representative isomorphism induces the natural edge certificate

$$H^{-1}(\mathcal{R}_{t_*}(C_{\tau,W}F)) = H_1((C_{\tau,W}F)^{t_*}) \xrightarrow{\sim} \mathrm{Eval}_{1,W,\tau}(F)(t_*) = \mathrm{Edge}_1(\mathrm{Eval}_{1,W,\tau}(F)).$$

Because $t_* > L$, both sides are zero. Therefore

$$\mathrm{Ext}^1(\mathcal{R}_{t_*}(C_{\tau,W}F), k) = 0,$$

while the repaired persistence profile itself remains nonzero. Thus a point-evaluation edge can miss a transient bar, and reverse recovery requires a separate zero-reflection or faithfulness theorem. $\square$

Remark 4.61 (What a reverse theorem would require). A reverse bridge would need a separately proved zero-reflection condition, for example

$$\mathrm{Edge}_1(P) = 0 \implies P = 0$$

on the declared repaired-profile class, together with compatibility showing that the realization edge is exactly that extractor. Such faithfulness is a new bridge theorem with new hypotheses; it cannot be inferred from Ext vanishing, amplitude, or eligibility alone.

4.12 Canonical audit-manifest extension

Definition 4.62 (Eligibility manifest extension). The Chapter 4 eligibility entry extends the Chapter 1 manifest and the Chapter 3 evaluation, detector, and representative records. For every invoked degree n , it records at least:

- (i) the pre-threshold and repaired degree- n profile identities;
- (ii) the exact route by which repaired-profile vanishing was established: direct profile proof, complete detector, exact comparison, or metric zero-transport;
- (iii) the detector stage, endpoint convention, stencil or birth/germ policy, completeness theorem, family scope, and artifact identity whenever a detector is consumed;
- (iv) the representative identifier and actual $\rho_{n,F,W,\tau}$;
- (v) the realization domain $\mathcal{E}_n$, functor $\mathcal{R}_n$, object $A_{n,F,W,\tau}$, versions, and immutable artifacts;
- (vi) the amplitude certificate or exact registered replacement;
- (vii) the fixed edge extractor E_n , its objectwise or natural scope, and the edge-realization certificate $\epsilon_{n,F,W,\tau}$;
- (viii) zero compatibility and, when a converse is used, the complete natural zero-reflection package;
- (ix) representative-change and realization-change compatibility;
- (x) the typed degree- n Ext-clause status and aggregate categorical status;
- (xi) the theorem or direct-computation identifiers supporting pass or reject;
- (xii) the exact/metric comparison records independently required by the topological zero statement;
- (xiii) a synthetic-versus-natural classification for every shifted detector object or encoded Ext result;
- (xiv) all Claim, CM, TB, proof, hash, and provenance links needed for independent replay.

For $n = 1$, this contains all fields of the inherited v19 eligibility entry.

Remark 4.63 (Manifest status and failure routing). An invoked Ext-clause with missing eligibility data is undefined; it is not a mathematical counterexample and not `not_invoked` unless scope exclusion was explicitly declared. Under Chapter 7, an incomplete manifest yields audit undefinedness unless another evidenced clause already forces rejection, while an inconsistent manifest yields Type V rejection. Only a complete and consistent manifest can support a categorical pass.

4.13 Bridge package

Theorem 4.64 (Eligible Ext-edge bridge package). *In the v20 Core, the following conservative package is available.*

- (i) *The complete v19 degree-one eligible bridge remains valid at its original strength.*
- (ii) *For every $n \geq 0$ and bounded derived object over the field,*

$$\mathrm{Ext}^n(A, k) \cong \mathrm{Hom}_k(H^{-n}(A), k)$$

canonically and naturally.

- (iii) *A standard n -eligible record supplies an independently meaningful realized object, conservative amplitude typing, a fixed edge extractor, and an artifact-backed edge-realization certificate.*
- (iv) *Under zero compatibility,*

$$\text{Eval}_{n,W,\tau}(F) = 0 \implies \text{Ext}^n(A_{n;F,W,\tau}, k) = 0.$$

- (v) *Under the separately proved natural zero-reflection package, the preceding implication upgrades to an equivalence of vanishing predicates.*
- (vi) *A complete fixed-stencil, birth, germ, or other Chapter 3 detector may supply zero-reflection only on its certified profile class and stage.*
- (vii) *A shifted detector object gives an encoded Ext duality but remains synthetic without an independent natural realization theorem.*
- (viii) *None of these clauses replaces metric, Type IV, transient-defect, manifest, claim-status, finite-to-infinite, or external Return-Theorem obligations.*

Proof. Item (i) is Theorem 4.36 and its inherited package. Item (ii) is Theorem 4.20. Item (iii) is Definition 4.17. Item (iv) is Theorem 4.29. Item (v) is Theorem 4.31. Item (vi) follows from Chapter 3 detector completeness together with Corollary 4.33 and its birth/germ specializations. Item (vii) is Proposition 4.57 and Remark 4.58. Item (viii) follows from the independent gate and interface contracts of Chapters 1, 5, 7, and 8. $\square$

Remark 4.65 (Package status). Theorem 4.64 is a conditional closure package. It separates unconditional field-coefficient derived algebra from declared realization hypotheses and from separately proved detector zero-reflection. It does not promote eligibility data, detector completeness, synthetic encoding, or a gate record into an external-domain theorem.

4.14 Explicit non-claims

The following are not asserted in this chapter.

- (i) No unconditional object-level equivalence $\text{PH}_n \cong \text{Ext}^n$ is asserted in any degree.
- (ii) No reverse implication from Ext^n -vanishing to repaired-profile vanishing is asserted without a separately proved natural zero-reflection package.
- (iii) No higher edge extractor is assumed faithful, conservative, or zero-reflecting merely because it is functorial or zero-compatible.
- (iv) No objectwise edge certificate is a natural transformation without a proved natural-family compatibility theorem.
- (v) No realization, edge extractor, stencil, or profile class may be chosen adaptively after inspecting the desired verdict.
- (vi) No invoked Ext clause is typed without the required representative, realization, amplitude policy, edge certificate, and manifest data.

- (vii) No missing eligibility, detector completeness, naturality, or artifact is interpreted as zero; the status is undefined.
- (viii) No finite detector supplies a reverse bridge outside its exact source stage, endpoint convention, coverage or birth/germ scope, and certified profile class.
- (ix) No soundness-only detector is treated as complete.
- (x) No synthetic shifted detector object is promoted to the principal natural higher Bridge merely because its encoded Ext group is dual to the detector output.
- (xi) No general-ring, sheaf, stack, motivic, dg, or spectral-sequence extension inherits the field theorem without separate hypotheses and proof.
- (xii) No standard amplitude certificate is said to be mathematically necessary for the field-coefficient higher Ext isomorphism; it is the conservative eligibility regime used by this release.
- (xiii) No global collapse functor C_τ , identity $C_{\tau,W}F = (C_\tau F)|_W$, global exactness, or Serre localization is used.
- (xiv) No threshold-gap, metric-zero transport, detector rank stability, or crop theorem is reproved by the higher Ext edge.
- (xv) No categorical discharge cancels a failed detector, metric, Type IV, transient-defect, manifest, or claim-status clause.
- (xvi) No E_1 , stencil, birth, germ, or other detector is a metric ledger component merely because its output is finite-dimensional.
- (xvii) No point-evaluation edge is assumed to detect transient bars; the inherited degree-one counterexample remains valid.
- (xviii) No representative-profile equivalence automatically implies realization equivalence in any degree.
- (xix) No arithmetic Ext group, Selmer group, PDE obstruction, motivic Ext, or external categorical invariant is automatically the Core group $\text{Ext}^n(A_n(F), k)$.
- (xx) No external mathematical conclusion follows without the separately proved interface and Return Theorem of Chapter 8.
- (xxi) No Search artifact, AI confidence value, spectral warning, ranking, or advisory invariant discharges a categorical clause.

4.15 Provenance and status

The constructible persistence input, repaired evaluation, finite detector certificates, and representative records are consumed from Chapters 1–3. The inherited degree-one bridge is the v19 theorem package. The new v20 mathematics consists of the canonical higher field-coefficient Ext edge, the standard n -eligible interface, the higher one-way discharge, and the conditional natural zero-reflecting vanishing equivalence. The persistence-to-derived edge remains an explicit realization hypothesis; a synthetic detector encoding remains separately classified.

Status labels for this chapter. *Inherited v19 Core definitions and results:* Definitions 4.3, 4.4, 4.6, 4.7, 4.8, 4.9, 4.43, 4.44, and 4.49; Lemma 4.22; Theorem 4.23; Corollaries 4.24 and 4.25; Theorem 4.36; Propositions 4.45, 4.50, 4.53, and 4.60.

New v20 Core derived results: Theorem 4.20, Corollary 4.21, Theorem 4.28, and Theorem 4.29.

Conditional v20 natural Bridge results: Definition 4.30, Theorem 4.31, Corollary 4.33, Proposition 4.32, Proposition 4.52, and Proposition 4.48.

Higher eligibility, gate, and manifest records: Definitions 4.12, 4.13, 4.14, 4.15, 4.16, 4.17, 4.39, 4.40, 4.42, 4.47, and 4.62; Proposition 4.41.

Synthetic/supporting algebra, not principal natural Bridge: Definition 4.56 and Proposition 4.57.

Conservative closure package: Theorem 4.64.

Explicit exclusions: Subsection 4.14.

5 Chapter 5: Failure Taxonomy, Terminal-Generic Type IV Diagnostics, and Transient Defect Detection

Chapter 5 contract. Chapter 5 preserves the v19 terminal-generic Type IV diagnostic and adds a strictly separate v20 finite detector for noncofinal defect bars. Type IV remains a typed comparison-morphism diagnostic, not a scalar proxy and not a synonym for every tower defect. A no-tower claim records `not_invoked`; an invoked but incomplete tower claim records `undefined`; only a valid terminal-generic diagnostic may be `certified_zero` or `obstructed`.

For a valid comparison morphism

$$\phi_{i,W,\tau} : \text{ColimProfile}_{i,W,\tau}(F_\bullet) \longrightarrow \text{ApexProfile}_{i,W,\tau}(F_\infty),$$

the terminal pair

$$(\mu_{i,W,\tau}, \nu_{i,W,\tau})$$

measures right-cofinal kernel and cokernel defects. A separately declared defect threshold $\tau_{\text{def}} > 0$ and a Chapter 3 complete detector, registered at one of the source stages

$$\text{kernel_defect}, \quad \text{cokernel_defect},$$

measure finite noncofinal defects that survive that defect threshold. The collapse threshold τ and defect threshold τ_{def} are independent typed parameters and are never silently identified.

Metric budgets enter this chapter only through the declared bottleneck scalarization

$$\text{val}_B : V \longrightarrow [0, +\infty].$$

Missing morphisms, source-semantic authority, terminal tameness, detector completeness, stage applicability, claim status, and manifest completeness remain non-scalar obligations. Zero terminal diagnostics together with zero transient detectors exclude long monitored defects, but do not imply full isomorphism unless shorter residual defects are separately excluded.

5.1 Scope, relocation, and standing regime

This chapter closes the v20 conservative Core failure taxonomy after the metric, record, finite-detector, representative, and higher Ext-edge repairs of Chapters 1–4. It consumes the following previously established

data:

- (i) the repaired windowed evaluation

$$\text{Eval}_{i,W,\tau}(F) = T_{\tau}^{\text{bar}}(W_W \mathbf{P}_i(F));$$

- (ii) the distinction between the pre-threshold profile

$$B_{i,W}(F) := W_W \mathbf{P}_i(F)$$

and the post-threshold repaired profile $\text{Eval}_{i,W,\tau}(F)$;

- (iii) single-profile threshold-gap readiness records and, whenever metric transport through hard deletion is used, two-sided threshold-gap comparison certificates from Chapter 3; exact comparisons instead carry their exact equality or profile-isomorphism evidence;
- (iv) admissible representative and realization records from Chapters 3–4;
- (v) the degreewise one-way categorical discharge

$$\text{Eval}_{n,W,\tau}(F) = 0 \implies \text{Ext}^n(A_n(F), k) = 0$$

in every invoked n -eligible regime of Chapter 4;

- (vi) the Chapter 3 source stages

$$\text{kernel_defect}, \quad \text{cokernel_defect},$$

together with detector soundness, completeness, endpoint, coverage, family, metric, and artifact requirements.

The purpose of Chapter 5 is:

- (i) to give a typed, nonexclusive failure taxonomy of Types I–V;
- (ii) to specify when a tower comparison is a mathematical morphism rather than a barcode-level suggestion;
- (iii) to define terminal-generic kernel/cokernel diagnostics on bounded and right-unbounded windows;
- (iv) to define full and interior defect profiles and complete finite detectors at a separately declared defect threshold;
- (v) to separate terminal/cofinal defects, long transient interior defects, and shorter residual defect noise;
- (vi) to prevent undefined or inapplicable Type IV or transient diagnostics from being reported as zero;
- (vii) to integrate the resulting diagnostics with B-Gate⁺ without upgrading thresholded defect zero to full comparison-map isomorphism.

Remark 5.1 (Relocation of the previous Chapter 5 material). The previous Chapter 5 material on Window Logic, Overlap Gate, Restart/Summability, and local-to-global certificate assembly is not discarded. In the v20 ordering it belongs after the present failure-and-diagnostics chapter and is therefore reserved for Chapter 6. The present chapter does not assert Restart or gluing theorems.

Remark 5.2 (Metric repair is consumed, not reproved). The sharp threshold-gap-safe stability theorem is not reproved here. Whenever a tower artifact or comparison record is built from perturbed repaired profiles, the relevant two-sided threshold-gap-certified records must already have been supplied under Chapters 1–3 and Appendix A. If the metric ledger is $\mathfrak{d}_{\text{met}} \in \mathbf{V}$, the gate-scoped real safety condition is

$$\varepsilon_{\mathbf{B}}(\mathfrak{d}_{\text{met}}) := \text{val}_{\mathbf{B}}(\mathfrak{d}_{\text{met}}) < \text{gap}_{\tau}(\mathfrak{B}) = \min_{B \in \mathfrak{B}} \text{clear}_{\tau}(B),$$

for the actual finite comparison family $\mathfrak{B}$ of pre-threshold profiles. The older notation $\Sigma\delta < \text{gap}_{\tau}$ is permitted only as an explicitly declared abbreviation for this scalarized real inequality. The metric bound protects the repaired source or apex comparison named in its record. It does not automatically induce any bound on a kernel or cokernel defect profile.

5.2 Typed failure landscape

Definition 5.3 (v20 failure types and transient-defect refinement). Let a Core claim be represented by a declared run record on a window W , collapse threshold τ , and finite set of monitored degrees I_{mon} . The five inherited failure types are nonexclusive. The v20 transient-defect verdict is an additional typed refinement attached to a valid tower artifact; it is not assigned a new Roman numeral and is not renamed terminal-generic Type IV.

Type I. Topological or persistence failure. For at least one invoked $n \in I_{\text{mon}}$,

$$\text{Eval}_{n,W,\tau}(F) \neq 0.$$

A certified complete Chapter 3 detector with value nonzero supplies a witness. An incomplete, inapplicable, or wrong-stage detector leaves the clause undefined; it does not prove Type I.

Type II. Categorical failure in an eligible regime. For at least one invoked n -eligible realization,

$$\text{Ext}^n(A_n(F), k) \neq 0.$$

Missing eligibility, edge realization, required naturality, or artifact data makes the degreewise clause undefined, not categorically obstructed.

Type III. Detector, operational, metric, comparison, or functorial instability. A required finite-level operation or safety certificate is absent, undefined, wrong-stage, or evidenced to fail. This includes failure of a repaired evaluation record, representative compatibility, crop budget, window-edge clearance, two-sided threshold-gap certificate, detector applicability or completeness, defect-threshold declaration, profile system morphism, source-semantic authority, terminal tameness, or declared comparison budget. Missing evidence is normally undefined; an evidenced false required condition is reject.

Type IV. Tower-level terminal-generic obstruction. A valid comparison artifact exists and

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) \neq (0, 0).$$

On a right-unbounded window this is an infinite-bar defect; on a bounded window it is a defect cofinal to the declared right endpoint. A certified nonzero transient interior detector with $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$ is recorded as a separate transient-defect obstruction and does not become terminal-generic Type IV by renaming.

Type V. Claim-status, manifest, provenance, or promotion failure. A run cites a rejected or deprecated claim, suppresses a required missing marker, reports undefined as pass or zero, overstates source provenance, relabels a detector certificate across profile stages, treats a sampled zero as complete, uses a Search/AI/advisory artifact as proof, treats a finite observation as an infinite theorem, or otherwise makes the canonical manifest incomplete or inconsistent.

When the governing claim requires absence of long transient defects, a valid transient detector with value `nonzero` is a mathematical gate obstruction even though the terminal Type IV status may remain `certified_zero`. Detector failure and detected defect are distinct: the former is Type III; the latter is a well-defined negative mathematical verdict for the stronger transient-defect clause.

Remark 5.4 (The taxonomy is nonexclusive). A single run may exhibit several failure types simultaneously. For example, a missing two-sided gap record is Type III; if the manifest nevertheless reports a certified pass, the same run also has Type V failure. The type number is not a severity ranking.

Definition 5.5 (Failure-check order). The Core audit order is:

- (1) claim and manifest well-formedness (Type V);
- (2) metric, operational, representative, and artifact well-formedness (Type III);
- (3) topological and eligible categorical verdicts (Types I–II);
- (4) invoked tower diagnostics (Type IV);
- (5) final B-Gate⁺ conjunction and budget decision.

A later numerical zero may not repair an earlier undefined or ill-typed record.

Remark 5.6 (Undefined is not zero). The absence of a tower comparison artifact, the failure of terminal tameness, or the absence of an actual morphism between the declared source and target profiles does not imply $\mu_{\text{Collapse}} = \nu_{\text{Collapse}} = 0$. It means that the Type IV diagnostic is undefined. Reporting that state as $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$ is a Type V manifest error.

Remark 5.7 (No scalar repair of non-scalar obligations). A larger metric ledger cannot manufacture a repaired tower-profile system, an actual comparison morphism, a terminal-tameness certificate, a kernel/cokernel artifact, a valid claim status, or a complete and consistent manifest. These are typed proof obligations, not numerical error terms. Consequently $\text{val}_B(\mathfrak{d}_{\text{met}}) < \text{gap}_\tau$ is necessary for metric safety when invoked, but it cannot turn undefined into `certified_zero`.

5.3 Structural and audited collapse admissibility

Definition 5.8 (Type IV scope status). Every Core run records exactly one of the following Type IV scope statuses:

`not_invoked`, `undefined`, `certified_zero`, `obstructed`.

They mean respectively:

- (i) no tower, apex, limit, or colimit comparison is part of the claim;
- (ii) tower semantics are invoked, but the diagnostic input is incomplete or ill-typed;
- (iii) an artifact valid for the claim’s declared source semantics exists and $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$;

(iv) an artifact valid for the claim's declared source semantics exists and $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) \neq (0, 0)$.

Definition 5.9 (Structural collapse admissibility). Fix a finite monitored degree set I_{mon} . A declared object $(F; W, \tau, I_{\text{mon}})$ is *structurally collapse-admissible* if:

- (i) $\text{Eval}_{n,W,\tau}(F) = 0$ for every invoked $n \in I_{\text{mon}}$;
- (ii) every invoked degree- n Ext clause has a complete n -eligibility record and is discharged;
- (iii) the Type IV scope status is either `not_invoked` or `certified_zero`;
- (iv) if absence of long transient defects is required, the full defect profiles, defect threshold, and complete kernel/cokernel detector records exist and the aggregate transient verdict is `certified_absent`;
- (v) no Type V claim-status defect is present in the structural record.

Definition 5.10 (Audited collapse admissibility). A structurally collapse-admissible run is *audited collapse-admissible* if, in addition:

- (i) every required repaired comparison declares exactly one comparison mode

$$\text{exact}, \quad \text{metric_gap_certified}, \quad \text{not_compared};$$
- (ii) every metric comparison transported through hard deletion has a two-sided threshold-gap certificate, while every exact comparison carries its exact equality or profile-isomorphism proof;
- (iii) every crop and window-edge effect used by the proof is controlled by the appropriate exact theorem or metric ledger component;
- (iv) whenever metric mode is invoked, the ledger aggregation law, scalarization val_B , actual discrepancy upper bound, and strict reserve

$$\text{val}_B(\mathbf{d}_{\text{met}}) < \text{gap}_\tau(\mathfrak{B})$$
 are established;
- (v) every detector is applied at its declared Chapter 3 source stage, and all endpoint, coverage, family, excess, completeness, and artifact obligations pass;
- (vi) every in-scope non-scalar proof obligation is present and well-typed;
- (vii) the canonical Chapter 7 manifest is complete and consistent;
- (viii) no Type III operational or metric failure remains.

Proposition 5.11 (Eligible monitored vanishing discharges the invoked Type II clauses). *Fix a finite set I_{mon} . For every invoked $n \in I_{\text{mon}}$, assume a complete n -eligibility record in the sense of Chapter 4. If*

$$\text{Eval}_{n,W,\tau}(F) = 0 \quad \text{for every invoked } n \in I_{\text{mon}},$$

then

$$\text{Ext}^n(A_n(F), k) = 0 \quad \text{for every invoked } n \in I_{\text{mon}}.$$

Consequently the invoked Type II clauses are discharged, but Type III, Type IV, transient-defect, Type V, and budget clauses remain independent.

Proof. Apply the Chapter 4 higher one-way discharge theorem degreewise and then use the aggregate higher categorical status rule. The independence of the remaining clauses is part of the Chapter 1 B-Gate⁺ contract. $\square$

5.4 Windowed tower-profile systems

Let

$$F_0 \longrightarrow F_1 \longrightarrow F_2 \longrightarrow \cdots$$

be a declared tower of filtered or persistence objects and let F_∞ be a declared apex, terminal object, or comparison target when one is supplied.

Definition 5.12 (Windowed repaired tower objects). For a monitored degree i , window W , and threshold τ , write

$$M_{n;i,W,\tau} := \text{Eval}_{i,W,\tau}(F_n).$$

These objects alone do not form a directed system, because barcode-level threshold deletion is not a global functor on arbitrary persistence morphisms.

Definition 5.13 (Repaired tower-profile system). A *repaired tower-profile system* for (i, W, τ) consists of:

- (i) the repaired objects $M_{n;i,W,\tau}$;
- (ii) actual persistence-profile morphisms

$$\psi_{n,m} : M_{n;i,W,\tau} \longrightarrow M_{m;i,W,\tau} \quad (n \leq m);$$

- (iii) the identities $\psi_{n,n} = \text{id}$ and composition law

$$\psi_{m,r} \circ \psi_{n,m} = \psi_{n,r};$$

- (iv) provenance showing that the morphisms are either explicitly constructed or supplied canonically by a declared τ -split/exact-admissible regime.

A sequence of barcode multisets or bottleneck matchings without these morphisms is not a repaired tower-profile system.

Definition 5.14 (Collapsed source profile). Given a repaired tower-profile system, a source profile

$$\text{ColimProfile}_{i,W,\tau}(F_\bullet)$$

is declared with exactly one provenance status:

$$\text{genuine_colimit}, \quad \text{verified_stabilization}, \quad \text{implementation_artifact}.$$

In the first case it is a genuine colimit in the declared constructible profile category. In the second it is a certified stabilized profile; it is identified with a categorical colimit only when a separate stabilization-to-colimit theorem is recorded. In the third it is an implementation object whose relation to the tower is part of the artifact and is not silently promoted either to a categorical colimit or to a verified stabilization.

Every claim also declares the source semantics it requires. A source artifact is admissible for that claim only when its provenance has sufficient semantic authority. In particular, an `implementation_artifact` may support an implementation-scoped comparison, but it cannot discharge a claim about a genuine tower colimit merely by being named `ColimProfile`.

Definition 5.15 (Collapsed apex profile). The repaired apex profile is

$$\text{ApexProfile}_{i,W,\tau}(F_\infty) := \text{Eval}_{i,W,\tau}(F_\infty),$$

provided the repaired evaluation record for F_∞ is declared.

Remark 5.16 (No automatic colimit–collapse interchange). No identity of the form

$$\lim_{\rightarrow n} \text{Eval}_{i,W,\tau}(F_n) \cong \text{Eval}_{i,W,\tau}(F_\infty)$$

or

$$\lim_{\rightarrow n} T_\tau^{\text{bar}}(M_n) \cong T_\tau^{\text{bar}}\left(\lim_{\rightarrow n} M_n\right)$$

is available in the general Core. Any such comparison is exactly the data that the tower artifact must supply or justify.

5.5 Tower comparison artifacts

Definition 5.17 (Tower comparison artifact). A *tower comparison artifact* for (i, W, τ) consists of the following declared data:

- (i) a repaired tower-profile system;
- (ii) a source profile $\text{ColimProfile}_{i,W,\tau}(F_\bullet)$ with provenance status;
- (iii) a repaired apex profile $\text{ApexProfile}_{i,W,\tau}(F_\infty)$;
- (iv) an actual morphism in the declared constructible profile category on W ,
$$\phi_{i,W,\tau} : \text{ColimProfile}_{i,W,\tau}(F_\bullet) \longrightarrow \text{ApexProfile}_{i,W,\tau}(F_\infty);$$
- (v) the construction or proof establishing that $\phi_{i,W,\tau}$ is well-typed;
- (vi) the claim-required source semantics and evidence that the source provenance has sufficient semantic authority for that claim;
- (vii) the repaired evaluation, comparison-mode, crop-budget, and representative records used in constructing the source, target, or comparison; every metric comparison transported through hard deletion carries a two-sided threshold-gap certificate, whereas an exact comparison carries its exact proof;
- (viii) the terminal mode and terminal-tameness certificate used by the diagnostic;
- (ix) an explicitly declared abelian constructible profile category on W , closed under the kernels and cokernels used here, together with kernel and cokernel computation artifacts or their categorical proof.
- (x) when transient detection is invoked, immutable full defect-profile identifiers sufficient to register new Chapter 3 detector records at `kernel_defect` and `cokernel_defect`; no pre-existing detector record is relabeled across stages.

Remark 5.18 (A bottleneck matching is not the comparison morphism). A bottleneck matching controls metric discrepancy between barcodes. It is not, by itself, a natural transformation of persistence modules and therefore does not define the kernel or cokernel needed by Type IV. The tower artifact must contain an actual morphism $\phi_{i,W,\tau}$ or a canonical construction of one. Moreover, the morphism must live in the declared abelian profile category and its source provenance must match the semantic strength required by the claim.

Definition 5.19 (Exact-admissible tower artifact). A tower comparison artifact is *exact-admissible* if its profile-system morphisms, source construction, comparison map, kernels, and cokernels are supplied by a declared regime in which the retained-long-bar assignment is functorial and preserves every exact sequence used by the diagnostic.

Remark 5.20 (Explicit artifacts remain allowed). Exact admissibility is sufficient but not necessary. An explicitly constructed morphism between declared repaired source and apex profiles is also admissible, provided all typing, source-semantics, provenance, ambient-abelian-category, kernel/cokernel, and terminal-tameness obligations are met. Artifact validity is always relative to the claim’s declared source semantics; a lower-authority implementation source does not become a categorical colimit by successful kernel/cokernel computation.

5.6 Terminal-generic dimension

The old shorthand “generic dimension equals infinite-bar multiplicity” is correct only on a right-unbounded window. On a bounded window, zero extension outside the window makes the ordinary $t \rightarrow +\infty$ dimension identically zero and would render the diagnostic vacuous. The following window-relative definition removes that ambiguity.

Definition 5.21 (Terminal-tame profile). Let $W = [u, v)$, where $v \in \mathbb{R} \cup \{+\infty\}$, and let M be a constructible persistence profile on W . We call M *terminal-tame on W* if:

- (i) every fiber near the terminal end is finite-dimensional;
- (ii) if $v < +\infty$, there exists $s < v$ such that every structure map $M(t) \rightarrow M(t')$ with $s \leq t \leq t' < v$ is an isomorphism;
- (iii) if $v = +\infty$, there exists $s \in \mathbb{R}$ such that every structure map $M(t) \rightarrow M(t')$ with $s \leq t \leq t'$ is an isomorphism.

Definition 5.22 (Terminal-generic dimension). For a terminal-tame profile M on $W = [u, v)$, define

$$\text{tgdim}_W(M) := \begin{cases} \lim_{t \uparrow v} \dim_k M(t), & v < +\infty, \\ \lim_{t \rightarrow +\infty} \dim_k M(t), & v = +\infty. \end{cases}$$

When $v = +\infty$, we may also write $\text{gdim}(M)$.

Proposition 5.23 (Well-definedness of terminal-generic dimension). *For every terminal-tame profile M on $W = [u, v)$, the limit in Definition 5.22 exists and is a finite nonnegative integer.*

Proof. By terminal tameness, there is a terminal subinterval on which every structure map is an isomorphism. Hence the finite-dimensional fibers on that subinterval all have the same dimension. The displayed dimension function is therefore eventually constant as $t \uparrow v$ when $v < +\infty$, or as $t \rightarrow +\infty$ when $v = +\infty$. Its eventual value is finite and is exactly the asserted limit. $\square$

Definition 5.24 (Cofinal bar in a window). A bar I in a barcode profile on $W = [u, v)$ is *cofinal in W* if:

- (i) $\sup I = v$ when $v < +\infty$; or
- (ii) I is unbounded above when $v = +\infty$.

For bounded W , a cofinal bar is a boundary-reaching bar relative to W , not a global infinite bar.

Remark 5.25 (Interval decomposition and terminal tameness are distinct inputs). In the declared constructible one-parameter regime, the monitored profile is interval-decomposable. Terminal tameness is an additional right-end condition; constructibility on bounded subintervals alone does not silently supply a common terminal tail. Proposition 5.26 uses both facts.

Proposition 5.26 (Terminal dimension counts cofinal bars). *For a terminal-tame, interval-decomposable object M on W , the terminal dimension equals the multiplicity of cofinal bars:*

$$\mathrm{tgdim}_W(M) = \#\{\text{bars cofinal in } W\},$$

where multiplicity is counted. On a right-unbounded window, the right-hand side counts the infinite bars.

Proof. Choose a terminal tail on which all structure maps are isomorphisms. A bar contributes one dimension to every fiber on that tail exactly when it is cofinal in W . Every noncofinal bar dies before the terminal tail and contributes zero there. Summing the surviving interval summands gives the stated equality. $\square$

Remark 5.27 (Interpretation of earlier gdim notation). The gdim -notation used schematically in Chapters 1–2 is interpreted by this chapter as the terminal-generic dimension appropriate to the declared window. For $W = [u, +\infty)$ it is the ordinary eventual dimension and counts infinite bars. For bounded $W = [u, v)$ the canonical local diagnostic is tgdim_W , which counts bars reaching v from the left.

Proposition 5.28 (Terminal tameness passes to defects). *Let $\phi : M \rightarrow N$ be a morphism in a declared abelian constructible profile category on W , closed under the pointwise kernels and cokernels used by the diagnostic. If M and N are terminal-tame, then $\ker \phi$ and $\mathrm{coker} \phi$ are terminal-tame.*

Proof. On a common terminal tail, the structure maps of M and N are isomorphisms and the naturality squares for ϕ commute. The induced structure maps on pointwise kernels and cokernels are therefore isomorphisms on the same tail. Finite-dimensionality is inherited pointwise. $\square$

5.7 Defect profiles and Type IV diagnostics

Definition 5.29 (Defect profiles). For a valid tower comparison artifact, define

$$\mathrm{Defect}_{i,W,\tau}^{\ker} := \ker(\phi_{i,W,\tau}), \quad \mathrm{Defect}_{i,W,\tau}^{\mathrm{coker}} := \mathrm{coker}(\phi_{i,W,\tau}).$$

These are objects of the declared constructible profile category on W .

Definition 5.30 (Degreewise Type IV diagnostics). Assume the defect profiles are terminal-tame. Define

$$\mu_{i,W,\tau} := \mathrm{tgdim}_W(\mathrm{Defect}_{i,W,\tau}^{\ker}), \quad \nu_{i,W,\tau} := \mathrm{tgdim}_W(\mathrm{Defect}_{i,W,\tau}^{\mathrm{coker}}).$$

Thus μ measures terminal-generic kernel defect and ν measures terminal-generic cokernel defect.

Definition 5.31 (Total collapse diagnostics). For a fixed declared window–threshold pair (W, τ) and a finite declared monitored-degree set I_{mon} , define

$$\mu_{\mathrm{Collapse}} := \sum_{i \in I_{\mathrm{mon}}} \mu_{i,W,\tau}, \quad \nu_{\mathrm{Collapse}} := \sum_{i \in I_{\mathrm{mon}}} \nu_{i,W,\tau}.$$

The legacy notations $u_{i,W,\tau}$ and u_{Collapse} are deprecated and have no normative use in v20.0. If the monitored-degree scope is infinite, neither total is defined without a separately proved aggregation or summability theorem and an explicit aggregation policy.

Definition 5.32 (Type IV diagnostic record). A Type IV diagnostic record contains the window, threshold, monitored degrees, declared tower scope, and exactly one status

`not_invoked,      undefined,      certified_zero,      obstructed.`

Its remaining required fields depend on that status.

- (i) For `not_invoked`, the record contains an explicit scope rationale showing that no tower, apex, limit, or colimit comparison belongs to the claim. No numerical μ, ν values are asserted.
- (ii) For `undefined`, the record contains either the identifier of a partial or invalid artifact, or an explicit missing-artifact marker, together with the list of missing or ill-typed obligations. No numerical μ, ν values are asserted as Core diagnostics.
- (iii) For `certified_zero` or `obstructed`, the record contains:
 - (a) a valid tower comparison artifact identifier, source-profile provenance status, claim-required source semantics, and semantic-authority check;
 - (b) the terminal mode `bounded_terminal` or `right_unbounded`;
 - (c) terminal-tameness certificates for the defect profiles;
 - (d) kernel and cokernel defect-profile artifacts;
 - (e) all degreewise and total μ, ν values over the fixed (W, τ) and declared finite set I_{mon} ;
 - (f) the full-interior-defect status, recorded separately from terminal-generic and thresholded transient verdicts, with exactly one value

`not_checked,      certified_absent,      certified_present;`

- (g) when transient detection is invoked, the separately declared defect threshold $\tau_{\text{def}} > 0$, detector component `full_profile` or `interior_noncofinal`, source stage, detector information class, certificate status, value status, and the aggregate long-transient status

`not_invoked,      undefined,      certified_absent,      certified_present.`

For the full-interior-defect field, `certified_absent` requires proof that both entire noncofinal defect profiles vanish. An explicit nonzero noncofinal witness is required for `certified_present`. By contrast, long-transient `certified_absent` means only that no monitored noncofinal defect bar has length greater than τ_{def} .

The status is `certified_zero` exactly when all invoked inputs are valid for the claim's declared source semantics and $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$. It is `obstructed` exactly when those inputs are valid and $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) \neq (0, 0)$. Both verdicts are relative to the declared source semantics and monitored scope. An absent artifact, insufficient-provenance source, or source-semantics mismatch is represented by a missing/invalid-obligation marker and yields `undefined`, never zero. The terminal Type IV status is determined only by (μ, ν) . A long-transient verdict is recorded in its own field and does not change `certified_zero` into `obstructed` when $(\mu, \nu) = (0, 0)$.

Definition 5.33 (Type IV obstruction and Type IV-free status). Relative to a valid tower comparison artifact whose source semantics match the claim:

- (i) the run has a *Type IV obstruction* if $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) \neq (0, 0)$;

- (ii) the run is *Type IV-free in the monitored terminal-generic sense* if $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$. Both statements are relative to the fixed (W, τ) , the finite monitored-degree set, and the source semantics recorded by the artifact.

Proposition 5.34 (Characterization of certified Type IV-free status). *A tower record is certified Type IV-free if and only if:*

- (i) *the tower comparison artifact is valid;*
- (ii) *its kernel and cokernel defect profiles are terminal-tame;*
- (iii) $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$.

Proof. This is the defining conjunction in Definitions 5.32 and 5.33. The validity and terminal-tameness clauses are necessary to ensure that the displayed zero is a defined Core diagnostic rather than a missing value. $\square$

Remark 5.35 (Zero diagnostics are not a full isomorphism theorem). The equality

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$$

says that no defect bar reaches the terminal end of the declared window. It does not imply

$$\ker \phi_{i,W,\tau} = 0 \quad \text{or} \quad \text{coker } \phi_{i,W,\tau} = 0.$$

Finite interior defect bars may remain. A full isomorphism conclusion requires a separate no-interior-defect proof establishing that the entire kernel and cokernel profiles vanish, not merely their terminal-generic parts.

Example 5.36 (Zero terminal diagnostics with a nonisomorphism). Let $W = [0, 10)$ and consider the morphism

$$0 \longrightarrow k_{[2,3)}.$$

Its cokernel is $k_{[2,3)}$, so the morphism is not an isomorphism. However, the cokernel has no bar cofinal in W , hence

$$\text{tgdim}_W(k_{[2,3)}) = 0.$$

Thus the terminal-generic diagnostic is zero while a finite interior defect remains.

Example 5.37 (Pure kernel, pure cokernel, and mixed Type IV modes). On a window $W = [u, v)$, let K_W denote one cofinal interval bar: $k_{[u,v)}$ when $v < +\infty$ and $k_{[u,+\infty)}$ when $v = +\infty$. Then:

$$\begin{aligned} K_W \rightarrow 0 & \quad \text{has} \quad (\mu, \nu) = (1, 0), \\ 0 \rightarrow K_W & \quad \text{has} \quad (\mu, \nu) = (0, 1), \\ K_W \oplus 0 \rightarrow 0 \oplus K_W & \quad \text{has} \quad (\mu, \nu) = (1, 1) \end{aligned}$$

for the evident zero maps. These are the pure-kernel, pure-cokernel, and mixed terminal-generic Type IV modes. The mixed map is explicitly the zero morphism from $K_W \oplus 0$ to $0 \oplus K_W$.

Remark 5.38 (Finite-layer checks do not determine Type IV). Finite-layer repaired evaluations do not logically determine the kernel or cokernel of an undeclared source-to-apex comparison. For example, a zero source profile together with a declared apex containing a cofinal bar produces a pure cokernel defect once the comparison $0 \rightarrow K_W$ is declared. This is an artifact-level mismatch witness; it is not a claim that an arbitrary apex is a categorical colimit.

5.8 Cofinal/interior decomposition and transient defect detectors

Definition 5.39 (Cofinal and interior barcode profiles). Let M be an interval-decomposable constructible profile on $W = [u, v]$. Define the *cofinal profile* $\text{Cof}_W(M)$ to be the barcode subprofile consisting of bars cofinal in W , and define the *interior profile* $\text{Int}_W(M)$ to be the barcode subprofile consisting of all noncofinal bars, with multiplicity. At barcode-profile isomorphism strength,

$$M \cong \text{Cof}_W(M) \oplus \text{Int}_W(M).$$

This decomposition is canonical on barcode isomorphism classes. It is not asserted to be a functor on arbitrary persistence morphisms.

Proposition 5.40 (Terminal/interior separation). *If M is terminal-tame and interval-decomposable, then*

$$\text{tgdim}_W(M) = \#\mathcal{B}(\text{Cof}_W(M))$$

with multiplicity, and

$$\text{tgdim}_W(M) = 0 \iff \text{Cof}_W(M) = 0 \iff M \cong \text{Int}_W(M).$$

Proof. The first assertion is Proposition 5.26. A finite direct sum of cofinal interval bars has zero terminal-generic dimension exactly when it is empty. The displayed decomposition then gives the final equivalence. $\square$

Definition 5.41 (Defect threshold and detector component). A transient-defect run declares a threshold

$$\tau_{\text{def}} > 0$$

independently of the collapse threshold τ . It also declares one component for each detector application:

$$\text{full_profile} \quad \text{or} \quad \text{interior_noncofinal}.$$

The full-profile component reads $K_{i,W,\tau}$ or $Q_{i,W,\tau}$. The interior component reads

$$K_{i,W,\tau}^\circ := \text{Int}_W(K_{i,W,\tau}), \quad Q_{i,W,\tau}^\circ := \text{Int}_W(Q_{i,W,\tau}).$$

Changing τ_{def} , the component, the endpoint convention, or the detector source stage creates a new detector claim and a new artifact.

Definition 5.42 (Kernel and cokernel defect detectors). Let D be a Chapter 3 finite detector with all applicability and completeness data declared at source stage `kernel_defect` or `cokernel_defect`. Define

$$D_{\tau_{\text{def}}}^{\text{ker,full}} := D_{\tau_{\text{def}}}(K_{i,W,\tau}), \quad D_{\tau_{\text{def}}}^{\text{coker,full}} := D_{\tau_{\text{def}}}(Q_{i,W,\tau}),$$

and, when the barcode-level interior component is declared,

$$D_{\tau_{\text{def}}}^{\text{ker},\circ} := D_{\tau_{\text{def}}}(K_{i,W,\tau}^\circ), \quad D_{\tau_{\text{def}}}^{\text{coker},\circ} := D_{\tau_{\text{def}}}(Q_{i,W,\tau}^\circ).$$

For a stencil detector, D is $D_{T,\tau_{\text{def}}}^{\text{stc}}$; birth, germ, and certified family-mesh detectors are also admissible at their exact registered strength.

Theorem 5.43 (Complete transient-defect characterization). *Assume a valid tower comparison artifact and full constructible defect profiles. Let the kernel and cokernel detector certificates be complete at the declared defect stages and components. Then*

$$\begin{aligned} D_{\tau_{\text{def}}}^{\text{ker}} = 0 & \iff T_{\tau_{\text{def}}}^{\text{bar}}(K^{\text{det}}) = 0, \\ D_{\tau_{\text{def}}}^{\text{coker}} = 0 & \iff T_{\tau_{\text{def}}}^{\text{bar}}(Q^{\text{det}}) = 0, \end{aligned}$$

where $K^{\text{det}}, Q^{\text{det}}$ denote the declared full or interior components. Nonzero detector value is equivalent to the existence of a retained bar in the corresponding component.

Proof. The interval-image, active-start, birth, germ, coverage, and mesh arguments of Chapter 3 apply to the supplied constructible defect profile with τ_{def} in place of the collapse threshold. The declared complete certificate therefore reflects zero of the corresponding hard threshold deletion. This theorem is the explicit authorization of those arguments at source stage `kernel_defect` or `cokernel_defect`; it is not a transport of a pre-threshold detector artifact. The component, endpoint, family, coverage, and artifact data remain hypotheses. $\square$

Remark 5.44 (Defect-stage registration). Theorem 5.43 is the stage-specific result anticipated by Definition 3.40. A detector proof or artifact registered at either stage

`windowed_prethreshold,` `repaired_postthreshold`

cannot be relabeled as a kernel/cokernel certificate. The full defect profiles and all certificate data are reconstructed and registered anew.

Corollary 5.45 (Full and interior detectors coincide under terminal zero). *If*

$$(\mu_{i,W,\tau}, \nu_{i,W,\tau}) = (0, 0),$$

then $K_{i,W,\tau} \cong K_{i,W,\tau}^\circ$ and $Q_{i,W,\tau} \cong Q_{i,W,\tau}^\circ$ at barcode-profile strength. Consequently a complete full-profile defect detector and the corresponding interior detector have the same zero/nonzero verdict.

Proof. Apply Proposition 5.40 to the kernel and cokernel defect profiles and then detector invariance under profile isomorphism from Chapter 3. $\square$

Definition 5.46 (Aggregate long-transient verdict). For fixed $(W, \tau, \tau_{\text{def}}, I_{\text{mon}})$, the aggregate long-transient verdict is exactly one of

`not_invoked,` `undefined,` `certified_absent,` `certified_present.`

It is `certified_absent` exactly when every required kernel and cokernel detector certificate is complete and every value is zero. It is `certified_present` when every detector needed to interpret the witness is well-typed and at least one required value is nonzero. Missing or wrong-stage evidence yields `undefined`, never absence.

Theorem 5.47 (Joint terminal/transient defect classification). *Assume a valid tower comparison artifact, terminal-tame defect profiles, and complete interior defect detectors at τ_{def} . Then exactly the following mathematical cases occur degreewise and after finite monitored aggregation:*

- (i) *if $(\mu, \nu) \neq (0, 0)$, a terminal/cofinal Type IV defect is present; the interior detector independently records whether a long noncofinal defect is also present;*

- (ii) if $(\mu, \nu) = (0, 0)$ and the long-transient verdict is *certified_present*, there is no terminal defect but there is a noncofinal kernel or cokernel bar of length greater than τ_{def} ;
- (iii) if $(\mu, \nu) = (0, 0)$ and the long-transient verdict is *certified_absent*, there is no terminal defect and no noncofinal defect bar longer than τ_{def} . Shorter nonzero defects may remain.

No case with an undefined detector certificate is assigned a mathematical zero verdict.

Proof. Terminal/cofinal presence is characterized by Proposition 5.40. Long noncofinal presence or absence is characterized by Theorem 5.43. The conclusions are independent conjunctions of these two classifications. $\square$

Definition 5.48 (Short-defect exclusion certificate). A short-defect exclusion certificate at τ_{def} proves that every nonzero bar in the declared interior kernel and cokernel profiles has length strictly greater than τ_{def} . It is a non-scalar structural theorem or artifact; it is not inferred from detector zero or from terminal-generic zero.

Theorem 5.49 (Conditional full-isomorphism upgrade). Let $\phi_{i,W,\tau}$ be a valid morphism in the declared abelian profile category. Suppose:

- (i) $(\mu_{i,W,\tau}, \nu_{i,W,\tau}) = (0, 0)$;
- (ii) complete interior kernel and cokernel detectors at τ_{def} are certified zero;
- (iii) a short-defect exclusion certificate holds.

Then

$$\ker \phi_{i,W,\tau} = 0, \quad \text{coker } \phi_{i,W,\tau} = 0,$$

and hence $\phi_{i,W,\tau}$ is an isomorphism.

Proof. Terminal zero excludes all cofinal kernel and cokernel bars. Complete interior detector zero excludes every interior bar of length greater than τ_{def} . The short-defect certificate excludes every remaining nonzero interior bar. Therefore both full defect profiles vanish. A morphism in an abelian category with zero kernel and zero cokernel is an isomorphism. $\square$

Example 5.50 (Zero terminal and transient diagnostics with a short residual defect). Let $W = [0, 10)$, choose $\tau_{\text{def}} > 0$, and consider

$$0 \longrightarrow k_{[2, 2+\tau_{\text{def}}/2)}.$$

The terminal pair is zero, and every complete defect detector at threshold τ_{def} is zero because the only defect bar is deleted by that threshold. The morphism is nevertheless not an isomorphism. Thus the short-defect exclusion in Theorem 5.49 is essential.

Remark 5.51 (No automatic metric transport of defect profiles). A bottleneck or interleaving bound between source profiles and a bound between apex profiles do not, by themselves, produce a bottleneck bound or an interleaving between their kernels or cokernels. Metric transport of a defect-detector verdict requires actual comparison data between the defect profiles and an independently proved kernel/cokernel stability theorem. Only then may the Chapter 2 threshold-gap and zero-separation machinery be applied at stage `kernel_defect` or `cokernel_defect`.

5.9 Artifact-coherent stable bands

Definition 5.52 (Artifact-coherent Type IV stable band). Fix a window W , monitored degrees, a tower-profile construction, and a terminal mode. An interval $J \subset (0, +\infty)$ is an *artifact-coherent Type IV stable band* if:

- (i) for every $\tau \in J$, a valid tower comparison artifact $\phi_{i,W,\tau}$ is supplied by one declared construction rule;
- (ii) the claim-required source semantics, source provenance, semantic authority, and terminal mode do not change silently across J ;
- (iii) all defect profiles are terminal-tame;
- (iv) every repaired comparison used by the artifact declares a comparison mode. In metric mode, the record carries the ledger $\mathfrak{d}_{\text{met},\tau}$, scalarization val_B , evidenced discrepancy bound, and strict inequality

$$\text{val}_B(\mathfrak{d}_{\text{met},\tau}) < \text{gap}_\tau(\mathfrak{B}_\tau);$$

in exact mode it carries the exact equality or profile-isomorphism proof and makes no automatic positive-radius robustness claim;

- (v) the associated Type IV record has status `certified_zero` for every $\tau \in J$.

Remark 5.53 (Stable band is an artifact family, not pointwise wishful thinking). Pointwise numerical zeros obtained from unrelated comparison maps do not define a stable band. The artifact family, source provenance, terminal convention, and comparison policy must remain auditable throughout the band. If a single budgeted perturbation is claimed uniformly on J , the manifest must provide a common ledger $\mathfrak{d}_{\text{met},J}$ and verify the uniform reserve condition

$$\text{val}_B(\mathfrak{d}_{\text{met},J}) < \inf_{\tau \in J} \text{gap}_\tau(\mathfrak{B}_\tau).$$

A pointwise family of positive reserves does not by itself imply that this infimum is positive.

Proposition 5.54 (No terminal-generic Type IV obstruction on a stable band). *If J is an artifact-coherent Type IV stable band, then every $\tau \in J$ is Type IV-free in the monitored terminal-generic sense relative to its declared artifact.*

Proof. By the stable-band definition, every threshold in J has a valid diagnostic record whose status is

`certified_zero`.

The certified-zero characterization then applies pointwise. □

Definition 5.55 (Artifact-coherent joint defect-clean band). An interval $J \subset (0, +\infty)$ is a joint defect-clean band at defect threshold τ_{def} if it is an artifact-coherent Type IV stable band and, for every $\tau \in J$:

- (i) the same defect threshold, detector stages, endpoint convention, component policy, and completeness rule are used;
- (ii) complete kernel and cokernel interior detectors are certified zero;
- (iii) the detector artifacts and source-semantic authority remain auditable across J .

Proposition 5.56 (Uniform terminal and long-transient cleanliness). *On an artifact-coherent joint defect-clean band, every threshold is free of monitored terminal defects and of monitored noncofinal defects longer than τ_{def} . Full isomorphism does not follow without a uniform short-defect exclusion theorem.*

Proof. Apply Proposition 5.54 and Theorem 5.47 pointwise on the band. □

5.10 Integration with B-Gate⁺

Definition 5.57 (Type IV and transient-defect gate discharge). The terminal Type IV and optional transient-defect clauses of B-Gate⁺ are discharged as follows.

- (i) if the claim has no tower/apex/limit semantics, the manifest records `not_invoked`;
- (ii) if tower semantics are invoked, a valid Type IV record with status `certified_zero` is required, and the source provenance must have sufficient authority for the claim;
- (iii) if the claim additionally requires absence of long transient defects, the aggregate long-transient verdict at the declared τ_{def} must be `certified_absent`;
- (iv) if full comparison-map isomorphism is required, either direct full defect-profile vanishing or the hypotheses of Theorem 5.49 must be certified.

The statuses `undefined`, `obstructed`, and a required long-transient `certified_present` fail their respective gate clauses. Terminal Type IV status and transient status remain separate fields.

Remark 5.58 (Meaning of $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$ in the gate). The shorthand

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$$

is used only in the invoked-tower case after the diagnostic has been shown to be defined. A no-tower claim uses `not_invoked`, not a fabricated numerical zero.

Proposition 5.59 (Audited admissibility implies the structural and safety clauses of B-Gate⁺). *If a run is audited collapse-admissible in the sense of Definition 5.10, then:*

- (i) *its topological clause passes;*
- (ii) *its invoked categorical clause passes;*
- (iii) *its Type IV clause is either correctly out of scope or certified zero;*
- (iv) *every required long-transient clause is certified absent, and any required full-isomorphism upgrade has its short-defect exclusion or direct full-defect proof;*
- (v) *every invoked metric threshold-gap and strict budget clause passes, and every exact comparison carries its exact proof;*
- (vi) *no Type III or Type V defect remains in those records.*

Hence the corresponding B-Gate⁺ record passes for every clause marked required by the run's declared Core scope. This does not by itself yield any domain-specific external conclusion; such a conclusion still requires its separately proved Interface or Bridge theorem.

Proof. Each item is one of the defining clauses of structural or audited collapse admissibility. The final statement is the B-Gate⁺ conjunction fixed in Chapter 1, with the Type IV scope refinement of Definition 5.57. □

Definition 5.60 (Failure routing record). A failure routing record contains:

- (i) every applicable failure type among I–V;

- (ii) the first failed audit stage in Definition 5.5;
- (iii) the exact failed clause and either the relevant artifact identifier or an explicit missing-artifact marker;
- (iv) whether the failure is mathematical, operational, diagnostic, transient-defect, or manifest-level;
- (v) the terminal Type IV status and the independent long-transient status when a tower artifact is in scope;
- (vi) the permitted repair action, if any, without changing the claim status.

No scalar score replaces this typed routing record.

5.11 Failure-and-diagnostics contract

Definition 5.61 (v20 failure-and-diagnostics contract). The normative Chapter 5 contract is the following.

- (i) Types I–V are nonexclusive and are checked in the declared audit order.
- (ii) Type II is meaningful only degreewise in an eligible Chapter 4 regime.
- (iii) Type III includes threshold-gap, detector applicability and completeness, crop-budget, representative, profile-system, artifact-typing, source-semantic, and terminal-tameness failures.
- (iv) Type IV is defined only relative to a valid comparison morphism and measures terminal-generic defect profiles through (μ, ν) .
- (v) Long transient defects are evaluated separately on full or noncofinal kernel/cokernel profiles at a declared $\tau_{\text{def}} > 0$.
- (vi) Type V includes false pass declarations, deprecated claims, stage relabeling, suppressed records, provenance overstatement, and undefined-as-zero substitutions.
- (vii) On bounded windows, Type IV counts defects cofinal to the right endpoint; on right-unbounded windows, it counts infinite-bar defects.
- (viii) $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$ together with transient certified absence excludes only terminal defects and noncofinal defects longer than τ_{def} .
- (ix) Full isomorphism requires full kernel/cokernel vanishing or a separately proved short-defect exclusion upgrade.
- (x) $u_{i,W,\tau}$ and u_{Collapse} are deprecated; $\nu_{i,W,\tau}$ and ν_{Collapse} are canonical.
- (xi) A no-tower claim records `not_invoked`; an invoked tower claim requires a valid `certified_zero` terminal record to pass.
- (xii) Search, AI, spectral, or advisory proxies cannot replace the actual comparison artifact or complete defect detector.

Remark 5.62 (Contract, not a mixed theorem bundle). Definition 5.61 is a typing and audit contract. Its mathematical content is supplied by the preceding definitions and propositions; it is not presented as a theorem combining definitions, policies, and proved implications into one undifferentiated statement.

5.12 Minimal manifest consequences

Remark 5.63 (Manifest hierarchy and canonical audit schema). The entries below extend the Chapter 1 Minimal Manifest and the Chapter 3 evaluation/representative records with failure-routing and Type IV data. They are not a competing manifest definition. Chapter 7 supplies the canonical v20.0 audit manifest into which these entries are embedded. An incomplete invoked Type IV entry yields `undefined`; an inconsistent entry is a Type V rejection.

Definition 5.64 (Type IV manifest entry). Every run records the Type IV scope status. The remaining manifest fields are status-dependent.

- (i) For `not_invoked`: the scope rationale and the claim clause declaring tower semantics out of scope.
- (ii) For `undefined`: the tower-profile, source, apex, or comparison identifiers that exist, together with an explicit missing/invalid-obligation marker for every absent mandatory object, source-semantics mismatch, insufficient-provenance source, absent ambient abelian category, failed typing, or failed terminal-tameness obligation. No Core numerical diagnostic is serialized as zero.
- (iii) For `certified_zero` or `obstructed`: the tower-profile system identifier; source and apex identifiers; source provenance; claim-required source semantics and semantic-authority verdict; valid comparison morphism artifact identifier; ambient abelian profile category; window, collapse threshold, monitored degrees, and evaluation order; terminal mode and terminal-tameness artifacts; full, cofinal, and interior defect-profile artifacts; all degree-wise and total μ, ν values; diagnostic status; and the separate full-interior-defect status. When transient detection is invoked, the entry also records τ_{def} , detector component, exact Chapter 3 source stage, information class, endpoint convention, stencil or birth/germ data, coverage/mesh/excess evidence, certificate and value statuses, aggregate long-transient verdict, and any short-defect exclusion certificate.

If metric perturbation enters the artifact construction, the entry also records $\mathbf{d}_{\text{met}}$, $\text{val}_{\mathbf{B}}$, the real budget $\varepsilon_{\mathbf{B}} = \text{val}_{\mathbf{B}}(\mathbf{d}_{\text{met}})$, and the gate-scoped comparison family $\mathfrak{B}$.

Definition 5.65 (Failure manifest entry). Every failed or conditionally accepted run records:

- (i) the applicable failure types;
- (ii) the first failed audit stage;
- (iii) the failed claim, theorem, gate, or artifact reference;
- (iv) whether the result is rejected, repairable, conditional, or out of scope;
- (v) the version boundary, including any successor-release candidate used in analysis but excluded from proof.

5.13 Explicit non-claims

The following are not asserted in Chapter 5.

- (i) No sequence of repaired barcode objects is automatically a directed system.
- (ii) No bottleneck matching is automatically a persistence-module morphism.

(iii) No automatic comparison

$$\lim_{\overrightarrow{n}} \text{Eval}_{i,w,\tau}(F_n) \longrightarrow \text{Eval}_{i,w,\tau}(F_\infty)$$

is asserted.

(iv) No exactness or global functoriality of T_τ^{bar} is used.

(v) No undefined diagnostic is interpreted as zero.

(vi) No bounded-window terminal bar is called a global infinite bar.

(vii) No equality $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$ is promoted to full comparison-map isomorphism without a separate no-interior-defect theorem.

(viii) No Type IV result is accepted without an actual comparison morphism, a source whose provenance has sufficient authority for the claim, a declared abelian profile category, terminal-tameness data, and defect-profile artifacts; their absence or mismatch is recorded explicitly and yields `undefined`.

(ix) No scalar ledger slack repairs a missing morphism, missing terminal-tameness certificate, invalid claim status, or incomplete manifest.

(x) No Type II verdict is issued without eligibility.

(xi) No scalar potential, Search artifact, AI confidence, spectral warning, or advisory invariant replaces Types I–V, B-Gate⁺, or the Type IV artifact.

(xii) No Restart/Summability, Čech gluing, or local-to-global certificate theorem is proved here; those are deferred to Chapter 6.

(xiii) No successor-release candidate, including any stronger NBT transport package, is used as a v20.0 theorem.

(xiv) No zero terminal-generic pair implies absence of finite noncofinal defects.

(xv) No complete transient detector at τ_{def} excludes defect bars of length at most that threshold.

(xvi) No full-profile defect detector isolates transient defects in the presence of cofinal bars unless an interior decomposition or separate witness is supplied.

(xvii) No detector certificate is relabeled from `windowed_prethreshold` or `repaired_postthreshold` to `kernel_defect` or `cokernel_defect`.

(xviii) No closeness of source and apex profiles automatically yields closeness of kernel or cokernel defect profiles.

(xix) No long-transient certified absence is a full-isomorphism theorem without direct full-defect vanishing or a short-defect exclusion certificate.

5.14 Provenance and status

Chapter 5 depends on Chapter 1 for the five-type failure landscape and B-Gate⁺, Chapter 2 and Appendix A for repaired persistence and metric safety, Chapter 3 for repaired evaluation, finite detector, stage, and representative records, and Chapter 4 for degreewise eligible Ext discharge. The inherited v19 mathematical clarification is the window-relative terminal-generic diagnostic. The v20 extension adds the barcode-level cofinal/interior decomposition, complete transient-defect characterization, joint terminal/transient classification, and conditional full-isomorphism upgrade. The record definitions are serialized through the canonical Chapter 7 audit manifest.

Inherited v19 Core mathematical propositions. Propositions [5.11](#), [5.23](#), [5.26](#), [5.28](#), [5.34](#), [5.54](#), and [5.59](#).

New v20 transient-defect results. Proposition [5.40](#), Theorem [5.43](#), Corollary [5.45](#), Theorem [5.47](#), Theorem [5.49](#), and Proposition [5.56](#).

Core definitions and contracts. Definitions [5.3](#), [5.5](#), [5.8](#), [5.9](#), [5.10](#), [5.13](#), [5.14](#), [5.17](#), [5.21](#), [5.22](#), [5.30](#), [5.31](#), [5.32](#), [5.33](#), [5.39](#), [5.41](#), [5.42](#), [5.46](#), [5.48](#), [5.52](#), [5.55](#), [5.57](#), and [5.61](#).

Operational examples and cautions. Examples [5.36](#), [5.37](#), and [5.50](#); Remarks [5.6](#), [5.7](#), [5.25](#), [5.63](#), [5.18](#), [5.16](#), [5.27](#), [5.35](#), [5.44](#), [5.51](#), and [5.58](#).

Explicit exclusions. Section [5.13](#).

6 Chapter 6: Window Logic, Overlap, κ -Restart, and Record-Level Gluing

Chapter 6 contract. This chapter is a conservative v20 extension of the v19 window, overlap, Restart, and record-atlas calculus. Its default conclusion is record-level coherence, not object-level descent. Exact simplex coherence is witnessed by actual isomorphisms satisfying a cocycle law. Metric simplex coherence is witnessed through declared reference profiles and discrepancy bounds; it does not manufacture transition morphisms or a cocycle.

When local detector zero verdicts are promoted to a covered-range zero statement, the record must name a global pre-threshold profile, its local restriction interfaces, complete local detectors, and a local-to-global coverage theorem proving that every globally retained bar is detected on at least one local or overlap window. Local zero verdicts, pairwise overlap passes, Restart reserve transport, and summable exposure are individually insufficient for that conclusion.

Terminal Type IV cleanliness and long-transient cleanliness are separate clauses. A source-profile overlap record does not transport kernel or cokernel defect profiles without a separately proved defect comparison. No numerical reserve, simplex budget, or Restart estimate repairs a missing morphism, detector-coverage artifact, higher-coherence record, descent hypothesis, or manifest field.

6.1 Scope, dependencies, and relocation discipline

This chapter works in the v20 conservative extension of the repaired Core established by Chapters 1–5 and Appendix A. Its inputs are:

- (i) the pre-threshold window profile

$$\mathcal{B}_{i,w}(F) := W_W \mathbf{P}_i(F);$$

(ii) the repaired post-threshold evaluation

$$\text{Eval}_{i,W,\tau}(F) := T_{\tau}^{\text{bar}}(\mathcal{B}_{i,W}(F));$$

(iii) the sharp two-sided threshold-gap theorem with closed band

$$[\tau - \varepsilon, \tau + \varepsilon];$$

(iv) the convention

$$\text{gap}_{\tau} = \text{clear}_{\tau}$$

as a ledger-facing alias;

(v) the representative and eligible-realization disciplines of Chapters 3–4;

(vi) the typed Type IV status and terminal-generic diagnostics of Chapter 5.

The purpose of this chapter is to define exactly when local certificate records may be compared, continued, and assembled without reintroducing an unsupported global theorem. It develops:

- (i) right-open partition and overlap-cover logic;
- (ii) sharp two-sided Overlap Gate certificates;
- (iii) finite Čech-depth coherence at the *record* level;
- (iv) conditional κ -admissible Restart;
- (v) the correct weighted summability condition for countable restart;
- (vi) countable certificate atlases under explicit local-finiteness and coherence hypotheses;
- (vii) exact-cocycle and metric-reference simplex coherence as distinct record types;
- (viii) detector-aware local-to-global coverage and its sharp failure modes;
- (ix) detector-safe Restart handoff and transient-defect participation;
- (x) a typed interface to, but not a proof of, object-level descent.

Remark 6.1 (Relocation of the former Chapter 6). The former v18.1 Chapter 6 material on the δ -ledger, Audit Mode, proposer–verifier separation, and Core sovereignty is not discarded. In the v19 main-text order it is relocated to Chapter 7, after the present chapter has fixed the window, overlap, restart, and gluing records that the audit chapter must consume. The present chapter uses ledger values but does not redefine the final audit-verdict semantics.

Remark 6.2 (No automatic local-to-global theorem). Neither a family of local B-Gate⁺ passes nor pairwise small bottleneck distance automatically produces a global persistence object, a global filtered representative, or a global theorem. This chapter certifies only the compatibility and propagation of declared certificate records unless an independent object-level gluing theorem is supplied.

6.2 Right-open domains, partitions, and overlap covers

Definition 6.3 (Right-open Core window). A *Core window* is a right-open interval

$$W = [u, v), \quad -\infty < u < v \leq +\infty.$$

It is *bounded* if $v < +\infty$, and *right-unbounded* if $v = +\infty$. All local profiles in this chapter are evaluated directly on the declared window before threshold deletion.

Definition 6.4 (Right-open partition cover). A family

$$\mathcal{W}_{\text{part}} = \{W_k = [u_k, u_{k+1})\}_{k \in K}$$

is a *right-open partition cover* of a declared range $X \subseteq \mathbb{R}$ if:

- (i) the windows are pairwise disjoint under the right-open endpoint convention;
- (ii) $\bigcup_{k \in K} W_k = X$;
- (iii) the indexing order is part of the data whenever sequential continuation or Restart is used.

The former term “MECE windowing” may be retained as an operational alias, but the mathematical object is a right-open partition cover.

Definition 6.5 (Overlap-admitting cover). A family

$$\mathcal{W}_{\text{ov}} = \{W_\alpha\}_{\alpha \in A}$$

of right-open windows is an *overlap-admitting cover* of X if

$$X = \bigcup_{\alpha \in A} W_\alpha$$

and every nonempty finite intersection used by the certificate logic is explicitly indexed and recorded. For a finite index set $\sigma = \{\alpha_0, \dots, \alpha_p\}$, write

$$W_\sigma := W_{\alpha_0} \cap \dots \cap W_{\alpha_p}.$$

Remark 6.6 (Partition and overlap covers have different jobs). A right-open partition cover is used for sequential Restart and accumulated-budget control. An overlap-admitting cover is used for compatibility and Čech-type coherence. A disjoint partition has no nontrivial overlap on which an Overlap Gate could be evaluated; conversely, an overlap cover does not by itself provide an ordered continuation path.

Definition 6.7 (Direct common-window profile). Let local objects F_α and F_β be declared on windows W_α and W_β , and suppose $W_{\alpha\beta} := W_\alpha \cap W_\beta \neq \emptyset$. Their pre-threshold common-window profiles are

$$\mathcal{B}_{i,\alpha\beta}^\alpha := W_{W_{\alpha\beta}} \mathbf{P}_i(F_\alpha), \quad \mathcal{B}_{i,\alpha\beta}^\beta := W_{W_{\alpha\beta}} \mathbf{P}_i(F_\beta).$$

For a declared common threshold $\tau_{\alpha\beta}$, the corresponding post-threshold profiles are

$$\text{Eval}_{i,\alpha\beta}^\alpha := T_{\tau_{\alpha\beta}}^{\text{bar}}(\mathcal{B}_{i,\alpha\beta}^\alpha), \quad \text{Eval}_{i,\alpha\beta}^\beta := T_{\tau_{\alpha\beta}}^{\text{bar}}(\mathcal{B}_{i,\alpha\beta}^\beta).$$

Remark 6.8 (No crop-after-collapse substitution). The profiles in Definition 6.7 are computed directly on the common window. They are not replaced by cropping a profile already collapsed on W_α or W_β , unless a separate window-adapted commutation certificate is supplied. This prevents the false identity

$$W_{W_{\alpha\beta}} T_\tau^{\text{bar}} \cong T_\tau^{\text{bar}} W_{W_{\alpha\beta}}$$

from entering the overlap proof through notation. When such a certificate is invoked, it must use the necessary-and-sufficient condition of Chapter 2 and Appendix A: every bar retained before cropping, including every infinite bar, is either disjoint from the common window or remains retained after nonempty clipping. A finite-bar-only check is insufficient.

6.3 Metric scalarization of the ledger

The threshold-gap theorem is stated in the real-valued bottleneck metric. Consequently, an abstract quantale-valued ledger cannot be compared directly with clear_τ unless a metric scalarization has been declared.

Definition 6.9 (Bottleneck scalarization). Let

$$(\mathcal{V}, \oplus, 0_{\mathcal{V}}, \preceq)$$

be the declared commutative ledger quantale. A *bottleneck scalarization* is a monotone, subadditive map

$$\text{val}_{\mathcal{B}} : \mathcal{V} \longrightarrow [0, +\infty]$$

satisfying

$$\text{val}_{\mathcal{B}}(0_{\mathcal{V}}) = 0, \quad \text{val}_{\mathcal{B}}(q \oplus q') \leq \text{val}_{\mathcal{B}}(q) + \text{val}_{\mathcal{B}}(q').$$

It converts a metric-ledger element into a real bottleneck-discrepancy upper bound. The manifest must also identify the actual profile comparison to which that upper bound applies.

Definition 6.10 (Scalarized metric ledger). For a gate-scoped metric ledger record

$$\begin{aligned} \mathbf{d}_{\text{met}} = & \delta^{\text{alg}} \oplus \delta^{\text{disc}} \oplus \delta^{\text{meas}} \oplus \delta^{\text{rep,met}} \oplus \delta^{\text{real,met}} \oplus \delta^{\text{cmp}} \\ & \oplus \delta^{\text{crop}} \oplus \delta^{\text{cont}} \oplus \delta^{\text{ov}} \oplus \delta^{\text{rst}}, \end{aligned}$$

define its scalarized bottleneck budget by

$$\varepsilon_{\mathcal{B}}(\mathbf{d}_{\text{met}}) := \text{val}_{\mathcal{B}}(\mathbf{d}_{\text{met}}).$$

Every displayed component is a metric-discrepancy contribution only. In particular, $\delta^{\text{rep,met}}$ and $\delta^{\text{real,met}}$ may measure proved profile distortion introduced by representative or realization transport; they do not encode the existence, typing, or compatibility of a representative or realization. Only components actually consumed by the audited comparison are included. Each component retains its provenance and may not be silently reused by an unrelated gate. Crop, overlap, continuation, and restart contributions remain separately addressable even when $\text{val}_{\mathcal{B}}$ eventually combines them into one real upper bound.

Definition 6.11 (Non-scalar continuation and coherence obligations). The following are typed proof obligations rather than metric-ledger components:

- (i) existence and typing of local representatives and realization data;
- (ii) actual overlap, face, continuation, and tower comparison morphisms when such morphisms are invoked;
- (iii) threshold reconciliation, κ_k -admissibility, terminal tameness, and Type IV certificates;
- (iv) higher-simplex compatibility and face coherence;
- (v) claim status, manifest completeness, consistency, and artifact readability.

No increase in $\varepsilon_{\mathcal{B}}$, no larger threshold gap, and no summability estimate can discharge a missing obligation in this list.

Remark 6.12 (Meaning of $\Sigma\delta < \text{gap}_\tau$). Whenever this chapter writes

$$\Sigma\delta < \text{gap}_\tau,$$

it means the gate-scoped typed inequality

$$\varepsilon_B(\mathbf{d}_{\text{met}}) = \text{val}_B(\mathbf{d}_{\text{met}}) < \text{gap}_\tau(\mathfrak{B}).$$

An unspecialized quantale element is not compared directly with the real number clear_τ , and a budget certified for one profile family $\mathfrak{B}$ is not silently spent again on another family.

Remark 6.13 (Sharp threshold band versus scalarization tightness). The word *sharp* in this chapter refers to the optimal half-width ε of the two-sided forbidden band

$$[\tau - \varepsilon, \tau + \varepsilon]$$

when ε is a valid bottleneck-distance upper bound. It does not assert that the chosen scalarization val_B is numerically tight. If $\text{val}_B(\mathbf{d}_{\text{met}})$ overestimates the true bottleneck error, the resulting certificate is conservative even though the underlying threshold theorem remains sharp.

Definition 6.14 (Gate-scoped threshold reserve). Let $\mathfrak{B} = \{B_j\}_{j \in J}$ be the finite family of pre-threshold profiles consumed by one declared local, overlap, or restart gate. Define

$$\text{gap}_\tau(\mathfrak{B}) := \min_{j \in J} \text{clear}_\tau(B_j),$$

and, for its scalarized metric budget ε , define the *threshold reserve*

$$\text{res}_\tau(\mathfrak{B}; \varepsilon) := \text{gap}_\tau(\mathfrak{B}) - \varepsilon.$$

The gate has positive reserve precisely when

$$\varepsilon < \text{gap}_\tau(\mathfrak{B}).$$

This reserve belongs to the declared gate and profile family; it is not a globally reusable balance.

Remark 6.15 (The gap is measured before thresholding). The family $\mathfrak{B}$ in Definition 6.14 consists of pre-threshold profiles. The post-threshold evaluations are used for the compatibility verdict, but they do not define the threshold clearance.

6.4 Sharp two-sided Overlap Gate

Definition 6.16 (Threshold reconciliation record). If two local records use different thresholds τ_α and τ_β , an overlap comparison is well-typed only after declaring a common overlap threshold $\tau_{\alpha\beta}$ and directly re-evaluating both local objects on $W_{\alpha\beta}$ at that threshold. The data establishing this choice are called a *threshold reconciliation record*. No comparison between differently thresholded outputs is silently interpreted as an Overlap Gate.

Definition 6.17 (Overlap comparison mode). Every overlap inference records exactly one comparison mode:

$$\text{exact}, \quad \text{metric_gap_certified}, \quad \text{not_compared}.$$

The modes mean:

- (i) `exact`: a direct equality or persistence-profile isomorphism on the directly evaluated common window is proved and artifact-backed;
- (ii) `metric_gap_certified`: a real pre-threshold distance bound and a two-sided threshold-gap certificate are supplied at the same budget;
- (iii) `not_compared`: no overlap-comparison conclusion is consumed by the claim.

An `exact` comparison proves the declared relation for the specified profiles. It does not, by itself, certify positive-radius robustness. The mode `not_compared` cannot discharge an overlap obligation. In particular, it is unavailable when the governing claim consumes the comparison.

Definition 6.18 (Exact overlap certificate). An *exact overlap certificate* on $(i, W_{\alpha\beta}, \tau_{\alpha\beta})$ contains:

- (i) the direct common-window pre-threshold and repaired profiles;
- (ii) the common threshold and threshold-reconciliation record;
- (iii) an actual equality or persistence-profile isomorphism

$$\text{Eval}_{i,\alpha\beta}^\alpha \xrightarrow{\sim} \text{Eval}_{i,\alpha\beta}^\beta;$$

- (iv) immutable source, target, proof, and artifact identifiers.

No threshold-gap reserve is inferred merely from this exact certificate.

Definition 6.19 (Two-sided overlap metric certificate). Fix a monitored degree i , an overlap $W_{\alpha\beta}$, a common threshold $\tau_{\alpha\beta}$, and a scalarized budget $\varepsilon_{\alpha\beta,i} \geq 0$. A *two-sided overlap metric certificate* consists of:

- (i) a pre-threshold bottleneck estimate

$$d_B(\mathcal{B}_{i,\alpha\beta}^\alpha, \mathcal{B}_{i,\alpha\beta}^\beta) \leq \varepsilon_{\alpha\beta,i};$$

- (ii) two-sided sharp clearance

$$\text{clear}_{\tau_{\alpha\beta}}(\mathcal{B}_{i,\alpha\beta}^\alpha) > \varepsilon_{\alpha\beta,i}, \quad \text{clear}_{\tau_{\alpha\beta}}(\mathcal{B}_{i,\alpha\beta}^\beta) > \varepsilon_{\alpha\beta,i};$$

- (iii) the ledger provenance whose bottleneck scalarization yields $\varepsilon_{\alpha\beta,i}$;
- (iv) the direct common-window and threshold-reconciliation records required above.

Proposition 6.20 (Sharp overlap transfer). Under a two-sided overlap metric certificate,

$$d_B(\text{Eval}_{i,\alpha\beta}^\alpha, \text{Eval}_{i,\alpha\beta}^\beta) \leq \varepsilon_{\alpha\beta,i}.$$

In the one-parameter constructible regime and under the infinite-bar convention of Appendix A, the same bound may be read using d_{int} .

Proof. Clause (i) of Definition 6.19 supplies the input distance bound. Clause (ii) says that neither pre-threshold barcode contains a finite bar whose length lies in the closed band

$$[\tau_{\alpha\beta} - \varepsilon_{\alpha\beta,i}, \tau_{\alpha\beta} + \varepsilon_{\alpha\beta,i}].$$

The sharp threshold-gap-safe stability theorem of Chapter 2 and Appendix A therefore applies and yields the post-threshold bottleneck bound. The bottleneck–interleaving isometry then gives the interleaving formulation in the declared constructible regime. $\square$

Definition 6.21 (Pairwise Overlap Gate). The *pairwise Overlap Gate* on $W_{\alpha\beta}$ passes if, for every monitored degree i :

- (i) exactly one overlap comparison mode is declared;
- (ii) in mode `exact`, a valid exact overlap certificate exists;
- (iii) in mode `metric_gap_certified`, a two-sided overlap metric certificate exists and the post-threshold discrepancy bound of Proposition 6.20 holds;
- (iv) mode `not_compared` is used only when the governing claim consumes no overlap inference;
- (v) all local representative or eligible-realization clauses used on the overlap have compatible provenance records;
- (vi) terminal Type IV is either explicitly `not_invoked` by scope or, when invoked, is `certified_zero`;
- (vii) when the governing overlap claim also requires absence of long transient kernel or cokernel defects, the Chapter 5 aggregate transient status is `certified_absent`;
- (viii) `undefined` is not treated as zero, `obstructed` is a terminal failure, and `certified_present` is a transient-defect failure when absence was required;
- (ix) every required crop, comparison, exact-proof, and threshold-reconciliation artifact is present.

Remark 6.22 (Two layers of the Overlap Gate). The Overlap Gate has two mathematically different layers:

- (i) the discrepancy verdict compares the *post-threshold* evaluations;
- (ii) threshold safety is certified on the *pre-threshold* profiles.

The second layer is not recoverable from the first. In particular, small post-threshold distance alone does not certify that threshold classification was stable. An exact overlap certificate may bypass metric transport for the declared exact comparison, but it does not supply positive-radius threshold robustness.

Remark 6.23 (Pairwise compatibility is not higher coherence). Passing every pairwise Overlap Gate does not by itself establish compatibility on triple or higher overlaps. Higher coherence must be declared through the nerve records below whenever the cover has nontrivial higher intersections.

6.5 Finite Čech depth and record-level coherence

Definition 6.24 (Čech depth). An overlap-admitting cover has *Čech depth at most K* if every intersection of $K + 2$ distinct windows is empty. Equivalently, its Čech nerve has simplicial dimension at most K . Thus all nonempty overlap records occur on at most $(K + 1)$ -fold intersections.

Definition 6.25 (Simplex coherence mode). Every nonempty nerve simplex used by a coherence claim declares exactly one simplex coherence mode:

`exact_cocycle`, `metric_reference`, `not_compared`.

The mode `exact_cocycle` consumes actual profile isomorphisms and a cocycle identity. The mode `metric_reference` consumes a common reference profile and certified metric comparisons to that reference. It certifies bounded record-level discrepancy but supplies no transition morphism or categorical cocycle. The mode `not_compared` is allowed only when the governing claim consumes no simplex coherence conclusion.

Definition 6.26 (Exact simplex cocycle record). Let $\sigma = \{\alpha_0, \dots, \alpha_p\}$ index a nonempty overlap W_σ , and fix a monitored degree and a common threshold. An *exact simplex cocycle record* contains actual repaired-profile isomorphisms

$$g_{\alpha\beta}^\sigma : \text{Eval}_{i,\sigma}^\alpha \xrightarrow{\sim} \text{Eval}_{i,\sigma}^\beta \quad (\alpha, \beta \in \sigma)$$

with immutable artifacts and the identities

$$g_{\alpha\alpha}^\sigma = \text{id}, \quad g_{\beta\alpha}^\sigma = (g_{\alpha\beta}^\sigma)^{-1}, \quad g_{\beta\gamma}^\sigma \circ g_{\alpha\beta}^\sigma = g_{\alpha\gamma}^\sigma.$$

Restriction to a face must agree with the previously declared face cocycle. These identities are mathematical equalities of actual morphisms, not merely small-distance statements.

Definition 6.27 (Metric-reference simplex record). For the same simplex σ , a *metric-reference simplex record* contains a declared pre-threshold reference profile $R_{i,\sigma}$ on W_σ , one common threshold τ_σ , and for every $\alpha \in \sigma$ a certified bound

$$d_B(\mathcal{B}_{i,\sigma}^\alpha, R_{i,\sigma}) \leq \varepsilon_{\alpha,\sigma,i},$$

together with a two-sided threshold-gap certificate for the pair $(\mathcal{B}_{i,\sigma}^\alpha, R_{i,\sigma})$ at that same budget. The record also contains the reference-profile identity, scalarization, threshold reconciliation, and face-restriction provance.

The resulting repaired comparisons are metric statements only. No map $\text{Eval}_{i,\sigma}^\alpha \rightarrow \text{Eval}_{i,\sigma}^\beta$, no cocycle, and no object-level descent datum is inferred from the reference profile.

Proposition 6.28 (Pairwise bound from a metric simplex reference). *Under a metric-reference simplex record, for every $\alpha, \beta \in \sigma$,*

$$d_B(\text{Eval}_{i,\sigma}^\alpha, \text{Eval}_{i,\sigma}^\beta) \leq \varepsilon_{\alpha,\sigma,i} + \varepsilon_{\beta,\sigma,i}.$$

Proof. Apply the Chapter 2 threshold-gap-safe theorem to each local-to-reference comparison. This gives repaired-profile bounds

$$d_B(\text{Eval}_{i,\sigma}^\alpha, T_{\tau_\sigma}^{\text{bar}}(R_{i,\sigma})) \leq \varepsilon_{\alpha,\sigma,i}$$

and the analogous bound for β . The triangle inequality gives the claim. $\square$

Remark 6.29 (Metric coherence is not an approximate cocycle). A family of pairwise distance bounds does not carry composable transition maps. The phrase *metric coherence* in this chapter means coherence of scope, reference identity, threshold, face records, and certified bounds. It does not mean that an exact cocycle exists up to an unspecified error.

Definition 6.30 (Simplex compatibility record). Let $\sigma = \{\alpha_0, \dots, \alpha_p\}$ index a nonempty overlap W_σ . A *simplex compatibility record* consists of:

- (i) direct pre-threshold profiles of every participating local object on W_σ ;
- (ii) a common threshold and threshold-reconciliation record;
- (iii) one simplex coherence mode from Definition 6.25;
- (iv) in mode `exact_cocycle`, an exact simplex cocycle record;
- (v) in mode `metric_reference`, a metric-reference simplex record and every local-to-reference threshold-gap artifact;
- (vi) an explicit runtime comparison mode for every pairwise or face comparison consumed by the record; no consumed comparison may use `not_compared`;

- (vii) a face-compatibility statement saying that restriction from σ to any face agrees with the already declared face record at the same coherence mode and reference or cocycle identity;
- (viii) typed terminal Type IV status and, whenever the claim requires long-transient cleanliness, the Chapter 5 aggregate transient status and its defect-detector artifacts;
- (ix) all immutable source, target, proof, scalarization, and artifact references.

Definition 6.31 (Čech-coherent certificate atlas). A family of local certificate records on $\mathcal{W}_{\text{ov}}$ is a *Čech-coherent certificate atlas* if every nonempty simplex of the nerve carries a simplex compatibility record and all face restrictions are mutually compatible.

Proposition 6.32 (Finite-depth verification bound). *If the cover has Čech depth at most K , then the coherence portion of the certificate atlas requires no records above simplicial degree K .*

Proof. By Definition 6.24, there are no nonempty intersections indexed by simplices of degree larger than K . Hence there are no higher overlap domains on which additional compatibility records could be evaluated. $\square$

Theorem 6.33 (Finite-cover record-level gluing). *Let $\mathcal{W}_{\text{ov}}$ be a finite overlap-admitting cover of finite Čech depth. Assume:*

- (i) *every window carries a valid local certificate record;*
- (ii) *every nonempty nerve simplex carries a simplex compatibility record in mode `exact_cocycle` or `metric_reference`; every consumed pairwise comparison uses runtime mode `exact` or `metric_gap_certified`;*
- (iii) *every required terminal Type IV scope has one permitted status,*

`not_invoked` or `certified_zero`,

and is never `undefined`; whenever the finite atlas claim also requires long-transient cleanliness, its aggregate transient status is `certified_absent`;

- (iv) *every metric-mode comparison satisfies its sharp two-sided threshold reserve, while every exact-mode comparison carries its exact proof artifact;*
- (v) *all face-compatibility statements and artifact references are audit-readable.*

Then the family is a Čech-coherent certificate atlas on the covered range.

Proof. The local records provide the vertices of the nerve. The simplex compatibility records provide all overlap data through the maximal nonempty nerve degree. In exact-cocycle mode, the actual isomorphisms satisfy the cocycle law. In metric-reference mode, the reference identity and certified bounds agree under face restriction, without being promoted to morphism-level descent. Finite depth and finiteness of the cover make the required record set finite. The threshold and Type IV clauses prevent undefined or unstable local comparisons from being reported as coherent. Therefore the family satisfies Definition 6.31. $\square$

Remark 6.34 (The theorem glues records, not mathematical objects). Theorem 6.33 does not assert that the filtered objects F_α , their representatives, or their derived realizations glue to a global object. It produces an auditable certificate atlas whose local verdicts and overlap records are coherent. Object-level descent requires an additional theorem in an explicitly declared category.

6.6 Detector-aware local-to-global coverage

Definition 6.35 (Global profile and local restriction interface). Fix a declared covered range X , a monitored degree i , and one global threshold $\tau > 0$. A *global profile interface* consists of a constructible pre-threshold profile $B_{i,X}$ and, for every atlas window W_α , the exact restricted profile

$$R_i^\alpha := W_{W_\alpha} B_{i,X}.$$

The observed local pre-threshold profile B_i^α is connected to R_i^α by exactly one of the following branches.

- (i) **exact**: an actual profile isomorphism $R_i^\alpha \xrightarrow{\sim} B_i^\alpha$ is supplied;
- (ii) **metric_zero_transport**: an artifact-supported bound

$$d_B(R_i^\alpha, B_i^\alpha) \leq \varepsilon_\alpha$$

is supplied, both profiles satisfy the Chapter 2 two-sided threshold-gap condition at ε_α , and

$$2\varepsilon_\alpha \leq \tau.$$

Thus repaired zero is preserved and reflected between the two local profiles. The operational manifest may retain the stricter reserve $2\varepsilon_\alpha < \tau$.

The interface is a declared comparison to a global profile. Pairwise local compatibility alone does not construct $B_{i,X}$.

Definition 6.36 (Certified local detector atlas). Relative to a global profile interface, a *certified local detector atlas* assigns to every W_α :

- (i) a finite witness set $T_\alpha \subseteq (W_\alpha)_\tau$, or the canonical witness family of a birth or right-germ detector;
- (ii) a Chapter 3 detector at source stage `windowed_prethreshold`;
- (iii) endpoint, critical-set, family, completeness, and artifact records;
- (iv) the exact or metric-zero-transport branch of Definition 6.35;
- (v) one mathematical detector value, zero or nonzero, and an independent certificate status.

A detector value may be used globally only at the strength of its local completeness theorem and interface branch.

Definition 6.37 (Global retained-bar coverage). A certified local detector atlas has *global retained-bar coverage* if for every bar I of $B_{i,X}$ retained by T_τ^{bar} , including every genuinely infinite retained bar, there exist an index α and a witness $t \in T_\alpha$ such that

$$t \in I \cap W_\alpha, \quad t + \tau \in I \cap W_\alpha.$$

For birth or right-germ detectors, the same clause is read using the corresponding canonical witness of the retained interval. The coverage artifact records how the universal quantifier over retained global bars is proved.

Theorem 6.38 (Detector-atlas local-to-global zero theorem). *Let a global profile interface, a certified local detector atlas, and global retained-bar coverage be given. Then*

$$T_\tau^{\text{bar}}(B_{i,X}) = 0 \iff D_\alpha^{\text{loc}}(B_i^\alpha) = 0 \text{ for every } \alpha.$$

Here every metric interface branch is required to satisfy its zero-transport hypotheses, and every local detector is required to be complete on its declared local profile class.

Proof. Assume first that the global repaired profile is zero. Window restriction cannot increase the length of a finite bar, and no infinite bar is present in a globally zero repaired profile. Hence $T_\tau^{\text{bar}}(R_i^\alpha) = 0$ for every α . Exact or metric-safe local zero transport gives $T_\tau^{\text{bar}}(B_i^\alpha) = 0$, and local detector completeness gives detector zero.

Conversely, suppose every local detector is zero and that a globally retained bar I exists. Global retained-bar coverage supplies α and a witness t with $t, t + \tau \in I \cap W_\alpha$. Therefore the exact restricted profile R_i^α is nonzero after thresholding. Exact or metric-safe zero reflection transfers this nonzero repaired predicate to B_i^α . Completeness of the local detector then forces a nonzero detector value, a contradiction. $\square$

Remark 6.39 (Coverage is the global proof obligation). The preceding theorem is valid even when coverage is verified by reading the complete global barcode. Such a post hoc verification, however, supplies no proof-compression advantage. A nontrivial finite-certificate claim must obtain coverage independently from geometry, endpoint lattices, bounded birth data, a family theorem, or another registered reduction.

Example 6.40 (Local zero and exact overlap coherence do not imply global zero). Fix $\tau > 0$, let $X = [0, 2\tau)$, and let

$$B_{i,X} = \{[0, 2\tau)\}.$$

Cover X by

$$W_1 = [0, \tau), \quad W_2 = [\frac{1}{2}\tau, \frac{3}{2}\tau), \quad W_3 = [\tau, 2\tau).$$

Every restricted bar has length exactly τ , so every local repaired profile is zero. All nonempty overlap repaired profiles are also zero and therefore admit exact cocycle records. Nevertheless the global bar has length $2\tau > \tau$ and survives globally. The missing hypothesis is global retained-bar coverage: no window contains a τ -span strictly inside the retained global bar after the hard $\leq \tau$ deletion convention is applied locally.

Definition 6.41 (Object-level descent interface). An object-level descent claim additionally declares a category with restriction functors, actual local objects, actual transition isomorphisms, an exact cocycle, an effective-descent theorem for that category and cover, and the artifact constructing the descended object. Metric-reference records, detector coverage, and a coherent certificate atlas are not substitutes for these data.

Remark 6.42 (Record gluing and object descent are separate). Theorem 6.38 proves a predicate about a declared global profile. It neither constructs that profile from local data nor descends filtered or derived objects. Definition 6.41 records the additional route needed for an object-level theorem.

6.7 κ -admissible Restart

Let

$$\mathcal{W}_{\text{part}} = \{W_k = [u_k, u_{k+1})\}_{k \geq 0}$$

be an ordered right-open partition cover. For each k , let $\mathfrak{B}_k$ be the finite pre-threshold profile family used by the local gate and choose a finite working gap

$$0 < g_k \leq \text{gap}_{\tau_k}(\mathfrak{B}_k).$$

If the exact clearance is $+\infty$, the manifest chooses a finite audit cap g_k for Restart arithmetic. Let $\varepsilon_k < +\infty$ be the gate-scoped scalarized local budget and define the finite local reserve

$$r_k := g_k - \varepsilon_k \in \mathbb{R}.$$

A local gate has positive working reserve exactly when $r_k > 0$. The use of a finite certified lower bound avoids undefined extended-real expressions in the Restart recursion.

Definition 6.43 (Continuation artifact). A *continuation artifact* from W_k to W_{k+1} records:

- (i) the source and target local record identifiers;
- (ii) the source and target thresholds;
- (iii) any endpoint-state, initialization, representative, or comparison data used to pass from the source record to the target record;
- (iv) a finite scalarized continuation loss $\eta_k \geq 0$, not already counted in the local target budget;
- (v) a declared degradation factor $\kappa_k \in (0, 1]$;
- (vi) a proof object or independently checkable artifact for the reserve inequality below;
- (vii) explicit confirmation that missing morphisms, representatives, eligibility records, Type IV data, or manifest fields have not been encoded numerically inside η_k .

Definition 6.44 (κ_k -admissible continuation). A continuation artifact is κ_k -*admissible* if it certifies

$$r_{k+1} \geq \kappa_k r_k - \eta_k.$$

The number κ_k is part of the certified continuation data; its existence is not inferred from window adjacency or from threshold deletion alone.

Remark 6.45 (Restart is conditional, not automatic). There is no Core theorem asserting that every admissible-looking continuation has some universal $\kappa > 0$. Window restriction may change bar lengths, thresholds may change, and the next pre-threshold profile may be unrelated to the preceding one. A Restart claim is licensed only by an explicit κ_k -admissible continuation artifact. This is the v19 downgrade of the former unconditional Restart lemma.

Remark 6.46 (Restart is a reserve-transport claim). A κ -Restart assertion transports a finite positive threshold reserve and is therefore stronger than an isolated exact equality at one transition. An exact continuation may be recorded as exact comparison data, but it does not become a κ_k -admissible Restart step unless finite working gaps, local budgets, r_k, r_{k+1} , and the certified recurrence

$$r_{k+1} \geq \kappa_k r_k - \eta_k$$

are all supplied.

Definition 6.47 (Restart transition status). Each transition has exactly one of the statuses

not_invoked, undefined, certified, failed.

They mean:

- (i) `not_invoked`: the governing scope explicitly declares that no continuation or inherited reserve claim is being made;
- (ii) `undefined`: Restart is in scope, but a required continuation artifact, scalarized budget, finite working reserve, or κ_k -certificate is missing or ill-typed;
- (iii) `certified`: the artifact is complete and the κ_k -admissible inequality is verified;

(iv) **failed**: the inputs are defined but the declared κ_k -inequality is false.

The transition status certifies the continuation law only. A successful Restart step additionally requires a passing target local record, in particular positive target reserve and no rejected or undefined non-reserve clause.

Theorem 6.48 (Finite-horizon Restart bound). *Assume the transitions $0 \rightarrow 1, \dots, n-1 \rightarrow n$ are respectively $\kappa_0, \dots, \kappa_{n-1}$ -admissible. Then*

$$r_n \geq \left(\prod_{j=0}^{n-1} \kappa_j \right) r_0 - \sum_{m=0}^{n-1} \left(\prod_{j=m+1}^{n-1} \kappa_j \right) \eta_m,$$

where an empty product is 1.

Proof. The case $n = 1$ is Definition 6.44. Assume the formula holds for n . Then

$$r_{n+1} \geq \kappa_n r_n - \eta_n.$$

Substituting the induction hypothesis and distributing κ_n gives

$$r_{n+1} \geq \left(\prod_{j=0}^n \kappa_j \right) r_0 - \sum_{m=0}^n \left(\prod_{j=m+1}^n \kappa_j \right) \eta_m.$$

Thus the formula follows by induction. □

Corollary 6.49 (Non-degrading continuation). *If $\kappa_k = 1$ for $0 \leq k < n$, then*

$$r_n \geq r_0 - \sum_{m=0}^{n-1} \eta_m.$$

Consequently, if

$$\sum_{m=0}^{n-1} \eta_m < r_0,$$

all reserves through level n are positive.

Definition 6.50 (Normalized Restart loss). Set

$$K_0 := 1, \quad K_n := \prod_{j=0}^{n-1} \kappa_j \quad (n \geq 1).$$

The normalized loss through level n is

$$L_n^{\text{norm}} := \sum_{m=0}^{n-1} \frac{\eta_m}{K_{m+1}}.$$

Corollary 6.51 (Normalized reserve criterion). *Under the hypotheses of Theorem 6.48,*

$$\frac{r_n}{K_n} \geq r_0 - L_n^{\text{norm}}.$$

Hence $L_n^{\text{norm}} < r_0$ implies $r_n > 0$.

Proof. Factor K_n from the lower bound in Theorem 6.48 and use

$$\prod_{j=m+1}^{n-1} \kappa_j = \frac{K_n}{K_{m+1}}.$$

□

Remark 6.52 (Ordinary summability is not enough). The condition $\sum_k \eta_k < \infty$ does not by itself guarantee positive reserves when the factors κ_k can contract. The correct inherited-reserve condition is normalized by the accumulated degradation:

$$\sum_{k \geq 0} \frac{\eta_k}{K_{k+1}} < r_0.$$

If $K_n \rightarrow 0$, even a certified infinite run need not possess a positive uniform lower bound on r_n .

Example 6.53 (Ordinary summability can coexist with reserve failure). Let $r_0 = 1$, $\kappa_k = \frac{1}{2}$, and $\eta_k = 2^{-k-2}$. Then

$$\sum_{k \geq 0} \eta_k = \frac{1}{2} < r_0,$$

but $K_{k+1} = 2^{-k-1}$, so

$$\frac{\eta_k}{K_{k+1}} = \frac{1}{2}$$

and the normalized loss diverges. If equality holds in the Restart recurrence, then

$$\frac{r_n}{K_n} = 1 - \frac{n}{2},$$

so the reserve is nonpositive from a finite stage onward. Thus ordinary summability is not a substitute for normalized reserve-safety.

Example 6.54 (Pointwise positive reserve need not be uniform). Let $\eta_k = 0$, $\kappa_k = \frac{1}{2}$, and take equality in every Restart step. Then

$$r_n = 2^{-n} r_0 > 0$$

for every finite n , but $\inf_n r_n = 0$.

6.8 Countable Restart and summability

Definition 6.55 (κ -normalized summability). A countable certified Restart run is κ -normalized summable if

$$\sum_{k=0}^{\infty} \frac{\eta_k}{K_{k+1}} < \infty.$$

It is *reserve-safe* if, in addition,

$$\sum_{k=0}^{\infty} \frac{\eta_k}{K_{k+1}} < r_0.$$

Theorem 6.56 (Conditional countable Restart principle). Let $\{W_k\}_{k \geq 0}$ be an ordered right-open partition cover. Assume:

(i) the initial local record has $r_0 > 0$;

- (ii) every invoked transition has status *certified*;
- (iii) the run is reserve-safe in the sense of Definition 6.55;
- (iv) every target window has a complete local record whose required non-reserve clauses pass, including representative, eligibility, and Type IV clauses when invoked;
- (v) no transition or target record is *undefined*.

Then every inherited reserve r_n is positive, and the sequence of local records forms an auditable sequential certificate record on the partition cover.

Proof. For every finite n , reserve-safety implies

$$L_n^{\text{norm}} \leq \sum_{k=0}^{\infty} \frac{\eta_k}{K_{k+1}} < r_0.$$

Corollary 6.51 therefore gives $r_n > 0$. The declared continuation artifacts and complete target records make every transition reconstructible. Thus the ordered family is an auditable sequential certificate record. $\square$

Corollary 6.57 (Pointwise reserve and uniform reserve are distinct). *Under the hypotheses of Theorem 6.56, set*

$$L_{\infty}^{\text{norm}} := \sum_{k=0}^{\infty} \frac{\eta_k}{K_{k+1}}, \quad c := r_0 - L_{\infty}^{\text{norm}} > 0.$$

Then every finite stage satisfies

$$r_n \geq K_n c > 0.$$

Consequently, pointwise positivity of all r_n follows from reserve-safety. A positive uniform lower bound follows if $\inf_n K_n > 0$; it does not follow merely from reserve-safety when $K_n \rightarrow 0$.

Proof. Corollary 6.51 gives

$$\frac{r_n}{K_n} \geq r_0 - L_n^{\text{norm}} \geq r_0 - L_{\infty}^{\text{norm}} = c.$$

Since every $K_n > 0$, the pointwise bound follows. Taking infima gives a uniform positive bound only when $\inf_n K_n > 0$. $\square$

Remark 6.58 (What the countable theorem does not say). Theorem 6.56 does not construct a single global filtered object or assert convergence of the local objects. It certifies the sequential inheritance of positive threshold reserve and the audibility of the record chain.

6.9 Detector-aware Restart handoff

Definition 6.59 (Detector-aware Restart handoff). A Restart transition that consumes a repaired-zero predicate records, in addition to reserve transport, exactly one target-zero route:

- (i) a fresh complete target-window detector certificate;
- (ii) an exact zero-transport isomorphism from a certified source profile to the target profile;
- (iii) a Chapter 2 metric-safe zero-transport record, including the post-threshold zero-separation condition.

The route identifies the detector source stage, target stage, threshold, endpoint convention, and artifact. A κ_k -admissible reserve inequality alone is not a zero-transport theorem.

Definition 6.60 (Boundary-bridge coverage for a partition). For an ordered partition $\{W_k\}$, let $\{V_k\}$ be declared bridge windows meeting both sides of the boundary between W_k and W_{k+1} . The augmented family

$$\mathcal{W}_{\text{aug}} := \{W_k\}_k \cup \{V_k\}_k$$

has *boundary-bridge coverage* at τ when every globally retained bar admits a certified τ -span witness in some partition or bridge window. Each bridge window carries its own direct evaluation and complete detector certificate.

Corollary 6.61 (Detector-safe partition globalization). Assume a global profile interface for the augmented partition/bridge family, boundary-bridge coverage, and zero complete detectors on every partition and bridge window. Then the global repaired profile is zero.

Proof. The augmented family is a certified local detector atlas with global retained-bar coverage. Apply Theorem 6.38. $\square$

Remark 6.62 (Reserve transport does not see crossing bars). A positive Restart reserve controls threshold classification inside the profiles named by the continuation record. It does not prove that a long bar crossing a partition boundary is detected in either disjoint cell. Fresh target detection or a zero-transport theorem controls the target predicate; boundary-bridge coverage controls the global predicate. These are distinct obligations.

6.10 Countable overlap atlases

Definition 6.63 (Audit-locally finite cover). A countable overlap cover $\{W_n\}_{n \geq 1}$ is *audit-locally finite* if every bounded subinterval of the declared range meets only finitely many windows and only finitely many recorded nerve simplices.

Definition 6.64 (Summable simplex exposure). Let N_{met} be the countable set of recorded nerve simplices whose simplex coherence mode is `metric_reference`. For $\sigma \in N_{\text{met}}$, let

$$e_\sigma := \sum_{\alpha \in \sigma} \varepsilon_{\alpha, \sigma}$$

be the declared scalarized exposure of all local-to-reference comparisons consumed on that simplex. More generally, a manifest may use any nonnegative e_σ that explicitly dominates every metric comparison charged to σ , provided its aggregation rule is fixed in advance. Define the unordered nonnegative sum by

$$\sum_{\sigma \in N_{\text{met}}} e_\sigma := \sup_{S \subseteq N_{\text{met}}, S \text{ finite}} \sum_{\sigma \in S} e_\sigma.$$

The atlas has *summable simplex exposure* when this quantity is finite. This definition is independent of enumeration. Exact-cocycle simplices require exact proof artifacts and are not assigned fictitious metric exposure merely to enter this sum.

Summability does not replace the separate local-to-reference sharp inequalities

$$\varepsilon_{\alpha, \sigma} < \text{gap}_{\tau_\sigma}(\mathcal{B}_{i, \sigma}^\alpha, R_{i, \sigma})$$

for every consumed metric comparison, nor any non-scalar coherence obligation. Repeated use of the same underlying discrepancy is charged according to the declared no-double-counting policy rather than silently reused for free.

Theorem 6.65 (Countable record-atlas principle). *Let $\{W_n\}_{n \geq 1}$ be a countable, audit-locally finite overlap cover of finite Čech depth. Assume:*

- (i) *every local window carries a complete certificate record;*
- (ii) *every nonempty nerve simplex is recorded and carries a simplex compatibility record in coherence mode `exact_cocycle` or `metric_reference`; every consumed comparison inside that record uses runtime mode `exact` or `metric_gap_certified`;*
- (iii) *every metric-reference simplex has positive local-to-reference threshold reserves, and every exact-cocycle simplex has actual cocycle morphisms and exact proof artifacts;*
- (iv) *the metric-reference simplex exposures are summable;*
- (v) *every terminal Type IV scope is either correctly `not_invoked` or `certified_zero`; whenever long-transient cleanliness is required, the corresponding aggregate transient status is `certified_absent`;*
- (vi) *all face restrictions and artifact references are audit-readable;*
- (vii) *if a global detector-zero conclusion is invoked, the atlas additionally carries a global profile interface, complete local detector certificates, exact or metric-safe local zero transport, global retained-bar coverage, and zero local detector values.*

Then the family forms an audit-locally finite Čech-coherent certificate atlas on the covered range, and its total declared scalarized metric-reference simplex exposure is finite. When the additional detector clause is invoked, the global repaired profile is zero.

Proof. Audit-local finiteness ensures that every bounded audit query touches only finitely many local and overlap records. Finite Čech depth bounds the order of intersections. The simplex compatibility records and face conditions provide coherence on every nonempty nerve simplex. Metric-reference local-to-reference reserves license their thresholded transports, while exact-cocycle records are licensed by their actual cocycle morphisms and exact proof artifacts. Definition 6.64 records finite total scalarized exposure across the metric-reference portion of the countable atlas. Therefore the resulting atlas is locally reconstructible and globally indexed as a coherent certificate record system with finite declared metric exposure. If the detector-globalization clause is invoked, Theorem 6.38 applies and yields the declared global repaired-zero predicate. $\square$

Remark 6.66 (Summability is not coherence). Simplex-budget summability is an audit-exposure condition. It neither creates missing face maps nor proves higher compatibility. Conversely, a coherent atlas may fail the chosen global exposure policy if its scalarized simplex budgets are not summable. The two obligations are logically independent and both are required by Theorem 6.65.

Remark 6.67 (No global pass from summability alone). A summable atlas is not automatically a single global B-Gate⁺ pass. The conclusion is a coherent atlas of local and overlap certificates. A single global pass requires an independently declared global object, global window, and global gate record.

Remark 6.68 (Canonical Type IV notation in Chapter 6). Whenever Chapter 6 cites aggregate Type IV diagnostics, the canonical pair is

$$(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}).$$

The legacy symbols $u_{i,W,\tau}$ and u_{Collapse} have no normative use. The coordinate u in a window $[u, \nu)$ is unrelated and is retained as a domain endpoint.

6.11 Integration with B-Gate⁺ and Type IV

Definition 6.69 (Local certificate status). A local window record has status

pass, reject, undefined

according to the B-Gate⁺, eligibility, Type IV, and manifest disciplines of Chapters 1–5. Undefined records cannot participate as successful vertices of an overlap atlas or as successful targets of Restart.

Proposition 6.70 (Compatibility does not repair a local failure). *Passing an Overlap Gate does not convert a locally rejected or undefined record into a local pass. Likewise, a certified Restart transition does not discharge a missing local B-Gate⁺, representative, or Type IV requirement on the target window.*

Proof. Overlap and Restart records certify compatibility or continuation between local records. They do not alter the internal gate clauses of either endpoint record. Hence local validity remains a separate necessary condition. □

Proposition 6.71 (Typed Type IV participation). *Whenever Type IV is in scope on a local window, overlap, or continuation target:*

- (i) *certified_zero permits the record to proceed;*
- (ii) *obstructed forces rejection;*
- (iii) *undefined prevents certification and is routed as Type III operational undefinedness; it becomes Type V only if the missing state is concealed, serialized as zero or pass, or makes the manifest inconsistent;*
- (iv) *not_invoked is admissible only when the governing policy explicitly places Type IV outside the scope of that record.*

Proof. This is the typed Type IV discipline of Chapter 5 applied to the local, overlap, and restart records defined in this chapter. □

Proposition 6.72 (Transient-defect participation and nontransport). *Suppose a local, overlap, simplex, Restart, or atlas claim requires absence of long transient kernel or cokernel defects. Then its Chapter 5 aggregate transient status must be certified_absent on the same monitored scope. Status certified_present forces failure of that stronger claim, and status undefined prevents certification.*

An exact or metric comparison of the underlying source or repaired profiles does not transport this transient status. Such transport requires actual comparison data for the kernel and cokernel defect profiles and a separately proved defect-stability or exact-identification theorem.

Proof. The first assertion is the Chapter 5 terminal/transient separation applied to the record types of this chapter. The second follows because kernel and cokernel are defined from the actual tower comparison morphism. Object-level or barcode-level closeness of its source and target does not determine the defect profiles without additional morphism-level comparison data. □

6.12 Failure routing

Definition 6.73 (Chapter 6 failure routing). Failures in this chapter are routed as follows; the types are not mutually exclusive.

- (i) an unstable threshold comparison or failed sharp gap condition is Type III;

- (ii) a false crop-commutation substitution, failed overlap comparison, failed exact cocycle, failed metric-reference record, failed detector coverage, or failed κ_k -inequality is Type III;
- (iii) a nonzero terminal-generic tower defect is Type IV; a certified long-transient defect fails any claim that required transient cleanliness, while missing transient evidence remains Type III undefinedness;
- (iv) a missing required mathematical or operational artifact yields an `undefined` clause and is Type III at the operational level. It is additionally Type V only when the missing-artifact marker is omitted or concealed, an unsupported pass or zero is reported, or the manifest is incomplete or inconsistent;
- (v) a local topological obstruction remains Type I, and an eligible categorical obstruction remains Type II.

Remark 6.74 (Why missing κ is not evidence for $\kappa = 1$). Absence of a degradation certificate does not justify setting $\kappa_k = 1$, $\eta_k = 0$, or declaring the transition harmless. It yields status `undefined` and therefore no certified Restart.

6.13 Minimal manifest consequences

Proposition 6.75 (Window/overlap/Restart manifest extension). *An audit-readable manifest using this chapter extends the Chapter 1 base manifest and the Chapter 3 profile/representative extension by recording, at minimum:*

- (i) *the cover type: `partition` or `overlap`;*
- (ii) *every window and every nonempty recorded overlap;*
- (iii) *the direct common-window evaluation policy;*
- (iv) *the local and overlap thresholds, including threshold-reconciliation records;*
- (v) *the pre-threshold profiles or their immutable artifact identifiers;*
- (vi) *the clearance and gap values used by each comparison;*
- (vii) *the ledger quantale and bottleneck scalarization;*
- (viii) *the comparison mode of every consumed local, overlap, and simplex comparison, together with every exact proof artifact or scalarized metric budget as applicable;*
- (ix) *every scalarized local, overlap, crop, and continuation budget actually consumed by a metric-mode inference;*
- (x) *every pairwise and higher simplex compatibility record, including the simplex coherence mode;*
- (xi) *every exact-cocycle morphism and cocycle check, or every metric-reference identity and local-to-reference certificate, as applicable;*
- (xii) *the Čech depth and audit-local-finiteness declaration, when applicable;*
- (xiii) *when detector globalization is invoked: the global profile identity, local restriction interfaces, detector stages, witness sets, completeness artifacts, local zero-transport branches, retained-bar coverage theorem, and local/global detector verdicts;*

- (xiv) every continuation artifact, κ_k , η_k , reserve r_k , Restart status, and any detector-aware target-zero route or boundary-bridge coverage record;
- (xv) all representative and eligible-realization records used by local clauses;
- (xvi) every typed terminal Type IV scope status and required comparison artifact;
- (xvii) whenever transient cleanliness is invoked, every full/interior defect component, detector record, and aggregate transient status;
- (xviii) any object-level descent interface and the external effective-descent theorem actually consumed;
- (xix) all local, overlap, restart, detector-globalization, and atlas verdicts;
- (xx) the failure-routing record for every rejected or undefined clause.

Proof. Each item is an input to, or a reconstructibility condition for, one of the definitions and theorems of this chapter. Omitting any invoked item prevents an independent verifier from reproducing the corresponding compatibility, restart, or gluing verdict. $\square$

Remark 6.76 (Canonical manifest location). Proposition 6.75 is an extension schema, not a competing canonical manifest. Chapter 7 integrates the Chapter 1 base fields, the Chapter 3 evaluation fields, the present window/overlap/Restart fields, and the remaining audit fields into the canonical v20 audit manifest with statuses complete, incomplete, and inconsistent.

6.14 Window, overlap, and Restart package

Theorem 6.77 (Conditional window/overlap/Restart closure package). *Within the v20 conservative extension of the repaired Core, the following package is available.*

- (i) Overlap profiles are evaluated directly on the common window before threshold deletion.
- (ii) Metric ledger data are compared with threshold clearance only after a declared bottleneck scalarization; the sharpness of the threshold band does not assert tightness of that scalarization.
- (iii) Overlap comparison is typed: exact mode consumes an exact profile proof, while metric-gap-certified mode uses a sharp two-sided certificate to transfer a pre-threshold ε -bound to the post-threshold evaluations.
- (iv) Pairwise Overlap Gates do not replace higher Čech coherence.
- (v) Finite-depth overlap data produce a coherent certificate atlas only at the record level.
- (vi) Restart is conditional on a declared κ_k -admissible continuation artifact.
- (vii) Finite-horizon reserves obey the product-weighted inequality of Theorem 6.48.
- (viii) Countable Restart requires κ -normalized reserve-safety; ordinary summability alone is insufficient, and pointwise positive reserves do not automatically have a positive uniform infimum.
- (ix) Exact simplex coherence is carried by actual cocycle isomorphisms; metric simplex coherence is carried by reference profiles and bounds and is not promoted to a cocycle.

- (x) *Local detector zeros globalize only through a declared global profile interface and retained-bar coverage theorem; exact overlap coherence alone does not suffice.*
- (xi) *Restart reserve transport does not transport detector zero and does not detect bars crossing a disjoint partition boundary without a target-zero route and boundary-bridge coverage.*
- (xii) *Countable overlap atlases additionally require audit-local finiteness, finite Čech depth, simplex coherence, exact proof artifacts on exact-cocycle simplices, and positive reserves plus summable exposure on metric-reference simplices.*
- (xiii) *Terminal Type IV cleanliness and long-transient cleanliness are separately typed and are not transported by source-profile comparison alone.*
- (xiv) *Record-level atlas coherence, detector-globalized zero, and object-level descent are three distinct conclusion strengths.*
- (xv) *Undefined artifacts or statuses are never promoted to successful compatibility, Restart, detector, or gluing verdicts.*

Proof. Items (i)–(iii) follow from Definitions 6.7, 6.9, 6.17, 6.18, and Proposition 6.20. Items (iv)–(v) follow from Remark 6.23 and Theorem 6.33. Items (vi)–(viii) follow from Definitions 6.44, Theorem 6.48, and Theorem 6.56. The exact/metric coherence distinction follows from Definitions 6.26 and 6.27. Detector globalization and its no-go boundary are Theorem 6.38 and Example 6.40. Detector-aware Restart follows from Definition 6.59 and Corollary 6.61. The countable statement is Theorem 6.65. Terminal/transient participation follows from Propositions 6.71 and 6.72. The conclusion-strength separation follows from Definition 6.41. The final nonpromotion rule follows from the typed status disciplines of Definitions 6.47 and 6.69. $\square$

Remark 6.78 (Closure package, not an additional mechanism). Theorem 6.77 is a closure package collecting the proved transfer bounds and the explicitly conditional record theorems of this chapter. It does not turn operational definitions into a new unconditional local-to-global theorem, and it does not supply any missing non-scalar artifact.

6.15 Explicit non-claims

The following are not asserted in this chapter.

- (i) No unconditional Restart lemma is asserted.
- (ii) No universal or automatically existing $\kappa > 0$ is asserted.
- (iii) No missing continuation artifact is interpreted as $\kappa = 1$ or zero loss.
- (iv) No ordinary summability condition $\sum_k \eta_k < \infty$ is claimed to preserve positive reserve under arbitrary contracting κ_k .
- (v) No pairwise Overlap Gate is claimed to imply higher Čech coherence.
- (vi) No exact overlap comparison is automatically a positive-radius robustness certificate, and no exact proof is forced into a fictitious metric budget merely for ledger summation.
- (vii) No comparison of differently thresholded outputs is made without threshold reconciliation and direct common-threshold re-evaluation.

- (viii) No crop-after-collapse comparison replaces direct common-window evaluation without a commutation certificate.
- (ix) No quantale-valued ledger is compared directly with a real threshold clearance without a declared scalarization.
- (x) No claim is made that the bottleneck scalarization is numerically tight merely because the two-sided threshold band is sharp.
- (xi) No metric reserve, large gap, or summability estimate repairs a missing representative, morphism, compatibility record, Type IV certificate, claim status, or manifest field.
- (xii) No pointwise conclusion $r_n > 0$ for every finite n is promoted to a uniform bound $\inf_n r_n > 0$ without an additional lower bound on K_n .
- (xiii) No coherent certificate atlas is automatically a global filtered object, global persistence module, or global B- Gate^+ pass.
- (xiv) No metric-reference simplex record is an exact cocycle, transition-map family, or descent datum.
- (xv) No family of local detector-zero verdicts, even with exact pairwise overlap coherence, implies a global repaired-zero predicate without a declared global profile interface and retained-bar coverage.
- (xvi) No Restart reserve inequality transports a detector-zero predicate or detects a bar crossing a partition boundary without a target-zero route and boundary-bridge coverage.
- (xvii) No summable simplex exposure creates detector completeness, face morphisms, higher coherence, or global retained-bar coverage.
- (xviii) No source-profile overlap or Restart comparison transports kernel or cokernel defect profiles without a separately proved defect comparison theorem.
- (xix) No terminal Type IV zero is identified with long-transient cleanliness, and no transient `certified_absent` verdict is promoted to full defect vanishing without the Chapter 5 short-defect exclusion hypotheses.
- (xx) No global profile is reconstructed merely from pairwise or simplex records; when a global profile is used by the detector theorem, it is declared input data.
- (xxi) No record-level gluing theorem is an effective object-level descent theorem.
- (xxii) No countable certificate atlas supplies a finite-to-infinite reduction or an external Return Theorem.
- (xxiii) No undefined Type IV, transient, Restart, detector, representative, or overlap status is treated as zero or pass.
- (xxiv) No Search-side map, score, or AI proposal supplies a missing overlap, continuation, detector, coverage, descent, or gluing certificate.

6.16 Provenance and status

The sharp overlap-transfer step is an application of the threshold-gap-safe stability theorem of Chapter 2 and Appendix A. The representative and eligible-realization clauses are inherited from Chapters 3–4. The typed terminal Type IV and transient-defect clauses are inherited from Chapter 5. The v19 inheritance consists of direct common-window evaluation, exact/metric overlap modes, κ -admissible Restart, normalized reserve bounds, and finite/countable record-atlas logic. The v20 contribution is the separation of exact-cocycle from metric-reference coherence, detector-aware global retained-bar coverage, explicit local-zero and Restart no-go boundaries, transient-defect participation, and the typed separation between record coherence, global zero predicates, and object-level descent.

Status labels for this chapter. *Core and conditional mathematical results.* Propositions 6.20 6.28, 6.32, 6.70, 6.71, 6.72, and 6.75. Theorems 6.33 6.38, 6.48, 6.56, 6.65, and 6.77. Corollaries 6.49 6.51, 6.57, and 6.61.

Counterexamples and sharpness witnesses. Examples 6.40 6.53, and 6.54.

Core, interface, and operational record definitions. Definitions 6.3 6.4, 6.5, 6.7, 6.9, 6.10, 6.11, 6.14, 6.16, 6.17, 6.18, 6.19, 6.21, 6.24, 6.25, 6.26, 6.27, 6.30, 6.31, 6.35, 6.36, 6.37, 6.41, 6.43, 6.44, 6.47, 6.50, 6.55, 6.59, 6.60, 6.63, 6.64, 6.69, and 6.73.

Operational cautions, nonpromotion rules, and integration remarks. Remarks 6.1 6.2, 6.6, 6.8, 6.12, 6.13, 6.15, 6.22, 6.23, 6.29, 6.34, 6.39, 6.42, 6.45, 6.46, 6.52, 6.58, 6.62, 6.66, 6.67, 6.68, 6.74, 6.76, and 6.78.

Explicit exclusions. Subsection 6.15.

7 Chapter 7: Typed Audit Semantics and Core Sovereignty

Chapter 7 contract. This chapter is the canonical audit layer of the v20.0 Core. It preserves the v19 atomic status algebra

pass, reject, undefined, not_invoked,

and integrates the detector, higher-obstruction, transient-defect, finite-to-infinite, atlas, and descent records introduced in Chapters 1–6. No new independent verdict algebra is created.

Every token is assigned a declared semantic sort. Runtime comparison modes, simplex-coherence modes, profile stages, interface branches, cover types, source-semantics tokens, and mathematical values are not atomic statuses and may not be serialized or consumed as though they were.

A Core verdict is a dependency-closed typed conjunction of independently evidenced clauses. Exact proof is not confused with positive-radius robustness. Numerical slack may bound a declared metric discrepancy, but it may not create detector completeness, zero reflection, a representative, morphism, realization, terminal or transient defect certificate, overlap cocycle, global retained-bar coverage, finite-to-infinite reduction, descent theorem, Return Theorem, claim status, or manifest field. The statuses reject, undefined, and justified not_invoked remain distinct.

7.1 Scope, dependency order, and relocation

This chapter closes the audit semantics of the v20 Core after the mathematical and record-theoretic constructions of Chapters 1–6. It consumes, but does not replace, the following previously established layers:

- (i) the claim discipline, comparison-mode distinction, and B-Gate⁺ contract of Chapter 1;
- (ii) the standard bottleneck metric, hard threshold deletion, and sharp two-sided stability theorem of Chapter 2 and Appendix A;
- (iii) the repaired-evaluation, finite detector, profile-stage, completeness, metric zero-transport, and representative disciplines of Chapter 3;
- (iv) the degree-one and higher eligible realization records, canonical Extⁿ-edge, one-way discharge, and conditional zero-reflection upgrade of Chapter 4;
- (v) the typed failure taxonomy, source-semantics discipline, terminal-generic Type IV diagnostics, transient-defect calculus, and finite-to-infinite reduction records of Chapter 5;
- (vi) the typed ledger scalarization, comparison-mode Overlap Gate, exact-cocycle/metric-reference simplex calculus, detector globalization, object-descent boundary, Čech atlas, and κ -admissible Restart discipline of Chapter 6.

The purpose of the present chapter is to determine when those inputs yield an auditable Core verdict. It therefore specifies:

- (i) which deviations may enter a numerical ledger, which comparisons are exact or metric, and which proof obligations may not be scalarized;
- (ii) the typed meanings of pass, reject, undefined, and not_invoked;
- (iii) the aggregation rule for atomic clause statuses in finite and countable scopes;
- (iv) the audit reading of B-Gate⁺, Type IV, Overlap, Restart, and atlas records;
- (v) manifest completeness, inconsistency, artifact closure, semantic replayability, schema migration, and repair traceability;
- (vi) detector, higher-edge, transient, global-zero, finite-to-infinite, descent, and release-acceptance audit readings;
- (vii) Core sovereignty and proposer–verifier separation.

Remark 7.1 (Relocation of the former Chapter 7). The former chapter entitled “Core Theorem Package and Explicit Non-Claims” is not discarded. In the v19 and v20 dependency order it follows the present audit chapter and is therefore relocated to Chapter 8. The present chapter precedes that closure package because claim status, auditability, and Core sovereignty are inputs to the final statement of what the Core does and does not assert.

Remark 7.2 (Normative rather than external-domain content). Most statements in this chapter are definitions, typing contracts, or consequences of those contracts. They govern admissible proof objects and verdicts; they are not new theorems about an external arithmetic, geometric, PDE, computational, or cryptographic domain.

7.2 Canonical token sorts and evidence graph

Definition 7.3 (Canonical token-sort registry). Every token occurrence consumed by a v20 audit record is assigned exactly one declared semantic sort. The canonical sorts relevant to the Main body are:

(i) *atomic audit status*:

pass, reject, undefined, not_invoked;

(ii) *claim evidence status* and *invocation role*, as in Definitions 7.23 and 7.24;

(iii) *runtime profile-comparison mode*:

exact, metric_gap_certified, not_compared;

(iv) *collapse policy*:

barcode_level, tau_split, exact_admissible;

(v) *gap-certificate status*:

gap_certified, gap_failed, gap_pending;

(vi) *detector profile stage*:

windowed_prethreshold, repaired_postthreshold,
kernel_defect, cokernel_defect;

(vii) *detector mode*:

birth, germ, certified_stencil, family_mesh;

(viii) *detector certificate status* and independent value:

detector_certified, detector_failed, detector_pending,
zero, nonzero, undefined;

(ix) *terminal Type IV status*:

not_invoked, undefined, certified_zero, obstructed;

(x) *full/interior inspection status* and *aggregate long-transient status*:

not_checked, certified_absent, certified_present,
not_invoked, undefined, certified_absent, certified_present;

The associated detector component is `full_profile` or `interior_noncofinal`.

(xi) *terminal mode* and *source semantics*:

bounded_terminal, right_unbounded,
genuine_colimit, verified_stabilization, implementation_artifact;

(xii) *cover type and simplex-coherence mode*:

partition, overlap,
exact_cocycle, metric_reference, not_compared;

(xiii) *derived interface branch*: `metric_zero_transport`. It abbreviates a successful `metric_gap_certified` comparison plus the separate repaired-zero separation hypotheses; it is neither a fourth comparison mode nor a status.

(xiv) *Restart-transition status*:

not_invoked, undefined, certified, failed;

(xv) *manifest status*:

complete, incomplete, inconsistent.

Every serialized occurrence records both its field name and token sort. Equal character strings in different sorts do not identify the underlying semantic values.

Proposition 7.4 (No cross-sort token substitution). *A token of one semantic sort cannot discharge a field of another sort. In particular, a simplex-coherence mode, cover type, profile stage, source semantics, interface branch, detector value, or manifest status is not an atomic audit verdict merely because its spelling resembles a verdict token. A cross-sort substitution makes the affected clause *undefined*; an intentional or concealed substitution is additionally a Type V inconsistency.*

Proof. The fields have different source types and governing predicates. A value of one type supplies no inhabitant or proof of another type. Concealment also contradicts the declared manifest semantics. $\square$

Definition 7.5 (Audit clause identity). An atomic audit clause is identified by

$c = (\text{ClauseID}, \text{kind}, \text{scope}, \text{predicate}, \text{dependencies}, \text{governing_claim}).$

Its scope includes every degree, window, threshold, detector stage, defect threshold, cover/simplex, tower source semantics, and realization policy that changes the meaning of the predicate. Two clauses differing in one of these fields are not silently merged.

Definition 7.6 (Typed evidence record). A typed evidence record for a clause contains an `EvidenceID`, evidence type, exact source and target, theorem or policy identifier, all hypotheses, profile stage or semantic branch, transformation history, immutable artifact identities, and one atomic status. Numerical output, a bare isomorphism symbol, a detector value without certificate status, or a citation without claim-use preflight is not a complete evidence record.

Definition 7.7 (Well-founded audit dependency graph). The dependency graph of a run has clauses and evidence records as typed nodes and has a directed edge from each conclusion to every prerequisite it consumes. It is *audit-well-founded* when every dependency chain reaches registered primitive evidence or an explicitly assumed interface hypothesis and no clause is used, directly or indirectly, to certify its own prerequisite. For a countable graph, the manifest additionally supplies a well-founded rank or an equivalent independently checkable termination certificate.

Proposition 7.8 (Determinacy on a complete well-founded graph). *Fix an immutable scope and canonical token-sort registry. If its clause registry is complete, its dependency graph is audit-well-founded, and every primitive record has a determined atomic status, then recursive evaluation by Definition 7.32 produces a unique headline verdict and preserves every lower-level status and failure route.*

Proof. Evaluate clauses in well-founded rank order. Each clause is then a fixed function of already evaluated prerequisites. The aggregate rule is deterministic, while the manifest separately retains all component statuses. $\square$

Definition 7.9 (Claim supersession and conditional override record). A v20 statement that narrows, repairs, or conditionally supersedes a v19 non-claim or weaker claim carries a new claim identity and records the predecessor statement, exact successor statement, added hypotheses, proof location, permitted stronger reading, still-forbidden readings, and immutable ancestry. The predecessor source and historical verdict are not mutated.

Proposition 7.10 (No silent claim supersession). *Absent a complete supersession record and successful claim-use preflight, the weaker predecessor reading governs. In particular, a conditional zero-reflecting vanishing equivalence does not silently become an unconditional object-level $\text{PH}_n \cong \text{Ext}^n$ claim.*

Proof. The stronger use has additional hypotheses and a distinct claim identity. Without their registered discharge, the weaker-safe interpretation applies. $\square$

7.3 Audit identity and immutable run scope

Definition 7.11 (Audit run identity). An *audit run identity* is a tuple

$$\mathfrak{R} = \left(\text{RunID}, \text{theory_version}, \text{schema_version}, \right. \\ \left. \text{claim_register_version}, \text{artifact_root} \right).$$

It identifies the exact theory release, manifest schema, claim-register snapshot, and content-addressed artifact root under which all verdicts in the run are interpreted.

Definition 7.12 (Immutable audit scope). The *immutable audit scope* of a run records, at minimum:

- (i) the object or object family under audit;
- (ii) the monitored degrees and, for Type IV totals, the finite set I_{mon} ;
- (iii) the windows, thresholds, terminal modes, and evaluation order;
- (iv) the complete registry of invoked clauses and explicitly non-invoked clauses;
- (v) the comparison mode of every consumed comparison, every simplex-coherence mode, cover type, derived interface branch, and token-sort declaration;
- (vi) every detector stage, mode, information class, certificate/value pair, defect threshold, global-profile interface, coverage family, finite-to-infinite reduction, and object-descent scope when invoked;
- (vii) the claim identifiers, invocation roles, theorem or policy locations, registered strengths, and every claim-supersession relation;
- (viii) the complete clause/evidence dependency graph and the immutable identifiers and content hashes of all source, profile, representative, comparison, detector, diagnostic, reduction, descent, and verdict artifacts.

A change to any of these data defines a new audit run rather than a silent mutation of the existing run.

Remark 7.13 (Why immutable identity matters). A previously rejected or undefined run may be repaired, but the repaired object receives a new RunID and ManifestID, optionally linked to the former run by an explicit parent or repair relation. Later evidence does not rewrite the historical meaning of the earlier verdict.

7.4 The typed δ -ledger

Definition 7.14 (Declared ledger algebra). A run using numerical error budgets declares a commutative unital ordered monoid

$$(V, \oplus, 0_V, \preceq)$$

whose operation is monotone in each variable. If the implementation also uses arbitrary joins and $\oplus$ distributes over those joins, the ledger algebra may be declared a commutative quantale. The word “quantale” is not used merely for a four-tuple lacking completeness and distributivity. Every component carries a declared semantic type, provenance, gate scope, and aggregation rule.

Definition 7.15 (Metric-bearing ledger components). The *metric-bearing ledger* is the declared element

$$\begin{aligned} \mathbf{d}_{\text{met}} = & \delta^{\text{alg}} \oplus \delta^{\text{disc}} \oplus \delta^{\text{meas}} \oplus \delta^{\text{rep,met}} \oplus \delta^{\text{real,met}} \oplus \delta^{\text{cmp}} \\ & \oplus \delta^{\text{crop}} \oplus \delta^{\text{cont}} \oplus \delta^{\text{ov}} \oplus \delta^{\text{rst}}, \end{aligned}$$

with only the components actually consumed by the audited metric comparison included. The terms represent implementation, discretization, measurement, comparison, crop, continuation, overlap, and Restart discrepancies, together with metric errors caused by representative choice or realization. A representative-related or realization-related term is included only when an actual comparison theorem supplies a quantitative bound. Existence, typing, and compatibility remain non-numerical obligations.

Definition 7.16 (Bottleneck scalarization). A bottleneck scalarization is a monotone subadditive map

$$\text{val}_B : V \longrightarrow [0, +\infty]$$

satisfying

$$\text{val}_B(0_V) = 0, \quad \text{val}_B(q \oplus q') \leq \text{val}_B(q) + \text{val}_B(q').$$

The scalarized metric budget is

$$\varepsilon_B(\mathbf{d}_{\text{met}}) := \text{val}_B(\mathbf{d}_{\text{met}}).$$

A metric gate requires this value to be finite and to be proved to dominate the actual bottleneck discrepancy consumed by that gate. A scalarization value without its metric upper-bound theorem is only a recorded number, not a certificate.

Remark 7.17 (Sharp threshold constant versus conservative scalarization). The adjective *sharp* refers to the optimal two-sided exclusion band $[\tau - \varepsilon, \tau + \varepsilon]$ for a valid bottleneck upper bound ε . It does not assert that a chosen scalarization is numerically tight. A conservative scalarization preserves soundness but may reduce the set of passing runs.

Definition 7.18 (Non-scalarizable proof obligations). The following are *non-scalarizable proof obligations*:

- (i) existence, typing, and equivalence strength of a required representative;
- (ii) eligibility, amplitude, edge-realization, and realization-equivalence certificates;

- (iii) existence and typing of an actual overlap, continuation, or tower comparison morphism;
- (iv) source semantics, terminal tameness, defect profiles, and interior-defect status;
- (v) threshold reconciliation, exact comparison, simplex coherence, continuation artifacts, and κ -admissibility certificates;
- (vi) detector stage applicability, soundness, completeness, birth/germ exhaustivity, stencil coverage, excess-bound source, endpoint convention, and family membership;
- (vii) naturality and zero reflection of a higher edge, representative and realization-change compatibility, and the non-synthetic classification of any claimed Bridge;
- (viii) local pass, target pass, retained-bar coverage, completeness of a countable registry, higher coherence, finite-to-infinite reduction, effective object descent, and Return-Theorem evidence;
- (ix) claim status, invocation role, supersession, repair records, and version boundaries;
- (x) manifest completeness, consistency, artifact closure, and identity obligations.

No increase in a numerical budget discharges an item in this list.

Proposition 7.19 (No scalar repair of missing evidence). *Let a required non-scalarizable proof obligation be missing or ill-typed. Then no value of $\mathfrak{d}_{\text{met}}$, no scalarized tolerance, and no positive threshold clearance can convert the corresponding clause into a certified pass.*

Proof. A metric ledger bounds a declared discrepancy. It does not construct the missing mathematical or governance object. Hence the obligation remains undefined, or a contradictory governance declaration remains rejected, independently of numerical slack. $\square$

Definition 7.20 (Gate-scoped threshold reserve). Let $\mathfrak{B} = \{B_j\}_{j \in J}$ be the finite family of pre-threshold profiles consumed by one comparison in `metric_gap_certified` mode. For finite $\varepsilon_B = \text{val}_B(\mathfrak{d}_{\text{met}})$, define

$$\text{gap}_\tau(\mathfrak{B}) := \min_{j \in J} \text{clear}_\tau(B_j)$$

and

$$\text{res}_\tau(\mathfrak{B}; \mathfrak{d}_{\text{met}}) := \text{gap}_\tau(\mathfrak{B}) - \varepsilon_B.$$

The metric clause has positive reserve precisely when

$$\varepsilon_B < \text{gap}_\tau(\mathfrak{B}).$$

If the scalarized budget is infinite, the metric clause does not pass. An exact comparison uses its exact proof artifact and receives no positive-radius robustness certificate merely from exactness.

Remark 7.21 (Meaning of $\Sigma\delta < \text{gap}_\tau$). The legacy notation $\Sigma\delta < \text{gap}_\tau$ denotes only the typed real inequality of Definition 7.20. An unscalarized ledger element is never compared directly with real clearance. A budget belongs to its declared gate and profile family and is not silently spent again elsewhere.

Definition 7.22 (Ledger completeness). A ledger is *complete and non-duplicative* for a metric clause if:

- (i) every metric perturbation affecting the clause is represented exactly once or by a documented joint bound;
- (ii) no perturbation is omitted merely because it is difficult to estimate;

- (iii) no component is counted twice under different names;
- (iv) every scalarized value is traceable to a proof or immutable artifact;
- (v) the aggregation law is the one declared by the run;
- (vi) the resulting real value is proved to bound the actual consumed discrepancy.

This definition is not invoked for a purely exact comparison unless a separate metric robustness claim is also made.

7.5 Claim-status and artifact preflight

Definition 7.23 (Claim-use status). Every registered item consumed by a run has exactly one status:

proved, conditional, operational, spec,
deprecated, rejected, unknown, superseded.

The statuses distinguish theorem-level evidence, conditional evidence, procedural rules, specifications, historical claims, unavailable claims, superseded claims, and unclassified items.

Definition 7.24 (Claim invocation role). Every registered item consumed by a run is also assigned exactly one role:

theorem_evidence, operational_rule,
specification_target, provenance_only.

The admissible combinations are:

- (i) `theorem_evidence`: status `proved`, or status `conditional` with every registered condition discharged;
- (ii) `operational_rule`: status `operational`, used only to govern a typed procedure;
- (iii) `specification_target`: status `spec`, used only as a schema, target, or acceptance criterion and never as proof of attainment;
- (iv) `provenance_only`: a `deprecated`, `rejected`, `unknown`, or `superseded` item cited only for history and contributing no passing theorem clause.

The role is fixed before verdict computation and may not be selected after seeing the desired result.

Definition 7.25 (Audit preflight). The *audit preflight* passes exactly when:

- (i) the run identity and immutable scope are complete;
- (ii) every consumed registered item has both a status and an invocation role;
- (iii) every status–role combination satisfies Definition 7.24;
- (iv) no `operational`, `spec`, `deprecated`, `rejected`, or `unknown` item is promoted to theorem evidence;
- (v) every relevant v19-to-v20 or earlier repair and every conditional claim supersession has an immutable ancestry record;
- (vi) every token inhabits the field’s declared sort, every detector is applied at a theorem-authorized stage, and every derived interface branch cites its component theorems;

- (vii) the clause registry and dependency graph are complete and audit-well-founded for the invoked finite or countable scope;
- (viii) every required artifact is present, typed, versioned, content-addressed, and linked to the run;
- (ix) no identifier, hash, version, role, status, stage, branch, or token sort makes incompatible assertions about the same input.

Definition 7.26 (Typed preflight verdict). The preflight has exactly one status:

`pass, reject, undefined.`

It is `pass` when Definition 7.25 is satisfied. It is `reject` when a well-typed record evidences forbidden theorem use, incompatible version or hash claims, an unauthorized scope exclusion, or another definite governance contradiction. It is `undefined` when a required identity, role, status, repair record, or artifact is missing, unresolved, or ill-typed but no evidenced contradiction yet forces rejection.

Remark 7.27 (Claim registration is not proof). A Claim Register records permitted strength and use. It does not prove an item by listing it. A proved entry points to an actual theorem and proof location; a `conditional` entry records all conditions; operational and specification entries retain their non-theorem roles.

Proposition 7.28 (Preflight failure blocks pass). *If the preflight status is not `pass`, the run cannot receive a Core pass verdict.*

Proof. A nonpassing preflight means that admissible theorem strength, scope, version, or artifact identity is either unresolved or contradicted. In neither case can every required Core clause be certified. $\square$

7.6 Typed atomic clause statuses

Definition 7.29 (Atomic audit status). Every atomic clause in the declared audit scope has exactly one status:

`pass, reject, undefined, not_invoked.`

They mean:

- (i) `pass`: the clause is well-typed, fully evidenced, and true;
- (ii) `reject`: the clause is well-typed and evidenced but false or obstructed, or a governed consistency clause is contradicted by incompatible declared evidence;
- (iii) `undefined`: the clause is required but incomplete, ill-typed, unresolved, or not computable from the declared evidence, with no evidenced contradiction yet forcing rejection;
- (iv) `not_invoked`: the governing immutable scope explicitly excludes the clause.

Remark 7.30 (Not invoked requires an explicit scope decision). A missing clause is not automatically `not_invoked`. The scope policy must positively declare that the clause is irrelevant to the claim; otherwise it is `undefined`.

Definition 7.31 (Canonical specialization maps). Specialized records embed into the atomic status algebra without creating a new algebra.

(i) Gap status maps by

`gap_certified` $\mapsto$ pass, `gap_failed` $\mapsto$ reject, `gap_pending` $\mapsto$ undefined.

(ii) Detector certificate status maps by

`detector_certified` $\mapsto$ pass,
`detector_failed` $\mapsto$ reject,
`detector_pending` $\mapsto$ undefined.

The independent detector value is consumed only after certificate pass. For a required zero predicate, certified zero passes, certified nonzero rejects, and value undefined is undefined. The interpretation is reversed only when the registered target predicate explicitly requires nonzero. An uncertified zero-looking output never passes.

(iii) Type IV status maps by

`certified_zero` $\mapsto$ pass, `obstructed` $\mapsto$ reject, `undefined` $\mapsto$ undefined.

The token `not_invoked` remains available only under an explicit no-tower scope decision.

(iv) For an invoked absence-of-long-transient clause, the map is

`certified_absent` $\mapsto$ pass,
`certified_present` $\mapsto$ reject,
`undefined`, `not_checked` $\mapsto$ undefined.

A justified out-of-scope record remains `not_invoked`. This field is independent of terminal Type IV status.

(v) Chapter 3 representative status maps by

`certified` $\mapsto$ pass.

Missing or ill-typed representative data map to undefined; an explicit scope exclusion remains `not_invoked`.

(vi) Chapter 4 degreewise Ext-clause statuses map identically to the atomic statuses with the same names. A one-way discharge passes only the invoked Ext-vanishing clause; a zero-reflection equivalence additionally requires its naturality and faithfulness package.

(vii) Chapter 6 Restart-transition status maps by

`failed` $\mapsto$ reject, `undefined` $\mapsto$ undefined.

Transition `certified` passes only the recurrence subclause, not the complete Restart step.

(viii) Manifest status maps by

`complete` $\mapsto$ pass, `incomplete` $\mapsto$ undefined, `inconsistent` $\mapsto$ reject.

- (ix) Runtime comparison modes, simplex-coherence modes, profile stages, cover types, source-semantics tokens, terminal modes, detector components, detector mathematical values, and `metric_zero_transport` are not statuses. Their own typed validation produces separate atomic clauses.

Definition 7.32 (Aggregate audit verdict). Let C_{all} be the finite or countable audit-indexed registry of potential clauses in one immutable scope, and let $C_{\text{req}} \subseteq C_{\text{all}}$ be the invoked clauses. Every clause outside C_{req} carries a justified `not_invoked` status. For a countable scope, the manifest certifies that the registry and its dependency closure are complete and audit-readable; a finite prefix cannot support a pass. The aggregate headline verdict is:

- (i) `reject` if at least one invoked clause is `reject`;
- (ii) `undefined` if no invoked clause is `reject` and at least one is `undefined`, or if a purported `not_invoked` clause lacks valid scope justification;
- (iii) `pass` if every invoked clause is `pass` and every excluded clause is justified.

The record retains all rejected, undefined, and non-invoked clause identifiers in addition to the headline.

Proposition 7.33 (Determinacy of complete records). *If every required clause is well-typed and evidenced and the scope registry is complete, then the aggregate verdict is either `pass` or `reject`; it is not `undefined`.*

Proof. Every required clause then has a truth-valued status `pass` or `reject`; apply Definition 7.32. $\square$

Proposition 7.34 (Undefined is neither success nor counterexample). *An `undefined` verdict is not a Core pass and does not, by itself, constitute a mathematical counterexample to the intended claim.*

Proof. Undefinedness records missing, ill-typed, unresolved, or unavailable evidence. It establishes neither the clause nor the negation of a well-typed mathematical statement. $\square$

Definition 7.35 (Comparison-mode audit record). Every consumed profile comparison declares exactly one mode:

`exact`, `metric_gap_certified`, `not_compared`.

The modes mean:

- (i) `exact`: an actual equality or persistence-profile isomorphism, together with every exact crop, restriction, source, target, and proof artifact used by the inference, is present;
- (ii) `metric_gap_certified`: an evidenced pre-threshold bottleneck upper bound, a complete scalarized metric ledger, and a two-sided threshold-gap certificate at the same budget are present;
- (iii) `not_compared`: the governing claim consumes no comparison inference between the profiles.

Exact mode proves only the declared exact relation and does not itself give positive-radius robustness. The mode `not_compared` cannot discharge a comparison, Overlap, Restart, or coherence obligation that is actually invoked.

7.7 Detector, higher-obstruction, defect, and globalization verdicts

Definition 7.36 (Detector-clause audit verdict). An invoked detector clause is pass exactly when the detector record has the correct source stage and scope, every soundness and completeness obligation passes, the certificate status is `detector_certified`, the value is defined, all transport branches pass, and the certified value satisfies the registered target predicate. It is `reject` when a well-typed required certificate condition is proved false or a certified value contradicts the target predicate. It is `undefined` when stage applicability, completeness, coverage, endpoint data, transport, value, or artifacts are missing, stale, circular, or ill-typed. It is `not_invoked` only by explicit scope exclusion.

Definition 7.37 (Degreewise higher-obstruction audit family). For every invoked $n \in I_{\text{mon}}$, the run records separate clauses for the repaired degree- n profile, admissible representative, n -eligibility, realized object and amplitude policy, edge extractor, edge-realization isomorphism, one-way Ext discharge, and any natural zero-reflection upgrade. The one-way conclusion

$$\text{Eval}_{n,W,\tau}(F) = 0 \implies \text{Ext}^n(A_n(F), k) = 0$$

may pass without a converse. A vanishing equivalence passes only when the registered natural zero-reflection package passes. An encoded synthetic object is classified as supporting detector duality and cannot discharge a principal natural-Bridge clause.

Definition 7.38 (Terminal and transient defect audit). An invoked tower record carries independent atomic clauses for source-semantic authority, actual comparison morphism, ambient abelian profile category, terminal tameness, terminal Type IV status, detector component, defect threshold τ_{def} , and long-transient status. Terminal `certified_zero` does not discharge transient absence, and transient `certified_absent` does not prove full kernel/cokernel vanishing. A full-isomorphism clause additionally requires direct full-defect vanishing or a valid short-defect-exclusion upgrade.

Definition 7.39 (Global detector-atlas verdict). A detector atlas has three separately audited conclusion levels:

- (i) *record coherence*: local vertices and every required simplex record pass at their declared modes;
- (ii) *global zero predicate*: a declared global profile interface, exact local restrictions, complete local detectors, exact or `metric_zero_transport` zero branches, and a global retained-bar coverage theorem pass;
- (iii) *object-level descent*: an external effective-descent theorem, actual transition isomorphisms and cocycle, and a descended-object artifact pass.

Passing a lower level does not impute either higher level.

Definition 7.40 (Finite-to-infinite reduction verdict). An invoked finite-to-infinite clause passes only when the manifest contains the actual directed-system morphisms, declared colimit/source semantics, the registered reduction theorem, its finite witness, a compatible apex cocone, the induced comparison morphism, apex agreement, and dependency-closed artifacts. A finite prefix, sampled stabilization, or eventually constant appearance without the reduction theorem is `undefined`, not pass. A well-typed counterexample or failed registered hypothesis is `reject`.

Definition 7.41 (Object-descent and Return-Theorem verdicts). Object descent and external Return are distinct interface clauses. Descent requires the category, restriction functors, actual local objects, exact transition data, effective-descent theorem, and descended-object artifact. Return requires a proved theorem from the exact AK-side conclusion to the stated external conclusion. Record coherence, global detector zero, finite-to-infinite reduction, and Core pass do not by themselves discharge either clause.

Proposition 7.42 (Separation of audit conclusion levels). *None of the following implications is available without the named additional clause:*

$$\begin{aligned} \text{record coherence} &\Rightarrow \text{global zero}, \\ \text{global zero} &\Rightarrow \text{object descent}, \\ \text{Core pass} &\Rightarrow \text{external theorem}. \end{aligned}$$

The required additional clauses are respectively retained-bar coverage, effective descent, and a Return Theorem.

Proof. Chapters 6 and 1 explicitly type the three conclusions differently and supply counterexamples or non-claims to the omitted implications. Chapter 8 governs the external return boundary. $\square$

7.8 Audit reading of B-Gate⁺

Definition 7.43 (B-Gate⁺ audit clause set). For a declared object, window, threshold, monitored degree family, and claim scope, the B-Gate⁺ registry contains every applicable clause below.

- (i) **preflight and identity:** run identity, token sorts, claim use, dependency graph, and artifacts pass;
- (ii) **evaluation typing:** every degree uses the fixed window-first repaired evaluation;
- (iii) **degreewise topological clauses:** each invoked $\text{Eval}_{n,W,\tau}(F) = 0$ is established directly or by an admitted complete detector;
- (iv) **degreewise categorical clauses:** each invoked higher edge has complete n -eligibility and the required Ext^n -vanishing; a converse or vanishing equivalence is consumed only through natural zero reflection;
- (v) **detector clauses:** stage, endpoint, soundness, completeness, coverage, certificate/value axes, metric transport, and artifact identity pass whenever detector evidence is consumed;
- (vi) **comparison clauses:** every runtime comparison has exactly one canonical mode; exact mode carries exact evidence, metric mode carries the complete two-sided chain, and `not_compared` is used only outside comparison scope;
- (vii) **metric budget clauses:** every metric ledger is complete, nonduplicative, scalarized, proved to dominate the actual discrepancy, and has the strict reserve required by its theorem;
- (viii) **terminal and transient clauses:** the terminal Type IV clause has a passing specialized record. When tower semantics are invoked, its status is `certified_zero`; otherwise `not_invoked` requires explicit scope exclusion. Every required long-transient clause is `certified_absent`, and any full-isomorphism claim carries direct full-defect or short-defect-exclusion evidence;
- (ix) **representative and realization clauses:** all actual profile isomorphisms, equivalence strengths, amplitude policies, edge certificates, and change-of-representative compatibilities pass;
- (x) **overlap and atlas clauses:** pairwise runtime comparisons pass, and every invoked simplex uses one declared coherence mode,

$$\text{exact_cocycle} \quad \text{or} \quad \text{metric_reference}.$$

Face-to-simplex compatibility, detector overlap certificates, and all countable-registry conditions also pass;

- (xi) **global detector clause:** when global zero is asserted, the global profile interface and retained-bar coverage theorem pass;
- (xii) **Restart clause:** every transition recurrence, target reserve, target local record, target-zero route, boundary-bridge coverage when required, and non-reserve obligation passes;
- (xiii) **finite-to-infinite, descent, and Return clauses:** each is included only when the governing claim uses that conclusion and must pass at its own type;
- (xiv) **manifest and failure routing:** the canonical manifest is complete, consistent, artifact-closed, semantically replayable, and every non-pass atomic clause has all applicable Type I–V routes.

Remark 7.44 (One-way bridge discharge only). Under a complete eligible realization record, the degree-one theorem 4.36 and its registered higher-degree successors may discharge the corresponding one-way categorical clauses from repaired vanishing. They do not discharge detector completeness, metric, terminal or transient defect, overlap, global coverage, Restart, reduction, descent, Return, manifest, or claim-status clauses. A converse requires the separately audited natural zero-reflection package.

Definition 7.45 (Core audit pass). A run receives a *Core audit pass* exactly when the aggregate verdict of its required B-Gate⁺ clause set is pass.

Definition 7.46 (Core audit reject). A run receives a *Core audit reject* exactly when the aggregate verdict is reject. The record identifies every evidenced failed clause and its failure route.

Definition 7.47 (Core audit undefined). A run receives a *Core audit undefined* verdict exactly when no evidenced clause forces rejection but at least one required clause remains undefined. The record identifies the missing, ill-typed, unresolved, or unavailable evidence.

Proposition 7.48 (Soundness of the pass reading). *A Core audit pass implies, for every invoked clause, the exact structural, metric, artifact, scope, and manifest conditions listed in Definition 7.43.*

Proof. Every required atomic clause has status pass; unpack the definitions. □

Proposition 7.49 (No pass by cancellation). *A passing clause cannot cancel a rejected or undefined clause. In particular, large clearance, a zero Ext group, or a zero Type IV diagnostic cannot repair a missing representative, failed exact comparison, failed metric reserve, undefined Restart transition, or forbidden claim use.*

Proof. The aggregate verdict is a typed conjunction, not a signed numerical sum. □

7.9 Local, overlap, Restart, and atlas audit composition

Definition 7.50 (Local certificate verdict). A local window certificate inherits pass, reject, or undefined from its local B-Gate⁺ audit. Only a local pass may serve as a successful atlas vertex or Restart endpoint.

Definition 7.51 (Compatibility verdict). Pairwise overlap comparison and higher-simplex coherence are audited at different token sorts.

- (i) A pairwise record passes in runtime mode *exact* or *metric_gap_certified* only when the directly evaluated common-window profiles, threshold reconciliation, applicable exact or metric evidence, detector overlap-scope certificate when consumed, face provenance, and terminal/transient obligations pass.

- (ii) A higher-simplex record passes in `exact_cocycle` mode only with actual profile isomorphisms, inverse/identity/cocycle equations, and face-to-simplex compatibility artifacts.
- (iii) It passes in `metric_reference` mode only with one immutable reference profile, all local-to-reference metric-gap certificates, face-to-simplex compatibility, and the declared simplex exposure. This mode produces no transition morphism or cocycle.
- (iv) `metric_zero_transport` may pass only as a derived branch after the runtime metric comparison and repaired-zero separation clauses pass.
- (v) A well-typed false exact relation, metric bound, reserve, cocycle, reference comparison, face condition, detector condition, or defect condition is `reject`; missing or ill-typed data are `undefined`; `not_invoked` requires explicit scope exclusion.

Definition 7.52 (Restart verdict). A complete Restart step receives:

- (i) `pass` exactly when the Chapter 6 transition status is `certified`, the target reserve is positive, the target local record passes, and every non-reserve artifact is present;
- (ii) `reject` when the declared κ_k -inequality fails, the defined target reserve is nonpositive, or the target local record rejects;
- (iii) `undefined` when a continuation artifact, finite working reserve, κ_k , η_k , scalarized budget, endpoint certificate, target record, or other required input is missing or ill-typed;
- (iv) `not_invoked` only when sequential inheritance is explicitly outside scope.

Transition status `certified` is necessary but not sufficient for a complete Restart pass.

Proposition 7.53 (Atlas composition rule). *A finite or countable certificate atlas is auditable as record-coherent only if every local vertex passes, every required nerve simplex is present, each simplex declares `exact_cocycle` or `metric_reference`, all consumed pairwise runtime comparisons pass, face-to-simplex compatibility passes, the declared Čech depth and audit-local-finiteness conditions pass, metric-reference exposure summability passes when invoked, all terminal/transient and Restart clauses pass, and the manifest is complete and artifact-closed. A rejected item rejects the atlas; if none rejects but one is undefined, the atlas is undefined.*

A global detector-zero verdict is added only if Definition 7.39(ii) passes. An object-level descent verdict is added only if item (iii) passes.

Proof. Apply the aggregate verdict rule to the Chapter 6 atlas clause registry. □

Remark 7.54 (Atlas pass is not a global-object pass). A coherent atlas certifies compatibility of local records. It does not create a single global filtered object, persistence module, or global B-Gate⁺ pass. Object-level descent and global audit require separate theorems and records.

Remark 7.55 (Pointwise reserve is not uniform reserve). Countable Restart may give $r_n > 0$ at every finite stage while $r_n \rightarrow 0$ when $K_n = \prod_{j < n} \kappa_j \rightarrow 0$. A claim $\inf_n r_n > 0$ requires its own lower-bound hypothesis and artifact.

7.10 Manifest completeness, consistency, and replayability

Remark 7.56 (Canonical manifest integration). The earlier manifest definitions form an extension chain rather than competing schemas: Chapter 1 supplies the base; Chapters 3–6 add evaluation, representative, eligibility, Type IV, overlap, Restart, and atlas fields. Definition 7.58 is their canonical semantic superset for v20.0. A chapter-local fragment may add fields, but it may not change shared meanings or final status semantics.

Definition 7.57 (Manifest status). A manifest has exactly one status:

complete, incomplete, inconsistent.

Here `complete` means every required field and artifact is present, typed, dependency-closed, and mutually consistent; `incomplete` means at least one required field or artifact is absent or unresolved; and `inconsistent` means declared fields, versions, hashes, statuses, roles, or artifacts make incompatible assertions about the same run.

Definition 7.58 (Canonical minimal v20 audit manifest). The canonical minimal v20 audit manifest contains, at minimum:

- (i) ManifestID, RunID, theory, SchemaID, claim-register versions, immutable scope, repair and supersession ancestry;
- (ii) the canonical token-sort registry and every field’s declared sort;
- (iii) source-object identities, content hashes, artifact root, complete finite or countable clause registry, evidence graph, dependency closure, and well-foundedness witness;
- (iv) monitored degrees, finite I_{mon} , windows, cover type, terminal modes, collapse and defect thresholds, evaluation order, hard deletion, infinite-bar and endpoint conventions;
- (v) every pre-threshold and repaired-profile identity and every detector source stage;
- (vi) every detector ID, theorem, information class, mode, sample/birth/germ set, endpoint convention, soundness, completeness, coverage/mesh/excess source, certificate size, two-axis status, value, and replay artifact;
- (vii) every runtime comparison mode; in exact mode the actual relation and crop/restriction artifacts; in metric mode the protected profile family, ledger algebra, scalarization, discrepancy theorem, two-sided gap record, zero-separation hypotheses when invoked, and strict reserve;
- (viii) representative, realization, degreewise n -eligibility, amplitude, edge extractor, edge-realization, naturality, zero-reflection, and records governing changes of representative and realization, together with the synthetic or principal Bridge classification;
- (ix) Type IV scope, tower system, source-semantics requirement and authority, ambient abelian category, actual comparison morphism, terminal tameness, degreewise and total μ , ν , and terminal status;
- (x) full/cofinal/interior defect profile identities, detector component, τ_{def} , transient detector record, long-transient status, and any short-defect-exclusion or direct full-defect proof;

- (xi) cover, local, overlap, threshold-reconciliation, and simplex records, with coherence mode recorded separately as

`exact_cocycle` or `metric_reference`;

the entry also contains the actual cocycle data or immutable reference profiles, face-to-simplex directions, Čech depth, audit-local finiteness, and exposure sums;

- (xii) when global detector zero is invoked: global profile identity, exact local restrictions, independent global coverage witnesses, local detector samples, zero-transport branches, retained-bar coverage theorem, and local/global verdicts;
- (xiii) every continuation artifact, working gap, κ_k , η_k , K_n , normalized loss, pointwise and claimed uniform reserves, target-zero route, bridge-window coverage, transition and complete Restart verdicts;
- (xiv) every invoked finite-to-infinite reduction theorem, actual directed system, finite witness, colimit, compatible apex cocone, induced comparison, apex agreement, and reduction verdict;
- (xv) every invoked object-descent category, restriction functor, actual local objects and transition isomorphisms, effective-descent theorem, descended object, and every external Return-Theorem statement and proof;
- (xvi) all atomic statuses, specialized-to-atomic maps, headline verdicts, Type I–V routes, missing-artifact markers, and justified scope exclusions;
- (xvii) every ClaimID, document role, evidence status, invocation role, theorem/policy locator, discharged condition, permitted use, forbidden stronger reading, supersession and repair record;
- (xviii) release-acceptance clause registry when the document set is advertised as a complete v20 release;
- (xix) provenance and semantic-replay data sufficient to reconstruct every deterministic computation and every logical verdict transition.

Definition 7.59 (Artifact closure). A canonical manifest has *artifact closure* when:

- (i) every referenced artifact has a content hash, type declaration, provenance, and immutable identifier;
- (ii) every transitive dependency needed to interpret or verify a consumed artifact is referenced or belongs to the registered theory release;
- (iii) no identifier is dangling, ambiguous, mutable under the same hash claim, or resolved differently by two fragments;
- (iv) the dependency graph contains no undeclared artifact replacement after the run identity was fixed.

Artifact closure is required for replayability but is not itself mathematical proof.

Definition 7.60 (Semantic schema-migration record). A migration from one manifest schema to another records source and target SchemaID, a total map on every consumed field, token-sort preservation, status and scope preservation, explicit defaults only for newly out-of-scope fields, semantic-diff evidence, artifact hashes, and a proof that verdict reconstruction at the old supported strength is preserved. Changed theorem strength, scope, or evidence creates a new run rather than a schema-only migration.

Proposition 7.61 (No lossy migration of theorem evidence). *A migration that drops a required hypothesis, dependency, token sort, missing-artifact marker, lower-level status, or forbidden stronger reading cannot preserve audit equivalence. The migrated record is *incomplete* or *inconsistent* until the loss is repaired by a new evidence-backed run.*

Proof. The omitted field participates in clause meaning or verdict reconstruction. Deleting it prevents reconstruction or changes the consumed claim. $\square$

Proposition 7.62 (Canonical manifest conformance). *A chapter-local manifest fragment is audit-admissible only if it embeds into the canonical manifest without changing the meaning of a shared field, status, role, identifier, comparison mode, or version boundary. Incompatible assignments make the canonical manifest *inconsistent*.*

Proof. The canonical manifest describes one immutable run; incompatible shared assignments violate that condition. $\square$

Definition 7.63 (Computational and semantic replayability). A manifest is *computationally replayable* if artifact closure lets an independent verifier reconstruct every deterministic profile, detector, ledger, diagnostic, dependency, and serialization computation. It is *semantically replayable* if the verifier can additionally reconstruct each clause’s source type, governing claim, hypotheses, token sort, specialization map, dependency transition, and headline aggregation. Core pass requires semantic replayability for every consumed machine-produced verdict.

Proposition 7.64 (Replay invariance under fixed inputs). *Two semantic replays using the same immutable artifacts, token registry, claim-register snapshot, schema semantics, and verifier version yield the same atomic and headline verdicts. A disagreement evidences an implementation, schema, artifact, or manifest inconsistency and blocks pass.*

Proof. Under fixed inputs every clause and aggregation operation is deterministic by Proposition 7.8. $\square$

Remark 7.65 (Replayability is not mathematical proof by itself). Replayability establishes reproducibility of the declared computation and record graph. It does not validate a false theorem, insufficient hypothesis, or semantically inadequate artifact.

Proposition 7.66 (Manifest effect on the headline verdict). *A manifest with status *incomplete* makes the manifest clause *undefined* unless another evidenced clause already forces rejection. A manifest with status *inconsistent* forces rejection as a Type V governance failure. Only a *complete*, artifact-closed manifest can support a Core pass.*

Proof. Incomplete data leave required evidence unresolved; inconsistent data contradict the single-run semantics. $\square$

7.11 Failure routing and preservation of diagnostic detail

Definition 7.67 (Audit failure-routing record). Every non-pass run records all applicable, nonexclusive failure types:

- (i) **Type I:** nonzero repaired topological profile;
- (ii) **Type II:** nonzero eligible categorical obstruction;

- (iii) **Type III:** any failed or undefined mathematical or operational obligation, including metric or crop control; detector stage, applicability, soundness, or completeness; overlap or simplex coherence; global coverage; exact comparison; continuation or Restart; representative or realization data; comparison morphisms or terminal tameness; transient defects; finite-to-infinite reduction; object descent; or a Return theorem;
- (iv) **Type IV:** a valid, semantically authorized tower comparison artifact whose terminal-generic diagnostic satisfies $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) \neq (0, 0)$;
- (v) **Type V:** forbidden theorem use, incomplete or inconsistent canonical manifest, missing repair/governance record, suppressed missing-artifact marker, unsupported zero or pass declaration, misrepresented scope/version/hash/provenance, cross-sort token substitution, silent stage relabeling, lossy schema migration, silent claim supersession, unauthorized scope exclusion, or proposer output treated as proof.

Mere absence or ill-typing of mathematical evidence first yields an undefined Type III clause. Type V is added when the governance record is itself incomplete, inconsistent, concealed, or semantically false.

Remark 7.68 (Headline compression does not erase details). The precedence of `reject` over `undefined` is only a headline rule. If one clause rejects and another is undefined, the manifest retains both states and both failure routes.

Proposition 7.69 (Repair creates a new verdict record). *Resolving an undefined clause or correcting a rejected clause produces a new run and manifest identity, optionally linked to the former run by an explicit repair relation. The original immutable RunID and historical verdict are not reused or overwritten.*

Proof. This is the immutable-scope rule of Definition 7.12. □

7.12 Core sovereignty and proposer–verifier separation

Definition 7.70 (Core sovereignty). Only Core-typed evidence evaluated under the registered verifier discipline may determine a Core verdict. No external proposal mechanism may overwrite, waive, or reinterpret an atomic Core clause. A human maintainer may revise theory, schema, or scope only by creating a new version or run identity.

Definition 7.71 (Proposer and verifier). A *proposer* is any human, algorithmic, Search-side, Platform-side, Companion-side, or AI mechanism that produces candidate objects, windows, thresholds, representatives, morphisms, claims, bridge programs, rankings, maps, or proof sketches. The *verifier* is the typed Core audit discipline of this chapter and the mathematical chapters it consumes.

Definition 7.72 (Re-entry protocol). A proposer output may influence a Core verdict only after:

- (i) normalization into a declared Core type;
- (ii) claim-status and invocation-role classification;
- (iii) production of required mathematical and artifact evidence;
- (iv) comparison-mode, ledger, and threshold-gap typing where applicable;
- (v) when the output is transformed, translated, summarized, formalized, or migrated, a semantic-diff or transformation-preservation certificate;
- (vi) manifest registration, token-sort validation, dependency closure, artifact closure, and immutable identity;

(vii) execution of every relevant verifier clause.

Proposition 7.73 (Search and AI outputs do not update verdicts). *A Search score, potential, terrain map, ranking, validity-map value, AI confidence, proof sketch, or agent recommendation does not alter a Core verdict merely by being produced.*

Proof. Such an output is proposer-side until it completes the re-entry protocol and passes the verifier. $\square$

Proposition 7.74 (Verifier independence). *Replacing the proposer mechanism does not change the mathematical meaning of a fixed verifier verdict computed from the same immutable Core inputs and registered theory versions.*

Proof. The verdict depends on registered inputs and clause semantics, not on the identity of the proposer. $\square$

7.13 v20 release-acceptance audit

Definition 7.75 (v20 release-acceptance record). A document set advertised as an *AK-HPDST v20.0 complete release* carries a release-level registry whose required clauses are exactly the mandatory Core mathematics, verification/governance, and external-demonstration obligations of Chapter 1 Policy 3. Each obligation has a claim identity, exact source strength, dependencies, artifacts, and one atomic status. Release status is distinct from the verdict of any individual Core run.

Definition 7.76 (Release headline status). The release headline is *pass* only when every mandatory obligation passes and all cross-document and machine-readable records are mutually consistent. It is *reject* if a well-typed mandatory claim or consistency requirement is evidenced false. It is *undefined* when no mandatory clause rejects but at least one remains missing, pending, unregistered, or ill-typed. Optional recommendations do not affect the headline unless the release policy explicitly promotes them.

Proposition 7.77 (No release completion by local success). *No individual detector theorem, higher-edge theorem, transient-defect theorem, Core pass, toy bridge, known-theorem calibration, or favorable external example can compensate for another missing mandatory release obligation.*

Proof. The release registry is a typed conjunction of independently mandatory clauses, governed by the same non-compensation rule as a run verdict. $\square$

7.14 Audit package

Proposition 7.78 (Typed audit-and-sovereignty contract consequences). *The v20 Core audit discipline has the following consequences:*

- (i) *metric discrepancies are controlled by a declared ledger, scalarization, and actual upper-bound proof;*
- (ii) *non-scalar obligations cannot be bought off by numerical slack;*
- (iii) *metric reserve is gate-scoped and pre-threshold only in `metric_gap_certified` mode, while exact mode is governed by its exact proof artifact;*
- (iv) *every atomic clause has one typed status;*
- (v) *pass requires every invoked clause to pass and every exclusion to be justified;*
- (vi) *false evidence yields rejection, while unresolved evidence yields undefinedness unless another clause rejects;*

- (vii) *detector, gap, Type IV, transient, and complete Restart records embed into the same atomic status algebra through typed specialization;*
- (viii) *local, compatibility, global-zero, Restart, reduction, descent, Return, release, and external verdicts remain distinct;*
- (ix) *only a complete, consistent, artifact-closed and semantically replayable manifest can support pass;*
- (x) *failure routing preserves Types I–V and all clause-level detail;*
- (xi) *proposer outputs remain non-authoritative until typed re-entry;*
- (xii) *the theorem-bearing verifier remains sovereign.*

Proof. Items (i)–(iii) follow from Definitions 7.15, 7.16, 7.18, 7.20, and 7.35. Items (iv)–(vi) follow from Definitions 7.29 and 7.32. Items (vii)–(viii) follow from Definitions 7.31, 7.51, and 7.52, together with Proposition 7.53. Items (ix)–(x) follow from Definitions 7.57, 7.59, and 7.67. Items (xi)–(xii) follow from Definitions 7.70, 7.71, and 7.72. $\square$

Remark 7.79 (Contract consequence list, not a new external theorem). Proposition 7.78 is a closure list for the typed audit contracts of this chapter. It is not a new theorem about persistence modules, external interfaces, or a problem-specific domain.

7.15 Explicit non-claims

The following are not asserted in this chapter.

- (i) No unscaled ledger element is directly compared with clear_τ .
- (ii) No scalarized budget repairs a missing representative, morphism, realization, claim status, or manifest field.
- (iii) No equality $\varepsilon_B = \text{gap}_\tau$ is a metric pass.
- (iv) No budget on one clause is reused elsewhere without an explicit joint-bound theorem.
- (v) No exact comparison is required to fabricate a threshold-gap reserve, and no exact equality or profile isomorphism is promoted to positive-radius robustness without a separate metric certificate.
- (vi) No `not_compared` mode discharges a comparison, overlap, Restart, or coherence obligation.
- (vii) No passing clause cancels a rejected or undefined clause.
- (viii) No `undefined` verdict is reported as pass or as a mathematical counterexample.
- (ix) No missing Type IV artifact is interpreted as $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$.
- (x) No missing κ -certificate is interpreted as $\kappa = 1$ or zero loss.
- (xi) No pointwise-positive Restart reserve is interpreted as a uniform positive lower bound.
- (xii) No finite prefix of a countable audit scope is treated as a complete pass certificate.
- (xiii) No pairwise overlap pass alone proves higher Čech coherence.

- (xiv) No coherent certificate atlas creates an unrestricted global object or global B-Gate⁺ pass.
- (xv) No zero terminal-generic Type IV diagnostic implies full comparison-map isomorphism on interior bars.
- (xvi) No use of $u_{i,W,\tau}$ or u_{Collapse} is normative; the cokernel diagnostics are $v_{i,W,\tau}$ and v_{Collapse} .
- (xvii) No total μ/ν diagnostic is aggregated across different windows, thresholds, or infinitely many monitored degrees without a separately proved aggregation or summability theorem.
- (xviii) No deprecated global non-expansiveness claim for hard threshold deletion is used.
- (xix) No band of half-width 2ε is called sharp for the two-sided threshold theorem; the sharp half-width is ε .
- (xx) No Claim Register entry proves itself by registration.
- (xxi) No `operational`, `spec`, `deprecated`, `rejected`, or `unknown` item is theorem evidence; its admissible non-theorem role remains explicitly typed.
- (xxii) No Search, Platform, Companion, or AI artifact updates a Core verdict outside the re-entry protocol.
- (xxiii) No successor-release candidate or unpromoted NBT statement is treated as a v20.0 theorem.
- (xxiv) No audit pass is a problem-specific external theorem without a separately proved Interface or Return theorem.
- (xxv) No specialized detector, gap, transient, Restart, manifest, or release token creates an independent status algebra outside the canonical four atomic statuses.
- (xxvi) No runtime comparison mode is replaced by `exact_cocycle`, `metric_reference`, or `metric_zero_transport`.
- (xxvii) No detector value is consumed without a passing certificate-status axis, and no certificate pass determines the mathematical value by itself.
- (xxviii) No detector theorem is moved by relabeling among the stages

`windowed_prethreshold,   repaired_postthreshold,`
`kernel_defect,   cokernel_defect.`
- (xxix) No terminal Type IV zero is transient absence, and no transient certified absence is full comparison-map isomorphism.
- (xxx) No local detector atlas proves global zero without an independent retained-bar coverage theorem.
- (xxxi) No record-coherent atlas proves object descent, and no Core pass proves an external conclusion without a Return Theorem.
- (xxxii) No finite prefix, observed stabilization, or finite sample proves an infinite tower statement without a finite-to-infinite reduction theorem.
- (xxxiii) No schema migration, summary, translation, formalization, or serialization is presumed semantics-preserving without a semantic-diff or preservation record.

(xxxiv) No conditional successor claim silently erases its predecessor non-claim, hypotheses, or forbidden stronger readings.

(xxxv) No complete-release designation is inferred from completion of only the Main-body mathematical chapters.

7.16 Provenance and status

This chapter introduces no new barcode stability theorem, Ext computation, detector characterization, or tower invariant. Its v19 inheritance is the four-valued atomic audit calculus, immutable run identity, non-compensating ledger discipline, canonical manifest, replay and sovereignty rules. Its v20 extension is the sorted token registry and the canonical integration of finite detectors, higher obstruction edges, terminal/-transient defects, detector atlases, finite-to-infinite reductions, descent/Return boundaries, semantic schema migration, and release-acceptance governance.

Status labels for this chapter. *Core audit definitions:*

Definitions [7.3](#), [7.5](#), [7.6](#), [7.7](#), [7.9](#), [7.11](#), [7.12](#), [7.14](#), [7.15](#), [7.16](#), [7.18](#), [7.20](#), [7.22](#), [7.23](#), [7.24](#), [7.25](#), [7.26](#), [7.29](#), [7.31](#), [7.32](#), [7.35](#), [7.36](#), [7.37](#), [7.38](#), [7.39](#), [7.40](#), [7.41](#), [7.43](#), [7.45](#), [7.46](#), [7.47](#), [7.50](#), [7.51](#), [7.52](#), [7.57](#), [7.58](#), [7.59](#), [7.60](#), [7.63](#), [7.67](#), [7.70](#), [7.71](#), [7.72](#), [7.75](#), [7.76](#).

Logical and audit consequences:

Results [7.4](#), [7.8](#), [7.10](#), [7.19](#), [7.28](#), [7.33](#), [7.34](#), [7.42](#), [7.48](#), [7.49](#), [7.53](#), [7.61](#), [7.62](#), [7.64](#), [7.66](#), [7.69](#), [7.73](#), [7.74](#), [7.77](#), [7.78](#).

Examples and precision remarks:

Examples/Remarks [7.1](#), [7.2](#), [7.13](#), [7.17](#), [7.21](#), [7.27](#), [7.30](#), [7.44](#), [7.54](#), [7.55](#), [7.56](#), [7.65](#), [7.68](#), [7.79](#).

Explicit non-claims:

Section [7.15](#).

8 Chapter 8: Core Closure, Return Architecture, and the Final AK Projection Principle

8.1 Final role of the chapter

Chapter 8 contract. This chapter closes the theorem-bearing main body of AK-HPDST v20.0. It introduces no theorem about an external arithmetic, geometric, analytic, PDE, or cryptographic domain merely by closure. It does, however, place the v20 finite-certificate, higher-obstruction, defect, atlas, and finite-to-infinite results into one final inference architecture and proves the internal eventually-constant reduction theorem used by that architecture.

The chapter fixes:

- (i) the proved mathematical nucleus and its exact hypotheses;
- (ii) the conditional and operational layers;
- (iii) the rejected shortcuts and immutable non-claims;
- (iv) the canonical Problem-Interface entry, realization, reduction, and Return-Theorem obligations;
- (v) the separation between main-body closure and release-package acceptance.

Projection supplies a Core-readable shadow. Verification certifies the shadow. Return alone carries a certified statement back to the source domain.

The governing inference rule is

The Core proves only what follows from declared types, hypotheses, and artifacts. A finite detector may compress evidence but may not invent completeness. A finite-to-infinite theorem may transport a certified finite witness but may not be replaced by sampled stabilization. An external conclusion requires its own proved Return Theorem.

The final dependency order is

metric repair, gate contract, and release contract	(Chs. 1–2; App. A)
→ finite detectors, proof compression, and representatives	(Ch. 3)
→ higher Ext-edges and zero-reflection boundary	(Ch. 4)
→ terminal/transient defects and Type IV	(Ch. 5)
→ overlap, detector atlases, and κ -Restart	(Ch. 6)
→ typed audit, evidence graphs, and release acceptance	(Ch. 7)
→ Core closure, reduction, Interface, and Return.	(Ch. 8)

Remark 8.1 (Closure is not a new external theorem). The strongest new mathematical statement in this chapter is an internal categorical reduction theorem for an eventually constant directed system with compatible apex agreement. No problem-specific external conclusion is asserted without a separately proved source-domain Return Theorem.

Remark 8.2 (Claim registration follows proof, not conversely). The v20 Claim Register records status, conformance, comparison policy, and permitted use. Registration proves nothing and cannot increase claim strength. If a register entry conflicts with a theorem statement, proof, explicit hypothesis, stage, or non-claim, the safer weaker reading governs until a new version records and proves the repair.

Remark 8.3 (Main-body closure and release-package closure are distinct). This chapter closes the inference paths of the theorem-bearing main body. The designation *AK-HPDST v20.0 complete release* is controlled by the Chapter 1 release contract and the Chapter 7 release registry. It additionally requires every mandatory appendix, Claim Register, toy-verification, counterexample, realization, known-theorem-recovery, and external-demonstration obligation to pass. Main-body closure does not fabricate those artifacts.

8.2 Canonical status vocabulary for the release

Definition 8.4 (Release-level claim classes). Every statement consumed by the release has exactly one evidence class:

`proved, conditional, operational, spec, deprecated, rejected, unknown.`

A proved item is used only under its theorem hypotheses. A conditional item becomes theorem evidence only after every registered condition is discharged. Operational and specification items remain non-theorem material.

Definition 8.5 (Claim invocation role). Every consumed item has exactly one invocation role:

`theorem_evidence, operational_rule, specification_target, provenance_only.`

Evidence class, conformance status, claim-level comparison policy, and runtime comparison mode are different fields and may not be substituted for one another.

Definition 8.6 (Safe-use order). A statement may be used only at or below its proved and registered strength. In particular:

- (i) a conditional theorem instance is unavailable until all conditions and dependencies pass;
- (ii) an operational rule does not become external mathematics;
- (iii) a specification target is not evidence of attainment;
- (iv) a pending-v20-repair or legacy-v20-only item cannot be used as v20 theorem evidence;
- (v) a successor claim supersedes a predecessor only through the immutable relation required by Chapter 7;
- (vi) ambiguity is resolved by the weaker class, role, conformance, and conclusion strength.

Definition 8.7 (Release token-sort conformance). Every semantic token used by the release belongs to the Chapter 7 canonical sort registry. Atomic status, claim class, invocation role, runtime comparison mode, claim comparison policy, detector stage, detector mode, detector certificate status, detector value, Type IV status, transient status, cover type, simplex-coherence mode, Restart status, and manifest status remain distinct sorts. A field name, label string, identifier, or derived branch such as `metric_zero_transport` is not silently promoted to an atomic status or runtime comparison mode.

Proposition 8.8 (Status cannot be promoted by prose). *A stronger verbal description, diagram, implementation success, numerical experiment, detector score, Search report, or AI-generated explanation does not promote the class, role, conformance, theorem strength, or conclusion level of a statement.*

Proof. Promotion requires a proof, complete typed artifacts, a conforming immutable registration change, and the Chapter 7 verifier. None of the listed objects supplies these requirements by itself. $\square$

8.3 The proved mathematical nucleus of v20.0

Definition 8.9 (Proved Core nucleus). The *proved Core nucleus* is the collection of theorem-level mathematical statements established in the final-audited Chapters 1–6, Appendix A, and the internal reduction theorem of this chapter, together with their proved counterexamples and exact boundaries. A theorem with hypotheses belongs to the nucleus at that conditional logical strength; a run may instantiate it only after the hypotheses pass. Operational status calculi, manifests, schemas, and acceptance policies belong to the verifier layer rather than the external mathematical nucleus.

Proposition 8.10 (Contents of the proved Core nucleus). *The following statements belong to the proved Core nucleus at exactly the stated strength.*

- (i) *Constructible one-parameter persistence modules over the coefficient field admit the declared barcode decomposition in the working regime.*
- (ii) *The standard bottleneck metric uses endpoint sup-cost for finite bars, half-length diagonal cost, and finite matching of infinite bars only with infinite bars using birth displacement.*
- (iii) *Hard barcode threshold deletion deletes precisely finite bars of length at most τ , retains longer finite bars and infinite bars, and is endpoint-independent, idempotent, and threshold-monotone at barcode-profile level.*
- (iv) *Positive-threshold short bars do not form a Serre subcategory in general; hence no global exact positive-threshold Serre reflector, quotient formula, or abelian localization is available in the general Core.*
- (v) *Hard positive-threshold deletion is not globally non-expansive for the standard bottleneck metric.*
- (vi) *If $d_B(B, C) \leq \varepsilon$ and both pre-threshold barcodes avoid $[\tau - \varepsilon, \tau + \varepsilon]$, then*

$$d_B(T_\tau^{\text{bar}} B, T_\tau^{\text{bar}} C) \leq \varepsilon.$$

The half-width ε is sharp.

- (vii) *Threshold clearance is measured after windowing and before deletion, and a numerical ledger enters a metric inference only through a scalarization proved to bound the actual consumed discrepancy.*
- (viii) *The repaired evaluation order is*

$$B_{i,W}(F) = W_W \mathbf{P}_i(F), \quad \text{Eval}_{i,W,\tau}(F) = T_\tau^{\text{bar}}(B_{i,W}(F)).$$

Window restriction and hard deletion do not commute automatically.

- (ix) *Admissible representatives of the same repaired profile give the same persistence-level vanishing verdict. Derived or Ext invariance requires separately declared realization-equivalence data.*
- (x) *The finite stencil detector is sound. Exhaustive birth sets, decoration-invariant right germs, certified coverage, and admissible augmented family meshes give the corresponding zero-reflecting characterizations under their exact stage, endpoint, family, and artifact hypotheses.*
- (xi) *Strict mesh separation is sharp at the equality boundary; the family-relative certificate bounds and the exact augmented-mesh stencil count hold at their declared information class. No uniform complete finite detector exists on the right-unbounded birth family without an additional escape hypothesis.*

- (xii) *The metric zero-verdict composition theorem transports a zero predicate only through the complete two-sided threshold-gap and zero-separation chain. It does not assert stability of detector rank.*
- (xiii) *For every $n \geq 0$ and bounded complex A of finite-dimensional vector spaces,*

$$\mathrm{Ext}^n(A, k) \cong \mathrm{Hom}_k(H^{-n}(A), k)$$

naturally contravariantly in A .

- (xiv) *Under a complete standard n -eligible realization,*

$$\mathrm{Eval}_{n,W,\tau}(F) = 0 \implies \mathrm{Ext}^n(A_{n,F,W,\tau}, k) = 0.$$

A converse or equivalence requires an independently meaningful, natural, zero-reflecting edge package.

- (xv) *A complete fixed-stencil detector may supply such zero reflection on its declared class when the detector functor, completeness theorem, natural edge realization, representative compatibility, and all artifacts pass. Encoded synthetic detector duality alone is not the principal natural Bridge.*
- (xvi) *For terminal-tame profiles, tgdim_W counts bars cofinal in the declared window. Applied to an actual tower comparison morphism it defines $\mu_{i,W,\tau}$ and $\nu_{i,W,\tau}$. Aggregation is only over the fixed finite monitored degree set.*
- (xvii) *The terminal/interior decomposition, complete transient-defect characterization, and joint terminal/transient classification hold under their barcode and detector hypotheses. Terminal zero does not exclude finite noncofinal defects.*
- (xviii) *Terminal zero, complete interior-detector zero, and a proved short-defect exclusion imply full kernel/cokernel vanishing and hence comparison-map isomorphism. The short-residual-defect example proves the necessity of an additional short-defect hypothesis.*
- (xix) *Exact and metric-reference simplex coherence are distinct. Finite-depth records glue at record level. A global detector-zero conclusion additionally requires a declared global profile interface, complete local detectors, valid local zero transport, and retained-bar coverage. Local zero plus exact overlap coherence alone is insufficient.*
- (xx) *Certified recurrences $r_{k+1} \geq \kappa_k r_k - \eta_k$ imply the product-weighted finite bound and normalized reserve criterion. Ordinary summability need not preserve reserve under contraction, and pointwise positive reserve need not have a uniform positive infimum.*
- (xxi) *A countable record atlas requires audit-local finiteness, finite Čech depth, complete simplex records, face-to-simplex compatibility, typed exposure summability, and a complete dependency closure. It does not create object-level descent.*
- (xxii) *The eventually-constant finite-to-infinite theorem of Theorem 8.18 identifies the colimit with a stabilized finite stage and, under compatible apex agreement, with the apex.*

Proof. Items (i)–(ix) are the repaired v20 persistence nucleus. Items (x)–(xii) are the Chapter 3 finite-detector and metric-composition results. Items (xiii)–(xv) are the Chapter 4 higher Ext-edge and zero-reflection boundary. Items (xvi)–(xviii) are the Chapter 5 terminal/transient results. Items (xix)–(xxi) are the Chapter 6 atlas and Restart results. Item (xxii) is proved below. Every item is included with its original hypotheses and non-claim boundary. $\square$

Remark 8.11 (The nucleus is small by design). The nucleus is narrower than the project architecture. Its strength comes from finite certificates with explicit completeness, natural higher edges, separation of terminal and transient defects, exact local-to-global coverage, and refusal to cross an infinite or external boundary without a proved reduction or Return Theorem.

8.4 Conditional mathematics and operational contracts

Definition 8.12 (Conditional Core layer). The *conditional Core layer* consists of proved implications whose run instances require separately supplied hypotheses or artifacts. It includes:

- (i) crop–collapse commutation after the exact all-retained-bars test;
- (ii) finite-detector completeness after stage, endpoint, coverage, family, mesh, and noncircular excess-source obligations;
- (iii) metric zero transport after the two-sided threshold and zero-separation chain, with one parent MetricUseID and no duplicated budget;
- (iv) realization and representative invariance beyond profile level;
- (v) eligible higher Ext-edges and natural zero-reflection packages;
- (vi) actual tower systems, colimits, comparison morphisms, terminal tameness, and complete transient-defect records;
- (vii) exact-cocycle or metric-reference simplex records, global retained-bar coverage, and object descent only through their separate theorems;
- (viii) κ -admissible continuation, normalized reserve-safety, and complete countable dependency closure;
- (ix) finite-to-infinite reduction and external Return Theorems.

Definition 8.13 (Operational Core layer). The *operational Core layer* contains the binding contracts determining what counts as an auditable proof object or verdict. It includes fixed evaluation order; token sorts; claim classes, conformance, policies, and roles; comparison and simplex modes; detector stage and two-axis status; gate-scoped metric-use identity; representative, realization, tower, overlap, atlas, Restart, reduction, descent, and Return records; Type I–V routing; the four atomic statuses; immutable run identity; well-founded finite or countable evidence graphs; semantic replay; release acceptance; and Core sovereignty.

Definition 8.14 (Canonical comparison mode). Every consumed runtime profile comparison has exactly one mode:

exact, metric_gap_certified, not_compared.

The claim-level comparison policy and the simplex-coherence modes `exact_cocycle` and `metric_reference` are different sorts. The derived branch `metric_zero_transport` references one parent metric comparison and is neither a fourth comparison mode nor a second spending of its budget.

Proposition 8.15 (Conditional and operational layers do not self-promote). *A conditional statement without discharged hypotheses is not an available theorem instance, and an operational contract is not an external mathematical theorem.*

Proof. The first assertion is implication logic. The second follows because verifier contracts classify and check evidence but do not supply the missing source-domain semantics or transport theorem. $\square$

Remark 8.16 (Canonical manifest location). The Chapter 1 base manifest and the Chapters 3–6 extensions are conforming fragments of the canonical Chapter 7 v20 schema. An incomplete manifest gives undefined unless another evidenced clause rejects; an inconsistent manifest gives Type V reject.

8.5 Internal finite-to-infinite reduction

Definition 8.17 (Eventually constant reduction package). Let C be a category and

$$M_0 \xrightarrow{f_0} M_1 \xrightarrow{f_1} M_2 \longrightarrow \dots$$

a directed sequence with a declared colimit (L, λ_n) . An *eventually constant reduction package* consists of:

- (i) an index N and proofs that f_n is an isomorphism for every $n \geq N$;
- (ii) the actual directed-system morphisms and colimit cocone;
- (iii) when an apex A_∞ is used, a compatible cocone $a_n : M_n \rightarrow A_\infty$;
- (iv) an apex-agreement proof that a_N is an isomorphism;
- (v) the induced comparison morphism $\phi : L \rightarrow A_\infty$ satisfying $\phi \lambda_n = a_n$;
- (vi) immutable theorem, witness, and artifact identities.

A finite prefix or sampled stabilization is not such a package.

Theorem 8.18 (Eventually constant tower with compatible apex agreement). *Under Definition 8.17, the canonical map*

$$\lambda_N : M_N \xrightarrow{\sim} L$$

is an isomorphism. If the apex cocone and apex agreement are included, then

$$\phi : L \xrightarrow{\sim} A_\infty$$

is an isomorphism.

Proof. For $n \leq N$, let $b_n : M_n \rightarrow M_N$ be the forward composite. For $n > N$, the composite $M_n \rightarrow M_N$ is an isomorphism; let b_n be its inverse. These maps form a cocone over the sequence with vertex M_N . By the universal property of L , there is a unique $\psi : L \rightarrow M_N$ with $\psi \lambda_n = b_n$. In particular $\psi \lambda_N = \text{id}_{M_N}$. Both $\lambda_N \psi$ and id_L agree after composition with every λ_n , so colimit uniqueness gives $\lambda_N \psi = \text{id}_L$. Hence λ_N is an isomorphism. For the apex comparison, $a_N = \phi \lambda_N$. Since a_N and λ_N are isomorphisms, so is $\phi = a_N \lambda_N^{-1}$. $\square$

Corollary 8.19 (Vanishing of full comparison defects under reduction). *If Theorem 8.18 is applied in the abelian profile category used by Chapter 5, then*

$$\ker \phi = 0, \quad \text{coker } \phi = 0.$$

Consequently all degreewise terminal diagnostics vanish, every complete full or interior defect detector has zero value, and the comparison is a full isomorphism on the declared profile scope.

Proof. An isomorphism in an abelian category has zero kernel and cokernel. The Chapter 5 diagnostics and detector statements then follow from their definitions and complete characterizations. $\square$

Example 8.20 (Finite prefixes do not determine an infinite tower). Fix N . In Vect_k , let two directed systems agree with the zero object through stage N . Let the first remain zero forever. Let the second have $M_{N+1} = k$, map $0 \rightarrow k$, and use identity maps on k thereafter. The two systems have identical data through stage N , but their colimits are respectively 0 and k . Thus no finite prefix alone determines the infinite colimit, eventual constancy, apex agreement, or comparison defect.

8.6 Deprecated and rejected claims closed by the repair

Definition 8.21 (Metric-repair and architecture rejection set). The following are unavailable for v20 theorem use:

- (i) global non-expansiveness of hard positive-threshold deletion;
- (ii) a general positive-threshold Serre localization or exact reflector;
- (iii) $\text{gap}_\tau = \frac{1}{2} \text{clear}_\tau$ or a sharp band of half-width 2ε ;
- (iv) unconditional crop–collapse commutation, automatic filtered collapse, or automatic tower comparison morphisms;
- (v) detector completeness inferred from soundness, finite sampling, observed success, stage relabeling, or a circular/full-barcode source of the claimed compression bound;
- (vi) detector pointwise rank stability inferred from bottleneck stability;
- (vii) synthetic shifted detector duality promoted to a natural higher PH–Ext Bridge;
- (viii) the reverse higher bridge without an independently proved natural zero-reflection theorem;
- (ix) terminal zero interpreted as full comparison isomorphism or absence of transient/short defects;
- (x) fabricated Type IV zero in a no-tower scope;
- (xi) a universal Restart constant, ordinary summability under arbitrary contraction, or pointwise reserve promoted to uniform reserve;
- (xii) pairwise overlap promoted to higher coherence, local detector zero promoted to global zero without coverage, or record coherence promoted to object descent;
- (xiii) finite-prefix or sampled stabilization promoted to an infinite tower, colimit, continuum, or external theorem;
- (xiv) duplicated spending of one metric discrepancy through a runtime comparison and its derived zero-transport branch;
- (xv) an operational, specification, pending-repair, legacy-only, deprecated, rejected, or unknown item promoted to theorem evidence;
- (xvi) a Core pass, known-theorem calibration, toy theorem, Search output, or AI proposal promoted to an external conclusion without Return.

Proposition 8.22 (Rejected claims cannot be repaired locally by a ledger). *No rejected structural claim in Definition 8.21 becomes valid by assigning it a larger numerical error budget.*

Proof. The rejected claims concern false global assertions, missing structure, circular proof sources, invalid type changes, duplicated accounting, or absent logical hypotheses. A scalar metric bound constructs none of them. $\square$

8.7 The final Core closure contract

Definition 8.23 (v20 Core closure contract). The v20 Core is the smallest versioned theorem-and-verifier package containing:

- (i) the proved nucleus of Definition 8.9;
- (ii) the conditional layer only at discharged hypotheses;
- (iii) the operational verifier layer and canonical Chapter 7 manifest;
- (iv) Appendix A at its registered metric-repair strength;
- (v) finite-detector, higher-edge, defect, atlas, and reduction theorems only at their exact stages and information classes;
- (vi) explicitly imported micro-theorems only at registered strength, role, conformance, and hypotheses;
- (vii) no external-domain semantics without a separately proved directional transport or Return Theorem.

The package is closed under valid mathematical consequence and proof-preserving versioned reuse. It is not enlarged by analogy, implementation, registration, projection, numerical success, or rhetoric.

Proposition 8.24 (Main-body Core closure). *Within the declared v20.0 main-body scope, every Core verdict is determined by the final-audited mathematical inputs, typed artifacts, detector certificates, comparison modes, gate-scoped metric uses, evidence graph, canonical manifest, and Chapter 7 aggregation. No proposer-side object changes that verdict without completing typed re-entry.*

Proof. The mathematical inputs are fixed by the proved and conditional layers. The verdict calculus and manifest are fixed by Chapter 7. Core sovereignty excludes direct updates by proposer-side material. $\square$

Proposition 8.25 (Final-audit closure criterion). *A v20 main-body run is audit-closed exactly when every invoked clause is well-typed and has a determined pass or reject status, every exclusion is justified, every finite or countable dependency closure is complete, every conditional and non-scalar obligation is discharged, every metric use is nonduplicative and bounded, and the canonical manifest is complete, consistent, artifact-closed, and semantically replayable. A fully evidenced rejection is a closed result. A run with unresolved required evidence remains undefined.*

Proof. Under the listed hypotheses every atomic clause has a truth-valued status and the Chapter 7 aggregate is determined. Conversely, an unresolved required clause or incomplete dependency closure prevents a determined audit closure. $\square$

Definition 8.26 (Main-body and release acceptance boundary). *Main-body closure* means that Chapters 1–8 form a closed typed inference architecture. *Release acceptance* means that the Chapter 7 release registry for every mandatory Chapter 1 obligation passes. These statuses are independent: the main body may be closed while a mandatory external calibration, appendix registration, toy verification, or Return artifact is still pending.

Remark 8.27 (This is a closure contract, not a universal completeness theorem). Closure does not claim that the Core decides every mathematical question, possesses a finite detector on every scope, realizes every external invariant, or supplies every Return Theorem. It states that every result claimed by the release must traverse the declared typed route.

8.8 High-dimensional projection as an Interface mechanism

The repaired AK route is

$$\begin{aligned} \text{external object} &\mapsto \text{Core-readable shadow} \\ &\mapsto \text{finite or structural Core evidence} \\ &\mapsto \text{audited Core conclusion.} \end{aligned}$$

A further proved arrow is required to return to the external source semantics.

Definition 8.28 (External source class). An *external source class* is a declared category, typed family, or solution space outside the minimal one-parameter persistence Core. Its objects, morphisms, equivalences, and target predicates are part of the source semantics and are not Core objects by default.

Definition 8.29 (AK projection datum). An *AK projection datum* consists of a source object X , a typed route

$$\mathfrak{R}_{AK} : \mathcal{D} \longrightarrow \text{FiltCh}(()k) \quad \text{or} \quad \mathfrak{R}_{AK} : \mathcal{D} \longrightarrow \text{Pers}_k^{\text{cons}},$$

the projected object F_X , the declared windows, thresholds, degrees, coefficient field, detector source stages, approximation and comparison data, constructibility evidence, purpose-faithfulness target, and immutable artifacts. Every invoked detector, realization, tower, atlas, reduction, and Return statement is separately identified with its direction, class, role, and hypotheses.

Definition 8.30 (Core-readable projected object). A projected object F_X is *Core-readable* on its declared scope when the window-first pre-threshold and repaired profiles are defined, every invoked detector has its exact stage and certificate, every comparison has a valid mode and nonduplicative metric-use identity when needed, every representative, realization, Type IV, overlap, Restart, reduction, and claim obligation is typed, every countable dependency registry is complete, and the Chapter 7 verifier can evaluate the canonical manifest.

Proposition 8.31 (Projection supplies input, not verdict). An *AK projection datum* determines neither a Core verdict nor an external-domain theorem by itself.

Proof. A Core verdict additionally requires all theorem hypotheses, artifacts, and audit clauses. An external theorem additionally requires a proved Return statement in the source semantics. $\square$

Remark 8.32 (Projection is structural). Projection may be geometric, arithmetic, analytic, categorical, spectral, computational, tower-theoretic, or hybrid. Its admissibility depends on exact typing and proof, not on conceptual elegance or numerical success.

8.9 Purpose-faithfulness and Interface entry

Definition 8.33 (Purpose-faithful realization). A route is *purpose-faithful* for a declared external claim only when it identifies the source feature, the Core-visible feature intended to represent it, the direction of use, and the theorem or controlled mismatch hypotheses connecting them. Purpose-faithfulness is relative to that claim and is not categorical faithfulness unless separately proved.

Definition 8.34 (Interface entry record). An *Interface entry record* contains the source object and semantics, projection datum and declared purpose, Core-readable target type, constructibility, filtration orientation, windows, thresholds, degrees, fixed evaluation order, every detector realization and completeness source, comparison and metric-use records, representative and higher-realization data, Type IV scope, tower and defect data, atlas and Restart data, finite-to-infinite package when invoked, complete finite or countable evidence graph, canonical manifest, immutable run identity, and the class, conformance, role, direction, and hypotheses of every transport statement.

Proposition 8.35 (No admissible Interface without typed entry). *An external object has no theorem-bearing access to the Core until its Interface entry record is complete and consistent and every invoked entry clause is typed.*

Proof. Without a complete typed entry, the verifier cannot reconstruct the Core input. Missing or unresolved entry evidence is undefined; evidenced false or inconsistent entry claims are reject. Neither is pass. $\square$

Remark 8.36 (Interface failure is not automatically Core rejection). A proposed route may be nonconstructible, not purpose-faithful, insufficiently evidenced, or ill-typed. This is an Interface failure or undefined entry. It is a Core rejection only when a properly invoked Core clause is well-typed and evidenced false or the canonical manifest is inconsistent.

8.10 Detector, higher-obstruction, and reduction realization interfaces

Definition 8.37 (Detector realization theorem interface). A detector realization theorem for an external source class identifies a finite external observable object Q_X and proves a natural isomorphism

$$Q_X \xrightarrow{\sim} D_{T,\tau}^{\text{stc}}(\text{Eval}_{n,w,\tau}(F_X))$$

or the corresponding birth, germ, family-mesh, kernel-defect, or cokernel-defect detector. It records the exact detector stage, endpoint convention, stencil or family, source of completeness, noncircular excess provenance, and preservation under the chosen realization. A numerical agreement without this theorem is only calibration.

Definition 8.38 (Detector soundness Bridge). A detector soundness Bridge is a proved directional theorem from a certified nonzero detector witness to a precisely stated external obstruction or failure predicate. Soundness consumes a well-typed nonzero witness and does not require detector completeness unless the external conclusion itself uses a zero equivalence.

Definition 8.39 (Detector completeness Bridge). A detector completeness Bridge is a proved theorem establishing the direction needed for an external zero conclusion, for example

$$D(P_X) = 0 \implies P_X = 0 \implies \mathcal{E}(X),$$

with the first implication supplied by an admitted complete detector and the second by a Return theorem. Sampling success, soundness, or full-barcode readout disguised as finite compression is insufficient.

Definition 8.40 (Higher Ext realization theorem interface). A higher Ext realization theorem identifies an independently meaningful external or geometric object $\mathcal{H}_n(X)$ with the natural negative cohomology edge or Ext obstruction of the eligible Core realization, through natural artifact-backed isomorphisms. Re-encoding a detector output into a shifted synthetic object may support computation but does not satisfy this principal realization interface by itself.

Definition 8.41 (Finite-to-infinite Interface record). Whenever a finite certificate is used for an infinite tower, continuum, unbounded window, or limit object, the Interface record identifies the exact finite witness, the registered reduction theorem, actual directed-system maps, colimit semantics, compatible apex cocone, apex agreement, and induced comparison. Theorem 8.18 is one admissible reduction route when its hypotheses are met; it is not a universal substitute for problem-specific compactness, convergence, or continuation.

8.11 The repaired evaluation path for external data

Definition 8.42 (Interface evaluation path). Every Core-relevant Interface run uses

$$\begin{aligned}
X &\mapsto F_X = \mathfrak{R}_{AK}(X) \\
&\mapsto \mathbf{P}_n(F_X) \\
&\mapsto B_{n,W}(F_X) = W_W \mathbf{P}_n(F_X) \\
&\mapsto \text{Eval}_{n,W,\tau}(F_X) = T_\tau^{\text{bar}}(B_{n,W}(F_X)) \\
&\mapsto \text{direct proof or admitted detector evidence} \\
&\mapsto \text{typed Core clauses and audit verdict.}
\end{aligned}$$

If the target is infinite, the declared reduction step follows the finite Core evidence. If the target is external, a Return Theorem follows the audited Core conclusion. No raw observable replaces a profile, detector, reduction, or Return arrow without a proved typed identification.

Proposition 8.43 (External origin does not relax Core hypotheses). *A projected object is subject to the same constructibility, detector, comparison, representative, eligibility, Type IV, atlas, Restart, dependency, claim, and manifest conditions as an internally specified Core object.*

Proof. Core theorems and verifier clauses are typed by inputs and hypotheses, not by data origin or intended application. $\square$

Remark 8.44 (Approximation must be visible). Sampling, truncation, discretization, smoothing, numerical solution, normalization, compression, learned representation, and adaptive detector selection must be explicit. Metric effects enter one identified ledger use only when scalarizable; semantic, naturality, completeness, morphism, and Return obligations remain non-scalar.

8.12 Distinct conclusion levels

Definition 8.45 (Interface outcome). The *Interface outcome* states whether the source was converted into a complete, consistent, Core-readable entry. Its statuses are *pass*, *reject*, and *undefined* under the declared entry contract.

Definition 8.46 (Core outcome for projected data). The *Core outcome for projected data* is the Chapter 7 verdict about the repaired projected data under the declared scope. It may concern record coherence, profile zero, higher Ext vanishing, terminal/transient defects, or another exactly typed Core predicate.

Definition 8.47 (Bridge outcome). A *Bridge outcome* is the registered status of a directional realization, soundness, completeness, higher-obstruction, reduction, or transport statement. Specification, toy theorem, conditional theorem, and proved transport are not interchangeable.

Definition 8.48 (External-domain outcome). An *external-domain outcome* is a theorem or refutation stated entirely in the original semantics of X .

Definition 8.49 (Finite-certificate outcome). A finite-certificate outcome records whether the finite detector or proof compression is applicable, certified, and sufficient for its exact Core predicate. It is not an infinite, descended, or external conclusion.

Definition 8.50 (Reduction and Return outcomes). A *reduction outcome* records whether a finite-to-infinite theorem applies to the declared limit target. A *Return outcome* records whether a proved Core-to-source theorem applies to the exact audited Core conclusion. They are separate clauses even when both are required by one external claim.

Proposition 8.51 (No outcome-layer collapse). *None of the following implications is available without the named additional clause:*

$$\begin{aligned} \text{Interface pass} &\Rightarrow \text{Core pass}, \\ \text{finite detector pass} &\Rightarrow \text{infinite conclusion}, \\ \text{record coherence} &\Rightarrow \text{global zero}, \\ \text{global zero} &\Rightarrow \text{object descent}, \\ \text{Core pass} &\Rightarrow \text{external theorem}. \end{aligned}$$

The missing clauses are respectively the Core verifier, finite-to-infinite reduction, retained-bar coverage, effective descent, and Return Theorem.

Proof. The antecedents and consequents inhabit different typed conclusion levels. Each omitted implication requires mathematical content not contained in its antecedent. $\square$

8.13 Transport theorems, Return, and Bridge promotion

Definition 8.52 (Interface-to-Core transport statement). An *Interface-to-Core transport statement* has direction

$$\text{external-domain structure} \Longrightarrow \text{specified Core-visible behavior}.$$

Definition 8.53 (Core-to-Interface transport statement). A *Core-to-Interface transport statement* has direction

$$\text{specified Core-visible behavior} \Longrightarrow \text{external-domain conclusion}.$$

Definition 8.54 (Transport theorem). A *transport theorem* is a proved directional statement with precise source and target semantics, purpose-faithful realization, Core-readable target, all detector and representative hypotheses, every comparison and metric-use record, all tower/atlas/reduction data, complete manifest, proof of the stated direction, and an explicit converse and non-claim boundary.

Definition 8.55 (External Return Theorem). Let $C_{AK}(X)$ be the exact Core conclusion produced for the projected object and let $\mathcal{E}(X)$ be an external-domain predicate. A *Return Theorem* is a proved theorem

$$C_{AK}(X) \Longrightarrow \mathcal{E}(X)$$

on a declared source class, together with every realization, preservation, finite-to-infinite, descent, and mismatch hypothesis needed by the proof. A bidirectional external characterization requires a separately proved reverse direction.

Definition 8.56 (Default status of transport statements). Every problem-specific realization, Bridge, reduction application, and Return statement is `spec` with role `specification_target` by default. It becomes theorem evidence only through explicit proof, conforming registration, artifact closure, and Chapter 7 verification. The two transport directions are independent.

Proposition 8.57 (No silent Bridge promotion). *A natural-looking projection, successful calibration, stable experiment, finite detector, Core pass, toy bridge, Search result, or AI-generated proof sketch does not silently promote a transport specification to theorem status.*

Proof. None supplies all source semantics, hypotheses, naturality, preservation, reduction, manifest, and proof obligations of Definitions 8.54 and 8.55. $\square$

Theorem 8.58 (Composed AK Return principle). *Let X be an external object. Assume:*

- (i) a proved typed Interface entry produces a Core-readable F_X ;
- (ii) the required Core theorem or certified detector establishes the exact Core predicate $C_{\text{fin}}(X)$;
- (iii) if the target is infinite, a proved reduction theorem establishes $C_{\text{fin}}(X) \Rightarrow C_{AK}(X)$, while otherwise $C_{AK} = C_{\text{fin}}$;
- (iv) a proved Return Theorem establishes $C_{AK}(X) \Rightarrow \mathcal{E}(X)$.

Then $\mathcal{E}(X)$ holds.

Proof. Compose the proved implications in their declared direction. The typed entry ensures that each intermediate predicate has the source and target required by the next implication. $\square$

Remark 8.59 (External Ext is not automatically the Core Ext). An arithmetic, sheaf-theoretic, deformation-theoretic, or categorical Ext group is not automatically $\text{Ext}^n(A_{n;F_X,W,\tau}, k)$. Any identification is itself higher-realization and Return mathematics and must be natural, directional, and proved.

8.14 Micro-theorems, toy bridges, Search, and Platform material

Definition 8.60 (Permitted adjacent support). Appendices and systems may provide registered micro-theorems, finite detector examples, endpoint/cofinal/mesh/metric counterexamples, toy bridge theorems, known-theorem-recovery calibrations, Interface definitions, Search proposals, Platform logs, proof-assistant artifacts, or Companion templates. Such material supports the main body only at its exact class, conformance, role, hypotheses, and transformation history.

Proposition 8.61 (Adjacent material does not enlarge the Core silently). *No appendix, Companion, Search report, Platform artifact, micro-theorem, toy theorem, or Interface schema enlarges the v20 Core merely by appearing in the project.*

Proof. Import requires a compatible theorem statement, proof, registered strength, discharged hypotheses, artifact closure, and all relevant verifier clauses. $\square$

Remark 8.62 (NBT release boundary). NBT remains a successor-release candidate program unless a later version proves and registers the required theorem package. Its possible control of apex-only Type IV cokernel modes does not strengthen v20.0.

8.15 Known-theorem recovery and release demonstration

Definition 8.63 (Known-theorem-recovery record). A *known-theorem-recovery record* selects an external theorem already proved independently on a benchmark class and reconstructs its conclusion through a complete AK Interface, Core, reduction, and Return chain. It records where the independent external proof is used only for validation and where the AK chain supplies its own proof obligations.

Proposition 8.64 (Calibration is not promotion). *Successful known-theorem recovery validates the advertised Interface and Return chain on the benchmark class but does not by itself prove a stronger external theorem, extend the source class, or discharge another release gate.*

Proof. The recovery theorem is scoped to its registered class and hypotheses. A stronger conclusion requires a stronger independently proved transport or Return theorem. $\square$

Definition 8.65 (v20 release-closure record). The final publication package contains one Chapter 7 release-acceptance record covering all mandatory Chapter 1 gates. In particular it separately records the finite detector theorems, higher Ext-edge and one-way discharge, any advertised natural zero-reflection instance, transient-defect results, finite-to-infinite theorem, CM/TB/counterexample and manifest synchronization, known-theorem recovery, and every advertised external realization and Return claim. Optional proof-assistant and compression comparisons remain optional unless promoted by policy.

Proposition 8.66 (Main closure does not predeclare release pass). *The completion of Chapters 1–8 does not by itself set the v20 release headline to pass. The headline is computed from the actual release registry and remains undefined whenever a mandatory external, appendix, registration, or artifact obligation is pending.*

Proof. Chapter 7 defines release acceptance as a conjunction of independently mandatory obligations. Main-body closure is only one of those obligations. □

Definition 8.67 (v20 release-gate reflection matrix). The final release registry reflects the Chapter 1 gates without changing their strength. The mandatory Core-mathematics block is:

- G1 birth-detector characterization;
- G2 germ or decorated-endpoint characterization;
- G3 certified-stencil and family-mesh completeness;
- G4 detector sharpness and boundary counterexamples;
- G5 right-unbounded finite-detector no-go;
- G6 metric zero-verdict composition;
- G7 family-relative minimal-certificate results;
- G8 canonical higher Ext-edge;
- G9 higher one-way repaired discharge;
- G10 an advertised non-synthetic natural zero-reflection instance on its declared class;
- G11 transient Type IV kernel/cokernel defect detection.

The mandatory verification/governance block is:

- G12 CM-v20 Core Micro-Theorem registration at exact theorem strength;
- G13 TB-v20 Toy Bridge verification;
- G14 endpoint, cofinal, mesh, metric, higher-edge, and transient-defect counterexamples;
- G15 manifest, token-sort, claim, and non-scalar-obligation consistency.

The mandatory external-demonstration block is:

- G16 at least one known-theorem recovery in the declared Navier–Stokes or Iwasawa route, at the exact recovered strength and with a complete Interface–Core–Reduction–Return chain;

G17 at least one proved natural realization or finite-to-infinite reduction theorem. Theorem 8.18 supplies the internal reduction option, but any advertised external application still requires its own entry and Return record.

The strongly recommended block is:

G18 partial proof-assistant formalization;

G19 computational demonstration of certificate reduction;

G20 comparison with full-barcode decision procedures.

Each gate remains an independently identified Chapter 7 release clause.

Proposition 8.68 (Mandatory and recommended release gates do not compensate). *A pass on one mandatory gate does not compensate for another mandatory gate that is rejected or undefined. Gates G18–G20 do not alter the headline unless a later release policy promotes them to mandatory status.*

Proof. The release headline is the typed conjunction defined in Chapter 7 and the Chapter 1 release contract. Optional recommendations are outside that mandatory conjunction at v20.0. $\square$

8.16 Explicit non-claims of the closed main body

The closed v20.0 main body asserts none of the following.

- (i) Global non-expansiveness of hard threshold deletion.
- (ii) Positive-threshold Serre localization, a global exact reflector, or a universal quotient formula.
- (iii) Sharpness of a band of half-width 2ε , or $\text{gap}_\tau = \frac{1}{2} \text{clear}_\tau$.
- (iv) Unconditional crop–collapse commutation or a global strict filtered collapse functor.
- (v) Automatic derived or Ext equivalence of profile representatives.
- (vi) Detector completeness from soundness, observed sampling, an unverified birth list, stage relabeling, or a circular/full-barcode source of the claimed finite proof.
- (vii) Pointwise detector-rank stability from bottleneck stability.
- (viii) A uniform complete finite detector on an unrestricted right-unbounded family without an escape theorem.
- (ix) A natural higher PH–Ext Bridge produced merely by shifting detector output into a synthetic derived object.
- (x) The reverse higher Bridge or global PH–Ext equivalence without independently proved natural zero reflection.
- (xi) An automatic tower comparison morphism from barcodes, sampled layers, or bottleneck matchings.
- (xii) Missing Type IV evidence interpreted as $(\mu_{\text{Collapse}}, \nu_{\text{Collapse}}) = (0, 0)$, or no-tower scope represented by fabricated zeros.
- (xiii) Terminal zero interpreted as full defect-profile vanishing, absence of transient defects, or full comparison isomorphism.

- (xiv) A universal $\kappa > 0$, default $\kappa = 1$, ordinary summability under contraction, or pointwise reserve promoted to uniform reserve.
- (xv) Pairwise overlap as higher Čech coherence, metric reference as an exact cocycle, or summability as a substitute for face maps.
- (xvi) Local detector zero as global zero without retained-bar coverage.
- (xvii) A coherent record atlas as a global filtered object or effective descent theorem.
- (xviii) A finite prefix, finite detector, numerical stabilization, or eventually constant appearance as an infinite theorem without a registered reduction package.
- (xix) A scalarization value as a certificate without an actual upper-bound proof, or duplicated spending of one metric use through a derived zero-transport branch.
- (xx) A numerical budget repairing a missing morphism, naturality, realization, eligibility, completeness, coherence, rank witness, status, role, or manifest field.
- (xxi) `not_compared` discharging a comparison-dependent clause, or `undefined` identified with `reject`, `pass`, `zero`, or `not_invoked`.
- (xxii) Claim status, conformance, or theorem strength created by registration, prose, implementation, Search, or AI output.
- (xxiii) A detector realization, higher Ext realization, finite-to-infinite theorem, or Return Theorem inferred from another one of those clauses.
- (xxiv) A Core pass, toy bridge, known-theorem calibration, or natural-looking projection proving an external theorem by itself.
- (xxv) A predeclared v20 complete-release pass before every mandatory release obligation is actually registered and verified.
- (xxvi) v20.0 theorem status for NBT or any other successor-release candidate.
- (xxvii) Completeness of AK theory for all mathematical problems, all realizations, all unbounded scopes, or all external source categories.

Remark 8.69 (Non-claims are part of the result). These exclusions are not rhetorical caution. They are typed boundaries that identify the exact theorem, artifact, or later version required to cross each remaining gap. The theory is defined by the routes it proves and by the shortcuts it makes impossible to report as proof.

8.17 Final architecture theorem

Proposition 8.70 (Final architecture of the v20 main body). *There are two distinct routes. A candidate route may begin with a specified Interface*

$$X \xrightarrow{\text{specified Interface}} F_X,$$

but produces only a candidate Core input. A theorem-bearing route has the form

$$\begin{array}{lcl}
X & \xrightarrow{\text{proved typed Interface}} & F_X = \mathfrak{R}_{AK}(X) \\
& \xrightarrow{\text{window-first persistence}} & B_{n,W}(F_X) \\
& \xrightarrow{\text{hard threshold deletion}} & \text{Eval}_{n,W,\tau}(F_X) \\
& \xrightarrow{\text{direct theorem or admitted finite certificate}} & C_{\text{fin}}(X) \\
& \xrightarrow{\text{proved reduction, when required}} & C_{AK}(X) \\
& \xrightarrow{\text{typed audit, evidence graph, and manifest}} & \text{Core verdict} \\
& \xrightarrow{\text{separately proved Return Theorem}} & \mathcal{E}(X).
\end{array}$$

Every arrow has declared source and target, direction, hypotheses, theorem class, role, artifacts, and failure boundary. No arrow may be omitted, reversed, merged, or replaced by numerical plausibility.

Proof. The first arrow is governed by purpose-faithful Interface entry. The next two are the repaired evaluation path. The finite evidence arrow is supplied by a Core theorem or admitted detector. Infinite targets require a reduction such as Theorem 8.18. Chapter 7 governs the verdict. The final external implication is exactly the additional content of a Return Theorem. $\square$

Theorem 8.71 (Final AK projection-and-return principle). *An external AK theorem is licensed precisely by a complete theorem-bearing chain of the form in Proposition 8.70. When every invoked arrow is proved and the final Core verdict passes, the declared external conclusion follows. If any required Interface, finite-certificate, reduction, audit, or Return arrow is absent or undefined, no external conclusion is licensed by the AK chain.*

Proof. The positive statement is Theorem 8.58 combined with the Chapter 7 pass semantics. The negative statement follows from the typed conjunction and no-outcome-collapse rule: a missing required arrow leaves the corresponding clause undefined and prevents a theorem-bearing composition. $\square$

Remark 8.72 (Final AK projection principle).

Projection reveals a structured shadow.
Finite certificates may compress its proof.
Higher Ext records a natural obstruction only through realization.
Defect calculus separates terminal failure from transient residue.
Atlases coordinate local evidence, and reduction alone crosses the finite–infinite boundary.
Only Return carries the certified statement home.

The asymmetry is deliberate. AK does not identify a shadow with its source; it makes the exact theorem needed for that identification visible.

Remark 8.73 (Closure of the main body). With this chapter, the AK-HPDST v20.0 main-body architecture is closed. The metric nucleus is sharp where proved; finite evidence is complete only where certified; higher obstruction is natural only where realized; terminal and transient defects remain separated; local evidence becomes global only through coverage; finite data become infinite only through reduction; and Core truth returns to an external domain only through a proved Return Theorem.

The theory therefore closes not by claiming that every projection is faithful, but by identifying, typing, and auditing every place where faithfulness must be proved. Future versions enlarge this closure only by supplying new proofs, new natural realizations, stronger reduction theorems, verified external Returns, and conforming immutable artifacts.

8.18 Provenance and status

This chapter consumes the final-audited Chapters 1–7 and Appendix A. It restates their mathematical results only at their registered strengths, proves the internal eventually-constant reduction theorem, and fixes the final Interface/Return boundary. The v20 Claim Register must register every item at the same or weaker strength and compatible role.

Release-level and closure definitions. Definitions [8.4](#), [8.5](#), [8.6](#), [8.7](#), [8.9](#), [8.12](#), [8.13](#), [8.14](#), [8.17](#), [8.21](#), [8.23](#), and [8.26](#).

Interface, realization, and Return definitions. Definitions [8.28](#), [8.29](#), [8.30](#), [8.33](#), [8.34](#), [8.37](#), [8.38](#), [8.39](#), [8.40](#), [8.41](#), [8.42](#), [8.45](#), [8.46](#), [8.47](#), [8.48](#), [8.49](#), [8.50](#), [8.52](#), [8.53](#), [8.54](#), [8.55](#), [8.56](#), [8.60](#), [8.63](#), [8.65](#), and [8.67](#).

Mathematical theorems and closure consequences. Propositions [8.8](#), [8.10](#), [8.15](#), [8.22](#), [8.24](#), [8.25](#), [8.31](#), [8.35](#), [8.43](#), [8.51](#), [8.57](#), [8.61](#), [8.64](#), [8.66](#), [8.68](#), and [8.70](#); Theorems [8.18](#), [8.58](#), and [8.71](#); Corollary [8.19](#); and Example [8.20](#).

Boundary and final-closure remarks. Remarks [8.1](#), [8.2](#), [8.3](#), [8.11](#), [8.16](#), [8.27](#), [8.32](#), [8.36](#), [8.44](#), [8.59](#), [8.62](#), [8.69](#), [8.72](#), and [8.73](#).

Explicit non-claims. Section [8.16](#).

Main-body closure. Proposition [8.24](#), Proposition [8.25](#), Proposition [8.70](#), and Theorem [8.71](#).